Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

3110 FOUNDATION DESIGN OF CONVEYOR 3110-CNV-002

DESIGN CALCULATION

AS BUILT

MD-216-3110-EG-CV-CAL-1002

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REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2007	Saudi Code for Concrete Structures

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
RL	:	Reduced Level
C.L.	:	Centre Line
CJ	:	Construction Joint
FDN	:	Foundation
TOC	:	Top of concrete
FGL	:	Finished Ground Level
FFL	:	Finished Floor Level
BOF	:	Bottom of footing
TYP	:	Typical
TOC	:	Top of concrete
IP	:	Insert Plate
RCC	:	Reinforced Cement Concrete
MKD.	:	Marked
THK.	:	Thick

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1. INTENT

The document is prepared for Concrete design calculation of Conveyor 3110-CNV-002.
This conveyor is between Secondary Crusher House and Screen building. Belt width of this conveyor is 1600 mm.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for foundation design of Conveyor 3110-CNV-002. All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
The column and beam reinforcement design have been performed in Staad-Pro Software. The foundation of Ground Portal GP-1 to GR-3; Trestle mkd. TR3, TR-4 & VGTU have been performed in MS Excel. Foundation of TR-1A, TR-1 & TR-2 has been designed by FEM analysis in Staad-Pro Software.

4. CONCRETE STRUCTURE

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundation for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 518 kN/m2 & 554 kN/m2 at 1m & 3m depth below FGL(ie. RL 911.500m & RL 909.500m respectively). Refer Table 4 - BH-17,18,19,25,29 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

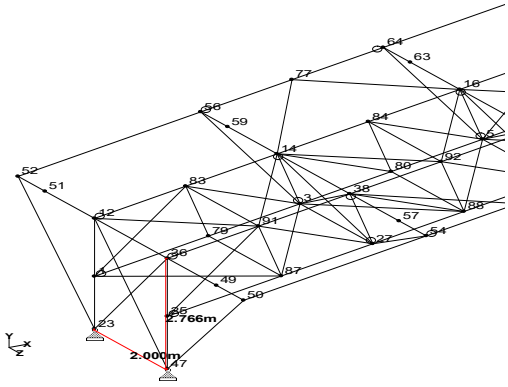
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 INPUT LOADS FROM SUPERSTRUCTURE

For load calculation of superstructure refer Document No: MD-216-3110-EG-ST-CAL-1002-RA01

5.1 Support Reaction from TR-1A

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
23	1 DL	0.0	61.3	-5.7
	2 LL	0.0	36.9	-1.3
	3 ML - CONVEYOR MATERIAL LOAD	0.0	42.1	-0.6
	4 WL (+Z)	0.0	-35.2	-12.6
	5 WL (-Z)	0.0	35.2	12.7
	6 TL1	0.0	0.2	5.9
	7 TL2	0.0	-0.2	-5.9
	8 EARTHQUAKE LOAD (+X)	-0.9	-0.4	0.0
	9 EARTHQUAKE LOAD (+Z)	0.0	-1.1	-0.4
47	1 DL	0.0	82.3	1.7
	2 LL	0.0	37.1	1.3
	3 ML - CONVEYOR MATERIAL LOAD	0.0	42.3	0.6
	4 WL (+Z)	0.0	35.2	-12.1
	5 WL (-Z)	0.0	-35.2	12.1
	6 TL1	0.0	-0.2	-5.9
	7 TL2	0.0	0.2	5.9
	8 EARTHQUAKE LOAD (+X)	-0.9	-0.4	0.0
	9 EARTHQUAKE LOAD (+Z)	0.0	1.1	-0.4



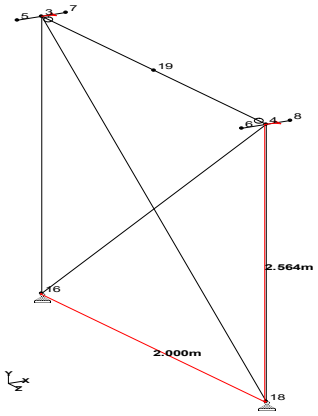
Staad model of 12m Gallery mkd. GR-1 End portal which is acting as TR1A

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5.2 Support Reaction from TR-1

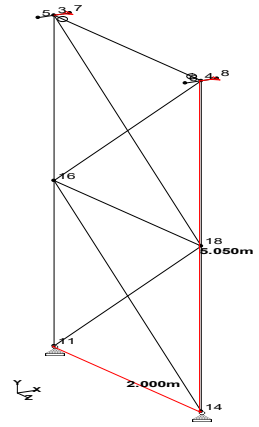
Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
16	1 DEAD LOAD	0.0	99.6	11.6
	2 LIVE LOAD	0.0	48.1	5.1
	3 MATERIAL LOAD	0.0	49.2	4.7
	4 WL (+Z)	-0.1	-110.9	-26.9
	5 WL (-Z)	0.1	110.9	26.9
	6 WL (+X)	-0.8	0.0	0.0
	7 WL (-X)	0.8	0.0	0.0
	8 TL1	0.0	-0.1	9.8
	9 TL2	0.0	0.1	-9.8
	10 EL (+X)	0.0	0.0	0.0
	11 EL (+Z)	0.0	-9.2	-3.0
18	1 DEAD LOAD	0.0	155.2	-19.6
	2 LIVE LOAD	0.0	47.9	-5.1
	3 MATERIAL LOAD	0.0	49.2	-4.7
	4 WL (+Z)	0.1	110.9	-27.4
	5 WL (-Z)	-0.1	-110.9	27.4
	6 WL (+X)	-0.8	0.0	0.0
	7 WL (-X)	0.8	0.0	0.0
	8 TL1	0.0	0.1	-9.8
	9 TL2	0.0	-0.1	9.8
	10 EL (+X)	0.0	0.1	0.0
	11 EL (+Z)	0.0	9.2	-2.9



Staad model of TR1 superstructure (along with dimension)

5.3 Support Reaction from TR-2

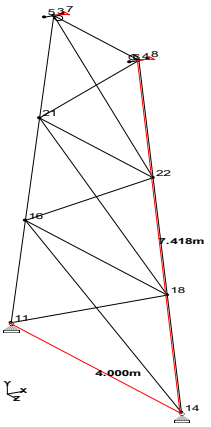
Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
11	1 DEAD LOAD	0.0	92.9	2.1
	2 LIVE LOAD	0.0	48.1	2.3
	3 MATERIAL LOAD	0.0	49.2	2.5
	4 WL (+Z)	-0.1	-180.9	-29.2
	5 WL (-Z)	0.1	180.9	29.2
	6 WL (+X)	-1.5	0.0	0.0
	7 WL (-X)	1.5	0.0	0.0
	8 TL1	0.0	-0.1	9.4
	9 TL2	0.0	0.1	-9.4
	10 EL (+X)	0.0	0.0	0.0
	11 EL (+Z)	0.0	-6.4	-1.0
14	1 DEAD LOAD	0.0	168.4	-10.1
	2 LIVE LOAD	0.0	47.9	-2.3
	3 MATERIAL LOAD	0.0	49.2	-2.5
	4 WL (+Z)	0.1	180.9	-29.1
	5 WL (-Z)	-0.1	-180.9	29.1
	6 WL (+X)	-1.5	0.0	0.0
	7 WL (-X)	1.5	0.0	0.0
	8 TL1	0.0	0.1	-9.4
	9 TL2	0.0	-0.1	9.4
	10 EL (+X)	0.0	0.1	0.0
	11 EL (+Z)	0.0	6.4	-0.9



Staad model of TR2 superstructure (along with dimension)

5.4 Support Reaction from TR-3

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
11	1 DEAD LOAD	0.0	109.3	16.7
	2 LIVE LOAD	0.0	48.1	7.6
	3 MATERIAL LOAD	0.0	49.2	7.8
	4 WL (+Z)	0.0	-125.1	-31.1
	5 WL (-Z)	0.0	125.1	31.1
	6 WL (+X)	-2.2	0.0	0.0
	7 WL (-X)	2.2	0.0	0.0
	8 TL1	0.0	0.0	18.9
	9 TL2	0.0	0.0	-18.9
	10 EL (+X)	0.0	0.0	0.0
	11 EL (+Z)	0.0	-8.7	-2.2
14	1 DEAD LOAD	0.0	156.3	-24.7
	2 LIVE LOAD	0.0	48.0	-7.6
	3 MATERIAL LOAD	0.0	49.2	-7.8
	4 WL (+Z)	0.0	125.1	-31.0
	5 WL (-Z)	0.0	-125.1	31.0
	6 WL (+X)	-2.2	0.0	0.0
	7 WL (-X)	2.2	0.0	0.0
	8 TL1	0.0	0.0	-18.9
	9 TL2	0.0	0.0	18.9
	10 EL (+X)	0.0	0.1	0.0
	11 EL (+Z)	0.0	8.7	-2.2



Staad model of TR3 superstructure (along with dimension)

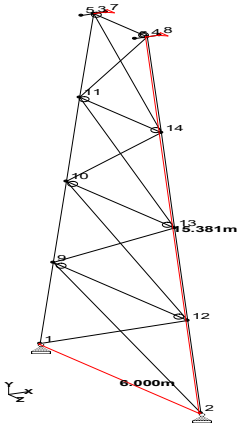
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5.5 Support Reaction from TR-4

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
1	1 DEAD LOAD	0.0	166.0	31.1
	2 LIVE LOAD	0.0	71.7	13.0
	3 MATERIAL LOAD	0.0	69.5	12.6
	4 WL (+Z)	0.0	-184.1	-24.7
	5 WL (-Z)	0.0	246.6	49.3
	6 WL (+X)	-4.6	0.0	0.0
	7 WL (-X)	4.6	0.0	0.0
	8 TL1	0.0	0.0	58.1
	9 TL2	0.0	0.0	-58.1
	10 EL (+X)	0.0	0.0	0.0
	11 EL (+Z)	0.0	-11.8	-2.2
2	1 DEAD LOAD	0.0	241.7	-42.3
	2 LIVE LOAD	0.0	72.3	-13.1
	3 MATERIAL LOAD	0.0	70.1	-12.7
	4 WL (+Z)	0.0	184.1	-24.7
	5 WL (-Z)	0.0	-246.6	49.3
	6 WL (+X)	-4.6	0.0	0.0
	7 WL (-X)	4.6	0.0	0.0
	8 TL1	0.0	0.0	-58.1
	9 TL2	0.0	0.0	58.1
	10 EL (+X)	0.0	0.0	0.0
	11 EL (+Z)	0.0	11.8	-2.2

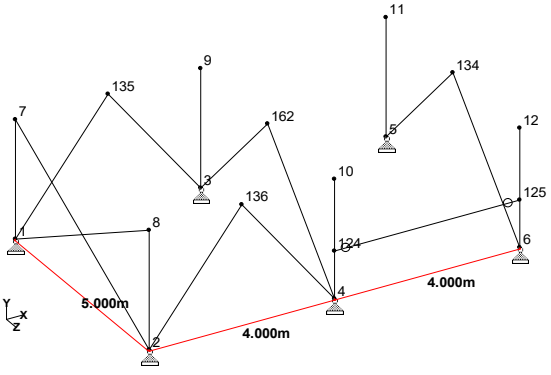


Staad model of TR4 superstructure (along with dimension)

5.6 Support Reaction from VGTU

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
1	1 DEAD LOAD	25.9	275.4	12.3
	2 LIVE LOAD	0.6	162.6	9.6
	3 ML - CONVEYOR MATERIAL LOAD	0.0	23.4	1.4
	4 WIND LOAD IN +Z DIRN	-0.3	-93.0	-25.2
	5 WIND LOAD IN -Z DIRN	0.3	92.9	25.2
	6 WIND LOAD IN +X DIRN	-51.9	-224.7	-8.7
	7 WIND LOAD IN -X DIRN	51.9	224.7	8.7
	8 TL1 (TEMP LOAD +25)	1.4	3.0	38.3
	9 TL2 (TEMP LOAD -25)	-1.4	-3.0	-38.3
	10 EL (+X)	-1.5	-7.2	-0.3
	11 EL (+Z)	0.1	-6.4	-1.6
2	1 DEAD LOAD	26.2	295.6	-16.2
	2 LIVE LOAD	0.6	157.4	-8.8
	3 ML - CONVEYOR MATERIAL LOAD	0.1	23.2	-1.3
	4 WIND LOAD IN +Z DIRN	0.3	93.0	-25.4
	5 WIND LOAD IN -Z DIRN	-0.3	-92.9	25.4
	6 WIND LOAD IN +X DIRN	-51.8	-225.9	9.1
	7 WIND LOAD IN -X DIRN	51.8	225.9	-9.1
	8 TL1 (TEMP LOAD +25)	1.1	-3.0	-37.4
	9 TL2 (TEMP LOAD -25)	-1.1	3.0	37.4
	10 EL (+X)	-1.6	-7.6	0.3
	11 EL (+Z)	-0.1	6.4	-1.6
3	1 DEAD LOAD	23.9	72.8	-1.3
	2 LIVE LOAD	-0.6	216.2	0.4
	3 ML - CONVEYOR MATERIAL LOAD	-0.1	35.8	-0.1
	4 WIND LOAD IN +Z DIRN	0.1	-107.0	-26.2
	5 WIND LOAD IN -Z DIRN	-0.1	106.9	26.2
	6 WIND LOAD IN +X DIRN	-51.9	224.0	-0.1
	7 WIND LOAD IN -X DIRN	51.9	-224.0	0.1
	8 TL1 (TEMP LOAD +25)	-1.0	-4.1	-0.3
	9 TL2 (TEMP LOAD -25)	1.0	4.1	0.3
	10 EL (+X)	-1.5	7.4	0.0
	11 EL (+Z)	0.1	-6.2	-1.3
4	1 DEAD LOAD	24.1	105.1	-4.2
	2 LIVE LOAD	-0.6	220.6	-2.0
	3 ML - CONVEYOR MATERIAL LOAD	0.0	36.6	-0.2
	4 WIND LOAD IN +Z DIRN	-0.1	107.0	-26.2
	5 WIND LOAD IN -Z DIRN	0.1	-106.9	26.2
	6 WIND LOAD IN +X DIRN	-53.2	226.6	-0.6
	7 WIND LOAD IN -X DIRN	53.2	-226.6	0.6
	8 TL1 (TEMP LOAD +25)	-0.9	4.1	-1.3
	9 TL2 (TEMP LOAD -25)	0.9	-4.1	1.3
	10 EL (+X)	-1.6	7.6	0.0
	11 EL (+Z)	-0.1	6.2	-1.3

L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN	
1 DEAD LOAD	0.0	17.1	0.3	
2 LIVE LOAD	0.0	33.7	0.4	
3 ML - CONVEYOR MATERIAL LOAD	0.0	0.2	0.1	
4 WIND LOAD IN +Z DIRN	0.0	-22.2	-10.5	
5 WIND LOAD IN -Z DIRN	0.0	22.2	10.5	
6 WIND LOAD IN +X DIRN	-1.2	0.7	0.2	5
7 WIND LOAD IN -X DIRN	1.2	-0.7	-0.2	
8 TL1 (TEMP LOAD +25)	-0.1	1.1	0.8	
9 TL2 (TEMP LOAD -25)	0.1	-1.1	-0.8	
10 EL (+X)	0.0	0.0	0.0	
11 EL (+Z)	0.0	-0.7	-0.2	
1 DEAD LOAD	-0.1	18.9	-0.5	
2 LIVE LOAD	0.0	30.7	0.3	
3 ML - CONVEYOR MATERIAL LOAD	0.0	-0.2	0.1	
4 WIND LOAD IN +Z DIRN	0.0	22.2	-10.5	
5 WIND LOAD IN -Z DIRN	0.0	-22.2	10.5	
6 WIND LOAD IN +X DIRN	0.1	-0.7	0.2	6
7 WIND LOAD IN -X DIRN	-0.1	0.7	-0.2	
8 TL1 (TEMP LOAD +25)	-0.5	-1.1	-0.1	
9 TL2 (TEMP LOAD -25)	0.5	1.1	0.1	
10 EL (+X)	0.0	0.0	0.0	
11 EL (+Z)	0.0	0.7	-0.2	



Staad model of VGTU superstructure (along with dimension)

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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5.7 Load Combinations for Design of Isolated and Combined footing

5.7.1 Load Combinations for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD COMB 301 1.0 DL	(Eq. 2.4.1-1 of SBC 301)
LOAD COMB 302 1.0 (DL + TL1) + 1.0 (LL+ML)	(Eq. 2.4.1-2 of SBC 301)
LOAD COMB 303 1.0 (DL + TL1) + 1.0 (LL+ML)	
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75 (LL+ML)	(Eq. 2.4.1-6 of SBC 301)
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75 (LL+ML)	
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75 (LL+ML)	
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75 (LL+ML)	
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75 (LL+ML)	
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75 (LL+ML)	
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75 (LL+ML)	
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75 (LL+ML)	
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)	(Eq. 2.4.1-7 of SBC 301)
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)	
LOAD COMB 314 0.6 DL + 1.0 WL(+X)	
LOAD COMB 315 0.6 DL + 1.0 WL(-X)	
LOAD COMB 316 0.6 DL + 0.7 EL(+X)	(Eq. 2.4.1-8 of SBC 301)
LOAD COMB 317 0.6 DL + 0.7 EL(-X)	
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)	
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)	

5.7.2 Load Combinations for Concrete design

Load Cases	Reference
LOAD COMB 401 1.4 DL	(Eq. 2.3.2-1 of SBC 301)
LOAD COMB 402 1.4 (DL + TL1) + 1.7 (LL+ML)	(Eq. 2.3.2-2 of SBC 301)
LOAD COMB 403 1.4 (DL + TL1) + 1.7 (LL+ML)	
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5 (LL+ML)	(Eq. 2.3.2-4 of SBC 301)
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5 (LL+ML)	
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5 (LL+ML)	
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5 (LL+ML)	
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5 (LL+ML)	(Eq. 2.3.2-5 of SBC 301)
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5 (LL+ML)	
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5 (LL+ML)	
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5 (LL+ML)	
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)	(Eq. 2.3.2-6 of SBC 301)
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)	
LOAD COMB 414 0.9 DL + 1.6 WL(+X)	
LOAD COMB 415 0.9 DL + 1.6 WL(-X)	
LOAD COMB 416 0.9 DL + 1.0 EL(+X)	(Eq. 2.3.2-7 of SBC 301)
LOAD COMB 417 0.9 DL + 1.0 EL(-X)	
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)	
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)	

5.8 Load Combinations for Isolated footing of Ground Portals

5.8.1 Load Combination for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD COMB 301 1.0 DL	(Eq. 2.4.1-1 of SBC 301)
LOAD COMB 302 1.0 (DL + TL1) + 1.0 (LL+ML)	(Eq. 2.4.1-2 of SBC 301)
LOAD COMB 303 1.0 (DL + TL1) + 1.0 (LL+ML)	
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75 (LL+ML)	(Eq. 2.4.1-6 of SBC 301)
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75 (LL+ML)	
LOAD COMB 306 0.6 DL + 1.0 WL(+Z)	(Eq. 2.4.1-7 of SBC 301)
LOAD COMB 307 0.6 DL + 1.0 WL(-Z)	

5.8.2 Load Combination for Concrete design

Load Case	Reference
LOAD COMB 401 1.4 DL	(Eq. 2.3.2-1 of SBC 301)
LOAD COMB 402 1.4 (DL + TL1) + 1.7 (LL+ML)	(Eq. 2.3.2-2 of SBC 301)
LOAD COMB 403 1.4 (DL + TL1) + 1.7 (LL+ML)	
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5 (LL+ML)	(Eq. 2.3.2-4 of SBC 301)
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5 (LL+ML)	
LOAD COMB 406 0.9 DL + 1.6 WL(+Z)	(Eq. 2.3.2-6 of SBC 301)
LOAD COMB 407 0.9 DL + 1.6 WL(-Z)	

Eq. 2.3.2-3 of SBC 301 - 2007 has not been considered as it would not produce worst load combination in the absence of Roof Live Load & Rain Load

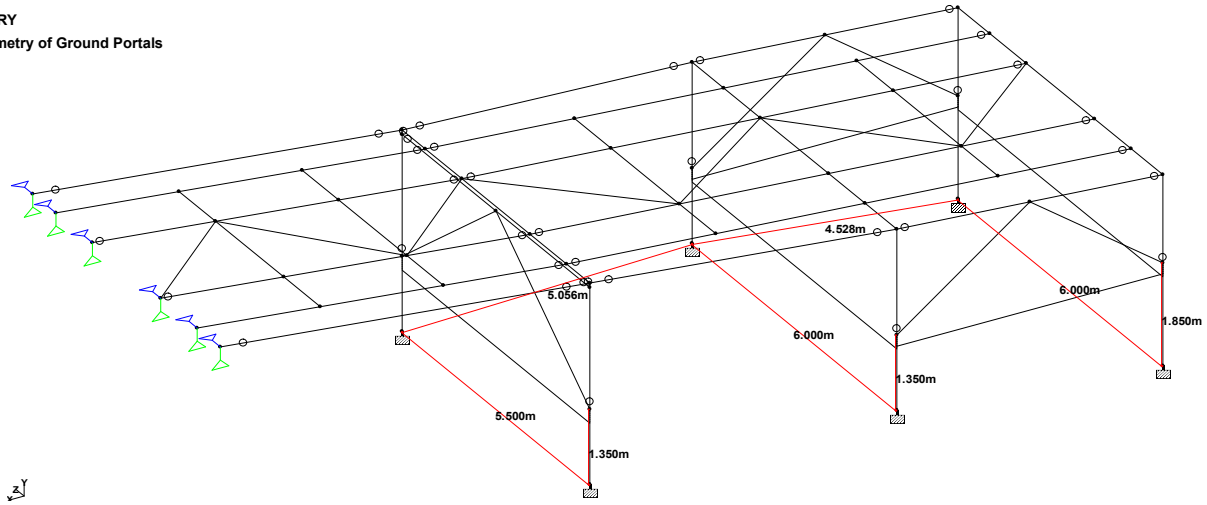
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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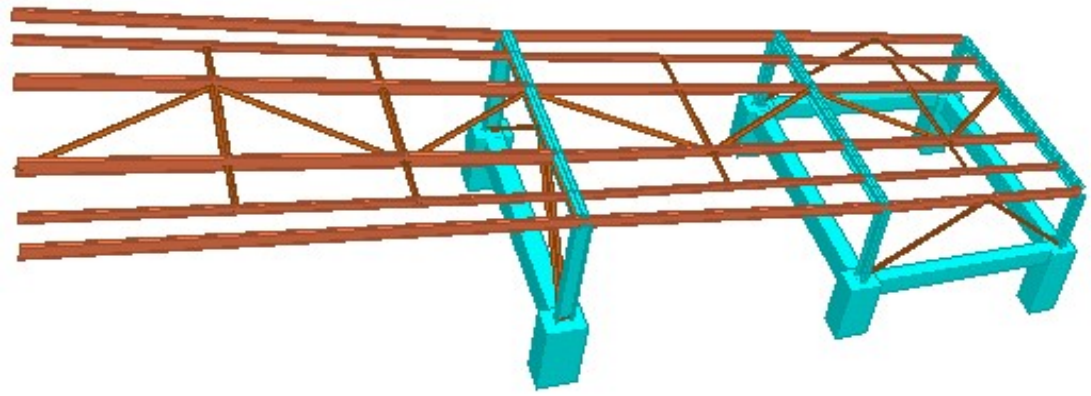
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6 GEOMETRY

6.1 Geometry of Ground Portals

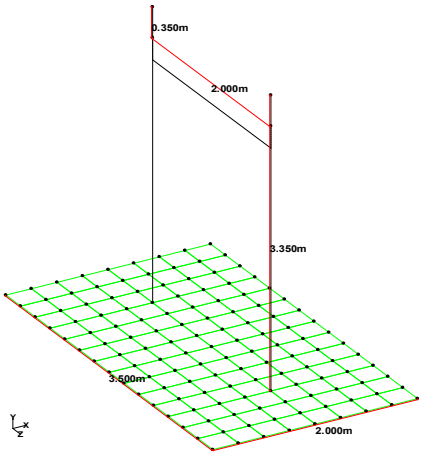


Staad model of Ground Portals along with dimension

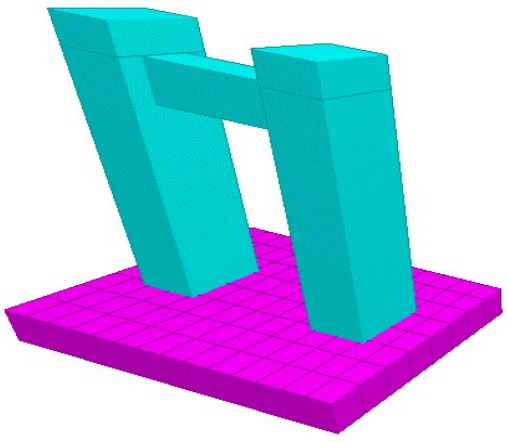


Isometric view of Ground portals (From Staad)

6.2 Geometry of combined footing mkd. F3 for trestle mkd. TR-1A



Staad Model of combined footing mkd. F3

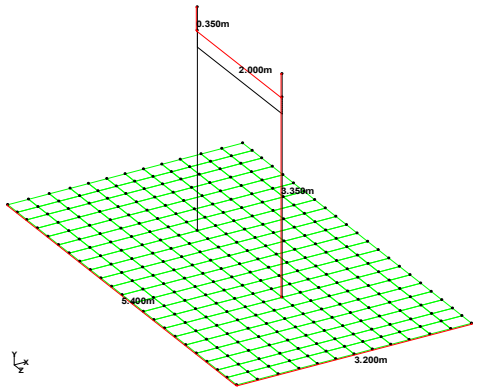


Isometric view of combined footing mkd. F3

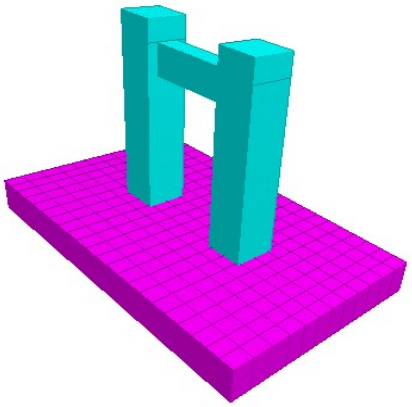
Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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Approved	PRI	PRI	PRI					Approved	PRI	PRI	PRI				

6.3 Geometry of combined footing mkd. F4 for trestle mkd. TR-1 & TR-2

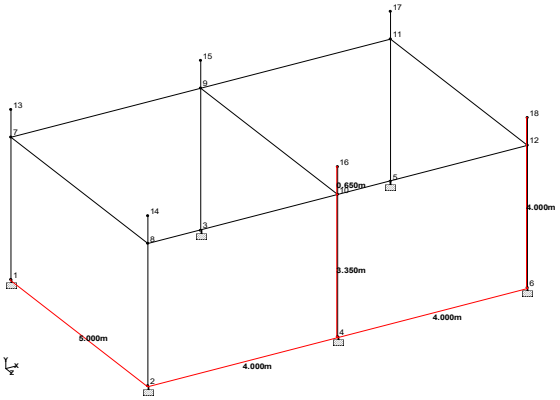


Staad Model of combined footing mkd. F4

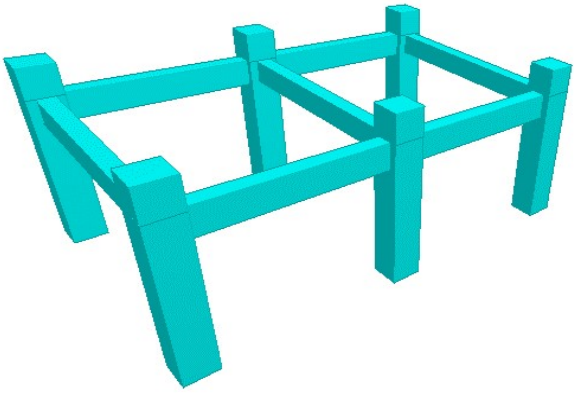


Isometric view of combined footing mkd. F4

6.4 Geometry of foundation arrangement of VGTU

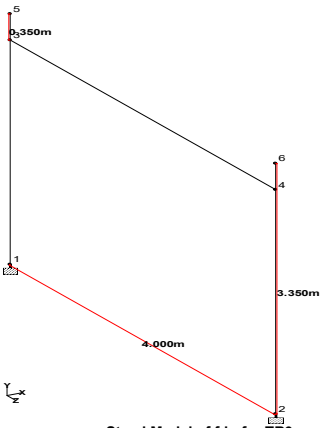


Staad Model of VGTU foundation

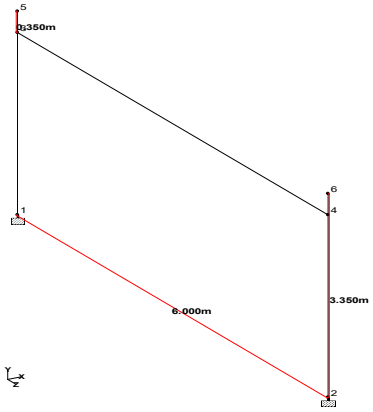


Isometric view of VGTU foundation

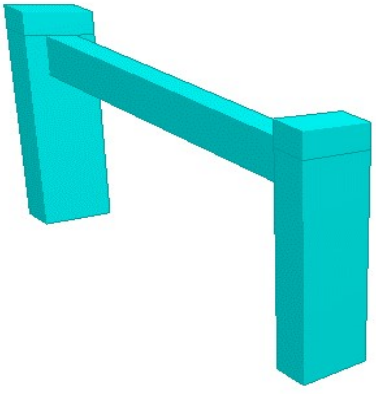
6.5 Geometry of foundation arrangement of TR3 & TR4



Staad Model of fdn for TR3



Staad Model of fdn for TR4



Isometric view of footing arrangement for TR3 & TR4

Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

7. DESIGN OF CONCRETE COMPONENTS

All foundations have been designed using worksheet in accordance to SBC-304.

All beams & columns have been designed in Staad-Pro by ACI 318M.

For Design Results, Refer Attachemnt 3 & 4 - Staad Output file.

7.1 Design Summary of Column

7.1.1 Column Reinforcement Summary

Column mark	Width (Along conv)	Depth (Across conv)	Reinforcement	Stirrup	Remarks	T.O.C. (RL)	Numbers	Pocket size for Shear Key
P1	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP1, GP2 & GP3, TR1A, TR1	912.800	10	250 x 80 x 150 Deep
P2	700	500	12 No.- D20	3 Nos D10 @ 200c/c	TR2 & TR3	912.800	4	250 x 100 x 150 Deep
P3	600	600	12 No.- D20	3 Nos D10 @ 200c/c	VGTU GRID B & C	913.450	4	300x80x150 Deep Star
P4	600	600	12 No.- D20	3 Nos D10 @ 200c/c	VGTU GRID A	913.450	2	200 x 80 x 150 Deep
P5	700	500	12 No.- D20	3 Nos D10 @ 200c/c	TR4	912.800	2	350 x 100 x 150 Deep
P6	450	450	8 No.- D20	2 Nos D10 @ 200c/c	VGTU Guide pipe Support	913.200	2	4 nos 80 x 80 x 50 Deep

7.2 Design Summary of Beam

7.2.1 Beam Reinforcement Summary

Beam mark	Width (mm)	Depth (mm)	T.O.B. (RL)	Trestle Mkd.	Top Reinforcement	Bottom Reinforcement	Stirrup	Numbers
TB1	300	500	912.500	GP1 to GP3	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	3
TB2	300	400	912.500		3 Nos D12	3 Nos D12	2L D8 STRPS @ 200c/c	2
TB3	300	500	912.500	TR1A to TR2	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	3
TB4	300	500	912.500	TR3	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	1
TB5	300	500	912.850	VGTU	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	3
TB6	300	600	912.850		3 Nos D20	3 Nos D20	2L D10 STRPS @ 200c/c	4
TB7	300	600	912.500	TR4	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	1

7.3 Design of Foundation

7.3.1 Foundation marking, Max. Base pressure and Reinforcement Summary

Structure Name	Foundation mark	Length (Along conv) (mm)	Width (Across conv) (mm)	Thickness (mm)	Max pressure (kN/m^2)	Top reinforcement	Bottom reinforcement	F.O.S.	Numbers
GP1	**F1**	**1200**	**1200**	**300**	**110**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.21**	**2**
GP2	**F2**	**1200**	**1200**	**300**	**152**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.29**	**2**
GP3	F2	1200	1200	300	90	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.17	2
TR1A	**F3**	**2000**	**3500**	**400**	**142**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.26**	**1**
TR1	F4	3200	5400	550	102	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.18	1
TR2	**F4**	**3200**	**5400**	**550**	**109**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.20**	**1**
TR3	F5	2500	2500	400	120	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.22	2
VGTU Grid-C	**F5**	**2500**	**2500**	**400**	**188**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.34**	**2**
VGTU Grid-B	**F6**	**3000**	**2500**	**400**	**139**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.25**	**2**
VGTU Grid-A	F5	2500	2500	400	110	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.20	2
TR4	F5	2500	2500	400	160	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.29	2

The design of the most critical foundations of the groups are attached in the following pages.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

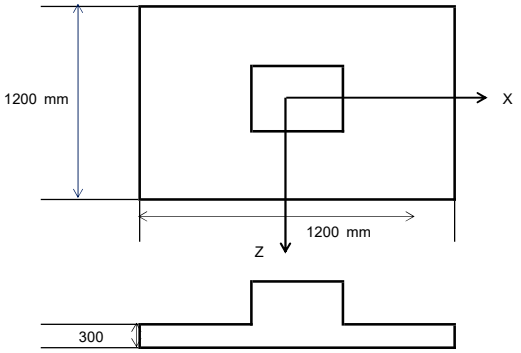
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7.3.2 Design of foundation F1 for supporting trestle mkd. GP1

Finished Ground level	F.G.L	=	-0.500	m	Staad Node no	=	86 & 88
Bottom of foundation	B.O.F	=	-2.000	m	Weight of soil	=	23.33 KN
Staad support reaction level		=	-2.000	m	Weight of foundation	=	10.37 KN
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00 KN
Surcharge	Ws	=	0.000	KN/m^2	Surcharge	=	0 KN
Length of footing	Lz	=	1.20	m	Pressure on top for tension	=	23.4 KN/m^2
Width of footing	Lx	=	1.20	m			
Thickness of footing	D	=	0.30	m			
Depth of pedestal	Bx	=	0.60	m			
Width of pedestal	Bz	=	0.60	m			
Height of Load point from bottom of footing	H	=	0.000	m			
Section Modulus of Pedestal	Zxx	=	0.29	m^3			
Section Modulus of Pedestal	Zzz	=	0.29	m^3			
Allowable Net Bearing Capacity		=	500	KN/m^2			
Gross Bearing Capacity		=	527	KN/m^2			
Grade of concrete		=	35	N/mm^2			
Grade of steel		=	420	N/mm^2			
Clear cover		=	75	mm			
Top Reinfocement		=	12	mm			
Bottom Reinforcement		=	16	mm			
Co-efficient of friction between soil & concrete		=	0.3				
Unit wt of soil		=	18	KN/m^3			
Unit wt of concrete		=	24	KN/m^3			

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	1.7	45.047	-6.6	-1.17	0.0	-7.4	78.7	84.51	24.85	1	0.16
302 1.0 (DL + TL1) +	1.8	56.437	-6.4	-3.01	0.077	-7.7	90.1	99.67	25.52	1	0.19
303 1.0 (DL + TL2) +	1.7	56.293	-6.3	-1.87	0.089	-7.4	90.0	94.58	30.41	1	0.18
304 1.0 DL+ 0.75 W	1.7	53.034	-5.9	-1.32	0.056	-7.5	86.7	90.72	29.74	1	0.17
305 1.0 DL+ 0.75 W	1.8	54.037	-6.9	-2.92	0.083	-7.5	87.7	97.22	24.63	1	0.18
306 0.6 DL + 1.0 WL	1.0	26.359	-3.3	0.36	0.0	-4.4	60.1	58.26	25.15	1	0.11
307 0.6 DL + 1.0 WL	1.1	27.697	-4.6	-1.77	0.036	-4.5	61.4	64.40	20.87	1	0.12
301 1.0 DL	1.8	48.71	8.3	3.91	-0.046	-7.5	82.4	96.81	17.64	1	0.18
302 1.0 (DL + TL1) +	1.7	60.123	7.6	4.91	-0.12	-7.6	93.8	108.45	21.85	1	0.21
303 1.0 (DL + TL2) +	1.9	60.348	8.3	5.05	-0.119	-7.6	94.0	109.12	21.50	1	0.21
304 1.0 DL+ 0.75 W	1.8	57.856	8.6	5.51	-0.1	-7.6	91.6	109.04	18.12	1	0.21
305 1.0 DL+ 0.75 W	1.8	56.852	7.5	3.91	-0.088	-7.5	90.5	102.54	23.22	1	0.19
306 0.6 DL + 1.0 WL	1.1	29.895	5.7	3.41	-0.045	-4.5	63.6	71.78	16.54	1	0.14
307 0.6 DL + 1.0 WL	1.0	28.557	4.3	1.28	0.0	-4.4	62.3	63.11	23.35	1	0.12

Max P_{max} = 109.12 KN/m^2
Max P_{min} = 16.54 KN/m^2
Resultant F.O.S = 0.21

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	2.4	63.1	-9.2	-1.64	0.0	-10.39	85.56	2.03
402 1.4 (DL + TL1) +	2.5	82.4	-8.9	-4.60	0.124	-10.76	110.55	3.90
403 1.4 (DL + TL2) +	2.4	82.2	-8.7	-3.00	0.141	-10.35	103.43	10.75
404 1.2 DL+ 1.6 WL	2.0	58.6	-6.7	-0.34	0.034	-8.88	72.71	8.74
405 1.2 DL+ 1.6 WL	2.1	60.8	-8.9	-3.75	0.091	-9.03	86.57	-2.15
406 0.9 DL + 1.6 WL	1.5	39.5	-4.8	0.65	-0.002	-6.60	52.59	2.23
407 0.9 DL + 1.6 WL	1.6	41.6	-7.0	-2.76	0.055	-6.76	61.94	-4.14
401 1.4 DL	2.5	68.2	11.6	5.48	-0.064	-10.48	102.78	-8.06
402 1.4 (DL + TL1) +	2.4	87.6	10.5	7.19	-0.191	-10.61	122.66	-0.95
403 1.4 (DL + TL2) +	2.6	87.9	11.5	7.39	-0.188	-10.62	123.60	-1.45
404 1.2 DL+ 1.6 WL	2.2	65.3	10.9	6.94	-0.12	-9.10	101.01	-10.34
405 1.2 DL+ 1.6 WL	2.1	63.1	8.7	3.52	-0.063	-8.95	87.15	0.55
406 0.9 DL + 1.6 WL	1.6	44.9	8.6	5.23	-0.07	-6.82	73.00	-10.63
407 0.9 DL + 1.6 WL	1.6	42.8	6.4	1.82	-0.012	-6.66	59.14	0.26

Max Net factored B.P = 123.60 KN/m^2

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-04-2020	18-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
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Outotec / L&T	0	1	2
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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.3 m
Moment about Z: Muz = 5.56 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 321.51 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 67.9 mm²

Length of cantilever along Z = 0.3 m
Moment about X: Mux = 5.56 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 275.84 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 73.3 mm²

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Effective Depth Along X = 217 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd

A_{stmin} = 390.6 mm²
A_{req} = 67.9 mm²
Required Spacing = **514.8** mm

Effective Depth Along X = 201 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd

A_{stmin} = 361.8 mm²
A_{req} = 73.3 mm²
Required Spacing = **555.7** mm
Ast prov = 1005.3 mm²

Top Reinforcement

Length of cantilever along X = 0.3 m
Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 327.46 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm²

Length of cantilever along Z = 0.3 m
Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 292.56 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm²

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Effective Depth Along X = 219 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 197.1 mm²
A_{req} = 0.0 mm²
Required Spacing = **573.8** mm

Effective Depth Along Z = 207 mm

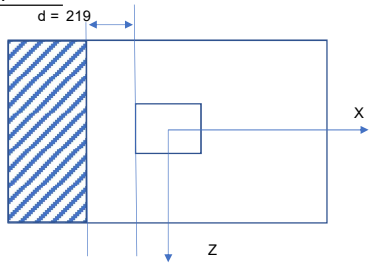
For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 186.3 mm²
A_{req} = 0.0 mm²
Required Spacing = **607.1** mm

Ast prov = 565.5 mm²

ONE WAY SHEAR CKECK



Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	06-04-2020	18-05-2020	12-10-2023	Date	06-04-2020	18-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	Prepared By	APL	APL	APL
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Along X Direction Cantilever

L	=	0.081	m		
Reinforcement ratio	ρ_w	=	0.0046		
Effective Depth Along X	d	=	219	mm	
Shear Force at d distance	V_u	=	10.01	KN	
Allowable shear strength	φVc	=	168.29	KN	O.K
Utility ratio		=	0.06		

Least of

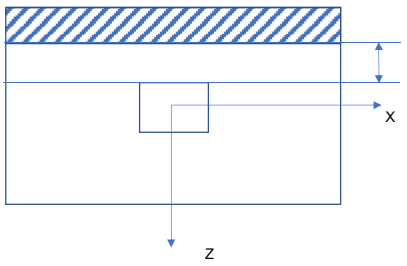
i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ — *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete
Mu = 0.41 KNm
φ = 0.75 for shear
φVc = 224.8 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$
φVc = 168.3 KN — *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$
= 281.8 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

L	=	0.093	m		
Reinforcement ratio	ρ_w	=	0.0049		
Effective Depth Along Z	d	=	207	mm	
Shear Force at d distance	V_u	=	11.49	KN	
Allowable shear strength	φVc	=	159.77	KN	O.K
Utility ratio		=	0.07		

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right) b_w d$ — *Table 22.5.5.1 SBC 304*

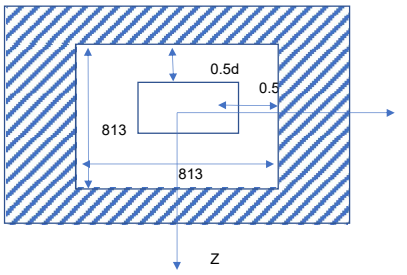
λ = 1 Normal concrete
Mu = 0.53 KNm
φ = 0.75 for shear
φVc = 204.0 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right) b_w d$
φVc = 159.8 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'} b_w d$
φVc = 266.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	1.44	m2
Punching area	=	0.66	m2
λ	=	1	
β	=	1	
αs	=	40	
b0	=	3252.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.27 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.25	m	
Depth at section	d	=	213	mm	
Shear Force at section	Vu	=	96.3	KN	
Allowable shear strength	φVc	=	1014.24	KN	O.K
Utility ratio			0.09		

Vc = $v_c b_0 d$
= 1352.32 KN
φ = 0.75 for shear
φV_n = φVc
φVc = 1014.2 KN

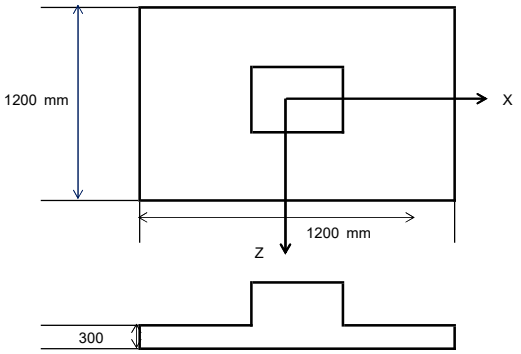
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
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7.3.3 Design of foundation F2 for supporting trestle mkd. GP2 & GP3

Finished Ground level	F.G.L	=	-0.500	m	Staad Node no	=	87 & 89	
Bottom of foundation	B.O.F	=	-1.500	m	Weight of soil	=	13.61 KN	*inputs
Staad support reaction level		=	-1.500	m	Weight of foundation	=	10.37 KN	
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00 KN	
Surcharge	Ws	=	0.000	KN/m^2	Surcharge	=	0 KN	
Length of footing	Lz	=	1.20	m	Pressure on top for tension	=	16.7 KN/m^2	
Width of footing	Lx	=	1.20	m				
Thickness of footing	D	=	0.30	m				
Depth of pedestal	Bx	=	0.60	m				
Width of pedestal	Bz	=	0.60	m				
Height of Load point from bottom of footing	H	=	0.000	m				
Section Modulus of Pedestal	Zxx	=	0.29	m^3				
Section Modulus of Pedestal	Zzz	=	0.29	m^3				
Allowable Net Bearing Capacity		=	500	KN/m^2				
Gross Bearing Capacity		=	518	KN/m^2				
Grade of concrete		=	35	N/mm^2				
Grade of steel		=	420	N/mm^2				
Clear cover		=	75	mm				
Top Reinfocement		=	12	mm				
Bottom Reinforcement		=	16	mm				
Co-efficient of friction between soil & concrete		=	0.3					
Unit wt of soil		=	18	KN/m^3				
Unit wt of concrete		=	24	KN/m^3				



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	8.9	63.775	-8.8	-1.81	0.0	-5.4	87.8	86.07	35.81	1	0.17
302 1.0 (DL + TL1) +	8.9	105.682	-11.2	-9.53	0.08	-5.4	129.7	141.94	38.14	1	0.27
303 1.0 (DL + TL2) +	8.9	105.442	-11.0	-8.53	0.097	-5.6	129.4	138.86	40.88	1	0.27
304 1.0 DL+ 0.75 W	8.9	93.889	-9.9	-6.53	0.083	-5.4	117.9	123.44	40.26	1	0.24
305 1.0 DL+ 0.75 W	8.9	96.341	-11.1	-7.92	0.07	-5.5	120.3	130.19	36.91	1	0.25
306 0.6 DL + 1.0 WL	5.3	36.63	-4.5	-0.16	0.0	-3.2	60.6	53.80	30.38	1	0.10
307 0.6 DL + 1.0 WL	5.4	39.899	-6.1	-2.01	0.016	-3.3	63.9	62.81	25.91	1	0.12
301 1.0 DL	8.9	67.212	10.1	3.36	-0.01	-5.5	91.2	94.05	32.60	1	0.18
302 1.0 (DL + TL1) +	8.9	110.536	12.7	11.35	-0.068	-5.3	134.5	151.38	35.45	1	0.29
303 1.0 (DL + TL2) +	9.0	110.608	12.8	10.73	-0.064	-5.7	134.6	150.62	36.30	1	0.29
304 1.0 DL+ 0.75 W	9.0	100.958	12.7	9.82	0.0	-5.6	124.9	140.14	33.38	1	0.27
305 1.0 DL+ 0.75 W	8.9	98.506	11.5	8.42	-0.059	-5.5	122.5	133.38	36.73	1	0.26
306 0.6 DL + 1.0 WL	5.4	41.962	6.8	2.95	0.003	-3.3	65.9	67.60	23.98	1	0.13
307 0.6 DL + 1.0 WL	5.3	38.693	5.3	1.09	0.0	-3.3	62.7	58.59	28.45	1	0.11

Max P_{max} = 151.38 KN/m^2
Max P_{min} = 23.98 KN/m^2
Resultant F.O.S = 0.29

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	12.4	89.3	-12.3	-2.53	0.1	-7.60	97.18	26.82
402 1.4 (DL + TL1) +	12.5	160.5	-16.4	-15.52	0.127	-7.60	191.70	31.21
403 1.4 (DL + TL2) +	12.4	160.2	-16.1	-14.11	0.151	-7.83	187.40	35.04
404 1.2 DL+ 1.6 WL	10.6	94.8	-10.5	-4.30	0.087	-6.48	103.26	28.42
405 1.2 DL+ 1.6 WL	10.7	100.0	-13.0	-7.27	0.059	-6.61	117.67	21.27
406 0.9 DL + 1.6 WL	7.9	54.8	-6.7	-0.14	0.051	-4.82	55.27	20.82
407 0.9 DL + 1.6 WL	8.1	60.0	-9.2	-3.11	0.022	-4.95	69.68	13.67
401 1.4 DL	12.5	94.1	14.1	4.71	-0.014	-7.68	108.36	22.33
402 1.4 (DL + TL1) +	12.4	167.8	18.6	18.20	-0.113	-7.49	205.70	27.29
403 1.4 (DL + TL2) +	12.6	167.9	18.7	17.32	-0.107	-8.05	204.65	28.49
404 1.2 DL+ 1.6 WL	10.8	105.0	14.7	9.36	-0.026	-6.68	128.56	17.20
405 1.2 DL+ 1.6 WL	10.7	99.7	12.2	6.39	-0.054	-6.55	114.15	24.35
406 0.9 DL + 1.6 WL	8.1	63.1	10.3	4.51	0.005	-5.01	76.87	10.78
407 0.9 DL + 1.6 WL	8.0	57.9	7.8	1.54	-0.023	-4.87	62.45	17.93

Max Net factored B.P = 205.70 KN/m^2

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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
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Ma'aden Revision No:	A01	C01	D01
Date	06-04-2020	18-05-2020	12-10-2023
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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.3 m
Moment about Z: Muz = 9.26 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 321.51 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 113.1 mm²

Effective Depth Along X = 217 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd

A_{stmin} = 390.6 mm²
A_{req} = 113.1 mm²
Required Spacing = **514.8** mm

Length of cantilever along Z = 0.3 m
Moment about X: Mux = 9.26 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 275.84 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 122.2 mm²

Effective Depth Along X = 201 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
A_{stmin} = 0.0014bd

A_{stmin} = 361.8 mm²
A_{req} = 122.2 mm²
Required Spacing = **555.7** mm

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***
Ast prov = 1005.3 mm²

Top Reinforcement

Length of cantilever along X = 0.3 m
Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 327.46 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section
Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm²

Effective Depth Along X = 219 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 197.1 mm²
A_{req} = 0.0 mm²
Required Spacing = **573.8** mm

Length of cantilever along Z = 0.3 m
Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 292.56 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section
Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm²

Effective Depth Along Z = 207 mm

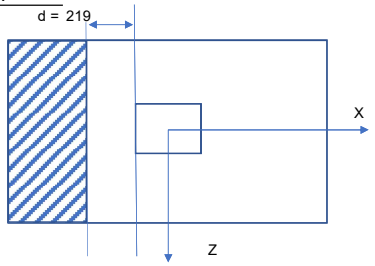
For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd *Table 24.4.3.2 SBC 304*

A_{stmin} = 186.3 mm²
A_{req} = 0.0 mm²
Required Spacing = **607.1** mm

Provide top Reinforcement = ***12 mm dia @ 200 c/c***
Ast prov = 565.5 mm²

ONE WAY SHEAR CKECK



Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	06-04-2020	18-05-2020	12-10-2023	Date	06-04-2020	18-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

Along X Direction Cantilever

L	=	0.081	m		
Reinforcement ratio	ρ_w	=	0.0046		
Effective Depth Along X	d	=	219	mm	
Shear Force at d distance	V_u	=	16.66	KN	
Allowable shear strength	φVc	=	168.29	KN	O.K
Utility ratio		=	0.10		

Least of

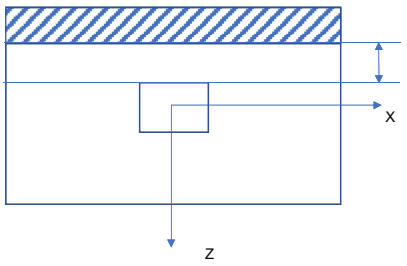
i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete; Mu = 0.67 KNm; φ = 0.75 for shear; φVc = 224.8 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$; φVc = 168.3 KN — *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$ = 281.8 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

L	=	0.093	m		
Reinforcement ratio	ρ_w	=	0.0049		
Effective Depth Along Z	d	=	207	mm	
Shear Force at d distance	V_u	=	19.13	KN	
Allowable shear strength	φVc	=	159.77	KN	O.K
Utility ratio		=	0.12		

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

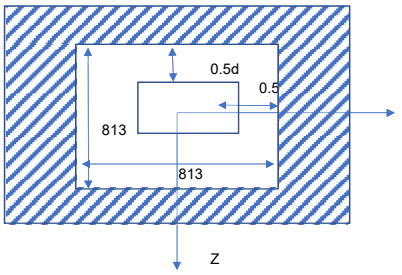
λ = 1 Normal concrete; Mu = 0.89 KNm; φ = 0.75 for shear; φVc = 204.0 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$; φVc = 159.8 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$; φVc = 266.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	1.44	m2
Punching area	=	0.66	m2
λ	=	1	
β	=	1	
αs	=	40	
b0	=	3252.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.27 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.25	m	
Depth at section	d	=	213	mm	
Shear Force at section	Vu	=	160.2	KN	
Allowable shear strength	φVc	=	1014.24	KN	O.K
Utility ratio			0.16		

Vc = $v_c b_0 d$ = 1352.32 KN

φ = 0.75 for shear

φV_n = φVc

φVc = 1014.2 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-04-2020	06-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

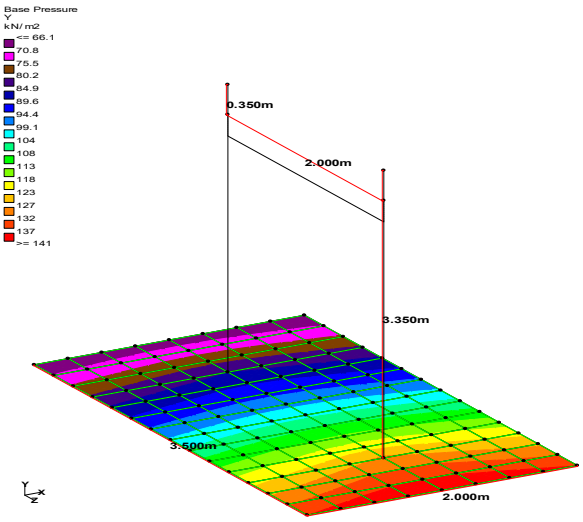
Outotec / L&T Revision No.:	0	1	2
Date	06-04-2020	06-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

7.3.4 Design of combined footing for Trestle mkd. TR1A and foundation mkd. F3

For Calculation of Base pressure and design, Refer STAAD File: 3110-CNV-002_TR1A FDN MKD_Combined

Staad Supoort Nodes are 5 & 6

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +913.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-3.5	m	
Top of pedestal (staad support)	T.O.F	=	-0.20	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	2.000	m	
Width of footing	Lx	=	3.500	m	
Thickness of footing	D	=	0.40	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	554	KN/m^2	
Soil load on footing in staad		=	46.8	KN/m^2	staad input



1) Check for Gross Base Pressure :

Maximum Base Pressur	=	141.4	KN/m^2	Load Case 304
Minimum Base Pressur	=	9.5	KN/m^2	Load Case 312

2) Design of foundation Raft :

Slab Thickness =	400 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	75 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

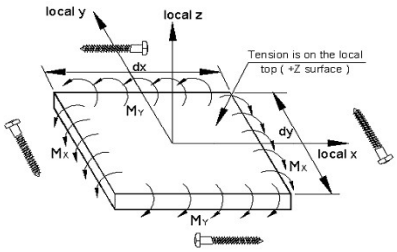
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	2.320	kN-m	Corresponding Mxy	-2.619	kN-m	*(Plate No. - 112 , L/C No. - 412)*
Min. Mx	-19.360	kN-m	Corresponding Mxy	2.916	kN-m	*(Plate No. - 16 , L/C No. - 403)*
Max. My	16.535	kN-m	Corresponding Mxy	0.777	kN-m	*(Plate No. - 66 , L/C No. - 403)*
Min. My	-14.458	kN-m	Corresponding Mxy	2.119	kN-m	*(Plate No. - 25 , L/C No. - 404)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	2.32 + 2.619 =	4.939 kN-m
My1 =	My + abs(Mxy) =	16.535 + 0.777 =	17.312 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	2.32 + 0.415 =	2.735 kN-m
My2 =	My+abs(Mxy^2/Mx) =	16.535 + 0.034 =	16.569 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 4.939 kN-m
Myd = 17.312 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 4.939 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 4.939X10^6/(1000X319^2) = 0.049 N/mm^2 deff. = 319 mm < 6.9 N/mm^2
Pt = 0.0001 Reqd. Ast = 0.0001X1000X319/100 = 0.32 cm^2
Provided 12 mm @ 200 C/C (Provide Ast = 5.65 cm^2)

Total Provide Ast = 5.65 cm^2 O.K.
Total Provide Pt = 0.0018

ALONG LOCAL Y - DIRECTION :

Mu = 17.312 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 17.312X10^6/(1000X307^2) = 0.17 N/mm^2 deff. = 307 mm < 6.9 N/mm^2
Pt = 0.0005 Reqd. Ast = 0.0005X1000X319/100 = 1.54 cm^2
Provided 12 mm @ 200 C/C (Provide Ast = 5.65 cm^2)

Total Provide Ast = 5.65 cm^2 O.K.
Total Provide Pt = 0.0018

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
Date	06-04-2020	06-05-2020	12-10-2023	Date	06-04-2020	06-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -19.36 - 2.916 = -22.276 kN-m
My1 = My - abs(Mxy) = -14.458 - 2.119 = -16.577 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -19.36 - 0.588 = -19.948 kN-m
My2 = My-abs(Mxy^2/Mx) = -14.458 - 0.232 = -14.226 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -22.276 kN-m
Myd = -16.577 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 22.276 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 22.276X10^6/(1000X317^2) = 0.222 N/mm^2 deff. = 317 mm < 6.9 N/mm^2
Pt = 0.0006 Reqd. Ast = 0.0006X1000X317/100 = 1.9 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0032

ALONG LOCAL Y - DIRECTION :

Mu = 16.577 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 16.577X10^6/(1000X301^2) = 0.183 N/mm^2 deff. = 301 mm < 6.9 N/mm^2
Pt = 0.0005 Reqd. Ast = 0.0005X1000X301/100 = 1.51 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0033

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.033 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 63 , L/C No. - 402)*
Vu = 10.46 kN

Provided Pt = 0.0032
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X10.461X10^3X317/22.276/10^6)X1000X317/10^3
= 302.631 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X317/10^3
= 317.308 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X317/10^3
= 543.865 kN
HENCE, Vc = 302.631 kN
Φt*Vc = 0.75X302.631 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 226.973 kN > 10.461T , Hence O.K.

ALONG LOCAL Y - DIRECTION :

SQy = -0.057 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 238 , L/C No. - 404)*
Vu = 17.157 kN

Pt provided = 0.0033
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0033X17.157X10^3X301/16.577/10^3)X1000X301/10^3
= 290.179 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0033)X1000X301/10^3
= 301.805 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X301/10^3
= 516.415 kN
HENCE, Vc = 290.179 kN
Φt*Vc = 0.75X290.179 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 217.634 kN > 17.157T , Hence O.K.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-04-2020	06-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
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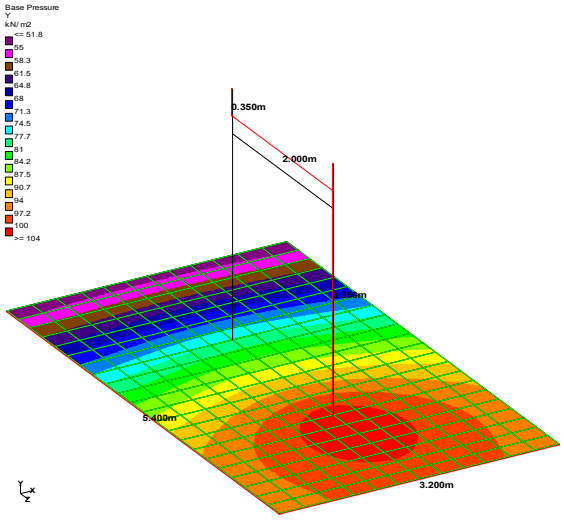
Outotec / L&T Revision No.:	0	1	2
Date	06-04-2020	06-05-2020	12-10-2023
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Approved	PRI	PRI	PRI

7.3.5 Design of combined footing for Trestle mkd. TR1 and foundation mkd. F4

For Calculation of Base pressure and design, Refer STAAD File: 3110-CNV-002_TR1 FDN MKD_Combined

Staad Supoort Nodes are 5 & 6

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +913.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-3.5	m	
Top of pedestal (staad support)	T.O.F	=	-0.20	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	3.200	m	
Width of footing	Lx	=	5.400	m	
Thickness of footing	D	=	0.55	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	554	KN/m^2	
Soil load on footing in staad		=	44.1	KN/m^2	staad input



1) Check for Gross Base Pressure :

Maximum Base Pressur	=	101.7	KN/m^2	Load Case 304
Minimum Base Pressur	=	14.8	KN/m^2	Load Case 312

2) Design of foundation Raft :

Slab Thickness =	550 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	75 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

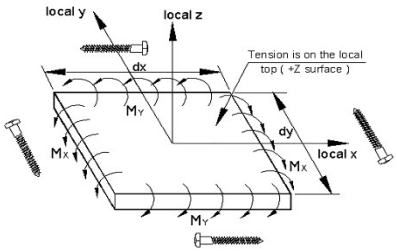
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	13.768	kN-m	Corresponding Mxy	-4.964	kN-m	*(Plate No. - 112 , L/C No. - 412)*
Min. Mx	-51.724	kN-m	Corresponding Mxy	6.752	kN-m	*(Plate No. - 24 , L/C No. - 404)*
Max. My	44.728	kN-m	Corresponding Mxy	-3.684	kN-m	*(Plate No. - 113 , L/C No. - 412)*
Min. My	-87.133	kN-m	Corresponding Mxy	1.820	kN-m	*(Plate No. - 25 , L/C No. - 404)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	13.768 + 4.964 =	18.732 kN-m
My1 =	My + abs(Mxy) =	44.728 + 3.684 =	48.412 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	13.768 + 0.551 =	14.319 kN-m
My2 =	My+abs(Mxy^2/Mx) =	44.728 + 0.756 =	45.484 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 18.732 kN-m
Myd = 48.412 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 18.732 kN-m
Φ*Kn = Mu/bd^2 deff. = 469 mm
Mu/bd^2 = 18.732X10^6/(1000X469^2) = 0.085 N/mm^2 < 6.9 N/mm^2
Pt = 0.0002 Reqd. Ast = 0.0002X1000X469/100 = 0.94 cm^2
Provided 12 mm @ 200 C/C (Provide Ast = 5.65)cm^2

Total Provide Ast = 5.65 cm^2 O.K.
Total Provide Pt = 0.0012

ALONG LOCAL Y - DIRECTION :

Mu = 48.412 kN-m
Φ*Kn = Mu/bd^2 deff. = 457 mm
Mu/bd^2 = 48.412X10^6/(1000X457^2) = 0.22 N/mm^2 < 6.9 N/mm^2
Pt = 0.0006 Reqd. Ast = 0.0006X1000X469/100 = 2.74 cm^2
Provided 12 mm @ 200 C/C (Provide Ast = 5.65)cm^2

Total Provide Ast = 5.65 cm^2 O.K.
Total Provide Pt = 0.0012

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-04-2020	06-05-2020	12-10-2023
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Approved	PRI	PRI	PRI

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -51.724 - 6.752 = -58.476 kN-m
My1 = My - abs(Mxy) = -87.133 - 1.82 = -88.953 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -51.724 - 0.523 = -52.247 kN-m
My2 = My-abs(Mxy^2/Mx) = -87.133 - 0.064 = -87.069 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -58.476 kN-m
Myd = -88.953 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 58.476 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 58.476X10^6/(1000X467^2) = 0.268 N/mm^2 deff. = 467 mm < 6.9 N/mm^2
Pt = 0.0007 Reqd. Ast = 0.0007X1000X467/100 = 3.27 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0022

ALONG LOCAL Y - DIRECTION :

Mu = 88.953 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 88.953X10^6/(1000X451^2) = 0.437 N/mm^2 deff. = 451 mm < 6.9 N/mm^2
Pt = 0.0012 Reqd. Ast = 0.0012X1000X451/100 = 5.41 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0022

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.118 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 288 , L/C No. - 404)*
Vu = 55.11 kN

Provided Pt = 0.0022
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0022X55.106X10^3X467/58.476/10^6)X1000X467/10^3
= 449.736 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0022)X1000X467/10^3
= 459.515 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X467/10^3
= 801.215 kN
HENCE, Vc = 449.736 kN
Φt*Vc = 0.75X449.736 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 337.302 kN > 55.106T , Hence O.K.

ALONG LOCAL Y - DIRECTION :

SQy = -0.171 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 169 , L/C No. - 404)*
Vu = 77.121 kN

Pt provided = 0.0022
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0022X77.121X10^3X451/88.953/10^3)X1000X451/10^3
= 433.5 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0022)X1000X451/10^3
= 443.772 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X451/10^3
= 773.764 kN
HENCE, Vc = 433.5 kN
Φt*Vc = 0.75X433.5 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 325.125 kN > 77.121T , Hence O.K.

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-04-2020	18-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

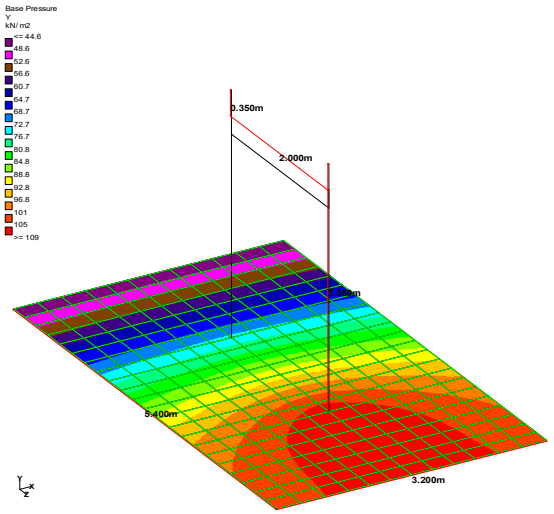
Outotec / L&T Revision No.:	0	1	2
Date	06-04-2020	18-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
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7.3.6 Design of combined footing for Trestle mkd. TR2 and foundation mkd. F4

For Calculation of Base pressure and design, Refer STAAD File: 3110-CNV-002_TR2 FDN MKD_Combined

Staad Supoort Nodes are 5 & 6

Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +913.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-3.5	m	
Top of pedestal (staad support)	T.O.F	=	-0.20	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	3.200	m	
Width of footing	Lx	=	5.400	m	
Thickness of footing	D	=	0.55	m	
Depth of pedestal	Bx	=	0.50	m	
Width of pedestal	Bz	=	0.70	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	554	KN/m^2	
Soil load on footing in staad		=	44.1	KN/m^2	staad input



1) Check for Gross Base Pressure :

Maximum Base Pressur	=	108.9	KN/m^2	Load Case 304
Minimum Base Pressur	=	5.2	KN/m^2	Load Case 312

2) Design of foundation Raft :

Slab Thickness =	550 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	75 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

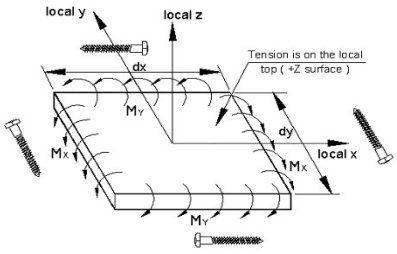
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	22.763	kN-m	Corresponding Mxy	-1.889	kN-m	*(Plate No. - 112 , L/C No. - 412)*
Min. Mx	-61.286	kN-m	Corresponding Mxy	10.191	kN-m	*(Plate No. - 24 , L/C No. - 404)*
Max. My	59.898	kN-m	Corresponding Mxy	-2.104	kN-m	*(Plate No. - 113 , L/C No. - 412)*
Min. My	-102.764	kN-m	Corresponding Mxy	3.604	kN-m	*(Plate No. - 25 , L/C No. - 404)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	22.763 + 1.889 =	24.652 kN-m
My1 =	My + abs(Mxy) =	59.898 + 2.104 =	62.002 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	22.763 + 0.06 =	22.823 kN-m
My2 =	My+abs(Mxy^2/Mx) =	59.898 + 0.132 =	60.03 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd = 24.652 kN-m
Myd = 62.002 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 24.652 kN-m
Φ*Kn = Mu/bd^2 deff. = 469 mm
Mu/bd^2 = 24.652X10^6/(1000X469^2) = 0.112 N/mm^2 < 6.9 N/mm^2
Pt = 0.0003 Reqd. Ast = 0.0003X1000X469/100 = 1.41 cm^2
Provided 12 mm @ 200 C/C (Provide Ast = 5.65)cm^2

Total Provide Ast = 5.65 cm^2 O.K.
Total Provide Pt = 0.0012

ALONG LOCAL Y - DIRECTION :

Mu = 62.002 kN-m
Φ*Kn = Mu/bd^2 deff. = 457 mm
Mu/bd^2 = 62.002X10^6/(1000X457^2) = 0.282 N/mm^2 < 6.9 N/mm^2
Pt = 0.0008 Reqd. Ast = 0.0008X1000X469/100 = 3.66 cm^2
Provided 12 mm @ 200 C/C (Provide Ast = 5.65)cm^2

Total Provide Ast = 5.65 cm^2 O.K.
Total Provide Pt = 0.0012

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Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2
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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -61.286 - 10.191 = -71.477 kN-m
My1 = My - abs(Mxy) = -102.764 - 3.604 = -106.368 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -61.286 - 1.011 = -62.297 kN-m
My2 = My-abs(Mxy^2/Mx) = -102.764 - 0.212 = -102.552 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -71.477 kN-m
Myd = -106.368 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 71.477 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 71.477X10^6/(1000X467^2) = 0.328 N/mm^2 deff. = 467 mm < 6.9 N/mm^2
Pt = 0.0009 Reqd. Ast =)0.0009X1000X467/100 = 4.2 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0022

ALONG LOCAL Y - DIRECTION :

Mu = 106.368 kN-m
Φ*Kn = Mu/bd^2
Mu/bd^2 = 106.368X10^6/(1000X451^2) = 0.523 N/mm^2 deff. = 451 mm < 6.9 N/mm^2
Pt = 0.0014 Reqd. Ast =)0.0014X1000X451/100 = 6.31 cm^2
Provided 16 mm @ 200 C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 O.K.
Total Provide Pt = 0.0022

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.146 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 288 , L/C No. - 404)*
Vu = 68.18 kN

Provided Pt = 0.0022
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0022X68.182X10^3X467/71.477/10^6)X1000X467/10^3
= 449.83 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0022)X1000X467/10^3
= 459.515 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X467/10^3
= 801.215 kN
HENCE, Vc = 449.83 kN
Φt*Vc = 0.75X449.83 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 337.373 kN > 68.182T , Hence O.K.

ALONG LOCAL Y - DIRECTION :

SQy = -0.207 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 169 , L/C No. - 404)*
Vu = 93.357 kN

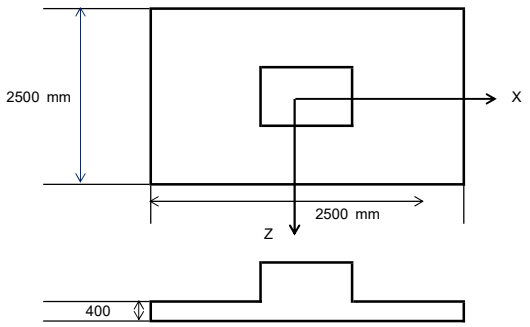
Pt provided = 0.0022
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0022X93.357X10^3X451/106.368/10^3)X1000X451/10^3
= 433.581 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0022)X1000X451/10^3
= 443.772 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X451/10^3
= 773.764 kN
HENCE, Vc = 433.581 kN
Φt*Vc = 0.75X433.581 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 325.186 kN > 93.357T , Hence O.K.

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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	06-04-2020	18-05-2020	12-10-2023	Date	06-04-2020	18-05-2020	12-10-2023
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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

7.3.7 Design of foundation F5 for VGTU Exterior columns

Finished Ground level	F.G.L	=	-0.500	m	Staad Node no	=	6	*inputs
Bottom of foundation	B.O.F	=	-3.500	m	Weight of soil	=	275.65 KN	
Staad support reaction level		=	-3.500	m	Weight of foundation	=	60.00 KN	
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00 KN	
Surcharge	Ws	=	0.000	KN/m^2	Surcharge	=	0 KN	
Length of footing	Lz	=	2.50	m	Pressure on top for tension	=	53.7 KN/m^2	
Width of footing	Lx	=	2.50	m				
Thickness of footing	D	=	0.40	m				
Depth of pedestal	Bx	=	0.60	m				
Width of pedestal	Bz	=	0.60	m				
Height of Load point from bottom of footing	H	=	0.000	m				
Section Modulus of Pedestal	Zxx	=	2.60	m^3				
Section Modulus of Pedestal	Zzz	=	2.60	m^3				
Allowable Net Bearing Capacity		=	500	KN/m^2				
Gross Bearing Capacity		=	554	KN/m^2				
Grade of concrete		=	35	N/mm^2				
Grade of steel		=	420	N/mm^2				
Clear cover		=	75	mm				
Top Reinfocement		=	12	mm				
Bottom Reinforcement		=	16	mm				
Co-efficient of friction between soil & concrete		=	0.3					
Unit wt of soil		=	18	KN/m^3				
Unit wt of concrete		=	24	KN/m^3				



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	-10.9	313.614	9.7	12.14	0.2	32.3	649.3	120.95	86.81	1	0.22
302 1.0 (DL + TL1) +	-9.7	502.827	25.1	23.32	0.681	31.0	838.5	155.04	113.28	1	0.28
303 1.0 (DL + TL1) +	-10.1	497.484	0.7	4.85	0.078	31.1	833.1	147.09	119.51	1	0.27
304 1.0 DL+ 0.75 W	-10.5	393.149	31.3	66.01	0.334	31.9	728.8	154.22	79.00	1	0.28
305 1.0 DL+ 0.75 W	-9.8	513.854	-7.0	-38.77	0.312	30.8	849.5	162.64	109.20	1	0.29
306 1.0 DL+ 0.75 W	11.0	306.539	9.8	11.61	0.3	-23.7	642.2	116.31	89.19	1	0.21
307 1.0 DL+ 0.75 W	-31.3	600.501	14.4	15.59	0.302	86.4	936.2	188.96	110.61	1	0.34
308 1.0 DL + 0.525	-9.7	450.148	12.1	13.56	0.319	30.3	785.8	142.56	108.89	1	0.26
309 1.0 DL + 0.525	-10.6	456.892	12.2	13.64	0.328	32.5	792.5	144.51	109.11	1	0.26
310 1.0 DL + 0.525	-10.2	450.553	12.9	15.88	0.291	31.5	786.2	143.98	107.61	1	0.26
311 1.0 DL + 0.525	-10.1	456.487	11.3	11.32	0.4	31.3	792.1	143.09	110.39	1	0.26
312 0.6 DL + 1.0 WL	-7.0	107.673	31.4	77.16	0.107	20.1	443.3	108.30	33.57	1	0.20
313 0.6 DL + 1.0 WL	-6.0	268.613	-19.7	-62.54	0.077	18.6	604.3	127.86	65.51	1	0.23
314 0.6 DL + 1.0 WL	21.6	-7.806	2.8	4.64	0.122	-54.0	327.8	74.98	29.93	1	0.14
315 0.6 DL + 1.0 WL	-34.7	384.143	8.9	9.93	0.064	92.8	719.8	154.62	75.71	1	0.28
316 0.6 DL + 0.7 EL	-6.0	183.673	5.8	7.23	0.1	17.9	519.3	92.76	73.42	1	0.17
317 0.6 DL + 0.7 EL	-7.1	192.664	5.9	7.34	0.099	20.8	528.3	95.35	73.71	1	0.17
318 0.6 DL + 0.7 EL	-6.6	184.213	7.0	10.32	0.049	19.5	519.9	94.64	71.71	1	0.17
319 0.6 DL + 0.7 EL	-6.5	192.124	4.7	4.25	0.1	19.2	527.8	93.47	75.42	1	0.17

Max P_{max} = 188.96 KN/m^2
Max P_{min} = 29.93 KN/m^2
Resultant F.O.S = 0.34

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	-15.2	439.1	13.6	17.00	0.2	45.24	94.15	46.35
402 1.4 (DL + TL1) +	-13.2	759.9	36.0	33.23	1.021	43.09	150.90	92.28
403 1.4 (DL + TL1) +	-13.9	752.4	2.0	7.37	0.177	43.11	139.78	101.01
404 1.2 DL+ 1.6 WL	-13.3	340.8	54.1	127.34	0.321	39.35	118.54	-9.48
405 1.2 DL+ 1.6 WL	-11.8	598.3	-27.5	-96.18	0.273	36.96	146.86	44.61
406 1.2 DL+ 1.6 WL	32.5	156.0	8.4	11.31	0.345	-79.32	59.77	-9.83
407 1.2 DL+ 1.6 WL	-57.6	783.2	18.1	19.78	0.252	155.62	192.66	57.95
408 1.2 DL + 1.0 EL	-11.8	463.2	13.2	15.47	0.29	36.08	93.90	54.32
409 1.2 DL + 1.0 EL	-13.4	476.0	13.4	15.62	0.307	40.22	97.61	54.72
410 1.2 DL + 1.0 EL	-12.6	464.0	14.9	19.88	0.236	38.36	96.60	51.87
411 1.2 DL + 1.0 EL	-12.5	475.3	11.7	11.20	0.362	37.95	94.91	57.17
412 0.9 DL + 1.6 WL	-10.5	153.5	49.6	122.73	0.162	30.29	83.31	-34.20
413 0.9 DL + 1.6 WL	-9.0	411.0	-32.0	-100.80	0.114	27.89	115.17	16.34
414 0.9 DL + 1.6 WL	35.3	-31.3	3.9	6.69	0.187	-88.38	31.50	-41.52
415 0.9 DL + 1.6 WL	-54.8	595.8	13.6	15.16	0.093	146.55	157.43	33.23
416 0.9 DL + 1.0 EL	-9.0	275.8	8.7	10.85	0.132	27.02	58.67	29.59
417 0.9 DL + 1.0 EL	-10.6	288.7	8.9	11.00	0.148	31.15	62.38	30.00
418 0.9 DL + 1.0 EL	-9.9	276.6	10.4	15.27	0.077	29.29	61.37	27.15
419 0.9 DL + 1.0 EL	-9.7	287.9	7.2	6.59	0.203	28.88	59.68	32.45

Max Net factored B.P = 192.66 KN/m^2

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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 86.94 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 686.10 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 736.7 mm^2

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 86.94 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 618.59 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 777.3 mm^2

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Effective Depth Along X = 317 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or A_{stmin} = 0.0014bd

A_{stmin} = 570.6 mm^2

A_{req} = 736.7 mm^2

Required Spacing = **272.9** mm

Effective Depth Along X = 301 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or A_{stmin} = 0.0014bd

A_{stmin} = 541.8 mm^2

A_{req} = 777.3 mm^2

Required Spacing = **258.7** mm

Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 0.95 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 694.79 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm^2

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 643.50 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm^2

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Effective Depth Along X = 319 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or A_{stmin} = 50% of 0.0014bd

Table 24.4.3.2 SBC 304

A_{stmin} = 287.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **393.9** mm

Effective Depth Along Z = 307 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or A_{stmin} = 50% of 0.0014bd

Table 24.4.3.2 SBC 304

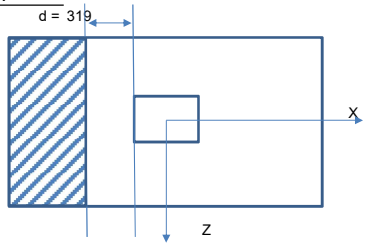
A_{stmin} = 276.3 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **409.3** mm

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T	0	1	2			
Date	06-04-2020	18-05-2020	12-10-2023				Date	06-04-2020	18-05-2020	12-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

Along X Direction Cantilever

Reinforcement ratio	ρ_w	=	0.0032			
Effective Depth Along X	d	=	319	mm		
Shear Force at d distance	V_u	=	121.57	KN		
Allowable shear strength	φVc	=	239.29	KN	O.K	
Utility ratio		=	0.51			

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ	=	1	Normal concrete	
Mu	=	38.35	KNm	
φ	=	0.75	for shear	
φVc	=	239.4	KN	

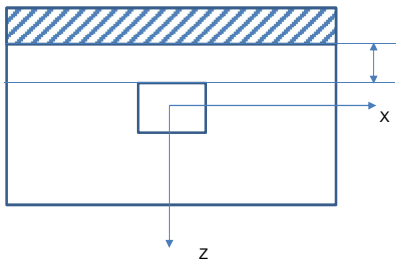
ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$

φVc	=	239.3	KN	*Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

=	410.5	KN	

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

Reinforcement ratio	ρ_w	=	0.0033			
Effective Depth Along Z	d	=	307	mm		
Shear Force at d distance	V_u	=	123.88	KN		
Allowable shear strength	φVc	=	230.19	KN	O.K	
Utility ratio		=	0.54			

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ	=	1	Normal concrete	
Mu	=	39.83	KNm	
φ	=	0.75	for shear	
φVc	=	230.2	KN	

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$

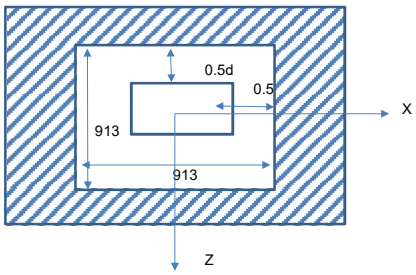
φVc	=	230.8	KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

φVc	=	395.0	KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	6.25	m2
Punching area	=	0.83	m2
λ	=	1	
β	=	1	
αs	=	40	
b0	=	3652.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.67 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.65	m		
Depth at section	d	=	313	mm		
Shear Force at section	Vu	=	1043.5	KN		
Allowable shear strength	φVc	=	1673.73	KN	O.K	
Utility ratio			0.62			

Vc = $v_c b_0 d$ = 2231.63 KN

φ = 0.75 for shear

φV_n = φVc

φVc = 1673.7 KN

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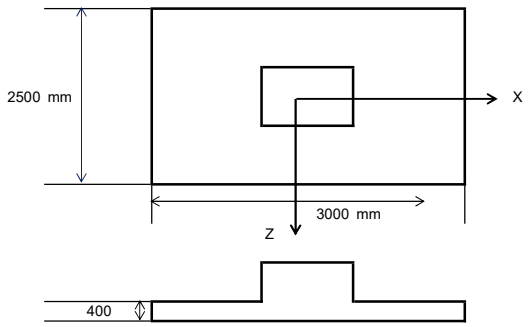
Ma'aden Revision No:	A01	C01	D01
Date	06-04-2020	18-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T	0	1	2
Date	06-04-2020	18-05-2020	12-10-2023
Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

7.3.8 Design of foundation F6 for VGTU Intermediate columns

Finished Ground level	F.G.L	=	-0.500	m	Staad Node no	=	3
Bottom of foundation	B.O.F	=	-3.500	m	Weight of soil	=	334.15 KN
Staad support reaction level		=	-3.500	m	Weight of foundation	=	72.00 KN
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00 KN
Surcharge	Ws	=	0.000	KN/m²	Surcharge	=	0 KN
Length of footing	Lz	=	2.50	m	Pressure on top for tension	=	54.2 KN/m²
Width of footing	Lx	=	3.00	m			
Thickness of footing	D	=	0.40	m			
Depth of pedestal	Bx	=	0.60	m			
Width of pedestal	Bz	=	0.60	m			
Height of Load point from bottom of footing	H	=	0.000	m			
Section Modulus of Pedestal	Zxx	=	3.13	m³			
Section Modulus of Pedestal	Zzz	=	3.75	m³			
Allowable Net Bearing Capacity		=	500	KN/m²			
Gross Bearing Capacity		=	554	KN/m²			
Grade of concrete		=	35	N/mm²			
Grade of steel		=	420	N/mm²			
Clear cover		=	75	mm			
Top Reinfocement		=	12	mm			
Bottom Reinforcement		=	16	mm			
Co-efficient of friction between soil & concrete		=	0.3				
Unit wt of soil		=	18	KN/m³			
Unit wt of concrete		=	24	KN/m³			

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	-19.4	169.292	-1.2	3.31	-0.2	39.5	575.4	88.33	65.12	1	0.16
302 1.0 (DL + TL1) +	-19.8	428.268	-1.2	5.59	-0.391	39.5	834.4	123.57	98.94	1	0.22
303 1.0 (DL + TL1) +	-19.6	420.347	-0.9	3.92	-0.159	39.4	826.5	121.97	98.43	1	0.22
304 1.0 DL+ 0.75 W	-19.5	431.111	17.2	54.69	-1.544	39.1	837.3	139.57	83.70	1	0.25
305 1.0 DL+ 0.75 W	-19.7	290.009	-19.4	-45.90	1.005	39.8	696.2	118.13	67.51	1	0.21
306 1.0 DL+ 0.75 W	10.1	521.797	-0.9	4.94	-0.3	-22.6	927.9	131.33	116.12	1	0.24
307 1.0 DL+ 0.75 W	-49.3	199.31	-1.3	3.85	-0.287	101.6	605.5	109.04	52.42	1	0.20
308 1.0 DL + 0.525	-19.0	364.336	-1.1	4.37	-0.27	38.2	770.5	114.31	91.15	1	0.21
309 1.0 DL + 0.525	-20.2	356.772	-1.1	4.42	-0.267	40.8	762.9	114.01	89.44	1	0.21
310 1.0 DL + 0.525	-19.6	363.473	-0.5	6.09	-0.357	39.4	769.6	115.07	90.17	1	0.21
311 1.0 DL + 0.525	-19.7	357.635	-1.7	2.69	-0.2	39.6	763.8	113.26	90.42	1	0.20
312 0.6 DL + 1.0 WL	-11.5	195.651	23.7	69.04	-1.85	23.3	601.8	108.54	51.94	1	0.20
313 0.6 DL + 1.0 WL	-11.7	7.516	-25.2	-65.07	1.548	24.2	413.7	82.43	27.88	1	0.15
314 0.6 DL + 1.0 WL	27.9	316.566	-0.5	2.71	-0.124	-59.0	722.7	112.97	79.75	1	0.20
315 0.6 DL + 1.0 WL	-51.2	-113.416	-1.0	1.25	-0.175	106.5	292.7	67.83	10.23	1	0.12
316 0.6 DL + 0.7 EL	-10.8	106.618	-0.7	1.95	-0.2	22.0	512.8	74.86	61.88	1	0.14
317 0.6 DL + 0.7 EL	-12.5	96.532	-0.7	2.02	-0.148	25.5	502.7	74.46	59.59	1	0.13
318 0.6 DL + 0.7 EL	-11.6	105.467	0.1	4.25	-0.267	23.6	511.6	75.86	60.57	1	0.14
319 0.6 DL + 0.7 EL	-11.7	97.683	-1.6	-0.28	0.0	23.9	503.8	73.63	60.72	1	0.13

Max P_{max} = 139.57 KN/m²
Max P_{min} = 10.23 KN/m²
Resultant F.O.S = 0.25

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	-27.1	237.0	-1.7	4.63	-0.3	55.37	47.85	15.36
402 1.4 (DL + TL1) +	-27.8	676.1	-1.6	8.26	-0.555	55.25	107.52	72.77
403 1.4 (DL + TL1) +	-27.5	665.0	-1.2	5.92	-0.23	55.20	105.28	72.05
404 1.2 DL+ 1.6 WL	-23.2	481.2	37.7	111.99	-3.032	46.67	112.44	15.88
405 1.2 DL+ 1.6 WL	-23.6	180.2	-40.5	-102.59	2.405	48.16	69.69	-21.65
406 1.2 DL+ 1.6 WL	39.9	674.6	-0.9	5.86	-0.272	-85.01	114.50	65.41
407 1.2 DL+ 1.6 WL	-86.7	-13.3	-1.8	3.52	-0.352	179.84	47.31	-50.86
408 1.2 DL + 1.0 EL	-22.2	337.9	-1.4	4.65	-0.314	44.95	58.52	31.58
409 1.2 DL + 1.0 EL	-24.6	323.5	-1.4	4.74	-0.309	49.88	57.94	28.31
410 1.2 DL + 1.0 EL	-23.3	336.2	-0.2	7.93	-0.48	47.21	59.95	29.70
411 1.2 DL + 1.0 EL	-23.5	325.1	-2.6	1.46	-0.143	47.63	56.51	30.18
412 0.9 DL + 1.6 WL	-17.2	302.9	38.0	110.27	-2.945	34.84	84.96	-4.19
413 0.9 DL + 1.6 WL	-17.6	1.9	-40.2	-104.31	2.492	36.34	43.32	-42.82
414 0.9 DL + 1.6 WL	45.9	496.3	-0.6	4.14	-0.184	-96.83	93.33	39.03
415 0.9 DL + 1.6 WL	-80.8	-191.6	-1.5	1.81	-0.264	168.02	19.83	-70.93
416 0.9 DL + 1.0 EL	-16.3	159.6	-1.1	2.93	-0.227	33.13	31.05	11.50
417 0.9 DL + 1.0 EL	-18.6	145.2	-1.1	3.02	-0.222	38.06	30.47	8.24
418 0.9 DL + 1.0 EL	-17.3	157.9	0.1	6.21	-0.393	35.38	32.48	9.63
419 0.9 DL + 1.0 EL	-17.5	146.8	-2.3	-0.26	-0.056	35.80	29.20	9.94

Max Net factored B.P = 114.50 KN/m²

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Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1.2 m

Moment about Z: Muz = 82.44 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 686.10 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 698.0 mm^2

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 51.67 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 618.59 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 458.5 mm^2

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Effective Depth Along X = 317 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or A_{stmin} = 0.0014bd

A_{stmin} = 570.6 mm^2

A_{req} = 698.0 mm^2

Required Spacing = **288.1** mm

Effective Depth Along X = 301 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or A_{stmin} = 0.0014bd

A_{stmin} = 541.8 mm^2

A_{req} = 458.5 mm^2

Required Spacing = **371.1** mm

Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 1.2 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 694.79 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm^2

Length of cantilever along Z = 0.95 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 643.50 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm^2

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Effective Depth Along X = 319 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or A_{stmin} = 50% of 0.0014bd

Table 24.4.3.2 SBC 304

A_{stmin} = 287.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **393.9** mm

Effective Depth Along Z = 307 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or A_{stmin} = 50% of 0.0014bd

Table 24.4.3.2 SBC 304

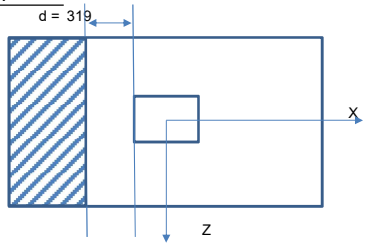
A_{stmin} = 276.3 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **409.3** mm

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	06-04-2020	18-05-2020	12-10-2023	Date	06-04-2020	18-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL	Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

Along X Direction Cantilever

L	=	0.881	m		
Reinforcement ratio ρ_w	=	0.0032			
Effective Depth Along X d	=	319	mm		
Shear Force at d distance V_u	=	100.87	KN		
Allowable shear strength φVc	=	235.75	KN	O.K	
Utility ratio	=	0.43			

Least of

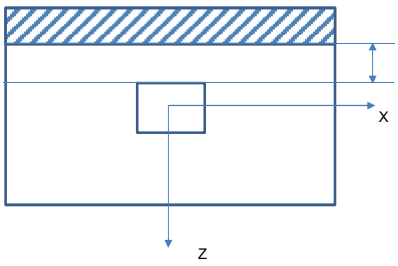
i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete
Mu = 44.43 KNm
φ = 0.75 for shear
φVc = 235.7 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 239.3 KN — *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
= 410.5 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

L	=	0.643	m		
Reinforcement ratio ρ_w	=	0.0033			
Effective Depth Along Z d	=	307	mm		
Shear Force at d distance V_u	=	73.62	KN		
Allowable shear strength φVc	=	230.19	KN	O.K	
Utility ratio	=	0.32			

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

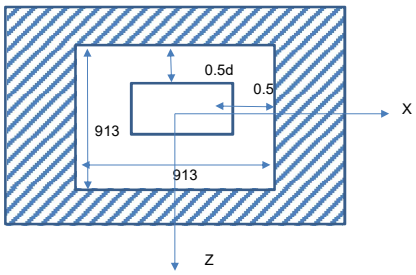
λ = 1 Normal concrete
Mu = 23.67 KNm
φ = 0.75 for shear
φVc = 230.2 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 230.8 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
φVc = 395.0 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	7.5	m2		
Punching area	=	0.83	m2		
λ	=	1			
β	=	1			
αs	=	40			
b0	=	3652.0	mm		

Allowable shear stress least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.67 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face u	=	3.65	m		
Depth at section d	=	313	mm		
Shear Force at section Vu	=	763.3	KN		
Allowable shear strength φVc	=	1673.73	KN	O.K	
Utility ratio		0.46			

Vc = $v_c b_0 d$
= 2231.63 KN
φ = 0.75 for shear
φV_n = φVc
φVc = 1673.7 KN

Global Project ID	900707
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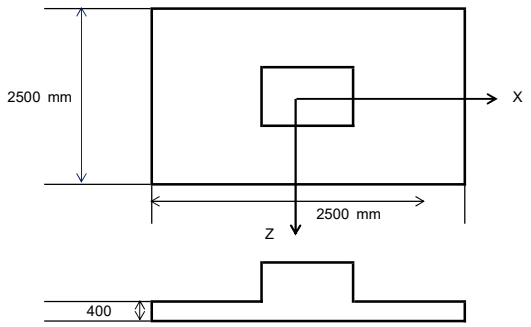
Ma'aden Revision No:	A01	C01	D01
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Outotec / L&T	0	1	2
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7.3.9 Design of foundation F5 for supporting trestle mkd. TR3

Finished Ground level	F.G.L	=	-0.500	m	Staad Node no	=	2
Bottom of foundation	B.O.F	=	-3.500	m	Weight of soil	=	276.12 KN
Staad support reaction level		=	-3.500	m	Weight of foundation	=	60.00 KN
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00 KN
Surcharge	Ws	=	0.000	KN/m^2	Surcharge	=	0 KN
Length of footing	Lz	=	2.50	m	Pressure on top for tension	=	53.8 KN/m^2
Width of footing	Lx	=	2.50	m			
Thickness of footing	D	=	0.40	m			
Depth of pedestal	Bx	=	0.50	m			
Width of pedestal	Bz	=	0.70	m			
Height of Load point from bottom of footing	H	=	0.000	m			
Section Modulus of Pedestal	Zxx	=	2.60	m^3			
Section Modulus of Pedestal	Zzz	=	2.60	m^3			
Allowable Net Bearing Capacity		=	500	KN/m^2			
Gross Bearing Capacity		=	554	KN/m^2			
Grade of concrete		=	35	N/mm^2			
Grade of steel		=	420	N/mm^2			
Clear cover		=	75	mm			
Top Reinfocement		=	12	mm			
Bottom Reinforcement		=	16	mm			
Co-efficient of friction between soil & concrete		=	0.3				
Unit wt of soil		=	18	KN/m^3			
Unit wt of concrete		=	24	KN/m^3			

*inputs



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	0.0	195.082	1.8	6.62	0.0	0.0	531.2	87.54	82.45	1	0.16
302 1.0 (DL + TL1) +	0.0	292.268	-5.3	1.40	0.005	0.0	628.4	101.09	99.99	1	0.18
303 1.0 (DL + TL1) +	0.0	292.196	2.6	7.19	0.002	0.0	628.3	103.29	97.77	1	0.19
304 1.0 DL+ 0.75 W	0.0	377.66	22.7	51.06	-0.005	0.1	713.8	133.84	94.57	1	0.24
305 1.0 DL+ 0.75 W	0.0	158.225	-23.9	-41.31	0.01	-0.1	494.3	94.99	63.20	1	0.17
306 1.0 DL+ 0.75 W	1.7	267.944	-0.6	4.88	0.0	-5.6	604.1	100.67	92.63	1	0.18
307 1.0 DL+ 0.75 W	-1.7	267.944	-0.6	4.88	0.003	5.6	604.1	100.66	92.64	1	0.18
308 1.0 DL + 0.525	0.0	267.975	-0.6	4.88	0.003	0.0	604.1	98.55	94.76	1	0.18
309 1.0 DL + 0.525	0.0	267.913	-0.6	4.87	0.003	0.0	604.0	98.52	94.77	1	0.18
310 1.0 DL + 0.525	0.0	273.311	0.6	7.13	0.003	0.0	609.4	100.25	94.76	1	0.18
311 1.0 DL + 0.525	0.0	262.578	-1.7	2.62	0.0	0.0	598.7	96.80	94.78	1	0.17
312 0.6 DL + 1.0 WL	0.0	263.337	32.1	65.56	-0.009	0.1	599.5	121.12	70.70	1	0.22
313 0.6 DL + 1.0 WL	0.0	-29.244	-29.9	-57.61	0.011	-0.1	306.9	71.26	26.94	1	0.13
314 0.6 DL + 1.0 WL	2.2	117.049	1.1	3.97	0.001	-7.5	453.2	76.90	68.12	1	0.14
315 0.6 DL + 1.0 WL	-2.2	117.049	1.1	3.97	0.001	7.5	453.2	76.89	68.12	1	0.14
316 0.6 DL + 0.7 EL	0.0	117.091	1.1	3.98	0.0	0.0	453.2	74.06	70.97	1	0.13
317 0.6 DL + 0.7 EL	0.0	117.007	1.1	3.97	0.001	0.0	453.1	74.04	70.96	1	0.13
318 0.6 DL + 0.7 EL	0.0	124.204	2.6	6.98	0.001	0.0	460.3	76.33	70.97	1	0.14
319 0.6 DL + 0.7 EL	0.0	109.894	-0.4	0.97	0.0	0.0	446.0	71.73	70.99	1	0.13

Max P_{max} = 133.84 KN/m^2
Max P_{min} = 26.94 KN/m^2
Resultant F.O.S = 0.24

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	0.0	273.1	2.5	9.27	0.0	0.00	47.26	40.14
402 1.4 (DL + TL1) +	0.0	438.3	-8.4	1.26	0.007	-0.05	70.63	69.63
403 1.4 (DL + TL1) +	0.0	438.2	2.6	9.36	0.003	-0.01	73.71	66.52
404 1.2 DL+ 1.6 WL	0.0	516.7	50.2	105.32	-0.014	0.15	123.17	42.18
405 1.2 DL+ 1.6 WL	0.1	48.6	-49.1	-91.75	0.019	-0.17	43.07	-27.52
406 1.2 DL+ 1.6 WL	3.6	282.7	0.6	6.78	0.003	-11.94	52.42	38.04
407 1.2 DL+ 1.6 WL	-3.6	282.7	0.6	6.78	0.003	11.92	52.41	38.05
408 1.2 DL + 1.0 EL	0.0	282.7	0.6	6.79	0.003	-0.07	47.87	42.60
409 1.2 DL + 1.0 EL	0.0	282.6	0.6	6.77	0.003	0.05	47.84	42.60
410 1.2 DL + 1.0 EL	0.0	292.9	2.7	11.08	0.003	-0.02	51.12	42.60
411 1.2 DL + 1.0 EL	0.0	272.5	-1.6	2.49	0.002	-0.01	44.55	42.64
412 0.9 DL + 1.6 WL	0.0	409.6	51.3	104.49	-0.015	0.15	105.73	25.36
413 0.9 DL + 1.6 WL	0.1	-58.5	-48.0	-92.57	0.018	-0.16	26.25	-44.97
414 0.9 DL + 1.6 WL	3.6	175.6	1.6	5.96	0.002	-11.93	34.96	21.22
415 0.9 DL + 1.6 WL	-3.6	175.6	1.6	5.96	0.002	11.92	34.96	21.22
416 0.9 DL + 1.0 EL	0.0	175.6	1.6	5.97	0.002	-0.06	30.42	25.78
417 0.9 DL + 1.0 EL	0.0	175.5	1.6	5.95	0.002	0.06	30.39	25.77
418 0.9 DL + 1.0 EL	0.0	185.8	3.8	10.26	0.002	-0.01	33.67	25.79
419 0.9 DL + 1.0 EL	0.0	165.4	-0.5	1.66	0.001	0.00	27.10	25.82

Max Net factored B.P = 123.17 KN/m^2

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Ma'aden Revision No:	A01	C01	D01
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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1 m

Moment about Z: Muz = 61.59 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 686.10 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 519.3 mm^2

Effective Depth Along X = 317 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or A_{stmin} = 0.0014bd

A_{stmin} = 570.6 mm^2

A_{req} = 519.3 mm^2

Required Spacing = **352.4** mm

Length of cantilever along Z = 0.9 m

Moment about X: Mux = 49.89 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 618.59 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 442.5 mm^2

Effective Depth Along X = 301 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or A_{stmin} = 0.0014bd

A_{stmin} = 541.8 mm^2

A_{req} = 442.5 mm^2

Required Spacing = **371.1** mm

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 1 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 694.79 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm^2

Effective Depth Along X = 319 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or A_{stmin} = 50% of 0.0014bd

Table 24.4.3.2 SBC 304

A_{stmin} = 287.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **393.9** mm

Length of cantilever along Z = 0.9 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 643.50 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm^2

Effective Depth Along Z = 307 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or A_{stmin} = 50% of 0.0014bd

Table 24.4.3.2 SBC 304

A_{stmin} = 276.3 mm^2

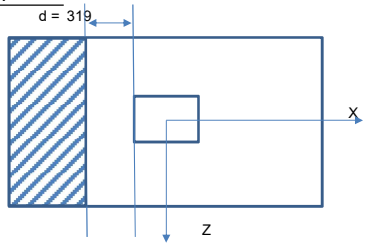
A_{req} = 0.0 mm^2

Required Spacing = **409.3** mm

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



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Along X Direction Cantilever

Reinforcement ratio	ρ_w	=	0.0032							
Effective Depth Along X	d	=	319	mm						
Shear Force at d distance	V_u	=	83.88	KN						
Allowable shear strength	φVc	=	238.48	KN	O.K					
Utility ratio		=	0.35							

L = 0.681 m

Least of

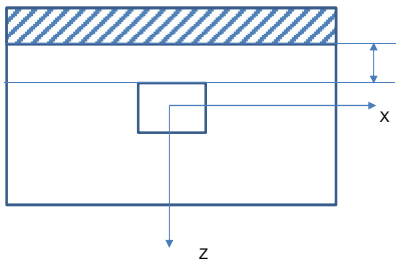
i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ = 1 Normal concrete
Mu = 28.56 KNm
φ = 0.75 for shear
φVc = 238.5 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 239.3 KN — *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
= 410.5 KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

Reinforcement ratio	ρ_w	=	0.0033							
Effective Depth Along Z	d	=	307	mm						
Shear Force at d distance	V_u	=	73.04	KN						
Allowable shear strength	φVc	=	230.77	KN	O.K					
Utility ratio		=	0.32							

L = 0.593 m

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

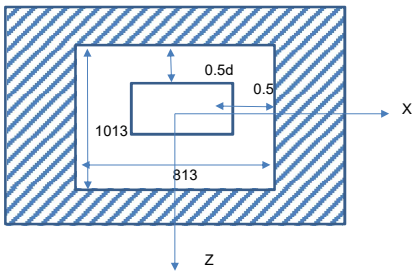
λ = 1 Normal concrete
Mu = 21.66 KNm
φ = 0.75 for shear
φVc = 231.2 KN

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$
φVc = 230.8 KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$
φVc = 395.0 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area = 6.25 m2
Punching area = 0.82 m2
λ = 1
β = 1.4
αs = 40
b0 = 3652.0 mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 2.44 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.67 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.65	m						
Depth at section	d	=	313	mm						
Shear Force at section	Vu	=	668.4	KN						
Allowable shear strength	φVc	=	1673.73	KN	O.K					
Utility ratio			0.40							

Vc = $v_c b_0 d$
= 2231.63 KN
φ = 0.75 for shear
φV_n = φVc
φVc = 1673.7 KN

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	06-04-2020	18-05-2020	12-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T	0	1	2
Date	06-04-2020	18-05-2020	12-10-2023
Revision Tex	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
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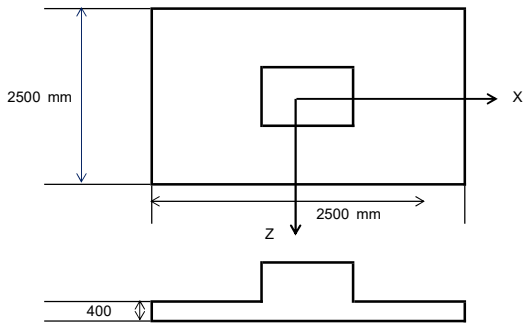
7.3.10 Design of foundation F5 for supporting trestle mkd. TR4

Finished Ground level	F.G.L	=	-0.500	m	Staad Node no	=	1
Bottom of foundation	B.O.F	=	-3.500	m	Weight of soil	=	276.12 KN
Staad support reaction level		=	-3.500	m	Weight of foundation	=	60.00 KN
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00 KN
Surcharge	Ws	=	0.000	KN/m²	Surcharge	=	0 KN
Length of footing	Lz	=	2.50	m	Pressure on top for tension	=	53.8 KN/m²
Width of footing	Lx	=	2.50	m			
Thickness of footing	D	=	0.40	m			
Depth of pedestal	Bx	=	0.50	m			
Width of pedestal	Bz	=	0.70	m			
Height of Load point from bottom of footing	H	=	0.000	m			
Section Modulus of Pedestal	Zxx	=	2.60	m³			
Section Modulus of Pedestal	Zzz	=	2.60	m³			
Allowable Net Bearing Capacity		=	500	KN/m²			
Gross Bearing Capacity		=	554	KN/m²			
Grade of concrete		=	35	N/mm²			
Grade of steel		=	420	N/mm²			
Clear cover		=	75	mm			
Top Reinfocement		=	12	mm			
Bottom Reinforcement		=	16	mm			
Co-efficient of friction between soil & concrete		=	0.3				
Unit wt of soil		=	18	KN/m³			
Unit wt of concrete		=	24	KN/m³			

*inputs

$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*



CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
1 DEAD LOAD\tDL	0.0	204.082	8.6	13.13	0.0	0.1	540.2	91.50	81.36	1	0.17
2 LIVE LOAD\tLL	0.0	71.725	3.0	2.08	0.001	0.0	407.8	66.06	64.45	1	0.12
3 MATERIAL LOAD\	0.0	69.492	2.9	2.02	0.001	0.0	405.6	65.68	64.11	1	0.12
4 WIND LOAD ALO	0.0	-194.213	24.7	52.41	-0.003	-0.1	141.9	42.85	2.56	1	0.08
5 WIND LOAD ALO	0.0	267	-49.3	-104.06	0.003	0.1	603.1	136.48	56.52	1	0.25
6 WIND LOAD ACR	4.6	0	0.0	0.00	0.0	-15.5	336.1	59.71	47.84	1	0.11
7 WIND LOAD ACR	-4.6	0	0.0	0.00	0	15.5	336.1	59.71	47.84	1	0.11
8 TEMP LOAD \tTL	0.0	0.013	13.0	8.80	0.001	0.0	336.1	57.16	50.40	1	0.10
9 TEMP LOAD \tTL	0.0	-0.013	-13.0	-8.80	-0.001	0.0	336.1	57.16	50.39	1	0.10
10 SEISMIC LOAD /	0.0	0.02	0.0	0.00	0	-0.1	336.1	53.82	53.75	1	0.10
11 SEISMIC LOAD /	0.0	-12.706	2.2	4.63	0.0	0.0	323.4	53.54	49.95	1	0.10
301 1.0 DL	0.0	204.082	8.6	13.13	0.003	0.1	540.2	91.50	81.36	1	0.17
302 1.0 (DL + TL1) +	0.0	345.312	27.4	26.03	0.006	0.1	681.4	119.07	98.98	1	0.21
303 1.0 (DL + TL1) +	0.0	345.286	1.4	8.43	0.005	0.1	681.4	112.30	105.75	1	0.20
304 1.0 DL+ 0.75 W	0.0	164.335	31.5	55.51	0.003	0.1	500.5	101.42	58.73	1	0.18
305 1.0 DL+ 0.75 W	0.0	510.245	-24.0	-61.84	0.0	0.1	846.4	159.22	111.62	1	0.29
306 1.0 DL+ 0.75 W	3.4	309.995	13.0	16.20	0.005	-11.5	646.1	114.01	92.75	1	0.21
307 1.0 DL+ 0.75 W	-3.5	309.995	13.0	16.20	0.005	11.7	646.1	114.09	92.66	1	0.21
308 1.0 DL + 0.525	0.0	310.005	13.0	16.20	0.0	0.1	646.1	109.63	97.13	1	0.20

Max P_{max} = 159.22 KN/m²
Max P_{min} = 2.56 KN/m²
Resultant F.O.S = 0.29

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
316 0.6 DL + 0.7 EL	0.0	122.5	5.2	7.88	0.0	-0.02	22.62	16.56
317 0.6 DL + 0.7 EL	0.0	122.4	5.2	7.88	0.002	0.11	22.66	16.52
318 0.6 DL + 0.7 EL	0.0	113.6	6.7	11.12	0.003	0.07	22.46	13.87
319 0.6 DL + 0.7 EL	0.0	131.3	3.6	4.64	0	0.02	22.80	19.23
405 1.2 DL+ 1.6 WL	-0.1	742.7	-65.7	-148.69	0.01	0.19	176.00	61.66
406 1.2 DL+ 1.6 WL	7.3	315.5	13.2	17.80	0.005	-24.61	66.77	34.19
407 1.2 DL+ 1.6 WL	-7.4	315.5	13.2	17.80	0.005	24.84	66.86	34.11
408 1.2 DL + 1.0 EL	0.0	315.5	13.2	17.80	0.005	0.02	57.33	43.64
409 1.2 DL + 1.0 EL	-0.1	315.5	13.2	17.80	0.005	0.20	57.39	43.56
410 1.2 DL + 1.0 EL	0.0	302.8	15.4	22.43	0.007	0.15	57.12	39.78
411 1.2 DL + 1.0 EL	0.0	328.2	11.0	13.18	0.003	0.07	57.60	47.43
412 0.9 DL + 1.6 WL	0.0	-127.1	47.2	95.68	-0.002	-0.02	16.41	-57.08
413 0.9 DL + 1.6 WL	0.0	610.9	-71.2	-154.68	0.007	0.15	157.19	38.29
414 0.9 DL + 1.6 WL	7.4	183.7	7.7	11.82	0.003	-24.66	43.39	15.38
415 0.9 DL + 1.6 WL	-7.4	183.7	7.7	11.82	0.003	24.79	43.45	15.33
416 0.9 DL + 1.0 EL	0.0	183.7	7.7	11.82	0.003	-0.02	33.94	24.85
417 0.9 DL + 1.0 EL	0.0	183.7	7.7	11.82	0.003	0.15	33.98	24.79
418 0.9 DL + 1.0 EL	0.0	171.0	9.9	16.44	0.005	0.11	33.71	21.00
419 0.9 DL + 1.0 EL	0.0	196.4	5.5	7.19	0.001	0.03	34.19	28.65

Max Net factored B.P = 176.00 KN/m²

Global Project ID	900707
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Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	06-04-2020	18-05-2020	12-10-2023	Date	06-04-2020	18-05-2020	12-10-2023
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Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 1 m

Moment about Z: Muz = 88.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 686.10 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 745.9 mm^2

Length of cantilever along Z = 0.9 m

Moment about X: Mux = 71.28 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 618.59 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 635.2 mm^2

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***

Effective Depth Along X = 317 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or A_{stmin} = 0.0014bd

A_{stmin} = 570.6 mm^2

A_{req} = 745.9 mm^2

Required Spacing = **269.6** mm

Effective Depth Along X = 301 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*

or A_{stmin} = 0.0014bd

A_{stmin} = 541.8 mm^2

A_{req} = 635.2 mm^2

Required Spacing = **316.5** mm

Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 1 m

Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uzlim} = 694.79 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm^2

Length of cantilever along Z = 0.9 m

Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

M_{uxlim} = 643.50 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

A_{st} = 0.0 mm^2

Provide top Reinforcement = ***12 mm dia @ 200 c/c***

Effective Depth Along X = 319 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or A_{stmin} = 50% of 0.0014bd

Table 24.4.3.2 SBC 304

A_{stmin} = 287.1 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **393.9** mm

Effective Depth Along Z = 307 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005}$ = 0.375

Min reinforcement greater of

A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom

or A_{stmin} = 50% of 0.0014bd

Table 24.4.3.2 SBC 304

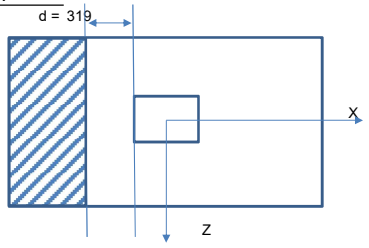
A_{stmin} = 276.3 mm^2

A_{req} = 0.0 mm^2

Required Spacing = **409.3** mm

Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	06-04-2020	18-05-2020	12-10-2023	Date	06-04-2020	18-05-2020	12-10-2023
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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

Along X Direction Cantilever

Reinforcement ratio	ρ_w	=	0.0032						
Effective Depth Along X	d	=	319	mm					
Shear Force at d distance	V_u	=	119.86	KN					
Allowable shear strength	φVc	=	238.48	KN	O.K				
Utility ratio		=	0.50						

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ	=	1	Normal concrete	
Mu	=	40.81	KNm	
φ	=	0.75	for shear	
φVc	=	238.5	KN	

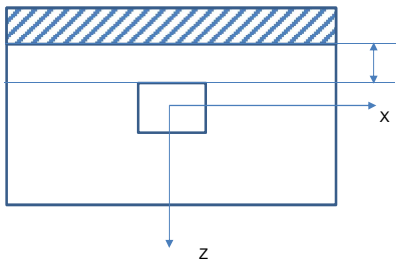
ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$

φVc	=	239.3	KN — *Table 22.5.5.1 SBC 304*

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

	=	410.5	KN

ONE WAY SHEAR CKECK



Along Z Direction Cantilever

Reinforcement ratio	ρ_w	=	0.0033						
Effective Depth Along Z	d	=	307	mm					
Shear Force at d distance	V_u	=	104.37	KN					
Allowable shear strength	φVc	=	230.77	KN	O.K				
Utility ratio		=	0.45						

Least of

i) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$ — *Table 22.5.5.1 SBC 304*

λ	=	1	Normal concrete	
Mu	=	30.95	KNm	
φ	=	0.75	for shear	
φVc	=	231.2	KN	

ii) $V_c = \left(0.16\lambda\sqrt{f_c'} + 17\rho_w\right)b_w d$

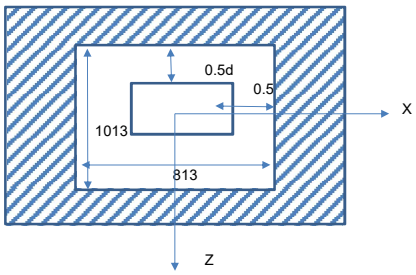
φVc	=	230.8	KN

iii) $V_c = 0.29\lambda\sqrt{f_c'}b_w d$

φVc	=	395.0	KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	6.25	m2
Punching area	=	0.82	m2
λ	=	1	
β	=	1.4	
αs	=	40	
b0	=	3652.0	mm

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

$v_c = 0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm²

$v_c = 0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 2.44 N/mm²

$v_c = 0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.67 N/mm²

Design vc = 1.95 N/mm²

Check at 0.5d from column face

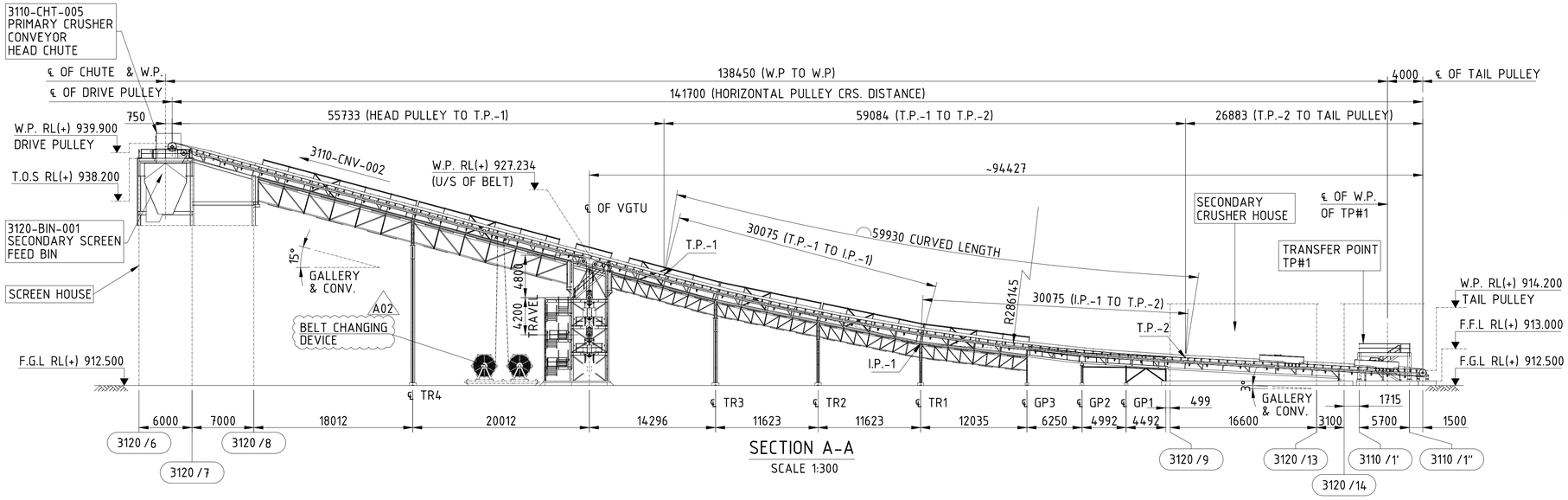
Perimeter at 0.5d from column face	u	=	3.65	m		Vc	=	$v_c b_0 d$
Depth at section	d	=	313	mm			=	2231.63 KN
Shear Force at section	Vu	=	955.1	KN		φ	=	0.75 for shear
Allowable shear strength	φVc	=	1673.73	KN	O.K	φV_n	=	φVc
Utility ratio			0.57			φVc	=	1673.7 KN

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	07-04-2020	18-05-2020	12-10-2023				Date	07-04-2020	18-05-2020	12-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTIO	AS BUILT			
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Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 1 **MECHANICAL GENERAL ARRANGEMENT DRAWINGS**

DRAWING NUMBER: MD-216-3110-EG-ME-LPX-1004 | REV: A02



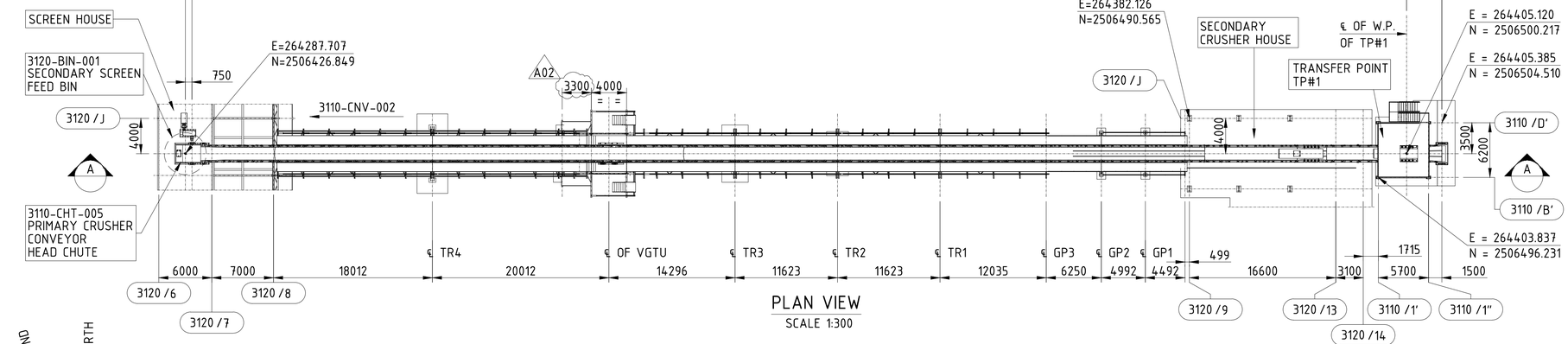
SECTION A-A
SCALE 1:300

ABBREVIATION

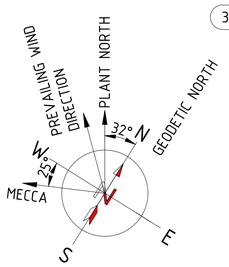
CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING
PROJ.:	PROJECTION		

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+) 913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+) 912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

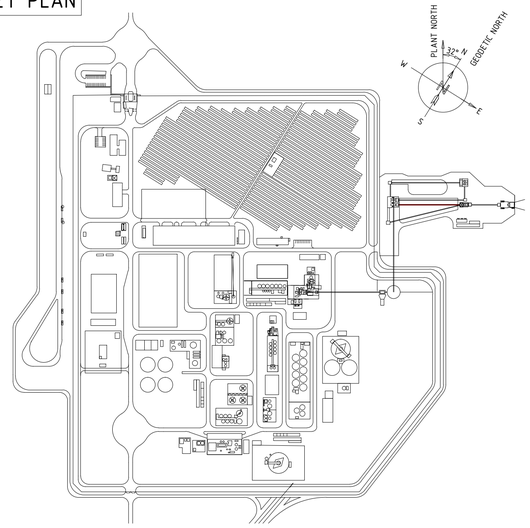
CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL



PLAN VIEW
SCALE 1:300



KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-ELV-1006	CONV. 3110-CNV-002 - ELEVATION	A02	16/03/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	SM	BRD	BD	BGR	
MD-216-3110-EG-ME-ELV-1005	CONV. 3110-CNV-002 - ELEVATION			COMMENTS DT. 20.02.2020 & REVISED AS MARKED A02					
MD-216-3110-EG-ME-ELV-1004	CONV. 3110-CNV-002 - ELEVATION	A01	22/01/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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LARSEN & TOUBRO DRAWING NUMBER	019045_100_H_G_16128_001		REVISION 1
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31135	DOCUMENT ID OU100669015	REVISION 1

DRAWN	HKD	18/01/2020
CHECKED	BRD	20/01/2020
ENGINEER	BD	20/01/2020
APPROVED	BGR	22/01/2020
EMPLOYER		
SCALE	1:300	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRIMARY CRUSHING
CONV. 3110-CNV-002 - PLAN VIEW

DRAWING NUMBER: MD-216-3110-EG-ME-LPX-1004 | REV: A02

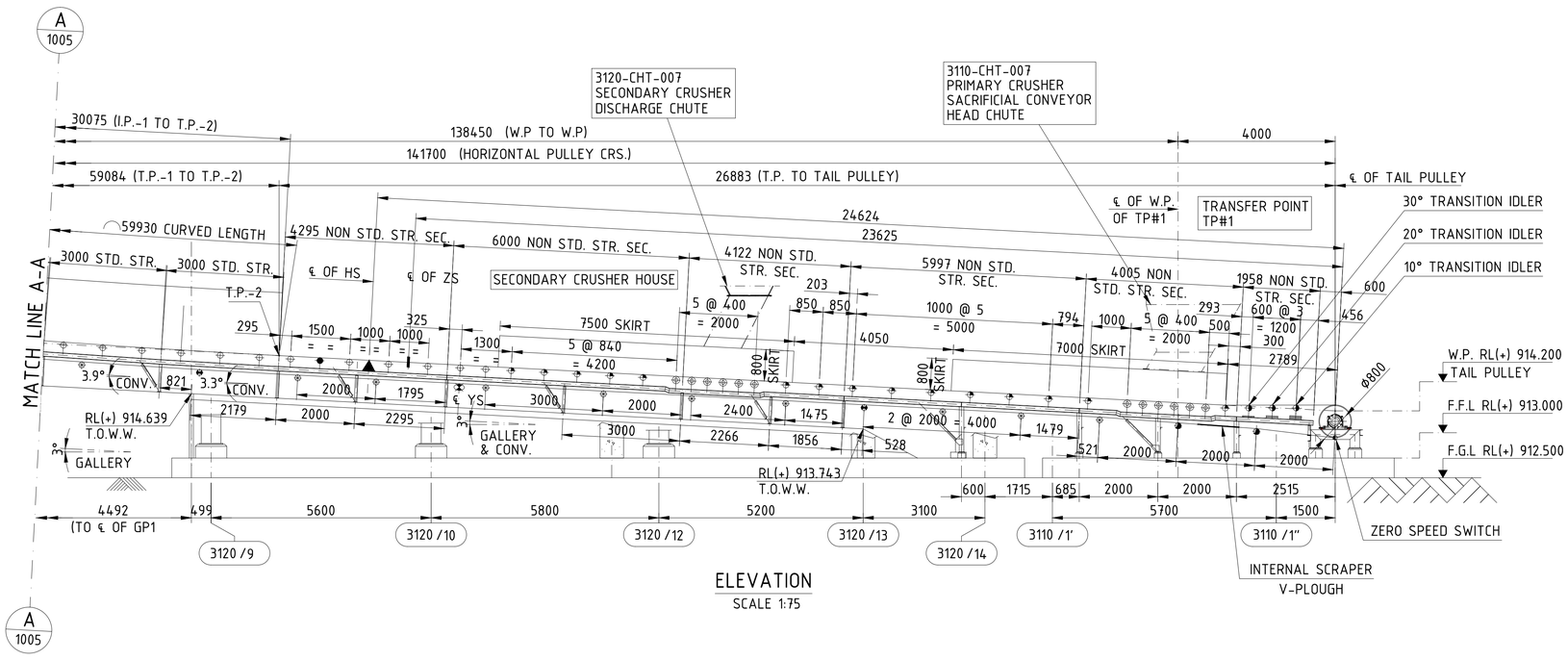
DRAWING NUMBER: MD-216-3110-EG-ME-ELV-1004 | REV: A02

ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING
PROJ.:	PROJECTION		

CONVEYOR CHARACTERISTICS

	PARTICULARS	SPECIFICATION	QUANTITY
MISCELLANEOUS EQUIPMENT	TAKE-UP / TRAVEL	VGTU / 4200 mm	
	TAKE-UP LOAD	~6760 kg.(APPROXIMATELY) RCC BLOCKS, 40 kg EACH	
	SCRAPER AT DISCHARGE PULLEY	EXTERNAL / MAIN & PRE CLEANER	1 NO. + 1 NO.
	SCRAPER AT TAKE-UP	INTERNAL V-PLOUGH	1 NO.
	SCRAPER AT TAIL PULLEY	INTERNAL V-PLOUGH	1 NO.
	BELT SWAY SWITCH (ZS)	@ 50 m INTERVAL	3 PAIRS
	PULL CORD SWITCH (HS)	@ 30 m INTERVAL	5 PAIRS
	ZERO SPEED SWITCH (SS)	AT TAIL PULLEY	1 NO.
	BELT RIP SWITCH (YS)	AFTER FEEDING ZONE	1 NO.
	DECK PLATE	3.15 mm THK. CONTINUOUS	1 LOT.
	BELT HOOD	GALVANIZED CORRUGATED SHEET	1 LOT.



ELEVATION
SCALE 1:75

CONVEYOR CHARACTERISTICS

		PARTICULARS	SPECIFICATION	QUANTITY
CONVEYOR		EQUIPMENT TAG NO.	3110-CNV-002	1 NO.
		EQUIPMENT DESCRIPTION	PRIMARY CRUSHER CONVEYOR	
		LOCATION	SECONDARY CRUSHER HOUSE TO SCREEN HOUSE	
		CAPACITY RATED / DESIGN (t/h)	1473 / 1630	
		BELT WIDTH (mm)	1600	
		BELT SPEED (m/sec)	1.668	
		TRIPPER FIXED / MOVABLE	N.A.	
		CONVEYOR LENGTH / LIFT (mm)	141700 / 25700	
MATERIAL		MATERIAL	GOLD ORE : FRESH ORE / OXIDE ORE	
		BULK DENSITY (t/m³)	1.6	
		MAX. LUMP SIZE (mm)	(-)340	
		MOISTURE CONTENT (%)	2	
		SURCHARGE ANGLE (deg.)	20	
BELTING		MAX. RUNNING TENSION (T1 kN)	137.319	
		SLACK SIDE TENSION (T2 kN)	43.313	
		TAIL TENSION (T0 kN)	42.627	
		BELT DESIGNATION	EP-1250 / 4 (HEAVY DUTY)	
		TOP / BOTTOM COVER THICKNESS (mm)	8 / 3	
		COVER GRADE / FABRIC	DIN-X / EP	
PULLEYS		DISCHARGE WITH DRIVE (MKD. X DIA.)	D-1 x 1000 mm	1 NO.
		DISCHARGE WITHOUT DRIVE (MKD. X DIA.)	N.A.	
		SNUB (MKD. X DIA.)	ND-2 x 630 mm	1 NO.
		TAIL (MKD. X DIA.)	ND-1 x 800 mm	1 NO.
		TAKE-UP (MKD. X DIA.)	ND-1 x 800 mm	1 NO.
		DEFLECTOR (MKD. X DIA.)	DR-1 x 273	1 NO.
		TRIPPER DISCHARGE (MKD. X DIA.)	N.A.	
		TRIPPER BEND (MKD. X DIA.)	N.A.	
		BEND (MKD. X DIA.)	ND-1 x 800 mm	2 NOS.
		LAGGING (mm)	DRIVE: 16thk CERAMIC	
			NON-DRIVE: 16thk PLAIN RUBBER	
BEARING/SLEEVE PLUMMER BLOCK		DISCHARGE WITH DRIVE	TBA	2 SETS
		DISCHARGE WITHOUT DRIVE	N.A.	
		SNUB	TBA	2 SETS
		TAIL	TBA	2 SETS
		TAKE-UP	TBA	2 SETS
		DEFLECTOR	TBA	2 SETS
		BEND	TBA	4 SETS
		TRIPPER DISCHARGE	N.A.	
		TRIPPER BEND	N.A.	
IDLER		TROUGHING (DIA. X deg.)	152.4 x 35°	113 NOS.
		TROUGHING RETRACTABLE (DIA. X deg.)	152.4 x 45°	14 NOS.
		TROUGHING TRAINER (DIA. X deg.)	152.4 x 35°	8 NOS.
		RETURN (DIA. X deg.)	152.4 x 10°	36 NOS.
		SELF ALIGNING RETURN TRAINER INVERTED IDLER (DIA. X deg.)	152.4 x 10°	7 NOS.
		IMPACT RETRACTABLE (DIA. X deg.)	190 x 45°	12 NOS.
		TRANSITION (ADJUSTABLE) (DIA. X deg.)	152.4 x 10°/20°/30°	(1+2+2) NOS.
		RETURN SELF CLEANING (DIA. X deg.)	190 x 0°	2 NOS.
		NON MAGNETIC TROUGHING (DIA. X deg.)	N.A.	
		WEIGHING IDLER (DIA. X deg.)	N.A.	
		SPECIAL MACHINED IDLER (DIA. X deg.)	N.A.	
		RETURN FLAT IDLER (DIA. X deg.)	152.4 x 0°	4 NOS.
DRIVE	MOTOR	RATING kW/rpm (SYN)	230 / 1800	1 NO.
		FRAME SIZE	TBA	
		TYPE	SQUIRREL CAGE	
		SUPPLY	4.16kV, 60Hz	
	REDUCER	TYPE	BEVEL HELICAL	1 NO.
		MODEL NO./COOLING/RATIO	TBA	
		RATIO	56	
		HOLD BACK	YES	
	BRAKE	TYPE	ELECTRO HYDRAULIC THRUSTER	1 NO.
		MODEL NO.	TBA	
		DRAM DIAMETER	TBA	
	COUPLING HIGH SPEED	TYPE	FLUID COUPLING (SCOOP TYPE)	1 NO.
		MODEL NO.	TBA	
	COUPLING LOW SPEED	TYPE	GEAR	1 NO.
		MODEL NO.	TBA	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-ELV-1006	CONV. 3110-CNV-002 - ELEVATION	A02	16/03/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	SM	BRD	BD	BRD	
MD-216-3110-EG-ME-ELV-1005	CONV. 3110-CNV-002 - ELEVATION			COMMENTS DT. 20.02.2020 & REVISED AS MARKED A02					
MD-216-3110-EG-ME-LPX-1004	CONV. 3110-CNV-002 - PLAN VIEW	A01	22/01/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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OUTOTEC DRAWING NUMBER: 900707 __GAU01 __ZC01 __MTL01 __31139 | DOCUMENT ID: OU100669019 | REVISION: 1

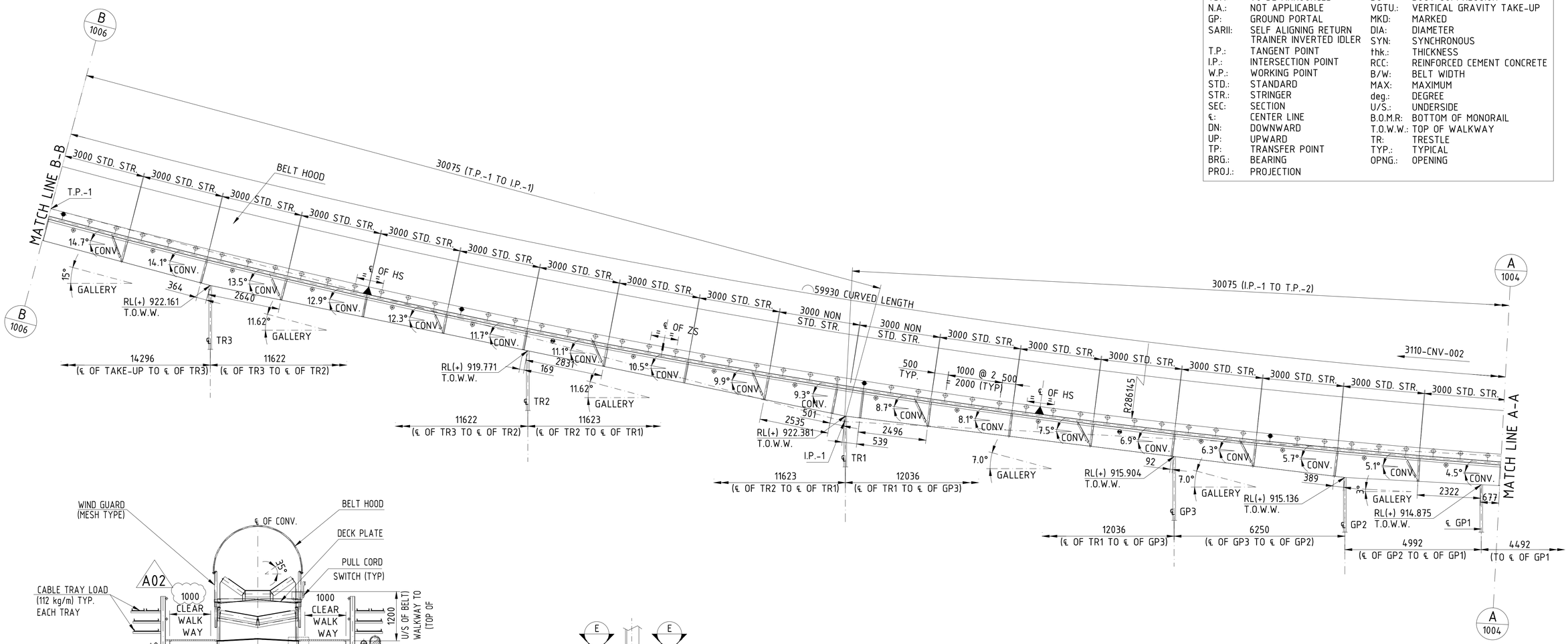
DRAWN	HKD	18/01/2020
CHECKED	BRD	20/01/2020
ENGINEER	BD	20/01/2020
APPROVED	BGR	22/01/2020
EMPLOYER		
SCALE	1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRIMARY CRUSHING
CONV. 3110-CNV-002 - ELEVATION

DRAWING NUMBER: MD-216-3110-EG-ME-ELV-1004 | REV: A02

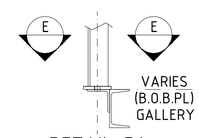
DRAWING NUMBER	REV
MD-216-3110-EG-ME-ELV-1005	A02

ABBREVIATION			
CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING
PROJ.:	PROJECTION		



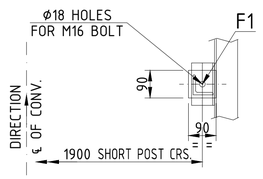
TYPICAL CONVEYOR SECTION ON GALLERY (1600 B/W)

SCALE 1:50



DETAIL-D1

SCALE 1:10



VIEW E-E

SCALE 1:10

TABLE-1 : LOAD FOR CONVEYOR SHORT POST

FOR CONV. 3110-CNV-002

POINT	VERTICAL LOAD IN "t"		HORIZONTAL LOAD IN "t"	
	DEAD LOAD	LIVE LOAD	ALONG	ACROSS
F1	(+)0.65	(+)0.80	(±)0.15	(±)0.08

SIGN CONVENTION
UPWARD (-)
DOWNWARD (+)
TOWARDS DRIVE PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

Note: Wind load & seismic to be considered during civil & structural design A02

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-ELV-1006	CONV. 3110-CNV-002 - ELEVATION	A02	16/03/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	SM	BRD	BD	BGR	
MD-216-3110-EG-ME-ELV-1004	CONV. 3110-CNV-002 - ELEVATION			COMMENTS DT. 20.02.2020 & REVISED AS MARKED A02					
MD-216-3110-EG-ME-LPX-1004	CONV. 3110-CNV-002 - PLAN VIEW	A01	22/01/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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LARSEN & TOUBRO DRAWING NUMBER	019045_100_H_G_16128_003
REVISION	1
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31138
DOCUMENT ID	OU100669018
REVISION	1

DRAWN	HKD	18/01/2020
CHECKED	BRD	20/01/2020
ENGINEER	BD	20/01/2020
APPROVED	BGR	22/01/2020
EMPLOYER		
SCALE	1:10, 1:50, 1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRIMARY CRUSHING
CONV. 3110-CNV-002 - ELEVATION

DRAWING NUMBER	REV
MD-216-3110-EG-ME-ELV-1005	A02

DRAWING NUMBER MD-216-3110-EG-ME-ELV-1006 REV A02

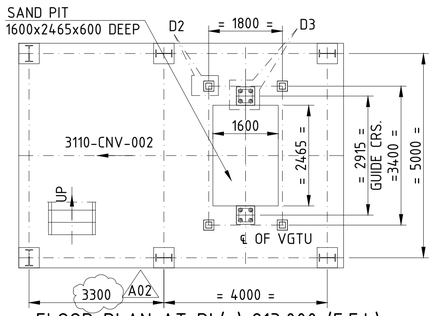
SIGN CONVENTION

UPWARD (-)
DOWNWARD (+)
TOWARDS DRIVE PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

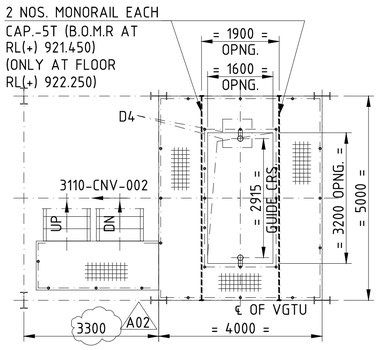
TABLE-2 : LOAD FOR TAKE UP BEND FRAME

FOR CONV. 3110-CNV-002

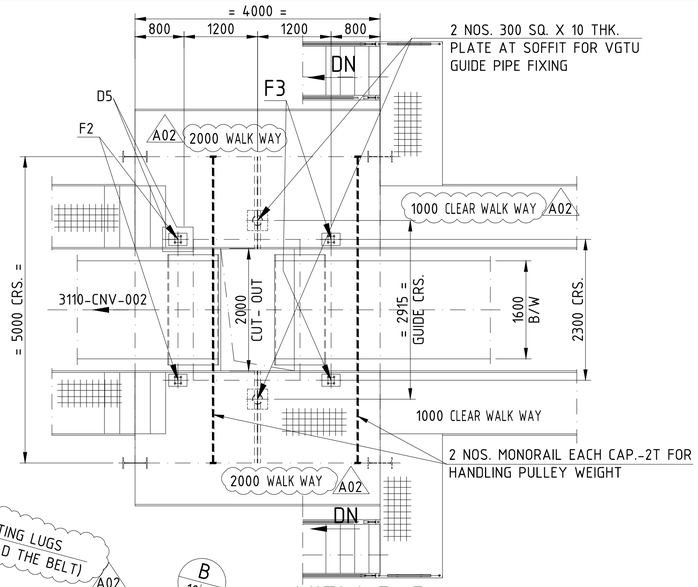
POINTS	VERTICAL LOAD (IN "t")	HORIZONTAL LOAD	
		ALONG (IN "t")	ACROSS (IN "t")
F2	(+) 4.00	(+) 1.50	(±) 0.40
F3	(+) 4.00	(-) 1.50	(±) 0.40



FLOOR PLAN AT RL(+) 913.000 (F.F.L)
(VGTU TOWER)
SCALE 1:75

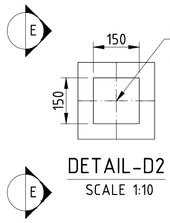


FLOOR PLAN AT RL(+) 922.250 T.O.S
FLOOR PLAN AT RL(+) 919.250 T.O.S
FLOOR PLAN AT RL(+) 916.250 T.O.S
(VGTU TOWER)
SCALE 1:75



VIEW B-B
SCALE 1::50

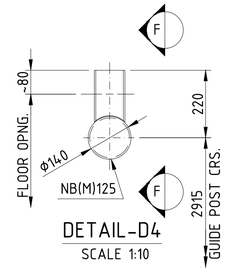
DETAIL-D5
SCALE 1:10



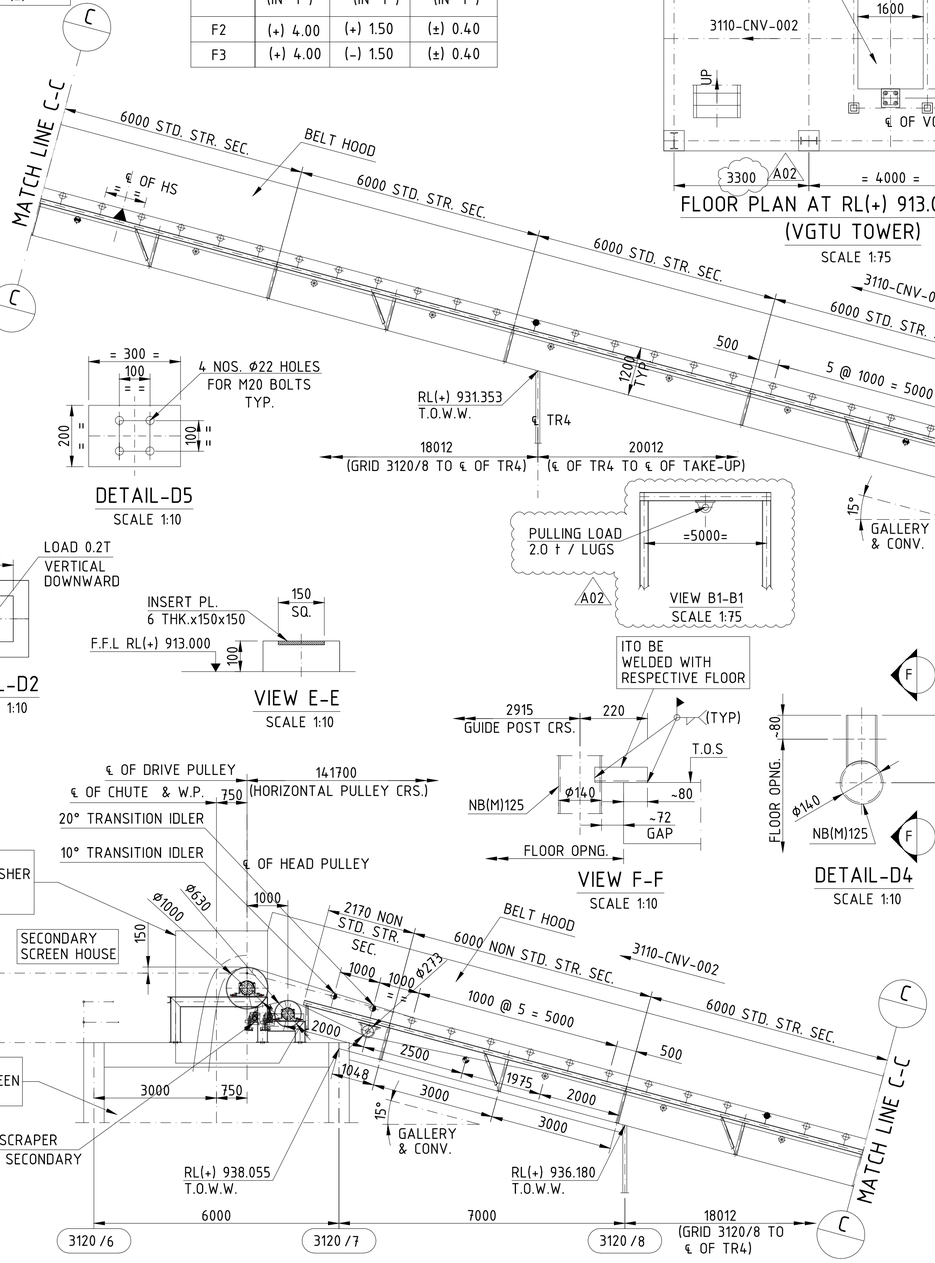
DETAIL-D2
SCALE 1:10

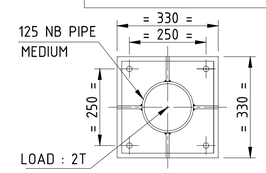
VIEW E-E
SCALE 1:10

VIEW F-F
SCALE 1:10

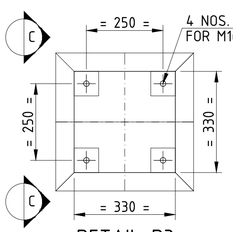


DETAIL-D4
SCALE 1:10

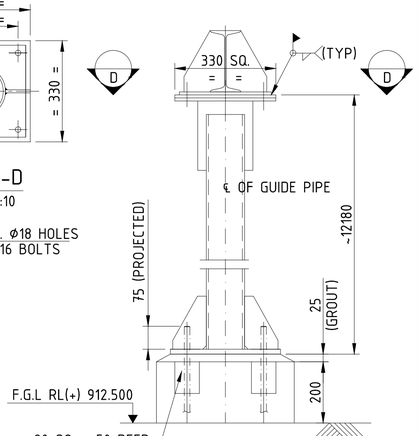




VIEW D-D
SCALE 1:10



DETAIL-D3
SCALE 1:10



VIEW C-C
SCALE 1:10

ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
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SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
T.P.:	TANGENT POINT	SYN:	SYNCHRONOUS
I.P.:	INTERSECTION POINT	thk.:	THICKNESS
W.P.:	WORKING POINT	RCC:	REINFORCED CEMENT CONCRETE
STD.:	STANDARD	B/W:	BELT WIDTH
STR.:	STRINGER	MAX:	MAXIMUM
SEC:	SECTION	deg.:	DEGREE
℄:	CENTER LINE	U/S.:	UNDERSIDE
DN:	DOWNWARD	B.O.M.R:	BOTTOM OF MONORAIL
UP:	UPWARD	T.O.W.W.:	TOP OF WALKWAY
TP:	TRANSFER POINT	TR:	TRESTLE
BRG.:	BEARING	TYP.:	TYPICAL
PROJ.:	PROJECTION	OPNG.:	OPENING

DOCUMENT NUMBER	REFERENCE DOCUMENTS			REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-ELV-1005	CONV. 3110-CNV-002 - ELEVATION	A02	16/03/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	SM	BRD	BD	BGR	
MD-216-3110-EG-ME-ELV-1004	CONV. 3110-CNV-002 - ELEVATION			COMMENTS DT. 20.02.2020 & REVISED AS MARKED △A02					
MD-216-3110-EG-ME-LPX-1004	CONV. 3110-CNV-002 - PLAN VIEW	A01	22/01/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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LARSEN & TOUBRO DRAWING NUMBER		REVISION
O19045_100_H_G_16128_004		1
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707__GAU01__ZC01__MTL01__31137	OU100669017	1

DRAWN	HKD	18/01/2020
CHECKED	BRD	20/01/2020
ENGINEER	BD	20/01/2020
APPROVED	BGR	22/01/2020
EMPLOYER		
SCALE	1:10, 1:75	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
PRIMARY CRUSHING
CONV. 3110-CNV-002 - ELEVATION

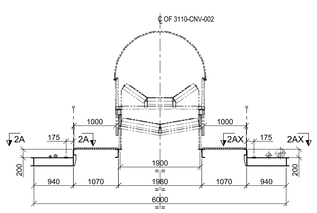
DRAWING NUMBER: MD-216-3110-EG-ME-ELV-1006 REV: A02

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

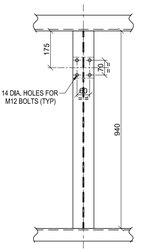
Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	07-04-2020	18-05-2020	12-10-2023				Date	07-04-2020	18-05-2020	12-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 2 **STRUCTURAL GENERAL ARRANGEMENT DRAWINGS CONTAINING BASE PLATE DETAILS**

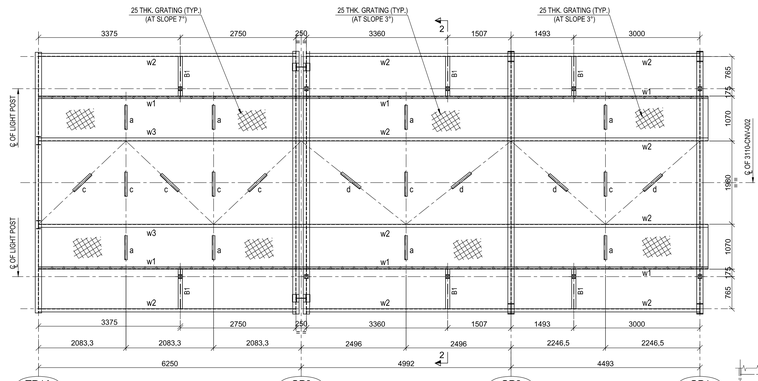
DRAWING NUMBER: MD-216-3110-EG-ST-GAD-1003 | REV: A01



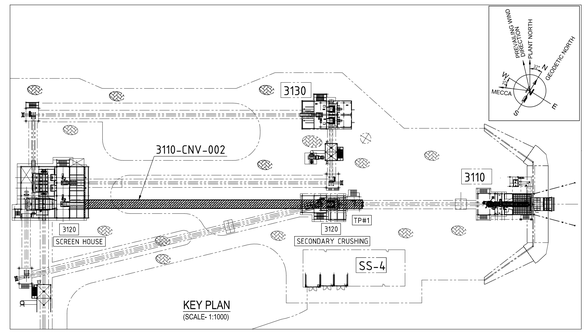
SECTION 2-2
SCALE - 1:50



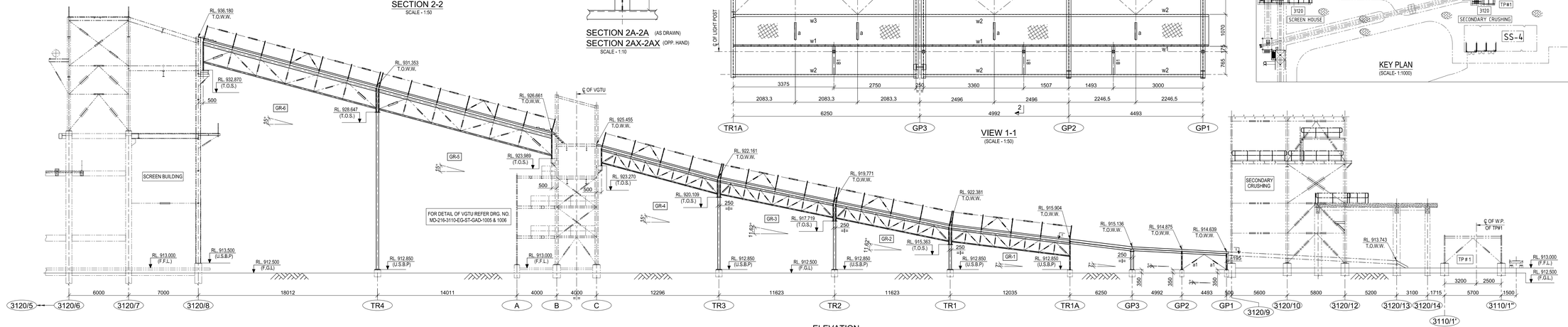
SECTION 2A-2A (AS DRAWN)
SECTION 2AX-2AX (OPP. HAND)
SCALE - 1:10



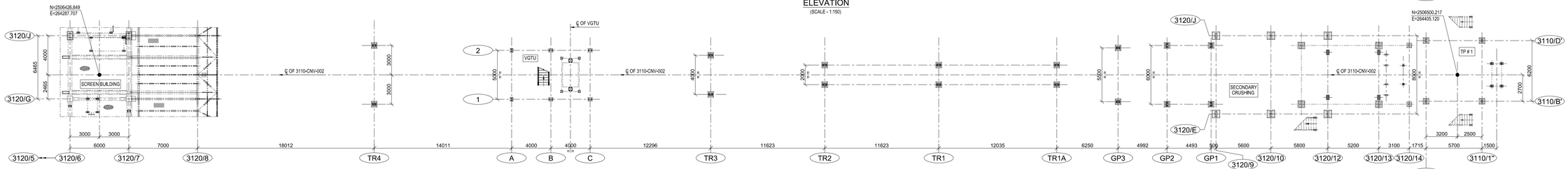
VIEW 1-1
(SCALE - 1:50)



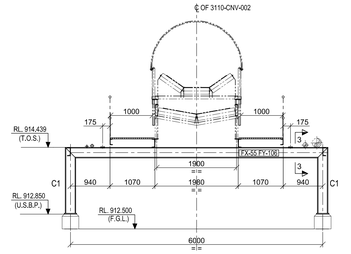
KEY PLAN
(SCALE- 1:1000)



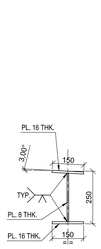
ELEVATION
(SCALE - 1:150)



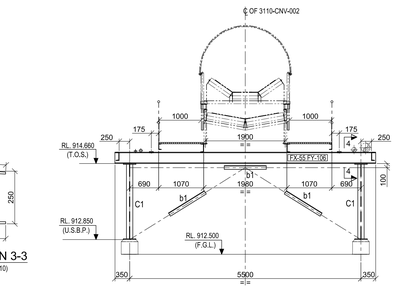
PLAN AT BASE PLATE LEVEL
(SCALE - 1:150)



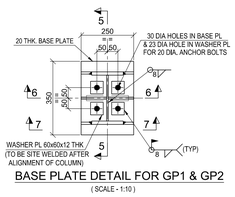
ELEVATION OF GROUND PORTAL MKD. GP1
(SCALE 1:50)



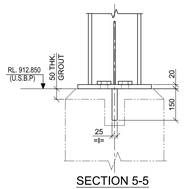
SECTION 3-3
(SCALE 1:10)



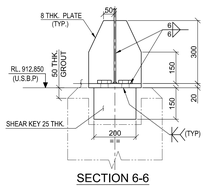
ELEVATION OF GROUND PORTAL MKD. GP3
(SCALE 1:50)



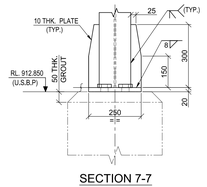
BASE PLATE DETAIL FOR GP1 & GP2
(SCALE - 1:10)



SECTION 5-5
(SCALE - 1:10)



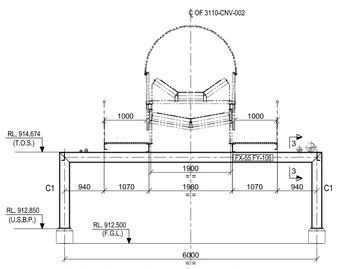
SECTION 6-6
(SCALE - 1:10)



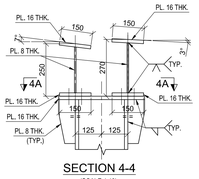
SECTION 7-7
(SCALE - 1:10)

SCHEDULE OF MEMBERS FOR GP-1 TO GP3					
MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	REMARKS
C1	1. PL. 150X16 2. PL. 218X8	I (2, 1)	-	-	GUSSET PLATE 8 THK.
a	L 50X50X6	L	-	-	
a1	2L 50X50X6	⊥	-	-	
b1	L 65X65X6	L	-	-	
c	L 75X75X8	L	-	-	
d	L 90X90X8	L	-	-	
w1	UPN 140	[	-	-	
w2	UPN 200	[	-	-	
w3	UPN 240	[	-	-	
B1	IPE 200	I	-	-	

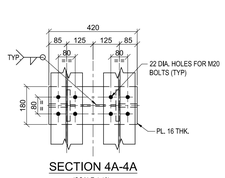
GRADE OF MATERIAL IS S275 JR
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
ALL MEMBER CONNECTIONS SHOULD BE DESIGN FOR A MINIMUM FORCE OF 50KN.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.



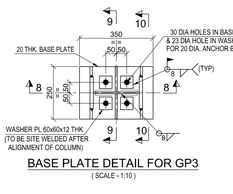
ELEVATION OF GROUND PORTAL MKD. GP2
(SCALE 1:50)



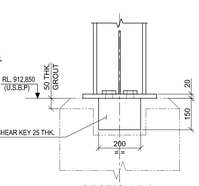
SECTION 4-4
(SCALE 1:10)



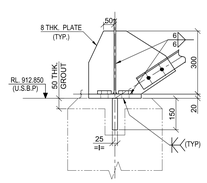
SECTION 4A-4A
(SCALE 1:10)



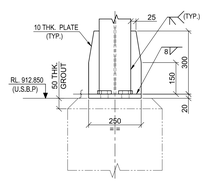
BASE PLATE DETAIL FOR GP3
(SCALE - 1:10)



SECTION 8-8
(SCALE - 1:10)



SECTION 9-9
(SCALE - 1:10)



SECTION 10-10
(SCALE - 1:10)

LEGENDS :-

F.G.L - FINISHED GROUND LEVEL
F.F.L - FINISHED FLOOR LEVEL
T.O.S. - TOP OF STEEL
R.L. - REDUCED LEVEL
F/S-FAR SIDE
N/S-NEAR SIDE
TYP.-TYPICAL
W.P.-WORKING POINT
MM-MILLIMETER
PL.-PLATE
MKD.-MARKED
℄ - CENTRE LINE
T.O.C. - TOP OF CONCRETE.
MR. - MONORAIL
B.O.M.R. - BOTTOM OF MONORAIL
B/S - BOTH SIDE
U.S.B.P. - UNDER SIDE OF BASE PLATE
THK. - THICK
DN - DOWN
MC - MOMENT CONNECTION
T.O.W.W-TOP OF WALKWAY
O/P - OVER PLATE
VGTU -VERTICAL GRAVITY TAKE UP UNIT
TP -TRANSFER POINT
OPP. - OPPOSITE
T.O.G. - TOP OF GRATING
SEC. - SECTION
T.O.W.W. - TOP OF WALKWAY

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF STEEL USED S275JR AS PER EN10025
3. THIS DRG. SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3110-EG-ST-GAD-1004 ,1005,1006 & 1007
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG - ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-LPX-1004	CONV. 3110-CNV-002 - PLAN VIEW								
MD-216-3110-EG-ME-ELV-1004	CONV. 3110-CNV-002 - ELEVATION								
MD-216-3110-EG-ME-ELV-1005	CONV. 3110-CNV-002 - ELEVATION								
MD-216-3110-EG-ME-ELV-1006	CONV. 3110-CNV-002 - ELEVATION	A01	06/04/2020	ISSUED FOR REFERENCE	HSK	SBY	RPK	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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MA'ADEN معادن
Saudi Arabian Mining Company شركة معادن العربية السعودية

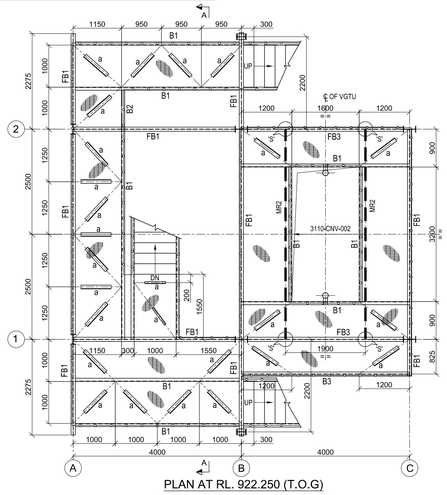
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LARSEN & TOUBRO DRAWING NUMBER: 019045_100_S_G_26013_003 | REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31143 | DOCUMENT ID: OU100670420 | REVISION: 0

DRAWN	HSK	03/04/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SBY	04/04/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	RPK	02/04/2020	STRUCTURAL GA FOR 3110-CNV-002, SH1
APPROVED	PRI	06/04/2020	PLAN, ELEVATIONS & DETAILS
EMPLOYER			
SCALE	1:50	A0	DRAWING NUMBER: MD - 216 - 3110 - EG - ST - GAD - 1003

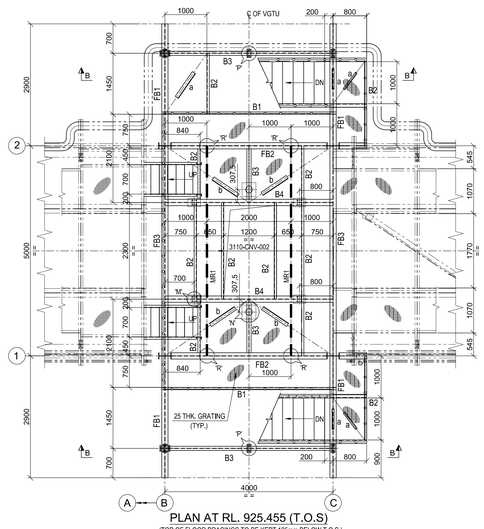
REV: A01

DRAWING NUMBER: MD-216-3110-EG-ST-GAD-1005 REV A01



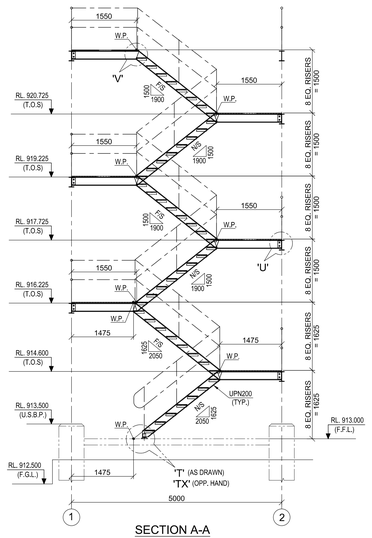
PLAN AT RL. 922.250 (T.O.G)

(TOP OF FLOOR BRACINGS TO BE KEPT 75mm BELOW T.O.S.)
(SCALE 1:50)



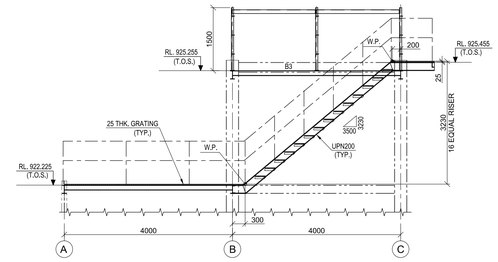
PLAN AT RL. 925.455 (T.O.S)

(TOP OF FLOOR BRACINGS TO BE KEPT 125mm BELOW T.O.S.)
(SCALE 1:50)



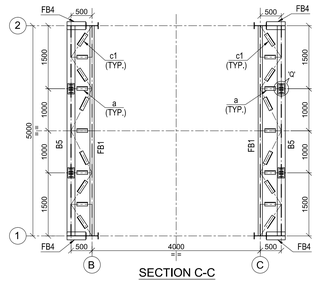
SECTION A-A

(SCALE 1:50)



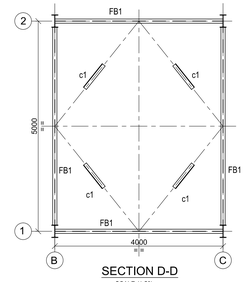
SECTION B-B

(SCALE 1:50)



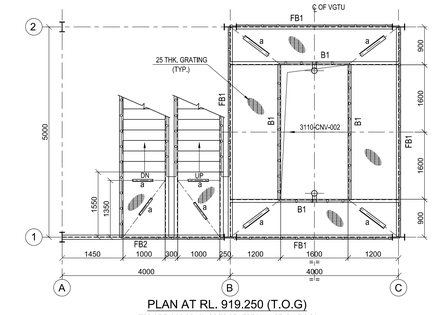
SECTION C-C

(SCALE 1:50)



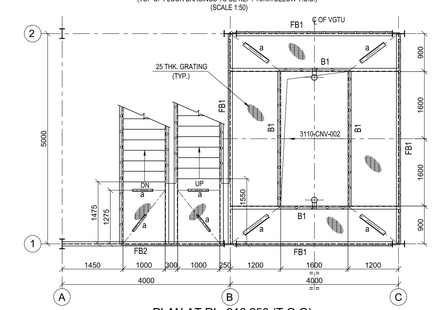
SECTION D-D

SCALE-(1:50)



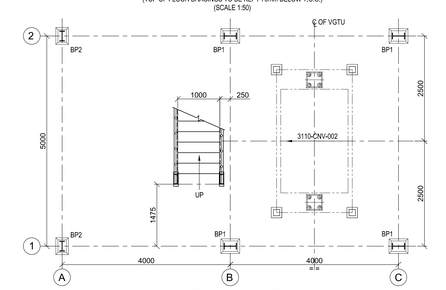
PLAN AT RL. 919.250 (T.O.G)

(TOP OF FLOOR BRACINGS TO BE KEPT 75mm BELOW T.O.S.)
(SCALE 1:50)



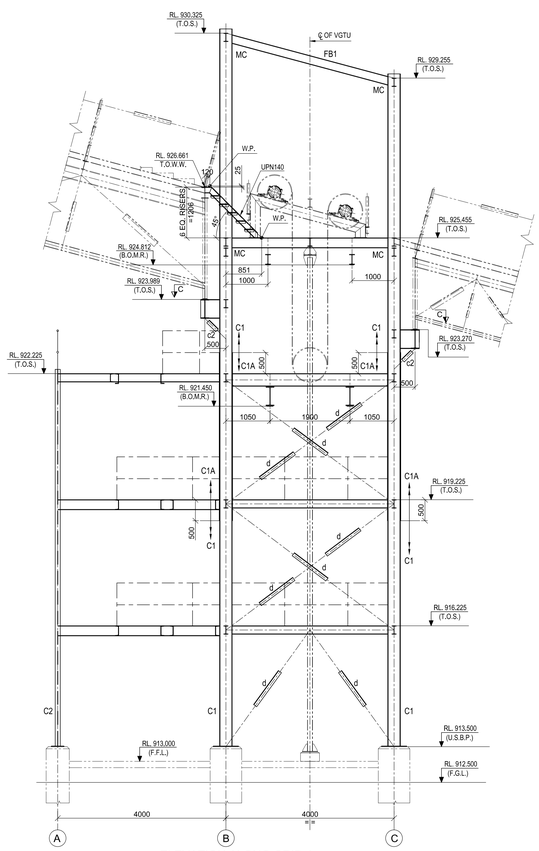
PLAN AT RL. 916.250 (T.O.G)

(TOP OF FLOOR BRACINGS TO BE KEPT 75mm BELOW T.O.S.)
(SCALE 1:50)



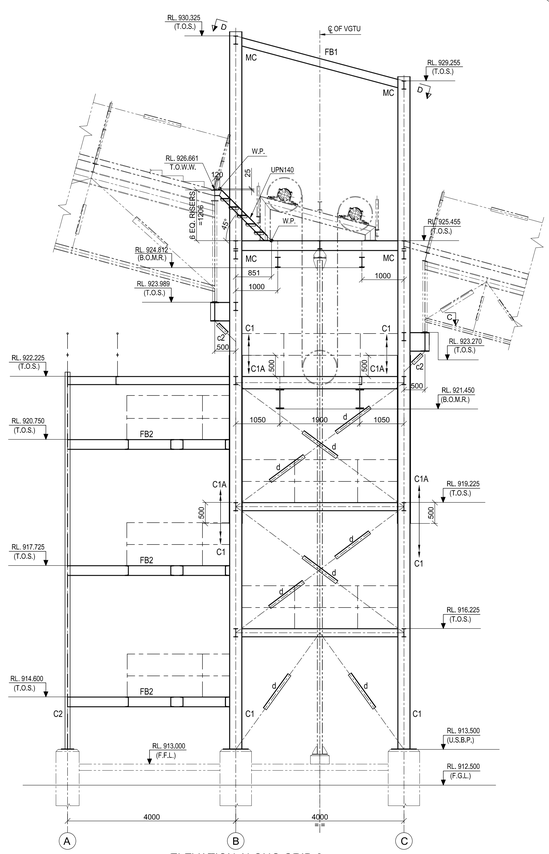
PLAN AT RL. 913.000 (F.F.L)

(SCALE 1:50)



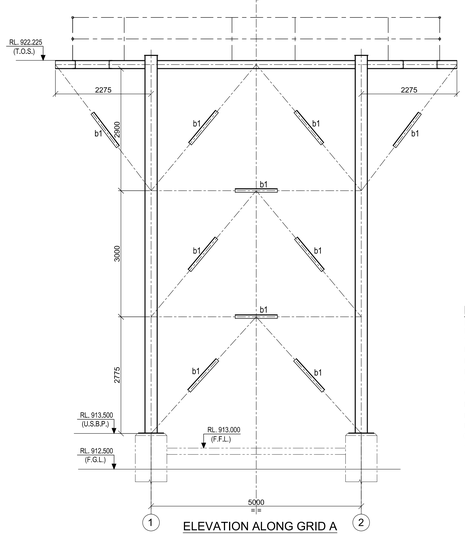
ELEVATION ALONG GRID 1

(SCALE 1:50)



ELEVATION ALONG GRID 2

(SCALE 1:50)



ELEVATION ALONG GRID A

(SCALE 1:50)

LEGENDS :-

F.G.L - FINISHED GROUND LEVEL
F.F.L - FINISHED FLOOR LEVEL
T.O.S. - TOP OF STEEL
R.L. - REDUCED LEVEL
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N/S-NEAR SIDE
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W.P.-WORKING POINT
MM-MILLIMETER
PL.-PLATE
MKD.-MARKED
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B.O.M.R. - BOTTOM OF MONORAIL
B/S - BOTH SIDE
U.S.B.P. - UNDER SIDE OF BASE PLATE
THK. - THICK
DN - DOWN
MC - MOMENT CONNECTION
T.O.WW-TOP OF WALKWAY
O/P - OVER PLATE
VGTU -VERTICAL GRAVITY TAKE UP UNIT
TP -TRANSFER POINT
OPP. - OPPOSITE
T.O.G. - TOP OF GRATING
SEC. - SECTION
T.O.W.W. - TOP OF WALKWAY

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. GRADE OF STEEL USED S275JR AS PER EN10025
3. THIS DRG. SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3110-EG-ST-GAD-1003, 1004 ,1006 & 1007
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-LPX-1004	CONV. 3110-CNV-002 - PLAN VIEW								
MD-216-3110-EG-ME-ELV-1004	CONV. 3110-CNV-002 - ELEVATION								
MD-216-3110-EG-ME-ELV-1005	CONV. 3110-CNV-002 - ELEVATION								
MD-216-3110-EG-ME-ELV-1006	CONV. 3110-CNV-002 - ELEVATION	A01	06/04/2020	ISSUED FOR REFERENCE	HSK	SBY	RPK	PRI	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

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Saudi Arabian Mining Company شركة معادن العربية السعودية

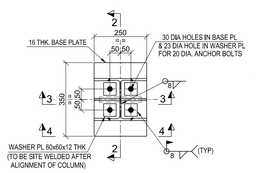
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LARSEN & TOUBRO DRAWING NUMBER	019045_100_S_G_26013_005		REVISION 0
OUTOTEC DRAWING NUMBER	900707__GA001__ZC01__MTL01__31145	DOCUMENT ID OU100670422	REVISION 0
DRAWN	HSK	03/04/2020	
CHECKED	SBY	04/04/2020	
ENGINEER	RPK	02/04/2020	
APPROVED	PRI	06/04/2020	
EMPLOYER			
SCALE	1:50	A0	

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
STRUCTURAL GA FOR 3110-CNV-002, SH3
PLAN, ELEVATIONS & DETAILS

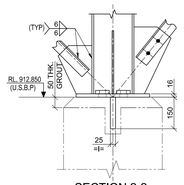
DRAWING NUMBER: MD-216-3110-EG-ST-GAD-1005 REV: A01

DRAWING NUMBER: MD-216-3110-EG-ST-GAD-1006 | REV: A01



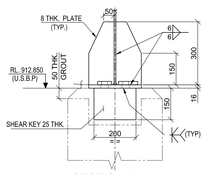
BASE PLATE DETAIL FOR TR1A

(SCALE - 1:10)



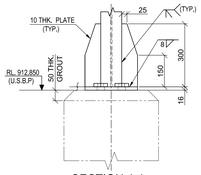
SECTION 2-2

(SCALE - 1:10)



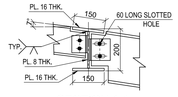
SECTION 3-3

(SCALE - 1:10)



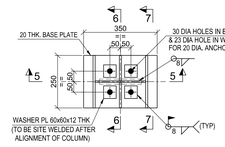
SECTION 4-4

(SCALE - 1:10)



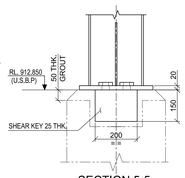
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(SCALE 1:10)



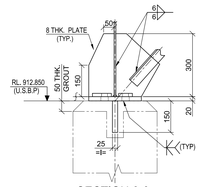
BASE PLATE DETAIL FOR TR1

(SCALE - 1:10)



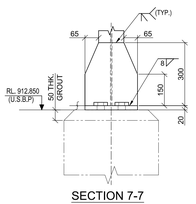
SECTION 5-5

(SCALE - 1:10)



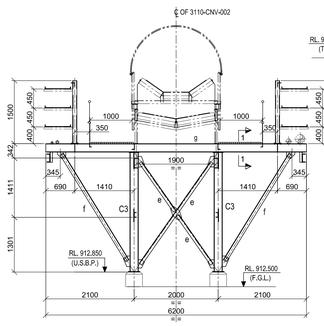
SECTION 6-6

(SCALE - 1:10)



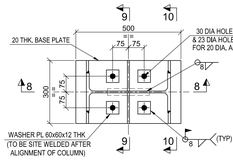
SECTION 7-7

(SCALE - 1:10)



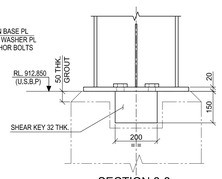
ELEVATION OF TRESTLE MKD. TR1A

(SCALE 1:50)



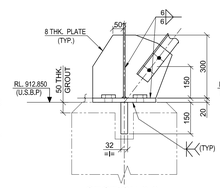
BASE PLATE DETAIL FOR TR2 & TR3

(SCALE - 1:10)



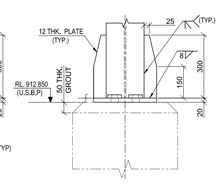
SECTION 8-8

(SCALE - 1:10)



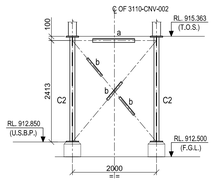
SECTION 9-9

(SCALE - 1:10)



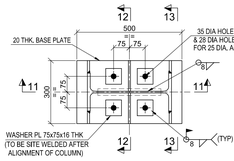
SECTION 10-10

(SCALE - 1:10)



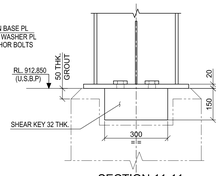
ELEVATION OF TRESTLE MKD. TR1

(SCALE 1:50)



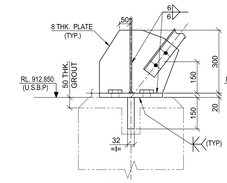
BASE PLATE DETAIL FOR TR4

(SCALE - 1:10)



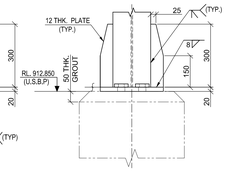
SECTION 11-11

(SCALE - 1:10)



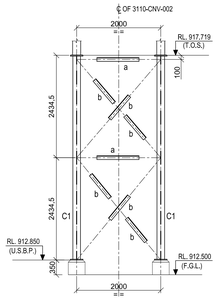
SECTION 12-12

(SCALE - 1:10)



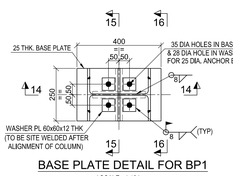
SECTION 13-13

(SCALE - 1:10)



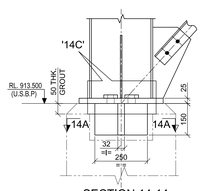
ELEVATION OF TRESTLE MKD. TR2

(SCALE 1:50)



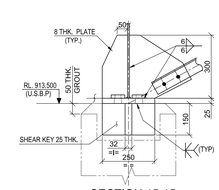
BASE PLATE DETAIL FOR BP1

(SCALE - 1:10)



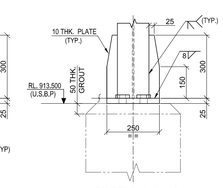
SECTION 14-14

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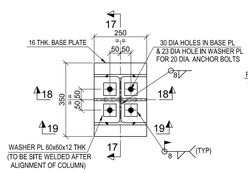
SECTION 15-15

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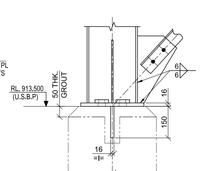
SECTION 16-16

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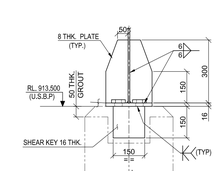
BASE PLATE DETAIL FOR BP2

(SCALE - 1:10)



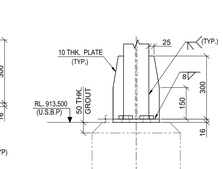
SECTION 17-17

(SCALE - 1:10)



SECTION 18-18

(SCALE - 1:10)

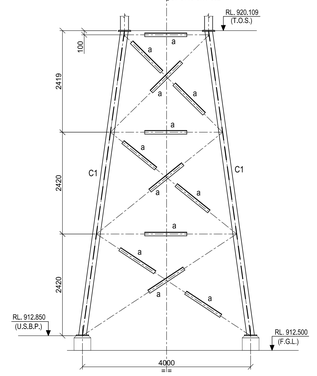


SECTION 19-19

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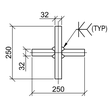
ELEVATION ALONG GRID B

(SCALE 1:50)



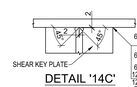
ELEVATION OF TRESTLE MKD. TR3

(SCALE 1:50)



VIEW 14A-14A

(SCALE - 1:10)

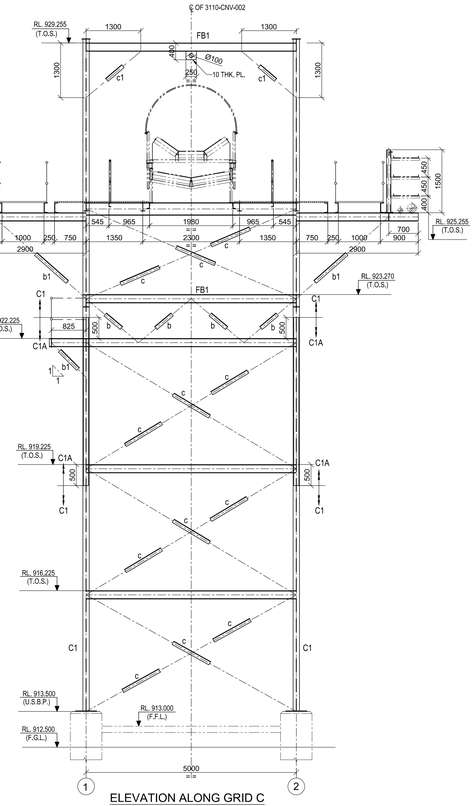


DETAIL '14C'

(SCALE - 1:10)

ELEVATION OF TRESTLE MKD. TR4

(SCALE 1:50)



ELEVATION ALONG GRID C

(SCALE 1:50)

SCHEDULE OF MEMBERS FOR TRESTLE MKD. TR1A TO TR4					
MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	REMARKS
C1	IPE 400	I	-	-	MATERIAL GRADE:S275 JR
C2	1. PL 120X10 2. PL. 240X8	1 2	-	-	
C3	IPE 200	I	-	-	
a	2L 50X50X6	⊤	91	-	
b	2L 65X65X6	⊤	115	-	
c	2L 75X75X8	⊤	113	-	
d	2L 90X90X8	⊤	171	-	
e	L 75X75X8	L	55	-	
f	L 90X90X8	L	-	-	
g	1. PL. 150X12 2. PL. 176X8	1 2	-	-	

GRADE OF MATERIAL IS S275 JR
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
ALL MEMBER CONNECTIONS SHOULD BE DESIGN FOR A MINIMUM FORCE OF 50KN.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED
FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos.
M20 BOLTS SHALL BE PROVIDED.

SCHEDULE OF MEMBERS FOR VGTU					
MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	REMARKS
C1	IPE 300	I	-	-	MATERIAL GRADE:S275 JR
C1A	1. IPE 300 2. PL. 120X8	2 1 2	-	-	
C2	IPE 240	I	-	-	
FB1	IPE 200	I	265	20	
FB2	IPE 240	I	30	66	
FB3	IPE 300	I	70	100	
FB4	IPE 450	I	235	-	
B1	UPN 120	L	-	-	
B2	UPN 100	L	-	-	
B3	IPE 200	I	-	51	
B4	IPE 300	I	-	60	
B5	IPE 450	I	90	235	
a	L 50X50X6	L	66	-	
b	L 65X65X6	L	-	-	
b1	2L 65X65X6	⊤	73	-	
c	2L 75X75X8	⊤	93	-	
c1	L 75X75X8	L	152	-	
c2	2L 75X75X8	L	331	-	
d	2L 90X90X8	⊤	193	-	
P1	IPE 200	I	-	-	
MR1	1. IPE 240 2. PL. 90X10	1 2	-	-	
MR2	1. IPE 450 2. PL. 140X12	1 2	-	-	

GRADE OF MATERIAL IS S275 JR
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
ALL MEMBER CONNECTIONS SHOULD BE DESIGN FOR A MINIMUM FORCE OF 50KN.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED
FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos.
M20 BOLTS SHALL BE PROVIDED.

LEGENDS :-

F.G.L - FINISHED GROUND LEVEL
F.F.L - FINISHED FLOOR LEVEL
T.O.S - TOP OF STEEL
R.L - REDUCED LEVEL
F/S-FAR SIDE
N/S-NEAR SIDE
TYP.-TYPICAL
W.P.-WORKING POINT
MM-MILLIMETER
PL - PLATE
MKD.-MARKED
℄- CENTRE LINE
T.O.C. - TOP OF CONCRETE.
MR. - MONORAIL
B.O.M.R. - BOTTOM OF MONORAIL
B/S - BOTH SIDE
U.S.B.P. - UNDER SIDE OF BASE PLATE
THK. - THICK
DN - DOWN
MC - MOMENT CONNECTION
T.O.W/W-TOP OF WALKWAY
O/P - OVER PLATE
VGTU -VERTICAL GRAVITY TAKE UP UNIT
TP -TRANSFER POINT
OPP. - OPPOSITE
T.O.G. - TOP OF GRATING
SEC. - SECTION
T.O.W.W. - TOP OF WALKWAY

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES, UNLESS OTHERWISE STATED.
2. GRADE OF STEEL USED S275JR AS PER EN10025
3. THIS DRG. SHALL BE READ IN CONJUNCTION WITH DRG. NO. MD-216-3110-EG-ST-GAD-1003, 1004 ,1005 & 1007
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3110-EG-ME-LPX-1004	CONV. 3110-CNV-002 - PLAN VIEW								
MD-216-3110-EG-ME-ELV-1004	CONV. 3110-CNV-002 - ELEVATION								
MD-216-3110-EG-ME-ELV-1005	CONV. 3110-CNV-002 - ELEVATION								
MD-216-3110-EG-ME-ELV-1006	CONV. 3110-CNV-002 - ELEVATION	A01	06/04/2020	ISSUED FOR REFERENCE	HSK	SBY	RPK	PRI	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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Outotec

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LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045_100_S_G_26013_006		0
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707__GAU01__ZC01__MTL01__31146	OU100670423	0

DRAWN	HSK	03/04/2020
CHECKED	SBY	04/04/2020
ENGINEER	RPK	02/04/2020
APPROVED	PRI	06/04/2020
EMPLOYER		
SCALE	1:50,1:10	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
STRUCTURAL GA FOR 3110-CNV-002, SH4
PLAN, ELEVATIONS & DETAILS

DRAWING NUMBER	REV
MD - 216 - 3110 - EG - ST - GAD - 1006	A01

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	07-04-2020	18-05-2020	12-10-2023				Date	07-04-2020	18-05-2020	12-10-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 3 **STAAD INPUT & OUTPUT FILES OF FOUNDATION FOR GROUND PORTALS**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 06-Apr-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART GROUND PORTAL
JOB REF CONVEYOR 3110-CNV-002
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 0 -0.25; 3 4.5 0 -0.25; 5 0 1.586 -0.25; 6 4.5 1.904 -0.25; 8 0 0 5.75;
9 4.5 0 5.75; 11 0 1.586 5.75; 12 4.5 1.904 5.75; 17 0 1.586 0.68;
18 4.5 1.904 0.68; 23 15.8 2.924 0.68; 24 0 1.586 1.75; 25 4.5 1.904 1.75;
26 15.8 2.924 1.75; 27 0 1.586 3.75; 28 4.5 1.904 3.75; 29 15.8 2.924 3.75;
30 0 1.586 4.82; 31 4.5 1.904 4.82; 32 15.8 2.924 4.82; 49 9.55 0 0;
50 9.55 2.189 0; 51 9.55 0 5.5; 52 9.55 2.189 5.5; 53 9.55 2.269 0;
54 9.55 2.269 0.68; 55 9.55 2.269 1.75; 56 9.55 2.269 3.75; 57 9.55 2.269 4.82;
58 9.55 2.269 5.5; 65 9.55 2.189 2.75; 66 2.25 1.745 0.68;
67 7.025 2.0865 0.68; 68 2.25 1.745 1.75; 69 7.025 2.0865 1.75;
70 2.25 1.745 3.75; 71 7.025 2.0865 3.75; 72 2.25 1.745 4.82;
73 7.025 2.0865 4.82; 74 11.6333 2.48733 0.68; 75 13.7167 2.70567 0.68;
76 11.6333 2.48733 4.82; 77 13.7167 2.70567 4.82; 78 11.6333 2.48733 1.75;
79 13.7167 2.70567 1.75; 80 11.6333 2.48733 3.75; 81 13.7167 2.70567 3.75;
82 15.8 2.924 3.53903e-008; 83 15.8 2.924 5.5; 84 2.25 1.745 5.75;
85 2.25 1.745 -0.25; 86 0 -1.85 -0.25; 87 4.5 -1.35 -0.25; 88 0 -1.85 5.75;
89 4.5 -1.35 5.75; 90 9.55 -1.35 0; 91 9.55 -1.35 5.5;
MEMBER INCIDENCES
2 2 5; 3 3 6; 5 8 11; 6 9 12; 8 5 17; 9 6 18; 14 17 24; 16 18 25; 17 17 66;
22 18 67; 23 24 27; 25 25 28; 26 24 68; 28 25 69; 29 27 30; 31 28 31; 32 27 70;
34 28 71; 35 30 11; 37 31 12; 38 30 72; 40 31 73; 59 49 50; 60 51 52; 61 53 54;
62 54 55; 63 55 56; 64 56 57; 65 57 58; 66 49 65; 67 51 65; 68 50 53; 69 52 58;
70 50 65; 77 54 74; 78 57 76; 91 55 78; 92 56 80; 97 65 52; 98 66 18; 99 67 54;
100 68 25; 101 69 55; 102 70 28; 103 71 56; 104 72 31; 105 73 57; 106 74 75;
107 75 23; 108 76 77; 109 77 32; 110 78 79; 111 79 26; 112 80 81; 113 81 29;
114 79 81; 115 78 80; 116 69 71; 117 68 70; 118 27 68; 119 68 28; 120 28 69;
121 69 56; 122 56 78; 123 78 81; 124 81 26; 125 75 79; 126 74 78; 127 80 76;
128 81 77; 129 67 69; 130 71 73; 131 66 68; 132 70 72; 133 5 85; 134 6 53;
135 11 84; 136 12 58; 137 53 82; 138 58 83; 139 84 12; 140 8 84; 141 9 84;
142 85 6; 143 2 85; 144 3 85; 145 86 2; 146 88 8; 147 87 3; 148 89 9;
149 90 49; 150 91 51; 151 2 8; 152 3 9; 153 49 51; 154 8 9; 155 2 3;
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
```

```
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
2 3 5 6 8 14 23 29 35 59 60 68 69 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
9 16 25 31 37 61 TO 65 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
17 22 38 40 77 78 98 99 104 TO 109 TABLE ST UPN140
26 28 32 34 100 TO 103 133 TO 139 142 TABLE ST UPN200
66 67 70 97 TABLE LD L65X65X6 SP 0.01
114 TO 117 122 TO 124 TABLE ST L75X75X8
125 TO 132 TABLE ST L50X50X6
91 92 110 TO 113 TABLE ST UPN240
140 141 143 144 TABLE LD L50X50X6 SP 0.01
118 TO 121 TABLE ST L90X90X8
MEMBER PROPERTY EUROPEAN
145 TO 150 PRIS YD 0.6 ZD 0.6
154 155 PRIS YD 0.4 ZD 0.3
151 TO 153 PRIS YD 0.5 ZD 0.3
CONSTANTS
BETA 90 MEMB 2 3 5 6 68 69
MATERIAL STEEL MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 -
38 40 59 TO 70 77 78 91 92 97 TO 144
MATERIAL CONCRETE MEMB 145 TO 155
SUPPORTS
23 26 29 32 82 83 FIXED BUT FX MX MY MZ
86 TO 91 FIXED
MEMBER TRUSS
66 67 114 TO 132 140 141 143 144
MEMBER RELEASE
2 3 5 6 17 26 32 38 59 TO 61 70 77 78 91 92 133 TO 138 START MP 0.99
65 97 99 101 103 105 111 113 134 136 TO 139 142 END MP 0.99
MEMBER OFFSET
151 TO 153 START 0 -0.25 0
151 TO 153 END 0 -0.25 0
154 155 START 0 -0.2 0
154 155 END 0 -0.2 0
LOAD 1 DL
SELFWEIGHT Y -1.1
*WALKWAY LOAD @50KG/SQ. M Including Seal Plate & H/R
MEMBER LOAD
17 22 38 40 77 78 98 99 104 TO 109 UNI GY -0.32
26 28 32 34 91 92 100 TO 103 110 TO 113 UNI GY -0.47
*PIPE DEAD LOAD
MEMBER LOAD
8 9 61 CON GY -4.08
35 37 65 CON GY -7.14
*CONVEYOR DEAD LOAD
```

```
MEMBER LOAD
26 32 CON GY -6.5 1.48
28 34 CON GY -6.5 0
101 103 CON GY -6.5 0.5
91 92 CON GY -6.5 1
110 112 CON GY -6.5 2
*
MEMBER LOAD
26 32 CON GZ 0.8 1.48
28 34 CON GZ 0.8 0
101 103 CON GZ 0.8 0.5
91 92 CON GZ 0.8 1
110 112 CON GZ 0.8 2
*
MEMBER LOAD
26 32 CON GX 1.5 1.48
28 34 CON GX 1.5 0
101 103 CON GX 1.5 0.5
91 92 CON GX 1.5 1
110 112 CON GX 1.5 2
LOAD 2 LL
*WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
MEMBER LOAD
17 22 26 28 32 34 38 40 77 78 91 92 98 99 101 TO 113 UNI GY -2
LOAD 3 ML
*CONVEYOR MATERIAL LOAD
MEMBER LOAD
26 32 CON GY -8 1.48
28 34 CON GY -8 0
101 103 CON GY -8 0.5
91 92 CON GY -8 1
110 112 CON GY -8 2
** Spillage Load @75kg/ Sq. m on seal plate
26 28 32 34 91 92 100 TO 103 110 TO 113 UNI GY -0.75
LOAD 4 WIND LOAD +Z
**WIND LOAD ON CONVEYOR HOOD
MEMBER LOAD
26 32 CON GZ 0.7 1.48
28 34 CON GZ 0.7 0
101 103 CON GZ 0.7 0.5
91 92 CON GZ 0.7 1
110 112 CON GZ 0.7 2
**
26 CON GY 1.6 1.48
32 CON GY -1.6 1.48
28 CON GY 1.6 0
34 CON GY -1.6 0
101 CON GY 1.6 0.5
103 CON GY -1.6 0.5
91 CON GY 1.6 1
92 CON GY -1.6 1
110 CON GY 1.6 2
112 CON GY -1.6 2
LOAD 5 WIND LOAD -Z
```

```
**WIND LOAD ON CONVEYOR HOOD
MEMBER LOAD
26 32 CON GZ -0.7 1.48
28 34 CON GZ -0.7 0
101 103 CON GZ -0.7 0.5
91 92 CON GZ -0.7 1
110 112 CON GZ -0.7 2
**
26 CON GY -1.6 1.48
32 CON GY 1.6 1.48
28 CON GY -1.6 0
34 CON GY 1.6 0
101 CON GY -1.6 0.5
103 CON GY 1.6 0.5
91 CON GY -1.6 1
92 CON GY 1.6 1
110 CON GY -1.6 2
112 CON GY 1.6 2
LOAD 6 TL1
TEMPERATURE LOAD
2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 59 TO 70 77 78 -
91 92 97 TO 132 TEMP 25
LOAD 7 TL2
TEMPERATURE LOAD
2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 59 TO 70 77 78 -
91 92 97 TO 132 TEMP -25
LOAD 11 G
REPEAT LOAD
1 1.0
LOAD 12 G + QN + MN
REPEAT LOAD
1 1.0 2 1.0 3 1.0
****LOAD COMBINATION for structural design
*LOAD COMB 201 1.4 DL
*1 1.4
*LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
*1 1.2 6 1.2 2 1.6 3 1.6
*LOAD COMB 203 1.2 (DL + TL2) + 1.6  (LL+ML)
*1 1.2 7 1.2 2 1.6 3 1.6
*LOAD COMB 204 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
*1 1.2 4 1.6 2 0.5 3 0.5
*LOAD COMB 205 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
*1 1.2 5 1.6 2 0.5 3 0.5
*LOAD COMB 206 0.9 DL + 1.6 WL(+Z)
*1 0.9 4 1.6
*LOAD COMB 207 0.9 DL + 1.6 WL(-Z)
*1 0.9 5 1.6
****LOAD COMBINATION FOR BEARING PRESSURE CHECK
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 6 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
1 1.0 7 1.0 2 1.0 3 1.0
```

```
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 307 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
************************************************************
****LOAD COMBINATION FOR FOUNDATION DESIGN
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 6 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
1 1.4 7 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 407 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
************************************************************
PERFORM ANALYSIS
*LOAD LIST 201 TO 206
*PARAMETER 1
*CODE LRFD
*FYLD 275000 MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 -
*40 59 TO 69 77 78 91 92 106 TO 113
*FU 410000 MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 -
*59 TO 69 77 78 91 92 106 TO 113
*TRACK 2 MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 59 -
*60 TO 69 77 78 91 92 106 TO 113
*BEAM 1 MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 59 -
*60 TO 69 77 78 91 92 106 TO 113
*TMAIN 300 MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 -
*59 TO 69 77 78 91 92 106 TO 113
*MAIN 200 MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 -
*59 TO 69 77 78 91 92 106 TO 113
*NSF 0.8 MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 59 -
*60 TO 70 77 78 91 92 97 TO 132
*UNT 1.9 MEMB 8 9 14 16 23 25 29 31 35 37 61 TO 65
*LY 1.9 MEMB 8 9 14 16 23 25 29 31 35 37 61 TO 65
*LX 1.9 MEMB 8 9 14 16 23 25 29 31 35 37 61 TO 65
*LZ 5.95 MEMB 8 9 14 16 23 25 29 31 35 37 61 TO 65
*KY 0.5 MEMB 134 136 TO 138
*KX 0.5 MEMB 134 136 TO 138
*CHECK CODE MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 -
*59 TO 70 77 78 91 92 97 TO 144
LOAD LIST 401 TO 407
START CONCRETE DESIGN
CODE ACI 2008
```

```
FC 35000 MEMB 145 TO 155
FYMAIN 420000 MEMB 145 TO 155
TRACK 1 MEMB 145 TO 155
CLB 0.05 MEMB 145 TO 155
CLT 0.05 MEMB 145 TO 155
CLS 0.05 MEMB 145 TO 155
MAXMAIN 20 MEMB 145 TO 150
MINMAIN 20 MEMB 145 TO 150
MAXMAIN 16 MEMB 151 TO 155
MINMAIN 12 MEMB 151 TO 155
MINSEC 8 MEMB 145 TO 155
REINF 0 MEMB 145 TO 150
DESIGN COLUMN 145 TO 150
DESIGN BEAM 151 TO 155
END CONCRETE DESIGN
FINISH
```

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**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  7, 2020                *
*           Time=    13:32: 2                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
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    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3110-CNV-002\Staad\3110-CNV-002 GRO.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 06-APR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART GROUND PORTAL
    9. JOB REF CONVEYOR 3110-CNV-002
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 2 0 0 -0.25; 3 4.5 0 -0.25; 5 0 1.586 -0.25; 6 4.5 1.904 -0.25; 8 0 0 5.75
   18. 9 4.5 0 5.75; 11 0 1.586 5.75; 12 4.5 1.904 5.75; 17 0 1.586 0.68
   19. 18 4.5 1.904 0.68; 23 15.8 2.924 0.68; 24 0 1.586 1.75; 25 4.5 1.904 1.75
   20. 26 15.8 2.924 1.75; 27 0 1.586 3.75; 28 4.5 1.904 3.75; 29 15.8 2.924 3.75
   21. 30 0 1.586 4.82; 31 4.5 1.904 4.82; 32 15.8 2.924 4.82; 49 9.55 0 0
   22. 50 9.55 2.189 0; 51 9.55 0 5.5; 52 9.55 2.189 5.5; 53 9.55 2.269 0
   23. 54 9.55 2.269 0.68; 55 9.55 2.269 1.75; 56 9.55 2.269 3.75; 57 9.55 2.269 4.82
   24. 58 9.55 2.269 5.5; 65 9.55 2.189 2.75; 66 2.25 1.745 0.68
   25. 67 7.025 2.0865 0.68; 68 2.25 1.745 1.75; 69 7.025 2.0865 1.75
   26. 70 2.25 1.745 3.75; 71 7.025 2.0865 3.75; 72 2.25 1.745 4.82
   27. 73 7.025 2.0865 4.82; 74 11.6333 2.48733 0.68; 75 13.7167 2.70567 0.68
   28. 76 11.6333 2.48733 4.82; 77 13.7167 2.70567 4.82; 78 11.6333 2.48733 1.75
   29. 79 13.7167 2.70567 1.75; 80 11.6333 2.48733 3.75; 81 13.7167 2.70567 3.75
   30. 82 15.8 2.924 3.53903E-008; 83 15.8 2.924 5.5; 84 2.25 1.745 5.75
   31. 85 2.25 1.745 -0.25; 86 0 -1.85 -0.25; 87 4.5 -1.35 -0.25; 88 0 -1.85 5.75
   32. 89 4.5 -1.35 5.75; 90 9.55 -1.35 0; 91 9.55 -1.35 5.5
   33. MEMBER INCIDENCES
   34. 2 2 5; 3 3 6; 5 8 11; 6 9 12; 8 5 17; 9 6 18; 14 17 24; 16 18 25; 17 17 66
   35. 22 18 67; 23 24 27; 25 25 28; 26 24 68; 28 25 69; 29 27 30; 31 28 31; 32 27 70
   36. 34 28 71; 35 30 11; 37 31 12; 38 30 72; 40 31 73; 59 49 50; 60 51 52; 61 53 54
   37. 62 54 55; 63 55 56; 64 56 57; 65 57 58; 66 49 65; 67 51 65; 68 50 53; 69 52 58
   38. 70 50 65; 77 54 74; 78 57 76; 91 55 78; 92 56 80; 97 65 52; 98 66 18; 99 67 54
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39. 100 68 25; 101 69 55; 102 70 28; 103 71 56; 104 72 31; 105 73 57; 106 74 75
40. 107 75 23; 108 76 77; 109 77 32; 110 78 79; 111 79 26; 112 80 81; 113 81 29
41. 114 79 81; 115 78 80; 116 69 71; 117 68 70; 118 27 68; 119 68 28; 120 28 69
42. 121 69 56; 122 56 78; 123 78 81; 124 81 26; 125 75 79; 126 74 78; 127 80 76
43. 128 81 77; 129 67 69; 130 71 73; 131 66 68; 132 70 72; 133 5 85; 134 6 53
44. 135 11 84; 136 12 58; 137 53 82; 138 58 83; 139 84 12; 140 8 84; 141 9 84
45. 142 85 6; 143 2 85; 144 3 85; 145 86 2; 146 88 8; 147 87 3; 148 89 9
46. 149 90 49; 150 91 51; 151 2 8; 152 3 9; 153 49 51; 154 8 9; 155 2 3
47. DEFINE MATERIAL START
48. ISOTROPIC STEEL
49. E 2.05E+008
50. POISSON 0.3
51. DENSITY 76.8195
52. ALPHA 1.2E-005
53. DAMP 0.03
54. TYPE STEEL
55. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
56. ISOTROPIC CONCRETE
57. E 2.17185E+007
58. POISSON 0.17
59. DENSITY 23.5616
60. ALPHA 1E-005
61. DAMP 0.05
62. TYPE CONCRETE
63. STRENGTH FCU 27579
64. END DEFINE MATERIAL
65. MEMBER PROPERTY EUROPEAN
66. 2 3 5 6 8 14 23 29 35 59 60 68 69 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
67. 9 16 25 31 37 61 TO 65 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
68. 17 22 38 40 77 78 98 99 104 TO 109 TABLE ST UPN140
69. 26 28 32 34 100 TO 103 133 TO 139 142 TABLE ST UPN200
70. 66 67 70 97 TABLE LD L65X65X6 SP 0.01
71. 114 TO 117 122 TO 124 TABLE ST L75X75X8
72. 125 TO 132 TABLE ST L50X50X6
73. 91 92 110 TO 113 TABLE ST UPN240
74. 140 141 143 144 TABLE LD L50X50X6 SP 0.01
75. 118 TO 121 TABLE ST L90X90X8
76. MEMBER PROPERTY EUROPEAN
77. 145 TO 150 PRIS YD 0.6 ZD 0.6
78. 154 155 PRIS YD 0.4 ZD 0.3
79. 151 TO 153 PRIS YD 0.5 ZD 0.3
80. CONSTANTS
81. BETA 90 MEMB 2 3 5 6 68 69
82. MATERIAL STEEL MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 -
83. 38 40 59 TO 70 77 78 91 92 97 TO 144
84. MATERIAL CONCRETE MEMB 145 TO 155
85. SUPPORTS
86. 23 26 29 32 82 83 FIXED BUT FX MX MY MZ
87. 86 TO 91 FIXED
88. MEMBER TRUSS
89. 66 67 114 TO 132 140 141 143 144
90. MEMBER RELEASE
91. 2 3 5 6 17 26 32 38 59 TO 61 70 77 78 91 92 133 TO 138 START MP 0.99
92. 65 97 99 101 103 105 111 113 134 136 TO 139 142 END MP 0.99
93. MEMBER OFFSET
94. 151 TO 153 START 0 -0.25 0
```

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 95. 151 TO 153 END 0 -0.25 0
 96. 154 155 START 0 -0.2 0
 97. 154 155 END 0 -0.2 0
 98. LOAD 1 DL
 99. SELFWEIGHT Y -1.1
100. *WALKWAY LOAD @50KG/SQ. M INCLUDING SEAL PLATE _H/R
101. MEMBER LOAD
102. 17 22 38 40 77 78 98 99 104 TO 109 UNI GY -0.32
103. 26 28 32 34 91 92 100 TO 103 110 TO 113 UNI GY -0.47
104. *PIPE DEAD LOAD
105. MEMBER LOAD
106. 8 9 61 CON GY -4.08
107. 35 37 65 CON GY -7.14
108. *CONVEYOR DEAD LOAD
109. MEMBER LOAD
110. 26 32 CON GY -6.5 1.48
111. 28 34 CON GY -6.5 0
112. 101 103 CON GY -6.5 0.5
113. 91 92 CON GY -6.5 1
114. 110 112 CON GY -6.5 2
115. *
116. MEMBER LOAD
117. 26 32 CON GZ 0.8 1.48
118. 28 34 CON GZ 0.8 0
119. 101 103 CON GZ 0.8 0.5
120. 91 92 CON GZ 0.8 1
121. 110 112 CON GZ 0.8 2
122. *
123. MEMBER LOAD
124. 26 32 CON GX 1.5 1.48
125. 28 34 CON GX 1.5 0
126. 101 103 CON GX 1.5 0.5
127. 91 92 CON GX 1.5 1
128. 110 112 CON GX 1.5 2
129. LOAD 2 LL
130. *WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
131. MEMBER LOAD
132. 17 22 26 28 32 34 38 40 77 78 91 92 98 99 101 TO 113 UNI GY -2
133. LOAD 3 ML
134. *CONVEYOR MATERIAL LOAD
135. MEMBER LOAD
136. 26 32 CON GY -8 1.48
137. 28 34 CON GY -8 0
138. 101 103 CON GY -8 0.5
139. 91 92 CON GY -8 1
140. 110 112 CON GY -8 2
141. ** SPILLAGE LOAD @75KG/ SQ. M ON SEAL PLATE
142. 26 28 32 34 91 92 100 TO 103 110 TO 113 UNI GY -0.75
143. LOAD 4 WIND LOAD +Z
144. **WIND LOAD ON CONVEYOR HOOD
145. MEMBER LOAD
146. 26 32 CON GZ 0.7 1.48
147. 28 34 CON GZ 0.7 0
148. 101 103 CON GZ 0.7 0.5
149. 91 92 CON GZ 0.7 1
150. 110 112 CON GZ 0.7 2
```

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151. **
152. 26 CON GY 1.6 1.48
153. 32 CON GY -1.6 1.48
154. 28 CON GY 1.6 0
155. 34 CON GY -1.6 0
156. 101 CON GY 1.6 0.5
157. 103 CON GY -1.6 0.5
158. 91 CON GY 1.6 1
159. 92 CON GY -1.6 1
160. 110 CON GY 1.6 2
161. 112 CON GY -1.6 2
162. LOAD 5 WIND LOAD -Z
163. **WIND LOAD ON CONVEYOR HOOD
164. MEMBER LOAD
165. 26 32 CON GZ -0.7 1.48
166. 28 34 CON GZ -0.7 0
167. 101 103 CON GZ -0.7 0.5
168. 91 92 CON GZ -0.7 1
169. 110 112 CON GZ -0.7 2
170. **
171. 26 CON GY -1.6 1.48
172. 32 CON GY 1.6 1.48
173. 28 CON GY -1.6 0
174. 34 CON GY 1.6 0
175. 101 CON GY -1.6 0.5
176. 103 CON GY 1.6 0.5
177. 91 CON GY -1.6 1
178. 92 CON GY 1.6 1
179. 110 CON GY -1.6 2
180. 112 CON GY 1.6 2
181. LOAD 6 TL1
182. TEMPERATURE LOAD
183. 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 59 TO 70 77 78 -
184. 91 92 97 TO 132 TEMP 25
185. LOAD 7 TL2
186. TEMPERATURE LOAD
187. 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 59 TO 70 77 78 -
188. 91 92 97 TO 132 TEMP -25
189. LOAD 11 G
190. REPEAT LOAD
191. 1 1.0
192. LOAD 12 G + QN + MN
193. REPEAT LOAD
194. 1 1.0 2 1.0 3 1.0
195. ****LOAD COMBINATION FOR STRUCTURAL DESIGN
196. *LOAD COMB 201 1.4 DL
197. *1 1.4
198. *LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
199. *1 1.2 6 1.2 2 1.6 3 1.6
200. *LOAD COMB 203 1.2 (DL + TL2) + 1.6  (LL+ML)
201. *1 1.2 7 1.2 2 1.6 3 1.6
202. *LOAD COMB 204 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
203. *1 1.2 4 1.6 2 0.5 3 0.5
204. *LOAD COMB 205 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
205. *1 1.2 5 1.6 2 0.5 3 0.5
206. *LOAD COMB 206 0.9 DL + 1.6 WL(+Z)
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207. *1 0.9 4 1.6
208. *LOAD COMB 207 0.9 DL + 1.6 WL(-Z)
209. *1 0.9 5 1.6
210. ****LOAD COMBINATION FOR BEARING PRESSURE CHECK
211. LOAD COMB 301 1.0 DL
212. 1 1.0
213. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
214. 1 1.0 6 1.0 2 1.0 3 1.0
215. LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
216. 1 1.0 7 1.0 2 1.0 3 1.0
217. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
218. 1 1.0 4 0.75 2 0.75 3 0.75
219. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
220. 1 1.0 5 0.75 2 0.75 3 0.75
221. LOAD COMB 306 0.6 DL + 1.0 WL(+Z)
222. 1 0.6 4 1.0
223. LOAD COMB 307 0.6 DL + 1.0 WL(-Z)
224. 1 0.6 5 1.0
225. ************************************************************
226. ****LOAD COMBINATION FOR FOUNDATION DESIGN
227. LOAD COMB 401 1.4 DL
228. 1 1.4
229. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
230. 1 1.4 6 1.4 2 1.7 3 1.7
231. LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
232. 1 1.4 7 1.4 2 1.7 3 1.7
233. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
234. 1 1.2 4 1.6 2 0.5 3 0.5
235. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
236. 1 1.2 5 1.6 2 0.5 3 0.5
237. LOAD COMB 406 0.9 DL + 1.6 WL(+Z)
238. 1 0.9 4 1.6
239. LOAD COMB 407 0.9 DL + 1.6 WL(-Z)
240. 1 0.9 5 1.6
241. ************************************************************
242. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**57**	**NUMBER OF MEMBERS**	**97**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**12**

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 9, TOTAL DEGREES OF FREEDOM = 294
TOTAL LOAD COMBINATION CASES = 14 SO FAR.

```
243. *LOAD LIST 201 TO 206
244. *PARAMETER 1
245. *CODE LRFD
246. *FYLD 275000 MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 -
247. *40 59 TO 69 77 78 91 92 106 TO 113
248. *FU 410000 MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 -
249. *59 TO 69 77 78 91 92 106 TO 113
250. *TRACK 2 MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 59 -
251. *60 TO 69 77 78 91 92 106 TO 113
252. *BEAM 1 MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 59 -
253. *60 TO 69 77 78 91 92 106 TO 113
254. *TMAIN 300 MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 -
255. *59 TO 69 77 78 91 92 106 TO 113
256. *MAIN 200 MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 -
257. *59 TO 69 77 78 91 92 106 TO 113
258. *NSF 0.8 MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 59 -
259. *60 TO 70 77 78 91 92 97 TO 132
260. *UNT 1.9 MEMB 8 9 14 16 23 25 29 31 35 37 61 TO 65
261. *LY 1.9 MEMB 8 9 14 16 23 25 29 31 35 37 61 TO 65
262. *LX 1.9 MEMB 8 9 14 16 23 25 29 31 35 37 61 TO 65
263. *LZ 5.95 MEMB 8 9 14 16 23 25 29 31 35 37 61 TO 65
264. *KY 0.5 MEMB 134 136 TO 138
265. *KX 0.5 MEMB 134 136 TO 138
266. *CHECK CODE MEMB 2 3 5 6 8 9 14 16 17 22 23 25 26 28 29 31 32 34 35 37 38 40 -
267. *59 TO 70 77 78 91 92 97 TO 144
268. LOAD LIST 401 TO 407
269. START CONCRETE DESIGN
270. CODE ACI 2008
271. FC 35000 MEMB 145 TO 155
272. FYMAIN 420000 MEMB 145 TO 155
273. TRACK 1 MEMB 145 TO 155
274. CLB 0.05 MEMB 145 TO 155
275. CLT 0.05 MEMB 145 TO 155
276. CLS 0.05 MEMB 145 TO 155
277. MAXMAIN 20 MEMB 145 TO 150
278. MINMAIN 20 MEMB 145 TO 150
279. MAXMAIN 16 MEMB 151 TO 155
280. MINMAIN 12 MEMB 151 TO 155
281. MINSEC 8 MEMB 145 TO 155
282. REINF 0 MEMB 145 TO 150
283. DESIGN COLUMN 145 TO 150
```

==========

COLUMN NO. 145 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	60.28	11.44	0.10257

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	60.28	23.62	0.21180

==========

COLUMN NO. 146 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37     68.09     11.42    0.09068
 ------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37     68.09     24.67    0.19584
 ------------------------------------------------------
```

==========

COLUMN NO. 147 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
------------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    110.23     14.13    0.09496
 ------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    110.23     21.69    0.14574
 ------------------------------------------------------
```

==========

COLUMN NO. 148 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	117.63	14.16	0.08916

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	117.63	22.06	0.13894

==========

COLUMN NO. 149 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	78.25	0.04	0.00035

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	78.25	21.60	0.20445

==========

COLUMN NO. 150 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	91.82	0.03	0.00028

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	91.82	23.26	0.18762

**********END OF COLUMN DESIGN RESULTS**********

284. DESIGN BEAM 151 TO 155

BEAM NO. 151 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	1148.	4602.	NO	NO

CRITICAL POS MOMENT= 9.25 KN-MET AT 3000.MM, LOAD 401
REQD STEEL= 75.MM2, RHO=0.0006, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 24968.6 cm^4

2	436.	2 - 12MM	0.	1400.	YES	NO

CRITICAL NEG MOMENT= 15.54 KN-MET AT 0.MM, LOAD 402
REQD STEEL= 126.MM2, RHO=0.0010, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 24968.6 cm^4

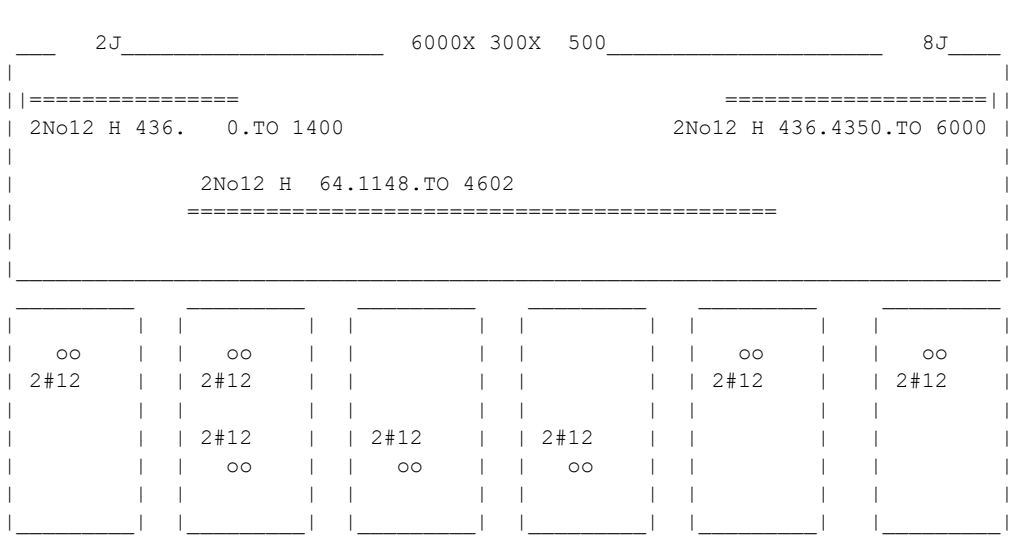
3	436.	2 - 12MM	4350.	6000.	NO	YES

CRITICAL NEG MOMENT= 16.02 KN-MET AT 6000.MM, LOAD 402
REQD STEEL= 130.MM2, RHO=0.0010, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 24968.6 cm^4

B E A M N O. 151 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   13.84 KNS  Vc=  123.06 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                    STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   14.09 KNS  Vc=  123.00 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                    STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 152 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

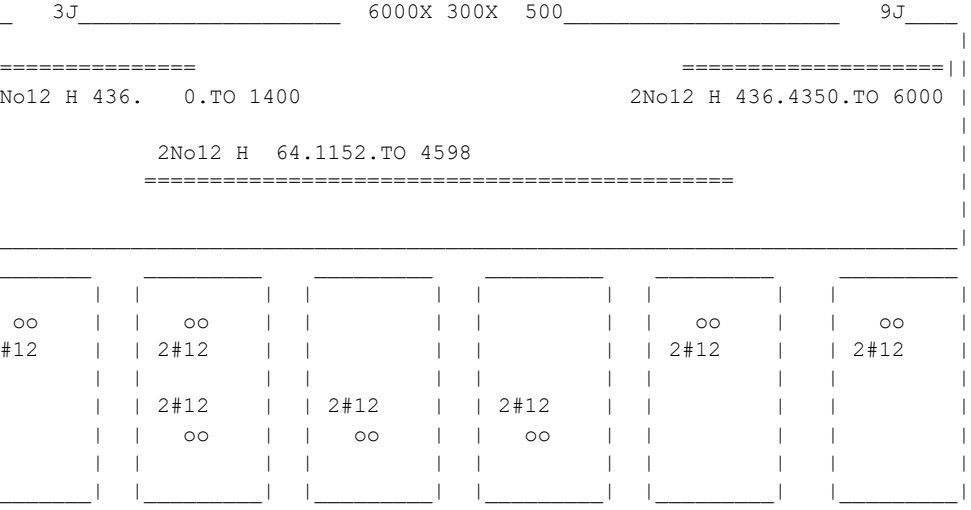
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	1152.	4598.	NO	NO
	CRITICAL POS MOMENT= 8.85 KN-MET AT 3000.MM, LOAD 401					
	REQD STEEL= 72.MM2, RHO=0.0005, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 24968.6 cm^4					
2	436.	2 - 12MM	0.	1400.	YES	NO
	CRITICAL NEG MOMENT= 16.77 KN-MET AT 0.MM, LOAD 402					
	REQD STEEL= 136.MM2, RHO=0.0010, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 24968.6 cm^4					
3	436.	2 - 12MM	4350.	6000.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     17.19 KN-MET  AT  6000.MM, LOAD  402|
|   REQD STEEL=   140.MM2, RHO=0.0011, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 24968.6 cm^4

B E A M N O. 152 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 13.91 KNS Vc= 127.20 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 14.02 KNS Vc= 127.20 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 153 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	1058.	4213.	NO	NO

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=      8.01 KN-MET  AT  2750.MM, LOAD  401|
|   REQD STEEL=    65.MM2, RHO=0.0005, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    24968.6 cm^4

  2        436.     2 - 12MM         0.         1317.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     12.33 KN-MET  AT     0.MM, LOAD  403|
|   REQD STEEL=   100.MM2, RHO=0.0008, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|

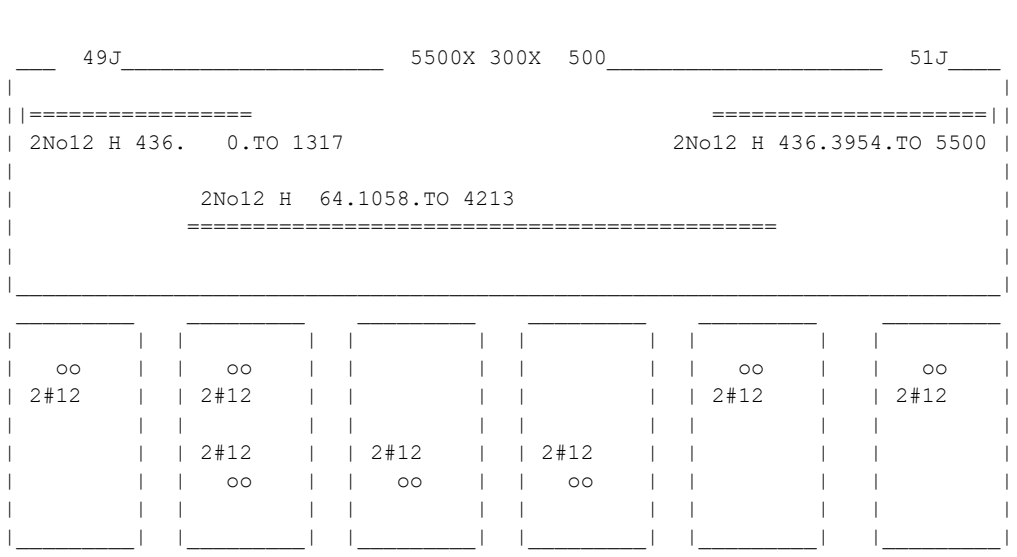
  Cracked Moment of Inertia Iz at above location =    24968.6 cm^4

  3        436.     2 - 12MM      3954.         5500.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     13.31 KN-MET  AT  5500.MM, LOAD  402|
|   REQD STEEL=   108.MM2, RHO=0.0008, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    24968.6 cm^4
```

B E A M N O. 153 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   12.41 KNS  Vc=  123.40 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    5.3 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   12.80 KNS  Vc=  123.22 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    5.3 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 154 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

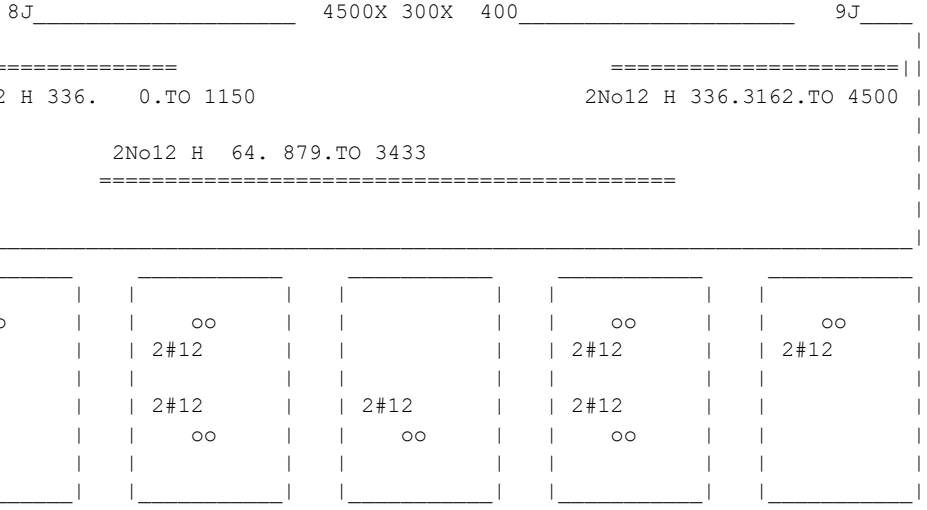
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	879.	3433.	NO	NO
	CRITICAL POS MOMENT= 4.07 KN-MET AT 2250.MM, LOAD 402					
	REQD STEEL= 43.MM2, RHO=0.0004, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 14415.4 cm^4					
2	336.	2 - 12MM	0.	1150.	YES	NO
	CRITICAL NEG MOMENT= 6.17 KN-MET AT 0.MM, LOAD 403					
	REQD STEEL= 65.MM2, RHO=0.0006, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 14415.4 cm^4					
3	336.	2 - 12MM	3162.	4500.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      7.86 KN-MET  AT  4500.MM, LOAD  401|
|   REQD STEEL=    83.MM2, RHO=0.0008, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    14415.4 cm^4
```

B E A M N O. 154 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=    7.95 KNS  Vc=   94.99 KNS  Vs=    0.00 KNS
 Tu=    0.06 KN-MET  Tc=    3.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=    8.74 KNS  Vc=   94.67 KNS  Vs=    0.00 KNS
 Tu=    0.06 KN-MET  Tc=    3.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 155 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	879.	3433.	NO	NO

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=      4.07 KN-MET  AT  2250.MM, LOAD  402|
|   REQD STEEL=    43.MM2, RHO=0.0004, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    14415.4 cm^4

  2        336.     2 - 12MM         0.         1150.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      6.19 KN-MET  AT     0.MM, LOAD  403|
|   REQD STEEL=    65.MM2, RHO=0.0006, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|

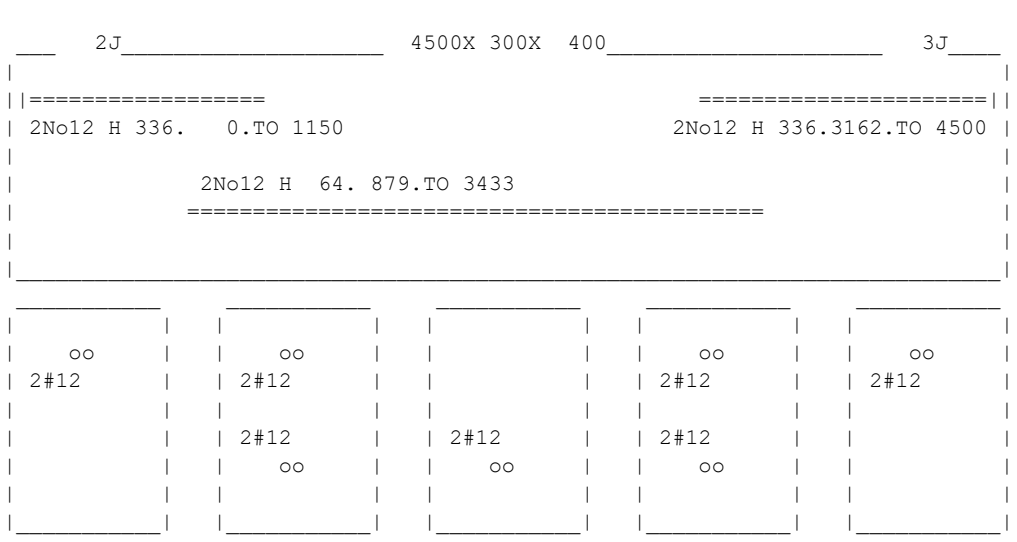
  Cracked Moment of Inertia Iz at above location =    14415.4 cm^4

  3        336.     2 - 12MM      3162.         4500.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      7.85 KN-MET  AT  4500.MM, LOAD  401|
|   REQD STEEL=    83.MM2, RHO=0.0008, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    14415.4 cm^4
```

B E A M N O. 155 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=    7.95 KNS  Vc=   94.99 KNS  Vs=    0.00 KNS
Tu=    0.08 KN-MET  Tc=    3.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=    8.73 KNS  Vc=   94.67 KNS  Vs=    0.00 KNS
Tu=    0.08 KN-MET  Tc=    3.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



**********END OF BEAM DESIGN**********

285. END CONCRETE DESIGN
286. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 13:32: 4 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	07-04-2020	18-05-2020	12-10-2023				Date	07-04-2020	18-05-2020	12-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 4 **STAAD INPUT & OUTPUT FILES OF COMBINED FOOTING MKD. F3 FOR TR1A**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 06-Apr-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART TRESTLE TR1A FOUNDATION KKD. F3
JOB REF FOUNDATIONS FOR CONVEYOR 3110-CNV-002
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 2; 3 0 0 0; 4 0 0 2; 5 0 0.35 0; 6 0 0.35 2; 14 -0.75 -3 2.75;
17 -0.5 -3 2.75; 19 -0.25 -3 2.75; 20 0 -3 2.75; 22 0.25 -3 2.75;
24 0.5 -3 2.75; 26 0.75 -3 2.75; 28 -0.75 -3 2.5; 30 -0.5 -3 2.5;
31 -0.25 -3 2.5; 32 0 -3 2.5; 33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5;
37 -0.75 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25;
42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 46 -0.75 -3 2; 48 -0.5 -3 2;
49 -0.25 -3 2; 50 0.25 -3 2; 51 0.5 -3 2; 52 0.75 -3 2; 54 -0.75 -3 1.75;
56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75;
60 0.5 -3 1.75; 61 0.75 -3 1.75; 63 -0.75 -3 1.5; 65 -0.5 -3 1.5;
66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5; 69 0.5 -3 1.5; 70 0.75 -3 1.5;
72 -0.75 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25;
77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 81 -0.75 -3 1; 83 -0.5 -3 1;
84 -0.25 -3 1; 85 0 -3 1; 86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1;
90 -0.75 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75;
95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 99 -0.75 -3 0.5;
101 -0.5 -3 0.5; 102 -0.25 -3 0.5; 103 0 -3 0.5; 104 0.25 -3 0.5;
105 0.5 -3 0.5; 106 0.75 -3 0.5; 108 -0.75 -3 0.25; 110 -0.5 -3 0.25;
111 -0.25 -3 0.25; 112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25;
115 0.75 -3 0.25; 117 -0.75 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0; 121 0.25 -3 0;
122 0.5 -3 0; 123 0.75 -3 0; 125 -0.75 -3 -0.25; 127 -0.5 -3 -0.25;
128 -0.25 -3 -0.25; 129 0 -3 -0.25; 130 0.25 -3 -0.25; 131 0.5 -3 -0.25;
132 0.75 -3 -0.25; 134 -0.75 -3 -0.5; 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5;
138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5; 141 0.75 -3 -0.5;
143 -0.75 -3 -0.75; 145 -0.5 -3 -0.75; 146 -0.25 -3 -0.75; 147 0 -3 -0.75;
148 0.25 -3 -0.75; 149 0.5 -3 -0.75; 150 0.75 -3 -0.75; 212 1 -3 2.75;
213 1 -3 2.5; 216 1 -3 2.25; 218 1 -3 2; 220 1 -3 1.75; 222 1 -3 1.5;
224 1 -3 1.25; 226 1 -3 1; 228 1 -3 0.75; 230 1 -3 0.5; 232 1 -3 0.25;
234 1 -3 0; 236 1 -3 -0.25; 238 1 -3 -0.5; 240 1 -3 -0.75; 260 -1 -3 2.75;
261 -1 -3 2.5; 263 -1 -3 2.25; 265 -1 -3 2; 267 -1 -3 1.75; 269 -1 -3 1.5;
271 -1 -3 1.25; 273 -1 -3 1; 275 -1 -3 0.75; 277 -1 -3 0.5; 279 -1 -3 0.25;
281 -1 -3 0; 283 -1 -3 -0.25; 285 -1 -3 -0.5; 287 -1 -3 -0.75;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
ELEMENT INCIDENCES SHELL
15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33;
20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41;
27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49;
34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56;
```

```
41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63;
48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69;
55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77;
60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85;
67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93;
74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99;
80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104;
84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111;
90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117;
96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121;
100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127;
105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131;
111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137;
114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140;
119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146;
122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149;
214 26 212 213 35; 217 35 213 216 44; 220 44 216 218 52; 223 52 218 220 61;
226 61 220 222 70; 229 70 222 224 79; 232 79 224 226 88; 235 88 226 228 97;
238 97 228 230 106; 241 106 230 232 115; 244 115 232 234 123;
247 123 234 236 132; 250 132 236 238 141; 253 141 238 240 150;
282 260 14 28 261; 285 261 28 37 263; 288 263 37 46 265; 291 265 46 54 267;
294 267 54 63 269; 297 269 63 72 271; 300 271 72 81 273; 303 273 81 90 275;
306 275 90 99 277; 309 277 99 108 279; 312 279 108 117 281;
315 281 117 125 283; 318 283 125 134 285; 321 285 134 143 287;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 214 217 220 223 226 229 232 235 238 241 -
244 247 250 253 282 285 288 291 294 297 300 303 306 309 312 315 318 321
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 214 217 226 229 232 235 238 241 250 253 282 285 294 -
297 300 303 306 309 318 321
_SHEAR 63 TO 68 71 TO 76 214 217 220 223 226 229 232 235 238 241 244 247 250 -
253 282 285 288 291 294 297 300 303 306 309 312 315 318 321
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 214 217 220 223 -
226 229 232 235 238 241 244 247 250 253 282 285 288 291 294 297 300 303 306 -
309 312 315 318 321 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.6 ZD 0.6
5 PRIS YD 0.5 ZD 0.3
```

```
CONSTANTS
MATERIAL CONCRETE ALL
*****************************************
SUPPORTS
1 2 14 17 19 20 22 24 26 28 30 TO 35 37 39 TO 44 46 48 TO 52 54 56 TO 61 63 -
65 TO 70 72 74 TO 79 81 83 TO 88 90 92 TO 97 99 101 TO 106 108 110 TO 115 -
117 119 TO 123 125 127 TO 132 134 136 TO 141 143 145 TO 150 212 213 216 218 -
220 222 224 226 228 230 232 234 236 238 240 260 261 263 265 267 269 271 273 -
275 277 279 281 283 285 287 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
*****************************************
MEMBER OFFSET
5 START 0 -0.25 0
5 END 0 -0.25 0
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
5 FX 0.01 FY -61.317 FZ 5.685
6 FX -0.01 FY -82.275 FZ -1.684
ELEMENT LOAD
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 214 217 220 223 -
226 229 232 235 238 241 244 247 250 253 282 285 288 291 294 297 300 303 306 -
309 312 315 318 321 PR GY -46.8
*****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
5 FX -0.006 FY -36.859 FZ 1.26
6 FX 0.006 FY -37.14 FZ -1.261
*****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
5 FX -0.006 FY -42.06 FZ 0.553
6 FX 0.006 FY -42.339 FZ -0.554
*****************************************
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
5 FX 0.033 FY 35.191 FZ 12.615
6 FX -0.033 FY -35.191 FZ 12.116
*****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
5 FX -0.033 FY -35.19 FZ -12.658
6 FX 0.033 FY 35.19 FZ -12.072
*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
5 FX 0.562 FY 0 FZ 0
6 FX 0.562 FY 0 FZ 0
*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
5 FX -0.562 FY 0 FZ 0
6 FX -0.562 FY 0 FZ 0
*****************************************
```

```
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
5 FX -0.005 FY -0.232 FZ -5.855
6 FX 0.005 FY 0.232 FZ 5.855
*****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
5 FX 0.005 FY 0.232 FZ 5.855
6 FX -0.005 FY -0.232 FZ -5.855
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
5 FX 0.92 FY 0.368 FZ 0.021
6 FX 0.92 FY 0.354 FZ -0.009
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
5 FX 0.001 FY 1.075 FZ 0.393
6 FX -0.001 FY -1.075 FZ 0.371
*****************************************
*****Load combination for bearing capacity check
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 6 0.75 2 0.75 3 0.75
LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 7 0.75 2 0.75 3 0.75
LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 0.525 2 0.75 3 0.75
LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 0.525 2 0.75 3 0.75
LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 -0.525 2 0.75 3 0.75
```

```
LOAD 312 0.6 DL + 1.0 WL(+Z)
REPEAT LOAD
1 0.6 4 1.0
LOAD 313 0.6 DL + 1.0 WL(-Z)
REPEAT LOAD
1 0.6 5 1.0
LOAD 314 0.6 DL + 1.0 WL(+X)
REPEAT LOAD
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
```

```
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 5 1.6
LOAD 414 0.9 DL + 1.6 WL(+X)
REPEAT LOAD
1 0.9 6 1.6
LOAD 415 0.9 DL + 1.6 WL(-X)
REPEAT LOAD
1 0.9 7 1.6
LOAD 416 0.9 DL + 1.0 EL(+X)
REPEAT LOAD
1 0.9 10 1.0
LOAD 417 0.9 DL + 1.0 EL(-X)
REPEAT LOAD
1 0.9 10 -1.0
LOAD 418 0.9 DL + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 11 1.0
LOAD 419 0.9 DL + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
*****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 5
FYMAIN 420000 MEMB 1 TO 5
TRACK 1 ALL
CLB 0.05 MEMB 1 TO 5
CLT 0.05 MEMB 1 TO 5
CLS 0.05 ALL
MAXMAIN 20 MEMB 1 TO 4
MAXMAIN 12 MEMB 5
MINMAIN 16 MEMB 1 TO 4
MINMAIN 12 MEMB 5
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  7, 2020                *
*           Time=    11:53:51                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3110-CNV-002\Staad\3110-CNV-002_TR1.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 06-APR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART TRESTLE TR1A FOUNDATION KKD. F3
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3110-CNV-002
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 -3 0; 2 0 -3 2; 3 0 0 0; 4 0 0 2; 5 0 0.35 0; 6 0 0.35 2; 14 -0.75 -3 2.75
   18. 17 -0.5 -3 2.75; 19 -0.25 -3 2.75; 20 0 -3 2.75; 22 0.25 -3 2.75
   19. 24 0.5 -3 2.75; 26 0.75 -3 2.75; 28 -0.75 -3 2.5; 30 -0.5 -3 2.5
   20. 31 -0.25 -3 2.5; 32 0 -3 2.5; 33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5
   21. 37 -0.75 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25
   22. 42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 46 -0.75 -3 2; 48 -0.5 -3 2
   23. 49 -0.25 -3 2; 50 0.25 -3 2; 51 0.5 -3 2; 52 0.75 -3 2; 54 -0.75 -3 1.75
   24. 56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75
   25. 60 0.5 -3 1.75; 61 0.75 -3 1.75; 63 -0.75 -3 1.5; 65 -0.5 -3 1.5
   26. 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5; 69 0.5 -3 1.5; 70 0.75 -3 1.5
   27. 72 -0.75 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25
   28. 77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 81 -0.75 -3 1; 83 -0.5 -3 1
   29. 84 -0.25 -3 1; 85 0 -3 1; 86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1
   30. 90 -0.75 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75
   31. 95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 99 -0.75 -3 0.5
   32. 101 -0.5 -3 0.5; 102 -0.25 -3 0.5; 103 0 -3 0.5; 104 0.25 -3 0.5
   33. 105 0.5 -3 0.5; 106 0.75 -3 0.5; 108 -0.75 -3 0.25; 110 -0.5 -3 0.25
   34. 111 -0.25 -3 0.25; 112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25
   35. 115 0.75 -3 0.25; 117 -0.75 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0; 121 0.25 -3 0
   36. 122 0.5 -3 0; 123 0.75 -3 0; 125 -0.75 -3 -0.25; 127 -0.5 -3 -0.25
   37. 128 -0.25 -3 -0.25; 129 0 -3 -0.25; 130 0.25 -3 -0.25; 131 0.5 -3 -0.25
   38. 132 0.75 -3 -0.25; 134 -0.75 -3 -0.5; 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5
```

```
39. 138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5; 141 0.75 -3 -0.5
40. 143 -0.75 -3 -0.75; 145 -0.5 -3 -0.75; 146 -0.25 -3 -0.75; 147 0 -3 -0.75
41. 148 0.25 -3 -0.75; 149 0.5 -3 -0.75; 150 0.75 -3 -0.75; 212 1 -3 2.75
42. 213 1 -3 2.5; 216 1 -3 2.25; 218 1 -3 2; 220 1 -3 1.75; 222 1 -3 1.5
43. 224 1 -3 1.25; 226 1 -3 1; 228 1 -3 0.75; 230 1 -3 0.5; 232 1 -3 0.25
44. 234 1 -3 0; 236 1 -3 -0.25; 238 1 -3 -0.5; 240 1 -3 -0.75; 260 -1 -3 2.75
45. 261 -1 -3 2.5; 263 -1 -3 2.25; 265 -1 -3 2; 267 -1 -3 1.75; 269 -1 -3 1.5
46. 271 -1 -3 1.25; 273 -1 -3 1; 275 -1 -3 0.75; 277 -1 -3 0.5; 279 -1 -3 0.25
47. 281 -1 -3 0; 283 -1 -3 -0.25; 285 -1 -3 -0.5; 287 -1 -3 -0.75
48. MEMBER INCIDENCES
49. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
50. ELEMENT INCIDENCES SHELL
51. 15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33
52. 20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41
53. 27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49
54. 34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56
55. 41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63
56. 48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69
57. 55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77
58. 60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85
59. 67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93
60. 74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99
61. 80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104
62. 84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111
63. 90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117
64. 96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121
65. 100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127
66. 105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131
67. 111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137
68. 114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140
69. 119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146
70. 122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149
71. 214 26 212 213 35; 217 35 213 216 44; 220 44 216 218 52; 223 52 218 220 61
72. 226 61 220 222 70; 229 70 222 224 79; 232 79 224 226 88; 235 88 226 228 97
73. 238 97 228 230 106; 241 106 230 232 115; 244 115 232 234 123
74. 247 123 234 236 132; 250 132 236 238 141; 253 141 238 240 150
75. 282 260 14 28 261; 285 261 28 37 263; 288 263 37 46 265; 291 265 46 54 267
76. 294 267 54 63 269; 297 269 63 72 271; 300 271 72 81 273; 303 273 81 90 275
77. 306 275 90 99 277; 309 277 99 108 279; 312 279 108 117 281
78. 315 281 117 125 283; 318 283 125 134 285; 321 285 134 143 287
79. START GROUP DEFINITION
80. ELEMENT
81. _MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
82. 63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
83. 108 111 112 115 116 119 120 123 124 214 217 220 223 226 229 232 235 238 241 -
84. 244 247 250 253 282 285 288 291 294 297 300 303 306 309 312 315 318 321
85. _MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
86. 111 TO 116 119 TO 124 214 217 226 229 232 235 238 241 250 253 282 285 294 -
87. 297 300 303 306 309 318 321
88. _SHEAR 63 TO 68 71 TO 76 214 217 220 223 226 229 232 235 238 241 244 247 250 -
89. 253 282 285 288 291 294 297 300 303 306 309 312 315 318 321
90. END GROUP DEFINITION
91. ELEMENT PROPERTY
92. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
93. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 214 217 220 223 -
94. 226 229 232 235 238 241 244 247 250 253 282 285 288 291 294 297 300 303 306 -
```

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 95. 309 312 315 318 321 THICKNESS 0.4
 96. DEFINE MATERIAL START
 97. ISOTROPIC CONCRETE
 98. E 2.17185E+007
 99. POISSON 0.17
100. DENSITY 23.5616
101. ALPHA 1E-005
102. DAMP 0.05
103. TYPE CONCRETE
104. STRENGTH FCU 27579
105. END DEFINE MATERIAL
106. MEMBER PROPERTY AMERICAN
107. 1 TO 4 PRIS YD 0.6 ZD 0.6
108. 5 PRIS YD 0.5 ZD 0.3
109. CONSTANTS
110. MATERIAL CONCRETE ALL
111. ****************************************
112. SUPPORTS
113. 1 2 14 17 19 20 22 24 26 28 30 TO 35 37 39 TO 44 46 48 TO 52 54 56 TO 61 63 -
114. 65 TO 70 72 74 TO 79 81 83 TO 88 90 92 TO 97 99 101 TO 106 108 110 TO 115 -
115. 117 119 TO 123 125 127 TO 132 134 136 TO 141 143 145 TO 150 212 213 216 218 -
116. 220 222 224 226 228 230 232 234 236 238 240 260 261 263 265 267 269 271 273 -
117. 275 277 279 281 283 285 287 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
```

JOINT	INFLUENCE AREA
1	0.062500
2	0.062500
14	0.031250
17	0.031250
19	0.031250
20	0.031250
22	0.031250
24	0.031250
26	0.031250
28	0.062500
30	0.062500
31	0.062500
32	0.062500
33	0.062500
34	0.062500
35	0.062500
37	0.062500
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500
44	0.062500
46	0.062500
48	0.062500
49	0.062500
50	0.062500
51	0.062500
52	0.062500

54	0.062500
56	0.062500
57	0.062500
58	0.062500
59	0.062500
60	0.062500
61	0.062500
63	0.062500
65	0.062500
66	0.062500
67	0.062500
68	0.062500
69	0.062500
70	0.062500
72	0.062500
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.062500
81	0.062500
83	0.062500
84	0.062500
85	0.062500
86	0.062500
87	0.062500
88	0.062500
90	0.062500
92	0.062500
93	0.062500
94	0.062500
95	0.062500
96	0.062500
97	0.062500
99	0.062500
101	0.062500
102	0.062500
103	0.062500
104	0.062500
105	0.062500
106	0.062500
108	0.062500
110	0.062500
111	0.062500
112	0.062500
113	0.062500
114	0.062500
115	0.062500
117	0.062500
119	0.062500
120	0.062500
121	0.062500
122	0.062500
123	0.062500
125	0.062500

127	0.062500
128	0.062500
129	0.062500
130	0.062500
131	0.062500
132	0.062500
134	0.062500
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
141	0.062500
143	0.031250
145	0.031250
146	0.031250
147	0.031250
148	0.031250
149	0.031250
150	0.031250
212	0.015625
213	0.031250
216	0.031250
218	0.031250
220	0.031250
222	0.031250
224	0.031250
226	0.031250
228	0.031250
230	0.031250
232	0.031250
234	0.031250
236	0.031250
238	0.031250
240	0.015625
260	0.015625
261	0.031250
263	0.031250
265	0.031250
267	0.031250
269	0.031250
271	0.031250
273	0.031250
275	0.031250
277	0.031250
279	0.031250
281	0.031250
283	0.031250
285	0.031250
287	0.015625
TOTAL	7.000000

```
118. ****************************************
119. MEMBER OFFSET
120. 5 START 0 -0.25 0
121. 5 END 0 -0.25 0
122. LOAD 1 DEAD LOAD DL
```

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123. SELFWEIGHT Y -1
124. JOINT LOAD
125. 5 FX 0.01 FY -61.317 FZ 5.685
126. 6 FX -0.01 FY -82.275 FZ -1.684
127. ELEMENT LOAD
128. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
129. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 214 217 220 223 -
130. 226 229 232 235 238 241 244 247 250 253 282 285 288 291 294 297 300 303 306 -
131. 309 312 315 318 321 PR GY -46.8
132. ****************************************
133. LOAD 2 LIVE LOAD LL
134. JOINT LOAD
135. 5 FX -0.006 FY -36.859 FZ 1.26
136. 6 FX 0.006 FY -37.14 FZ -1.261
137. ****************************************
138. LOAD 3 MATERIAL LOAD LL
139. JOINT LOAD
140. 5 FX -0.006 FY -42.06 FZ 0.553
141. 6 FX 0.006 FY -42.339 FZ -0.554
142. ****************************************
143. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
144. JOINT LOAD
145. 5 FX 0.033 FY 35.191 FZ 12.615
146. 6 FX -0.033 FY -35.191 FZ 12.116
147. ****************************************
148. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
149. JOINT LOAD
150. 5 FX -0.033 FY -35.19 FZ -12.658
151. 6 FX 0.033 FY 35.19 FZ -12.072
152. ****************************************
153. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
154. JOINT LOAD
155. 5 FX 0.562 FY 0 FZ 0
156. 6 FX 0.562 FY 0 FZ 0
157. ****************************************
158. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
159. JOINT LOAD
160. 5 FX -0.562 FY 0 FZ 0
161. 6 FX -0.562 FY 0 FZ 0
162. ****************************************
163. LOAD 8 TEMP LOAD   TL1
164. JOINT LOAD
165. 5 FX -0.005 FY -0.232 FZ -5.855
166. 6 FX 0.005 FY 0.232 FZ 5.855
167. ****************************************
168. LOAD 9 TEMP LOAD   TL2
169. JOINT LOAD
170. 5 FX 0.005 FY 0.232 FZ 5.855
171. 6 FX -0.005 FY -0.232 FZ -5.855
172. ****************************************
173. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
174. JOINT LOAD
175. 5 FX 0.92 FY 0.368 FZ 0.021
176. 6 FX 0.92 FY 0.354 FZ -0.009
177. ****************************************
178. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
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179. JOINT LOAD
180. 5 FX 0.001 FY 1.075 FZ 0.393
181. 6 FX -0.001 FY -1.075 FZ 0.371
182. ****************************************
183. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
184. LOAD 301 1.0 DL
185. REPEAT LOAD
186. 1 1.0
187. LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
188. REPEAT LOAD
189. 1 1.0 8 1.0 2 1.0 3 1.0
190. LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
191. REPEAT LOAD
192. 1 1.0 9 1.0 2 1.0 3 1.0
193. LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
194. REPEAT LOAD
195. 1 1.0 4 0.75 2 0.75 3 0.75
196. LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
197. REPEAT LOAD
198. 1 1.0 5 0.75 2 0.75 3 0.75
199. LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
200. REPEAT LOAD
201. 1 1.0 6 0.75 2 0.75 3 0.75
202. LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
203. REPEAT LOAD
204. 1 1.0 7 0.75 2 0.75 3 0.75
205. LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
206. REPEAT LOAD
207. 1 1.0 10 0.525 2 0.75 3 0.75
208. LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
209. REPEAT LOAD
210. 1 1.0 10 -0.525 2 0.75 3 0.75
211. LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
212. REPEAT LOAD
213. 1 1.0 11 0.525 2 0.75 3 0.75
214. LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
215. REPEAT LOAD
216. 1 1.0 11 -0.525 2 0.75 3 0.75
217. LOAD 312 0.6 DL + 1.0 WL(+Z)
218. REPEAT LOAD
219. 1 0.6 4 1.0
220. LOAD 313 0.6 DL + 1.0 WL(-Z)
221. REPEAT LOAD
222. 1 0.6 5 1.0
223. LOAD 314 0.6 DL + 1.0 WL(+X)
224. REPEAT LOAD
225. 1 0.6 6 1.0
226. LOAD 315 0.6 DL + 1.0 WL(-X)
227. REPEAT LOAD
228. 1 0.6 7 1.0
229. LOAD 316 0.6 DL + 0.7 EL(+X)
230. REPEAT LOAD
231. 1 0.6 10 0.7
232. LOAD 317 0.6 DL + 0.7 EL(-X)
233. REPEAT LOAD
234. 1 0.6 10 -0.7
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235. LOAD 318 0.6 DL + 0.7 EL(+Z)
236. REPEAT LOAD
237. 1 0.6 11 0.7
238. LOAD 319 0.6 DL + 0.7 EL(-Z)
239. REPEAT LOAD
240. 1 0.6 11 -0.7
241. ****************************************
242. *****LOAD COMBINATION FOR FOUNDATION DESIGN
243. LOAD 401 1.4 DL
244. REPEAT LOAD
245. 1 1.4
246. LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
247. REPEAT LOAD
248. 1 1.4 8 1.4 2 1.7 3 1.7
249. LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
250. REPEAT LOAD
251. 1 1.4 9 1.4 2 1.7 3 1.7
252. LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
253. REPEAT LOAD
254. 1 1.2 4 1.6 2 0.5 3 0.5
255. LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
256. REPEAT LOAD
257. 1 1.2 5 1.6 2 0.5 3 0.5
258. LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
259. REPEAT LOAD
260. 1 1.2 6 1.6 2 0.5 3 0.5
261. LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
262. REPEAT LOAD
263. 1 1.2 7 1.6 2 0.5 3 0.5
264. LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
265. REPEAT LOAD
266. 1 1.2 10 1.0 2 0.5 3 0.5
267. LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
268. REPEAT LOAD
269. 1 1.2 10 -1.0 2 0.5 3 0.5
270. LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
271. REPEAT LOAD
272. 1 1.2 11 1.0 2 0.5 3 0.5
273. LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
274. REPEAT LOAD
275. 1 1.2 11 -1.0 2 0.5 3 0.5
276. LOAD 412 0.9 DL + 1.6 WL(+Z)
277. REPEAT LOAD
278. 1 0.9 4 1.6
279. LOAD 413 0.9 DL + 1.6 WL(-Z)
280. REPEAT LOAD
281. 1 0.9 5 1.6
282. LOAD 414 0.9 DL + 1.6 WL(+X)
283. REPEAT LOAD
284. 1 0.9 6 1.6
285. LOAD 415 0.9 DL + 1.6 WL(-X)
286. REPEAT LOAD
287. 1 0.9 7 1.6
288. LOAD 416 0.9 DL + 1.0 EL(+X)
289. REPEAT LOAD
290. 1 0.9 10 1.0
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```
291. LOAD 417 0.9 DL + 1.0 EL(-X)
292. REPEAT LOAD
293. 1 0.9 10 -1.0
294. LOAD 418 0.9 DL + 1.0 EL(+Z)
295. REPEAT LOAD
296. 1 0.9 11 1.0
297. LOAD 419 0.9 DL + 1.0 EL(-Z)
298. REPEAT LOAD
299. 1 0.9 11 -1.0
300. ****************************************
301. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**139**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**112**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**135**

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 429
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

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302. ****************************************
303. LOAD LIST 401 TO 419
304. START CONCRETE DESIGN
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305. CODE ACI 2008
306. FC 35000 MEMB 1 TO 5
307. FYMAIN 420000 MEMB 1 TO 5
308. TRACK 1 ALL
309. CLB 0.05 MEMB 1 TO 5
310. CLT 0.05 MEMB 1 TO 5
311. CLS 0.05 ALL
312. MAXMAIN 20 MEMB 1 TO 4
313. MAXMAIN 12 MEMB 5
314. MINMAIN 16 MEMB 1 TO 4
315. MINMAIN 12 MEMB 5
316. MINSEC 8 ALL
317. REINF 0 MEMB 1 TO 4
318. DESIGN COLUMN 1 TO 4
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COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	139.89	0.00	0.00001

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	139.89	0.70	0.00166

==

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    132.07      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    132.07      0.00    0.00000
 ---------------------------------------------------------
```

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COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    197.40      0.00    0.00001
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    197.40     10.05    0.01697
 ---------------------------------------------------------
```

==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	177.21	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	177.21	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

319. DESIGN BEAM 5

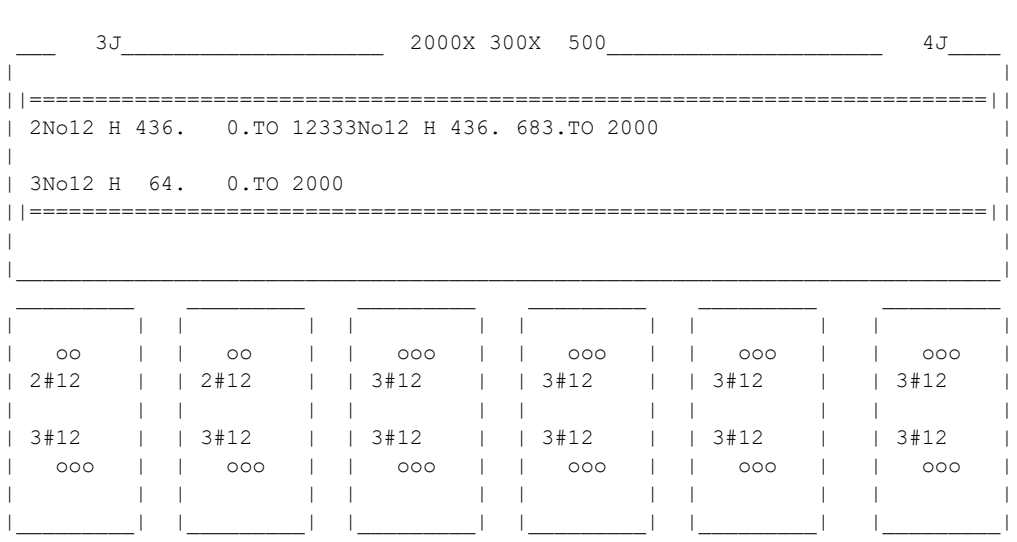
BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	3 - 12MM	0.	2000.	YES	YES
	CRITICAL POS MOMENT= 33.59 KN-MET AT 0.MM, LOAD 404					
	REQD STEEL= 274.MM2, RHO=0.0021, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 86. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 35784.0 cm^4					
2	436.	2 - 12MM	0.	1233.	YES	NO
	CRITICAL NEG MOMENT= 27.74 KN-MET AT 0.MM, LOAD 413					
	REQD STEEL= 226.MM2, RHO=0.0017, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 24968.6 cm^4					
3	436.	3 - 12MM	683.	2000.	NO	YES
	CRITICAL NEG MOMENT= 33.49 KN-MET AT 2000.MM, LOAD 404					
	REQD STEEL= 273.MM2, RHO=0.0021, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 86. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 35784.0 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 1.22 KNS Vc= 123.06 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 6.82 KNS Vc= 226.15 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

320. END CONCRETE DESIGN
321. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 11:53:57 ****

```
*************************************************************
*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
*                                                           *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.        *
*                http://www.bentley.com                     *
*************************************************************
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	07-04-2020	18-05-2020	12-10-2023				Date	07-04-2020	18-05-2020	12-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 5 **STAAD INPUT & OUTPUT FILES OF COMBINED FOOTING MKD. F4 FOR TR1**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 06-Apr-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART TRESTLE TR1 FOUNDATION MKD. F4
JOB REF FOUNDATIONS FOR CONVEYOR 3110-CNV-002
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 2; 3 0 1.49012e-008 0; 4 0 1.49012e-008 2; 5 0 0.35 0;
6 0 0.35 2; 7 0 -3 -1.7; 8 0 -3 3.7; 9 1.6 -3 -1.7; 10 1.6 -3 3.7;
11 -1.6 -3 -1.7; 12 -1.6 -3 3.7; 13 -0.75 -3 3.7; 14 -0.75 -3 2.75;
15 -1.6 -3 2.75; 16 -0.5 -3 3.7; 17 -0.5 -3 2.75; 18 -0.25 -3 3.7;
19 -0.25 -3 2.75; 20 0 -3 2.75; 21 0.25 -3 3.7; 22 0.25 -3 2.75; 23 0.5 -3 3.7;
24 0.5 -3 2.75; 25 0.75 -3 3.7; 26 0.75 -3 2.75; 27 1.6 -3 2.75;
28 -0.75 -3 2.5; 29 -1.6 -3 2.5; 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5;
33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5; 36 1.6 -3 2.5; 37 -0.75 -3 2.25;
38 -1.6 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25;
42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.6 -3 2.25;
46 -0.75 -3 2; 47 -1.6 -3 2; 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2;
51 0.5 -3 2; 52 0.75 -3 2; 53 1.6 -3 2; 54 -0.75 -3 1.75; 55 -1.6 -3 1.75;
56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75;
60 0.5 -3 1.75; 61 0.75 -3 1.75; 62 1.6 -3 1.75; 63 -0.75 -3 1.5;
64 -1.6 -3 1.5; 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5;
69 0.5 -3 1.5; 70 0.75 -3 1.5; 71 1.6 -3 1.5; 72 -0.75 -3 1.25;
73 -1.6 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25;
77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 80 1.6 -3 1.25;
81 -0.75 -3 1; 82 -1.6 -3 1; 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1;
86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1; 89 1.6 -3 1; 90 -0.75 -3 0.75;
91 -1.6 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75;
95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 98 1.6 -3 0.75;
99 -0.75 -3 0.5; 100 -1.6 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5;
103 0 -3 0.5; 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 107 1.6 -3 0.5;
108 -0.75 -3 0.25; 109 -1.6 -3 0.25; 110 -0.5 -3 0.25; 111 -0.25 -3 0.25;
112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25; 115 0.75 -3 0.25;
116 1.6 -3 0.25; 117 -0.75 -3 0; 118 -1.6 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0;
121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 124 1.6 -3 0; 125 -0.75 -3 -0.25;
126 -1.6 -3 -0.25; 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25;
130 0.25 -3 -0.25; 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 133 1.6 -3 -0.25;
134 -0.75 -3 -0.5; 135 -1.6 -3 -0.5; 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5;
138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5; 141 0.75 -3 -0.5;
142 1.6 -3 -0.5; 143 -0.75 -3 -0.75; 144 -1.6 -3 -0.75; 145 -0.5 -3 -0.75;
146 -0.25 -3 -0.75; 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75;
150 0.75 -3 -0.75; 151 1.6 -3 -0.75; 152 -0.75 -3 -1.7; 153 -0.5 -3 -1.7;
154 -0.25 -3 -1.7; 155 0.25 -3 -1.7; 156 0.5 -3 -1.7; 157 0.75 -3 -1.7;
158 -0.75 -3 -1; 159 -1.6 -3 -1; 160 -0.75 -3 -1.25; 161 -1.6 -3 -1.25;
162 -0.5 -3 -1; 163 -0.5 -3 -1.25; 164 -0.25 -3 -1; 165 -0.25 -3 -1.25;
```

166 0 -3 -1; 167 0 -3 -1.25; 168 0.25 -3 -1; 169 0.25 -3 -1.25; 170 0.5 -3 -1;
171 0.5 -3 -1.25; 172 0.75 -3 -1; 173 0.75 -3 -1.25; 174 1.6 -3 -1;
175 1.6 -3 -1.25; 176 -0.75 -3 3.25; 177 -1.6 -3 3.25; 178 -0.75 -3 3;
179 -1.6 -3 3; 180 -0.5 -3 3.25; 181 -0.5 -3 3; 182 -0.25 -3 3.25;
183 -0.25 -3 3; 184 0 -3 3.25; 185 0 -3 3; 186 0.25 -3 3.25; 187 0.25 -3 3;
188 0.5 -3 3.25; 189 0.5 -3 3; 190 0.75 -3 3.25; 191 0.75 -3 3;
192 1.6 -3 3.25; 193 1.6 -3 3; 194 -0.75 -3 -1.475; 195 -1.6 -3 -1.475;
196 -0.5 -3 -1.475; 197 -0.25 -3 -1.475; 198 0 -3 -1.475; 199 0.25 -3 -1.475;
200 0.5 -3 -1.475; 201 0.75 -3 -1.475; 202 1.6 -3 -1.475; 203 -0.75 -3 3.475;
204 -1.6 -3 3.475; 205 -0.5 -3 3.475; 206 -0.25 -3 3.475; 207 0 -3 3.475;
208 0.25 -3 3.475; 209 0.5 -3 3.475; 210 0.75 -3 3.475; 211 1.6 -3 3.475;
212 1.03333 -3 2.75; 213 1.03333 -3 2.5; 214 1.31667 -3 2.75;
215 1.31667 -3 2.5; 216 1.03333 -3 2.25; 217 1.31667 -3 2.25; 218 1.03333 -3 2;
219 1.31667 -3 2; 220 1.03333 -3 1.75; 221 1.31667 -3 1.75; 222 1.03333 -3 1.5;
223 1.31667 -3 1.5; 224 1.03333 -3 1.25; 225 1.31667 -3 1.25; 226 1.03333 -3 1;
227 1.31667 -3 1; 228 1.03333 -3 0.75; 229 1.31667 -3 0.75; 230 1.03333 -3 0.5;
231 1.31667 -3 0.5; 232 1.03333 -3 0.25; 233 1.31667 -3 0.25; 234 1.03333 -3 0;
235 1.31667 -3 0; 236 1.03333 -3 -0.25; 237 1.31667 -3 -0.25;
238 1.03333 -3 -0.5; 239 1.31667 -3 -0.5; 240 1.03333 -3 -0.75;
241 1.31667 -3 -0.75; 242 1.03333 -3 -1; 243 1.31667 -3 -1;
244 1.03333 -3 -1.25; 245 1.31667 -3 -1.25; 246 1.03333 -3 3.25;
247 1.03333 -3 3; 248 1.31667 -3 3.25; 249 1.31667 -3 3; 250 1.03333 -3 -1.475;
251 1.31667 -3 -1.475; 252 1.03333 -3 -1.7; 253 1.31667 -3 -1.7;
254 1.03333 -3 3.7; 255 1.03333 -3 3.475; 256 1.31667 -3 3.7;
257 1.31667 -3 3.475; 258 -1.31667 -3 2.75; 259 -1.31667 -3 2.5;
260 -1.03333 -3 2.75; 261 -1.03333 -3 2.5; 262 -1.31667 -3 2.25;
263 -1.03333 -3 2.25; 264 -1.31667 -3 2; 265 -1.03333 -3 2;
266 -1.31667 -3 1.75; 267 -1.03333 -3 1.75; 268 -1.31667 -3 1.5;
269 -1.03333 -3 1.5; 270 -1.31667 -3 1.25; 271 -1.03333 -3 1.25;
272 -1.31667 -3 1; 273 -1.03333 -3 1; 274 -1.31667 -3 0.75;
275 -1.03333 -3 0.75; 276 -1.31667 -3 0.5; 277 -1.03333 -3 0.5;
278 -1.31667 -3 0.25; 279 -1.03333 -3 0.25; 280 -1.31667 -3 0;
281 -1.03333 -3 0; 282 -1.31667 -3 -0.25; 283 -1.03333 -3 -0.25;
284 -1.31667 -3 -0.5; 285 -1.03333 -3 -0.5; 286 -1.31667 -3 -0.75;
287 -1.03333 -3 -0.75; 288 -1.31667 -3 -1; 289 -1.03333 -3 -1;
290 -1.31667 -3 -1.25; 291 -1.03333 -3 -1.25; 292 -1.31667 -3 3.25;
293 -1.31667 -3 3; 294 -1.03333 -3 3.25; 295 -1.03333 -3 3;
296 -1.31667 -3 -1.475; 297 -1.03333 -3 -1.475; 298 -1.31667 -3 -1.7;
299 -1.03333 -3 -1.7; 300 -1.31667 -3 3.7; 301 -1.31667 -3 3.475;
302 -1.03333 -3 3.7; 303 -1.03333 -3 3.475;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
ELEMENT INCIDENCES SHELL
15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33;
20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41;
27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49;
34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56;
41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63;
48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69;
55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77;
60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85;
67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93;
74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99;
80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104;

84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111;
90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117;
96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121;
100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127;
105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131;
111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137;
114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140;
119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146;
122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149;
137 143 145 162 158; 138 158 162 163 160; 140 145 146 164 162;
141 162 164 165 163; 143 146 147 166 164; 144 164 166 167 165;
146 147 148 168 166; 147 166 168 169 167; 149 148 149 170 168;
150 168 170 171 169; 152 149 150 172 170; 153 170 172 173 171;
162 176 180 181 178; 163 178 181 17 14; 165 180 182 183 181; 166 181 183 19 17;
168 182 184 185 183; 169 183 185 20 19; 171 184 186 187 185; 172 185 187 22 20;
174 186 188 189 187; 175 187 189 24 22; 177 188 190 191 189; 178 189 191 26 24;
184 160 163 196 194; 185 194 196 153 152; 186 163 165 197 196;
187 196 197 154 153; 188 165 167 198 197; 189 197 198 7 154;
190 167 169 199 198; 191 198 199 155 7; 192 169 171 200 199;
193 199 200 156 155; 194 171 173 201 200; 195 200 201 157 156;
200 13 16 205 203; 201 203 205 180 176; 202 16 18 206 205; 203 205 206 182 180;
204 18 8 207 206; 205 206 207 184 182; 206 8 21 208 207; 207 207 208 186 184;
208 21 23 209 208; 209 208 209 188 186; 210 23 25 210 209; 211 209 210 190 188;
214 26 212 213 35; 215 212 214 215 213; 216 214 27 36 215; 217 35 213 216 44;
218 213 215 217 216; 219 215 36 45 217; 220 44 216 218 52; 221 216 217 219 218;
222 217 45 53 219; 223 52 218 220 61; 224 218 219 221 220; 225 219 53 62 221;
226 61 220 222 70; 227 220 221 223 222; 228 221 62 71 223; 229 70 222 224 79;
230 222 223 225 224; 231 223 71 80 225; 232 79 224 226 88; 233 224 225 227 226;
234 225 80 89 227; 235 88 226 228 97; 236 226 227 229 228; 237 227 89 98 229;
238 97 228 230 106; 239 228 229 231 230; 240 229 98 107 231;
241 106 230 232 115; 242 230 231 233 232; 243 231 107 116 233;
244 115 232 234 123; 245 232 233 235 234; 246 233 116 124 235;
247 123 234 236 132; 248 234 235 237 236; 249 235 124 133 237;
250 132 236 238 141; 251 236 237 239 238; 252 237 133 142 239;
253 141 238 240 150; 254 238 239 241 240; 255 239 142 151 241;
256 150 240 242 172; 257 240 241 243 242; 258 241 151 174 243;
259 172 242 244 173; 260 242 243 245 244; 261 243 174 175 245;
262 190 246 247 191; 263 246 248 249 247; 264 248 192 193 249;
265 191 247 212 26; 266 247 249 214 212; 267 249 193 27 214;
268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251;
271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253;
274 25 254 255 210; 275 254 256 257 255; 276 256 10 211 257;
277 210 255 246 190; 278 255 257 248 246; 279 257 211 192 248;
280 15 258 259 29; 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38;
284 259 261 263 262; 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264;
288 263 37 46 265; 289 47 264 266 55; 290 264 265 267 266; 291 265 46 54 267;
292 55 266 268 64; 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73;
296 268 269 271 270; 297 269 63 72 271; 298 73 270 272 82; 299 270 271 273 272;
300 271 72 81 273; 301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275;
304 91 274 276 100; 305 274 275 277 276; 306 275 90 99 277;
307 100 276 278 109; 308 276 277 279 278; 309 277 99 108 279;
310 109 278 280 118; 311 278 279 281 280; 312 279 108 117 281;
313 118 280 282 126; 314 280 281 283 282; 315 281 117 125 283;
316 126 282 284 135; 317 282 283 285 284; 318 283 125 134 285;

```
319 135 284 286 144; 320 284 285 287 286; 321 285 134 143 287;
322 144 286 288 159; 323 286 287 289 288; 324 287 143 158 289;
325 159 288 290 161; 326 288 289 291 290; 327 289 158 160 291;
328 177 292 293 179; 329 292 294 295 293; 330 294 176 178 295;
331 179 293 258 15; 332 293 295 260 258; 333 295 178 14 260;
334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297;
337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299;
340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303;
343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
214 TO 345
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
_SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 345
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.6 ZD 0.6
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
*****************************************
SUPPORTS
1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
*****************************************
MEMBER OFFSET
5 START 0 -0.25 0
5 END 0 -0.25 0
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
```

```
5 FX -0.029 FY -99.633 FZ -11.556
6 FX 0.005 FY -155.245 FZ 19.556
ELEMENT LOAD
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
*****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
5 FX -0.005 FY -48.1 FZ -5.074
6 FX 0.005 FY -47.9 FZ 5.074
*****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
5 FX -0.01 FY -49.2 FZ -4.718
6 FX 0.01 FY -49.2 FZ 4.718
*****************************************
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
5 FX 0.102 FY 110.886 FZ 26.896
6 FX -0.102 FY -110.886 FZ 27.407
*****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
5 FX -0.103 FY -110.896 FZ -26.878
6 FX 0.103 FY 110.896 FZ -27.424
*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
5 FX 0.769 FY 0 FZ 0
6 FX 0.769 FY 0 FZ 0
*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
5 FX -0.769 FY 0 FZ 0
6 FX -0.769 FY 0 FZ 0
*****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
5 FX -0.014 FY 0.134 FZ -9.799
6 FX 0.014 FY -0.134 FZ 9.779
*****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
5 FX 0.014 FY -0.134 FZ 9.799
6 FX -0.014 FY 0.134 FZ -9.779
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
5 FX 0.033 FY 0.013 FZ 0.001
6 FX 0.034 FY -0.063 FZ 0.009
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
```

```
JOINT LOAD
5 FX -0.005 FY 9.167 FZ 2.995
6 FX 0.005 FY -9.167 FZ 2.884
*****************************************
*****Load combination for bearing capacity check
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 6 0.75 2 0.75 3 0.75
LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 7 0.75 2 0.75 3 0.75
LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 0.525 2 0.75 3 0.75
LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 0.525 2 0.75 3 0.75
LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD 312 0.6 DL + 1.0 WL(+Z)
REPEAT LOAD
1 0.6 4 1.0
LOAD 313 0.6 DL + 1.0 WL(-Z)
REPEAT LOAD
1 0.6 5 1.0
LOAD 314 0.6 DL + 1.0 WL(+X)
REPEAT LOAD
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
```

```
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
****************************************
*****Load combination for foundation design
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 5 1.6
LOAD 414 0.9 DL + 1.6 WL(+X)
REPEAT LOAD
1 0.9 6 1.6
LOAD 415 0.9 DL + 1.6 WL(-X)
REPEAT LOAD
```

```
1 0.9 7 1.6
LOAD 416 0.9 DL + 1.0 EL(+X)
REPEAT LOAD
1 0.9 10 1.0
LOAD 417 0.9 DL + 1.0 EL(-X)
REPEAT LOAD
1 0.9 10 -1.0
LOAD 418 0.9 DL + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 11 1.0
LOAD 419 0.9 DL + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 11 -1.0
**************************************
PERFORM ANALYSIS
**************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 5
FYMAIN 420000 MEMB 1 TO 5
TRACK 1 ALL
CLB 0.05 MEMB 1 TO 5
CLT 0.05 MEMB 1 TO 5
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  7, 2020                *
*           Time=    12:11:28                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3110-CNV-002\Staad\3110-CNV-002_TR1.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 06-APR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART TRESTLE TR1 FOUNDATION MKD. F4
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3110-CNV-002
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 -3 0; 2 0 -3 2; 3 0 1.49012E-008 0; 4 0 1.49012E-008 2; 5 0 0.35 0
   18. 6 0 0.35 2; 7 0 -3 -1.7; 8 0 -3 3.7; 9 1.6 -3 -1.7; 10 1.6 -3 3.7
   19. 11 -1.6 -3 -1.7; 12 -1.6 -3 3.7; 13 -0.75 -3 3.7; 14 -0.75 -3 2.75
   20. 15 -1.6 -3 2.75; 16 -0.5 -3 3.7; 17 -0.5 -3 2.75; 18 -0.25 -3 3.7
   21. 19 -0.25 -3 2.75; 20 0 -3 2.75; 21 0.25 -3 3.7; 22 0.25 -3 2.75; 23 0.5 -3 3.7
   22. 24 0.5 -3 2.75; 25 0.75 -3 3.7; 26 0.75 -3 2.75; 27 1.6 -3 2.75
   23. 28 -0.75 -3 2.5; 29 -1.6 -3 2.5; 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5
   24. 33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5; 36 1.6 -3 2.5; 37 -0.75 -3 2.25
   25. 38 -1.6 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25
   26. 42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.6 -3 2.25
   27. 46 -0.75 -3 2; 47 -1.6 -3 2; 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2
   28. 51 0.5 -3 2; 52 0.75 -3 2; 53 1.6 -3 2; 54 -0.75 -3 1.75; 55 -1.6 -3 1.75
   29. 56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75
   30. 60 0.5 -3 1.75; 61 0.75 -3 1.75; 62 1.6 -3 1.75; 63 -0.75 -3 1.5
   31. 64 -1.6 -3 1.5; 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5
   32. 69 0.5 -3 1.5; 70 0.75 -3 1.5; 71 1.6 -3 1.5; 72 -0.75 -3 1.25
   33. 73 -1.6 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25
   34. 77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 80 1.6 -3 1.25
   35. 81 -0.75 -3 1; 82 -1.6 -3 1; 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1
   36. 86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1; 89 1.6 -3 1; 90 -0.75 -3 0.75
   37. 91 -1.6 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75
   38. 95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 98 1.6 -3 0.75
```

39. 99 -0.75 -3 0.5; 100 -1.6 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5
40. 103 0 -3 0.5; 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 107 1.6 -3 0.5
41. 108 -0.75 -3 0.25; 109 -1.6 -3 0.25; 110 -0.5 -3 0.25; 111 -0.25 -3 0.25
42. 112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25; 115 0.75 -3 0.25
43. 116 1.6 -3 0.25; 117 -0.75 -3 0; 118 -1.6 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0
44. 121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 124 1.6 -3 0; 125 -0.75 -3 -0.25
45. 126 -1.6 -3 -0.25; 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25
46. 130 0.25 -3 -0.25; 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 133 1.6 -3 -0.25
47. 134 -0.75 -3 -0.5; 135 -1.6 -3 -0.5; 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5
48. 138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5; 141 0.75 -3 -0.5
49. 142 1.6 -3 -0.5; 143 -0.75 -3 -0.75; 144 -1.6 -3 -0.75; 145 -0.5 -3 -0.75
50. 146 -0.25 -3 -0.75; 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75
51. 150 0.75 -3 -0.75; 151 1.6 -3 -0.75; 152 -0.75 -3 -1.7; 153 -0.5 -3 -1.7
52. 154 -0.25 -3 -1.7; 155 0.25 -3 -1.7; 156 0.5 -3 -1.7; 157 0.75 -3 -1.7
53. 158 -0.75 -3 -1; 159 -1.6 -3 -1; 160 -0.75 -3 -1.25; 161 -1.6 -3 -1.25
54. 162 -0.5 -3 -1; 163 -0.5 -3 -1.25; 164 -0.25 -3 -1; 165 -0.25 -3 -1.25
55. 166 0 -3 -1; 167 0 -3 -1.25; 168 0.25 -3 -1; 169 0.25 -3 -1.25; 170 0.5 -3 -1.
56. 171 0.5 -3 -1.25; 172 0.75 -3 -1; 173 0.75 -3 -1.25; 174 1.6 -3 -1.
57. 175 1.6 -3 -1.25; 176 -0.75 -3 3.25; 177 -1.6 -3 3.25; 178 -0.75 -3 3
58. 179 -1.6 -3 3; 180 -0.5 -3 3.25; 181 -0.5 -3 3; 182 -0.25 -3 3.25
59. 183 -0.25 -3 3; 184 0 -3 3.25; 185 0 -3 3; 186 0.25 -3 3.25; 187 0.25 -3 3
60. 188 0.5 -3 3.25; 189 0.5 -3 3; 190 0.75 -3 3.25; 191 0.75 -3 3
61. 192 1.6 -3 3.25; 193 1.6 -3 3; 194 -0.75 -3 -1.475; 195 -1.6 -3 -1.475
62. 196 -0.5 -3 -1.475; 197 -0.25 -3 -1.475; 198 0 -3 -1.475; 199 0.25 -3 -1.475
63. 200 0.5 -3 -1.475; 201 0.75 -3 -1.475; 202 1.6 -3 -1.475; 203 -0.75 -3 3.475
64. 204 -1.6 -3 3.475; 205 -0.5 -3 3.475; 206 -0.25 -3 3.475; 207 0 -3 3.475
65. 208 0.25 -3 3.475; 209 0.5 -3 3.475; 210 0.75 -3 3.475; 211 1.6 -3 3.475
66. 212 1.03333 -3 2.75; 213 1.03333 -3 2.5; 214 1.31667 -3 2.75
67. 215 1.31667 -3 2.5; 216 1.03333 -3 2.25; 217 1.31667 -3 2.25; 218 1.03333 -3 2
68. 219 1.31667 -3 2; 220 1.03333 -3 1.75; 221 1.31667 -3 1.75; 222 1.03333 -3 1.5
69. 223 1.31667 -3 1.5; 224 1.03333 -3 1.25; 225 1.31667 -3 1.25; 226 1.03333 -3 1
70. 227 1.31667 -3 1; 228 1.03333 -3 0.75; 229 1.31667 -3 0.75; 230 1.03333 -3 0.5
71. 231 1.31667 -3 0.5; 232 1.03333 -3 0.25; 233 1.31667 -3 0.25; 234 1.03333 -3 0
72. 235 1.31667 -3 0; 236 1.03333 -3 -0.25; 237 1.31667 -3 -0.25
73. 238 1.03333 -3 -0.5; 239 1.31667 -3 -0.5; 240 1.03333 -3 -0.75
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75. 244 1.03333 -3 -1.25; 245 1.31667 -3 -1.25; 246 1.03333 -3 3.25
76. 247 1.03333 -3 3; 248 1.31667 -3 3.25; 249 1.31667 -3 3; 250 1.03333 -3 -1.475
77. 251 1.31667 -3 -1.475; 252 1.03333 -3 -1.7; 253 1.31667 -3 -1.7
78. 254 1.03333 -3 3.7; 255 1.03333 -3 3.475; 256 1.31667 -3 3.7
79. 257 1.31667 -3 3.475; 258 -1.31667 -3 2.75; 259 -1.31667 -3 2.5
80. 260 -1.03333 -3 2.75; 261 -1.03333 -3 2.5; 262 -1.31667 -3 2.25
81. 263 -1.03333 -3 2.25; 264 -1.31667 -3 2; 265 -1.03333 -3 2
82. 266 -1.31667 -3 1.75; 267 -1.03333 -3 1.75; 268 -1.31667 -3 1.5
83. 269 -1.03333 -3 1.5; 270 -1.31667 -3 1.25; 271 -1.03333 -3 1.25
84. 272 -1.31667 -3 1; 273 -1.03333 -3 1; 274 -1.31667 -3 0.75
85. 275 -1.03333 -3 0.75; 276 -1.31667 -3 0.5; 277 -1.03333 -3 0.5
86. 278 -1.31667 -3 0.25; 279 -1.03333 -3 0.25; 280 -1.31667 -3 0
87. 281 -1.03333 -3 0; 282 -1.31667 -3 -0.25; 283 -1.03333 -3 -0.25
88. 284 -1.31667 -3 -0.5; 285 -1.03333 -3 -0.5; 286 -1.31667 -3 -0.75
89. 287 -1.03333 -3 -0.75; 288 -1.31667 -3 -1; 289 -1.03333 -3 -1.
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91. 293 -1.31667 -3 3; 294 -1.03333 -3 3.25; 295 -1.03333 -3 3
92. 296 -1.31667 -3 -1.475; 297 -1.03333 -3 -1.475; 298 -1.31667 -3 -1.7
93. 299 -1.03333 -3 -1.7; 300 -1.31667 -3 3.7; 301 -1.31667 -3 3.475
94. 302 -1.03333 -3 3.7; 303 -1.03333 -3 3.475

95. MEMBER INCIDENCES
96. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
97. ELEMENT INCIDENCES SHELL
98. 15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33
99. 20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41
100. 27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49
101. 34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56
102. 41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63
103. 48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69
104. 55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77
105. 60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85
106. 67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93
107. 74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99
108. 80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104
109. 84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111
110. 90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117
111. 96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121
112. 100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127
113. 105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131
114. 111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137
115. 114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140
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122. 162 176 180 181 178; 163 178 181 17 14; 165 180 182 183 181; 166 181 183 19 17
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129. 200 13 16 205 203; 201 203 205 180 176; 202 16 18 206 205; 203 205 206 182 180
130. 204 18 8 207 206; 205 206 207 184 182; 206 8 21 208 207; 207 207 208 186 184
131. 208 21 23 209 208; 209 208 209 188 186; 210 23 25 210 209; 211 209 210 190 188
132. 214 26 212 213 35; 215 212 214 215 213; 216 214 27 36 215; 217 35 213 216 44
133. 218 213 215 217 216; 219 215 36 45 217; 220 44 216 218 52; 221 216 217 219 218
134. 222 217 45 53 219; 223 52 218 220 61; 224 218 219 221 220; 225 219 53 62 221
135. 226 61 220 222 70; 227 220 221 223 222; 228 221 62 71 223; 229 70 222 224 79
136. 230 222 223 225 224; 231 223 71 80 225; 232 79 224 226 88; 233 224 225 227 226
137. 234 225 80 89 227; 235 88 226 228 97; 236 226 227 229 228; 237 227 89 98 229
138. 238 97 228 230 106; 239 228 229 231 230; 240 229 98 107 231
139. 241 106 230 232 115; 242 230 231 233 232; 243 231 107 116 233
140. 244 115 232 234 123; 245 232 233 235 234; 246 233 116 124 235
141. 247 123 234 236 132; 248 234 235 237 236; 249 235 124 133 237
142. 250 132 236 238 141; 251 236 237 239 238; 252 237 133 142 239
143. 253 141 238 240 150; 254 238 239 241 240; 255 239 142 151 241
144. 256 150 240 242 172; 257 240 241 243 242; 258 241 151 174 243
145. 259 172 242 244 173; 260 242 243 245 244; 261 243 174 175 245
146. 262 190 246 247 191; 263 246 248 249 247; 264 248 192 193 249
147. 265 191 247 212 26; 266 247 249 214 212; 267 249 193 27 214
148. 268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251
149. 271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253
150. 274 25 254 255 210; 275 254 256 257 255; 276 256 10 211 257

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151. 277 210 255 246 190; 278 255 257 248 246; 279 257 211 192 248
152. 280 15 258 259 29; 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38
153. 284 259 261 263 262; 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264
154. 288 263 37 46 265; 289 47 264 266 55; 290 264 265 267 266; 291 265 46 54 267
155. 292 55 266 268 64; 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73
156. 296 268 269 271 270; 297 269 63 72 271; 298 73 270 272 82; 299 270 271 273 272
157. 300 271 72 81 273; 301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275
158. 304 91 274 276 100; 305 274 275 277 276; 306 275 90 99 277
159. 307 100 276 278 109; 308 276 277 279 278; 309 277 99 108 279
160. 310 109 278 280 118; 311 278 279 281 280; 312 279 108 117 281
161. 313 118 280 282 126; 314 280 281 283 282; 315 281 117 125 283
162. 316 126 282 284 135; 317 282 283 285 284; 318 283 125 134 285
163. 319 135 284 286 144; 320 284 285 287 286; 321 285 134 143 287
164. 322 144 286 288 159; 323 286 287 289 288; 324 287 143 158 289
165. 325 159 288 290 161; 326 288 289 291 290; 327 289 158 160 291
166. 328 177 292 293 179; 329 292 294 295 293; 330 294 176 178 295
167. 331 179 293 258 15; 332 293 295 260 258; 333 295 178 14 260
168. 334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297
169. 337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299
170. 340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303
171. 343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294
172. START GROUP DEFINITION
173. ELEMENT
174. _MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
175. 63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
176. 108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
177. 165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
178. 214 TO 345
179. _MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
180. 111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
181. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
182. 214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
183. _SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
184. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
185. 214 TO 345
186. END GROUP DEFINITION
187. ELEMENT PROPERTY
188. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
189. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
190. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
191. 178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
192. DEFINE MATERIAL START
193. ISOTROPIC CONCRETE
194. E 2.17185E+007
195. POISSON 0.17
196. DENSITY 23.5616
197. ALPHA 1E-005
198. DAMP 0.05
199. TYPE CONCRETE
200. STRENGTH FCU 27579
201. END DEFINE MATERIAL
202. MEMBER PROPERTY AMERICAN
203. 1 TO 4 PRIS YD 0.6 ZD 0.6
204. 5 PRIS YD 0.5 ZD 0.3
205. CONSTANTS
206. MATERIAL CONCRETE ALL
```

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207. ****************************************
208. SUPPORTS
209. 1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
```

JOINT	INFLUENCE AREA
1	0.062500
2	0.062500
7	0.028125
8	0.028125
9	0.015937
10	0.015937
11	0.015937
12	0.015937
13	0.030000
14	0.066666
15	0.035416
16	0.028125
17	0.062500
18	0.028125
19	0.062500
20	0.062500
21	0.028125
22	0.062500
23	0.028125
24	0.062500
25	0.030000
26	0.066666
27	0.035416
28	0.066666
29	0.035416
30	0.062500
31	0.062500
32	0.062500
33	0.062500
34	0.062500
35	0.066666
36	0.035416
37	0.066666
38	0.035416
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500
44	0.066666
45	0.035416
46	0.066666
47	0.035416
48	0.062500
49	0.062500
50	0.062500
51	0.062500
52	0.066666
53	0.035416

54	0.066666
55	0.035416
56	0.062500
57	0.062500
58	0.062500
59	0.062500
60	0.062500
61	0.066666
62	0.035416
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64	0.035416
65	0.062500
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67	0.062500
68	0.062500
69	0.062500
70	0.066666
71	0.035416
72	0.066666
73	0.035416
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.066666
80	0.035416
81	0.066666
82	0.035416
83	0.062500
84	0.062500
85	0.062500
86	0.062500
87	0.062500
88	0.066666
89	0.035416
90	0.066666
91	0.035416
92	0.062500
93	0.062500
94	0.062500
95	0.062500
96	0.062500
97	0.066666
98	0.035416
99	0.066666
100	0.035416
101	0.062500
102	0.062500
103	0.062500
104	0.062500
105	0.062500
106	0.066666
107	0.035416
108	0.066666
109	0.035416

110	0.062500
111	0.062500
112	0.062500
113	0.062500
114	0.062500
115	0.066666
116	0.035416
117	0.066666
118	0.035416
119	0.062500
120	0.062500
121	0.062500
122	0.062500
123	0.066666
124	0.035416
125	0.066666
126	0.035416
127	0.062500
128	0.062500
129	0.062500
130	0.062500
131	0.062500
132	0.066666
133	0.035416
134	0.066666
135	0.035416
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
141	0.066666
142	0.035416
143	0.066666
144	0.035416
145	0.062500
146	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.066666
151	0.035416
152	0.030000
153	0.028125
154	0.028125
155	0.028125
156	0.028125
157	0.030000
158	0.066666
159	0.035416
160	0.063333
161	0.033645
162	0.062500
163	0.059375
164	0.062500
165	0.059375

166	0.062500
167	0.059375
168	0.062500
169	0.059375
170	0.062500
171	0.059375
172	0.066666
173	0.063333
174	0.035416
175	0.033645
176	0.063333
177	0.033645
178	0.066666
179	0.035416
180	0.059375
181	0.062500
182	0.059375
183	0.062500
184	0.059375
185	0.062500
186	0.059375
187	0.062500
188	0.059375
189	0.062500
190	0.063333
191	0.066666
192	0.033645
193	0.035416
194	0.060000
195	0.031875
196	0.056250
197	0.056250
198	0.056250
199	0.056250
200	0.056250
201	0.060000
202	0.031875
203	0.060000
204	0.031875
205	0.056250
206	0.056250
207	0.056250
208	0.056250
209	0.056250
210	0.060000
211	0.031875
212	0.070834
213	0.070834
214	0.070834
215	0.070834
216	0.070834
217	0.070834
218	0.070834
219	0.070834
220	0.070834
221	0.070834

222	0.070834
223	0.070834
224	0.070834
225	0.070834
226	0.070834
227	0.070834
228	0.070834
229	0.070834
230	0.070834
231	0.070834
232	0.070834
233	0.070834
234	0.070834
235	0.070834
236	0.070834
237	0.070834
238	0.070834
239	0.070834
240	0.070834
241	0.070834
242	0.070834
243	0.070834
244	0.067292
245	0.067292
246	0.067292
247	0.070834
248	0.067292
249	0.070834
250	0.063750
251	0.063750
252	0.031875
253	0.031875
254	0.031875
255	0.063750
256	0.031875
257	0.063750
258	0.070834
259	0.070834
260	0.070834
261	0.070834
262	0.070834
263	0.070834
264	0.070834
265	0.070834
266	0.070834
267	0.070834
268	0.070834
269	0.070834
270	0.070834
271	0.070834
272	0.070834
273	0.070834
274	0.070834
275	0.070834
276	0.070834
277	0.070834

278	0.070834
279	0.070834
280	0.070834
281	0.070834
282	0.070834
283	0.070834
284	0.070834
285	0.070834
286	0.070834
287	0.070834
288	0.070834
289	0.070834
290	0.067292
291	0.067292
292	0.067292
293	0.070834
294	0.067292
295	0.070834
296	0.063750
297	0.063750
298	0.031875
299	0.031875
300	0.031875
301	0.063750
302	0.031875
303	0.063750
TOTAL	17.280014

```
210. ****************************************
211. MEMBER OFFSET
212. 5 START 0 -0.25 0
213. 5 END 0 -0.25 0
214. LOAD 1 DEAD LOAD DL
215. SELFWEIGHT Y -1
216. JOINT LOAD
217. 5 FX -0.029 FY -99.633 FZ -11.556
218. 6 FX 0.005 FY -155.245 FZ 19.556
219. ELEMENT LOAD
220. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
221. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
222. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
223. 178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
224. ****************************************
225. LOAD 2 LIVE LOAD LL
226. JOINT LOAD
227. 5 FX -0.005 FY -48.1 FZ -5.074
228. 6 FX 0.005 FY -47.9 FZ 5.074
229. ****************************************
230. LOAD 3 MATERIAL LOAD LL
231. JOINT LOAD
232. 5 FX -0.01 FY -49.2 FZ -4.718
233. 6 FX 0.01 FY -49.2 FZ 4.718
234. ****************************************
235. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
236. JOINT LOAD
237. 5 FX 0.102 FY 110.886 FZ 26.896
238. 6 FX -0.102 FY -110.886 FZ 27.407
```

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239. ****************************************
240. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
241. JOINT LOAD
242. 5 FX -0.103 FY -110.896 FZ -26.878
243. 6 FX 0.103 FY 110.896 FZ -27.424
244. ****************************************
245. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
246. JOINT LOAD
247. 5 FX 0.769 FY 0 FZ 0
248. 6 FX 0.769 FY 0 FZ 0
249. ****************************************
250. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
251. JOINT LOAD
252. 5 FX -0.769 FY 0 FZ 0
253. 6 FX -0.769 FY 0 FZ 0
254. ****************************************
255. LOAD 8 TEMP LOAD   TL1
256. JOINT LOAD
257. 5 FX -0.014 FY 0.134 FZ -9.799
258. 6 FX 0.014 FY -0.134 FZ 9.779
259. ****************************************
260. LOAD 9 TEMP LOAD   TL2
261. JOINT LOAD
262. 5 FX 0.014 FY -0.134 FZ 9.799
263. 6 FX -0.014 FY 0.134 FZ -9.779
264. ****************************************
265. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
266. JOINT LOAD
267. 5 FX 0.033 FY 0.013 FZ 0.001
268. 6 FX 0.034 FY -0.063 FZ 0.009
269. ****************************************
270. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
271. JOINT LOAD
272. 5 FX -0.005 FY 9.167 FZ 2.995
273. 6 FX 0.005 FY -9.167 FZ 2.884
274. ****************************************
275. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
276. LOAD 301 1.0 DL
277. REPEAT LOAD
278. 1 1.0
279. LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
280. REPEAT LOAD
281. 1 1.0 8 1.0 2 1.0 3 1.0
282. LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
283. REPEAT LOAD
284. 1 1.0 9 1.0 2 1.0 3 1.0
285. LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
286. REPEAT LOAD
287. 1 1.0 4 0.75 2 0.75 3 0.75
288. LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
289. REPEAT LOAD
290. 1 1.0 5 0.75 2 0.75 3 0.75
291. LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
292. REPEAT LOAD
293. 1 1.0 6 0.75 2 0.75 3 0.75
294. LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
```

```
295. REPEAT LOAD
296. 1 1.0 7 0.75 2 0.75 3 0.75
297. LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
298. REPEAT LOAD
299. 1 1.0 10 0.525 2 0.75 3 0.75
300. LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
301. REPEAT LOAD
302. 1 1.0 10 -0.525 2 0.75 3 0.75
303. LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
304. REPEAT LOAD
305. 1 1.0 11 0.525 2 0.75 3 0.75
306. LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
307. REPEAT LOAD
308. 1 1.0 11 -0.525 2 0.75 3 0.75
309. LOAD 312 0.6 DL + 1.0 WL(+Z)
310. REPEAT LOAD
311. 1 0.6 4 1.0
312. LOAD 313 0.6 DL + 1.0 WL(-Z)
313. REPEAT LOAD
314. 1 0.6 5 1.0
315. LOAD 314 0.6 DL + 1.0 WL(+X)
316. REPEAT LOAD
317. 1 0.6 6 1.0
318. LOAD 315 0.6 DL + 1.0 WL(-X)
319. REPEAT LOAD
320. 1 0.6 7 1.0
321. LOAD 316 0.6 DL + 0.7 EL(+X)
322. REPEAT LOAD
323. 1 0.6 10 0.7
324. LOAD 317 0.6 DL + 0.7 EL(-X)
325. REPEAT LOAD
326. 1 0.6 10 -0.7
327. LOAD 318 0.6 DL + 0.7 EL(+Z)
328. REPEAT LOAD
329. 1 0.6 11 0.7
330. LOAD 319 0.6 DL + 0.7 EL(-Z)
331. REPEAT LOAD
332. 1 0.6 11 -0.7
333. ****************************************
334. *****LOAD COMBINATION FOR FOUNDATION DESIGN
335. LOAD 401 1.4 DL
336. REPEAT LOAD
337. 1 1.4
338. LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
339. REPEAT LOAD
340. 1 1.4 8 1.4 2 1.7 3 1.7
341. LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
342. REPEAT LOAD
343. 1 1.4 9 1.4 2 1.7 3 1.7
344. LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
345. REPEAT LOAD
346. 1 1.2 4 1.6 2 0.5 3 0.5
347. LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
348. REPEAT LOAD
349. 1 1.2 5 1.6 2 0.5 3 0.5
350. LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
```

```
351. REPEAT LOAD
352. 1 1.2 6 1.6 2 0.5 3 0.5
353. LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
354. REPEAT LOAD
355. 1 1.2 7 1.6 2 0.5 3 0.5
356. LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
357. REPEAT LOAD
358. 1 1.2 10 1.0 2 0.5 3 0.5
359. LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
360. REPEAT LOAD
361. 1 1.2 10 -1.0 2 0.5 3 0.5
362. LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
363. REPEAT LOAD
364. 1 1.2 11 1.0 2 0.5 3 0.5
365. LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
366. REPEAT LOAD
367. 1 1.2 11 -1.0 2 0.5 3 0.5
368. LOAD 412 0.9 DL + 1.6 WL(+Z)
369. REPEAT LOAD
370. 1 0.9 4 1.6
371. LOAD 413 0.9 DL + 1.6 WL(-Z)
372. REPEAT LOAD
373. 1 0.9 5 1.6
374. LOAD 414 0.9 DL + 1.6 WL(+X)
375. REPEAT LOAD
376. 1 0.9 6 1.6
377. LOAD 415 0.9 DL + 1.6 WL(-X)
378. REPEAT LOAD
379. 1 0.9 7 1.6
380. LOAD 416 0.9 DL + 1.0 EL(+X)
381. REPEAT LOAD
382. 1 0.9 10 1.0
383. LOAD 417 0.9 DL + 1.0 EL(-X)
384. REPEAT LOAD
385. 1 0.9 10 -1.0
386. LOAD 418 0.9 DL + 1.0 EL(+Z)
387. REPEAT LOAD
388. 1 0.9 11 1.0
389. LOAD 419 0.9 DL + 1.0 EL(-Z)
390. REPEAT LOAD
391. 1 0.9 11 -1.0
392. ****************************************
393. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**303**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**264**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**299**

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 921
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

394. **
395. LOAD LIST 401 TO 419
396. START CONCRETE DESIGN

```
397. CODE ACI 2008
398. FC 35000 MEMB 1 TO 5
399. FYMAIN 420000 MEMB 1 TO 5
400. TRACK 1 ALL
401. CLB 0.05 MEMB 1 TO 5
402. CLT 0.05 MEMB 1 TO 5
403. CLS 0.05 ALL
404. MAXMAIN 25 ALL
405. MINMAIN 16 ALL
406. MINSEC 8 ALL
407. REINF 0 MEMB 1 TO 4
408. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	218.46	0.02	0.00003

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	218.46	18.74	0.02859

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	214.59	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	214.59	0.00	0.00000

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	358.52	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	358.52	4.50	0.00418

==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	334.37	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	334.37	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

409. DESIGN BEAM 5

BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

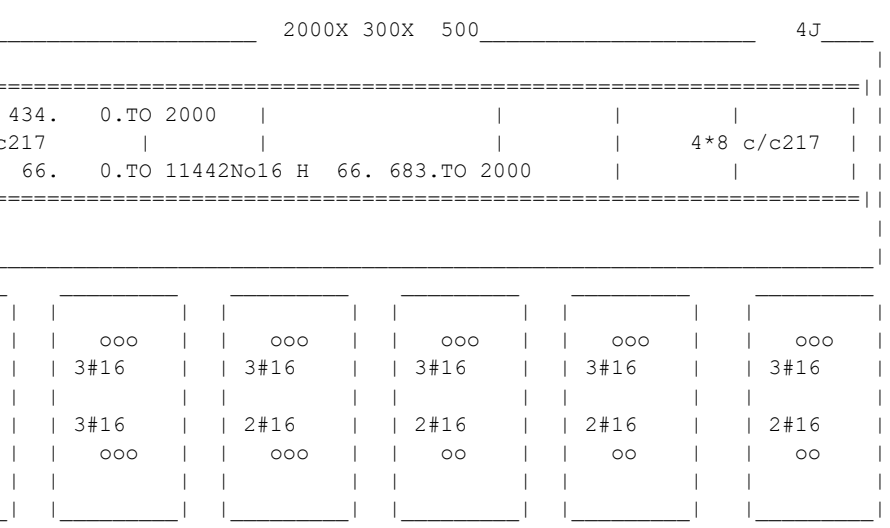
LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	1144.	YES	NO
	CRITICAL POS MOMENT= 55.97 KN-MET AT 0.MM, LOAD 412					
	REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
2	66.	2 - 16MM	683.	2000.	NO	YES
	CRITICAL POS MOMENT= 47.63 KN-MET AT 2000.MM, LOAD 413					
	REQD STEEL= 392.MM2, RHO=0.0030, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 41100.8 cm^4					
3	434.	3 - 16MM	0.	2000.	YES	YES
	CRITICAL NEG MOMENT= 66.66 KN-MET AT 2000.MM, LOAD 404					
	REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 58.56 KNS Vc= 121.98 KNS Vs= 0.00 KNS
Tu= 0.04 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM

AT END SUPPORT - Vu= 63.36 KNS Vc= 121.98 KNS Vs= 0.00 KNS
Tu= 0.04 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM



**********END OF BEAM DESIGN**********

410. END CONCRETE DESIGN
411. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 12:11:36 ****

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************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
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Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	07-04-2020	18-05-2020	12-10-2023				Date	07-04-2020	18-05-2020	12-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 6 **STAAD INPUT & OUTPUT FILES OF COMBINED FOOTING MKD. F4 FOR TR2**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 06-Apr-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART TRESTLE TR2 FOUNDATION MKD. F4
JOB REF FOUNDATIONS FOR CONVEYOR 3110-CNV-002
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 2; 3 0 1.49012e-008 0; 4 0 1.49012e-008 2; 5 0 0.35 0;
6 0 0.35 2; 7 0 -3 -1.7; 8 0 -3 3.7; 9 1.6 -3 -1.7; 10 1.6 -3 3.7;
11 -1.6 -3 -1.7; 12 -1.6 -3 3.7; 13 -0.75 -3 3.7; 14 -0.75 -3 2.75;
15 -1.6 -3 2.75; 16 -0.5 -3 3.7; 17 -0.5 -3 2.75; 18 -0.25 -3 3.7;
19 -0.25 -3 2.75; 20 0 -3 2.75; 21 0.25 -3 3.7; 22 0.25 -3 2.75; 23 0.5 -3 3.7;
24 0.5 -3 2.75; 25 0.75 -3 3.7; 26 0.75 -3 2.75; 27 1.6 -3 2.75;
28 -0.75 -3 2.5; 29 -1.6 -3 2.5; 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5;
33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5; 36 1.6 -3 2.5; 37 -0.75 -3 2.25;
38 -1.6 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25;
42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.6 -3 2.25;
46 -0.75 -3 2; 47 -1.6 -3 2; 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2;
51 0.5 -3 2; 52 0.75 -3 2; 53 1.6 -3 2; 54 -0.75 -3 1.75; 55 -1.6 -3 1.75;
56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75;
60 0.5 -3 1.75; 61 0.75 -3 1.75; 62 1.6 -3 1.75; 63 -0.75 -3 1.5;
64 -1.6 -3 1.5; 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5;
69 0.5 -3 1.5; 70 0.75 -3 1.5; 71 1.6 -3 1.5; 72 -0.75 -3 1.25;
73 -1.6 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25;
77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 80 1.6 -3 1.25;
81 -0.75 -3 1; 82 -1.6 -3 1; 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1;
86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1; 89 1.6 -3 1; 90 -0.75 -3 0.75;
91 -1.6 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75;
95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 98 1.6 -3 0.75;
99 -0.75 -3 0.5; 100 -1.6 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5;
103 0 -3 0.5; 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 107 1.6 -3 0.5;
108 -0.75 -3 0.25; 109 -1.6 -3 0.25; 110 -0.5 -3 0.25; 111 -0.25 -3 0.25;
112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25; 115 0.75 -3 0.25;
116 1.6 -3 0.25; 117 -0.75 -3 0; 118 -1.6 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0;
121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 124 1.6 -3 0; 125 -0.75 -3 -0.25;
126 -1.6 -3 -0.25; 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25;
130 0.25 -3 -0.25; 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 133 1.6 -3 -0.25;
134 -0.75 -3 -0.5; 135 -1.6 -3 -0.5; 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5;
138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5; 141 0.75 -3 -0.5;
142 1.6 -3 -0.5; 143 -0.75 -3 -0.75; 144 -1.6 -3 -0.75; 145 -0.5 -3 -0.75;
146 -0.25 -3 -0.75; 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75;
150 0.75 -3 -0.75; 151 1.6 -3 -0.75; 152 -0.75 -3 -1.7; 153 -0.5 -3 -1.7;
154 -0.25 -3 -1.7; 155 0.25 -3 -1.7; 156 0.5 -3 -1.7; 157 0.75 -3 -1.7;
158 -0.75 -3 -1; 159 -1.6 -3 -1; 160 -0.75 -3 -1.25; 161 -1.6 -3 -1.25;
162 -0.5 -3 -1; 163 -0.5 -3 -1.25; 164 -0.25 -3 -1; 165 -0.25 -3 -1.25;
```

166 0 -3 -1; 167 0 -3 -1.25; 168 0.25 -3 -1; 169 0.25 -3 -1.25; 170 0.5 -3 -1;
171 0.5 -3 -1.25; 172 0.75 -3 -1; 173 0.75 -3 -1.25; 174 1.6 -3 -1;
175 1.6 -3 -1.25; 176 -0.75 -3 3.25; 177 -1.6 -3 3.25; 178 -0.75 -3 3;
179 -1.6 -3 3; 180 -0.5 -3 3.25; 181 -0.5 -3 3; 182 -0.25 -3 3.25;
183 -0.25 -3 3; 184 0 -3 3.25; 185 0 -3 3; 186 0.25 -3 3.25; 187 0.25 -3 3;
188 0.5 -3 3.25; 189 0.5 -3 3; 190 0.75 -3 3.25; 191 0.75 -3 3;
192 1.6 -3 3.25; 193 1.6 -3 3; 194 -0.75 -3 -1.475; 195 -1.6 -3 -1.475;
196 -0.5 -3 -1.475; 197 -0.25 -3 -1.475; 198 0 -3 -1.475; 199 0.25 -3 -1.475;
200 0.5 -3 -1.475; 201 0.75 -3 -1.475; 202 1.6 -3 -1.475; 203 -0.75 -3 3.475;
204 -1.6 -3 3.475; 205 -0.5 -3 3.475; 206 -0.25 -3 3.475; 207 0 -3 3.475;
208 0.25 -3 3.475; 209 0.5 -3 3.475; 210 0.75 -3 3.475; 211 1.6 -3 3.475;
212 1.03333 -3 2.75; 213 1.03333 -3 2.5; 214 1.31667 -3 2.75;
215 1.31667 -3 2.5; 216 1.03333 -3 2.25; 217 1.31667 -3 2.25; 218 1.03333 -3 2;
219 1.31667 -3 2; 220 1.03333 -3 1.75; 221 1.31667 -3 1.75; 222 1.03333 -3 1.5;
223 1.31667 -3 1.5; 224 1.03333 -3 1.25; 225 1.31667 -3 1.25; 226 1.03333 -3 1;
227 1.31667 -3 1; 228 1.03333 -3 0.75; 229 1.31667 -3 0.75; 230 1.03333 -3 0.5;
231 1.31667 -3 0.5; 232 1.03333 -3 0.25; 233 1.31667 -3 0.25; 234 1.03333 -3 0;
235 1.31667 -3 0; 236 1.03333 -3 -0.25; 237 1.31667 -3 -0.25;
238 1.03333 -3 -0.5; 239 1.31667 -3 -0.5; 240 1.03333 -3 -0.75;
241 1.31667 -3 -0.75; 242 1.03333 -3 -1; 243 1.31667 -3 -1;
244 1.03333 -3 -1.25; 245 1.31667 -3 -1.25; 246 1.03333 -3 3.25;
247 1.03333 -3 3; 248 1.31667 -3 3.25; 249 1.31667 -3 3; 250 1.03333 -3 -1.475;
251 1.31667 -3 -1.475; 252 1.03333 -3 -1.7; 253 1.31667 -3 -1.7;
254 1.03333 -3 3.7; 255 1.03333 -3 3.475; 256 1.31667 -3 3.7;
257 1.31667 -3 3.475; 258 -1.31667 -3 2.75; 259 -1.31667 -3 2.5;
260 -1.03333 -3 2.75; 261 -1.03333 -3 2.5; 262 -1.31667 -3 2.25;
263 -1.03333 -3 2.25; 264 -1.31667 -3 2; 265 -1.03333 -3 2;
266 -1.31667 -3 1.75; 267 -1.03333 -3 1.75; 268 -1.31667 -3 1.5;
269 -1.03333 -3 1.5; 270 -1.31667 -3 1.25; 271 -1.03333 -3 1.25;
272 -1.31667 -3 1; 273 -1.03333 -3 1; 274 -1.31667 -3 0.75;
275 -1.03333 -3 0.75; 276 -1.31667 -3 0.5; 277 -1.03333 -3 0.5;
278 -1.31667 -3 0.25; 279 -1.03333 -3 0.25; 280 -1.31667 -3 0;
281 -1.03333 -3 0; 282 -1.31667 -3 -0.25; 283 -1.03333 -3 -0.25;
284 -1.31667 -3 -0.5; 285 -1.03333 -3 -0.5; 286 -1.31667 -3 -0.75;
287 -1.03333 -3 -0.75; 288 -1.31667 -3 -1; 289 -1.03333 -3 -1;
290 -1.31667 -3 -1.25; 291 -1.03333 -3 -1.25; 292 -1.31667 -3 3.25;
293 -1.31667 -3 3; 294 -1.03333 -3 3.25; 295 -1.03333 -3 3;
296 -1.31667 -3 -1.475; 297 -1.03333 -3 -1.475; 298 -1.31667 -3 -1.7;
299 -1.03333 -3 -1.7; 300 -1.31667 -3 3.7; 301 -1.31667 -3 3.475;
302 -1.03333 -3 3.7; 303 -1.03333 -3 3.475;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
ELEMENT INCIDENCES SHELL
15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33;
20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41;
27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49;
34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56;
41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63;
48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69;
55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77;
60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85;
67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93;
74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99;
80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104;

84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111;
90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117;
96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121;
100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127;
105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131;
111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137;
114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140;
119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146;
122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149;
137 143 145 162 158; 138 158 162 163 160; 140 145 146 164 162;
141 162 164 165 163; 143 146 147 166 164; 144 164 166 167 165;
146 147 148 168 166; 147 166 168 169 167; 149 148 149 170 168;
150 168 170 171 169; 152 149 150 172 170; 153 170 172 173 171;
162 176 180 181 178; 163 178 181 17 14; 165 180 182 183 181; 166 181 183 19 17;
168 182 184 185 183; 169 183 185 20 19; 171 184 186 187 185; 172 185 187 22 20;
174 186 188 189 187; 175 187 189 24 22; 177 188 190 191 189; 178 189 191 26 24;
184 160 163 196 194; 185 194 196 153 152; 186 163 165 197 196;
187 196 197 154 153; 188 165 167 198 197; 189 197 198 7 154;
190 167 169 199 198; 191 198 199 155 7; 192 169 171 200 199;
193 199 200 156 155; 194 171 173 201 200; 195 200 201 157 156;
200 13 16 205 203; 201 203 205 180 176; 202 16 18 206 205; 203 205 206 182 180;
204 18 8 207 206; 205 206 207 184 182; 206 8 21 208 207; 207 207 208 186 184;
208 21 23 209 208; 209 208 209 188 186; 210 23 25 210 209; 211 209 210 190 188;
214 26 212 213 35; 215 212 214 215 213; 216 214 27 36 215; 217 35 213 216 44;
218 213 215 217 216; 219 215 36 45 217; 220 44 216 218 52; 221 216 217 219 218;
222 217 45 53 219; 223 52 218 220 61; 224 218 219 221 220; 225 219 53 62 221;
226 61 220 222 70; 227 220 221 223 222; 228 221 62 71 223; 229 70 222 224 79;
230 222 223 225 224; 231 223 71 80 225; 232 79 224 226 88; 233 224 225 227 226;
234 225 80 89 227; 235 88 226 228 97; 236 226 227 229 228; 237 227 89 98 229;
238 97 228 230 106; 239 228 229 231 230; 240 229 98 107 231;
241 106 230 232 115; 242 230 231 233 232; 243 231 107 116 233;
244 115 232 234 123; 245 232 233 235 234; 246 233 116 124 235;
247 123 234 236 132; 248 234 235 237 236; 249 235 124 133 237;
250 132 236 238 141; 251 236 237 239 238; 252 237 133 142 239;
253 141 238 240 150; 254 238 239 241 240; 255 239 142 151 241;
256 150 240 242 172; 257 240 241 243 242; 258 241 151 174 243;
259 172 242 244 173; 260 242 243 245 244; 261 243 174 175 245;
262 190 246 247 191; 263 246 248 249 247; 264 248 192 193 249;
265 191 247 212 26; 266 247 249 214 212; 267 249 193 27 214;
268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251;
271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253;
274 25 254 255 210; 275 254 256 257 255; 276 256 10 211 257;
277 210 255 246 190; 278 255 257 248 246; 279 257 211 192 248;
280 15 258 259 29; 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38;
284 259 261 263 262; 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264;
288 263 37 46 265; 289 47 264 266 55; 290 264 265 267 266; 291 265 46 54 267;
292 55 266 268 64; 293 266 267 269 268; 294 267 54 63 269; 295 64 268 270 73;
296 268 269 271 270; 297 269 63 72 271; 298 73 270 272 82; 299 270 271 273 272;
300 271 72 81 273; 301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275;
304 91 274 276 100; 305 274 275 277 276; 306 275 90 99 277;
307 100 276 278 109; 308 276 277 279 278; 309 277 99 108 279;
310 109 278 280 118; 311 278 279 281 280; 312 279 108 117 281;
313 118 280 282 126; 314 280 281 283 282; 315 281 117 125 283;
316 126 282 284 135; 317 282 283 285 284; 318 283 125 134 285;

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319 135 284 286 144; 320 284 285 287 286; 321 285 134 143 287;
322 144 286 288 159; 323 286 287 289 288; 324 287 143 158 289;
325 159 288 290 161; 326 288 289 291 290; 327 289 158 160 291;
328 177 292 293 179; 329 292 294 295 293; 330 294 176 178 295;
331 179 293 258 15; 332 293 295 260 258; 333 295 178 14 260;
334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297;
337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299;
340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303;
343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
214 TO 345
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
_SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
214 TO 345
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.7 ZD 0.5
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
*****************************************
SUPPORTS
1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
*****************************************
MEMBER OFFSET
5 START 0 -0.25 0
5 END 0 -0.25 0
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
```

```
5 FX -0.015 FY -92.861 FZ -2.092
6 FX 0.002 FY -168.361 FZ 10.092
ELEMENT LOAD
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
*****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
5 FX -0.002 FY -48.1 FZ -2.342
6 FX 0.002 FY -47.9 FZ 2.342
*****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
5 FX -0.005 FY -49.2 FZ -2.456
6 FX 0.005 FY -49.2 FZ 2.456
*****************************************
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
5 FX 0.052 FY 180.856 FZ 29.169
6 FX -0.052 FY -180.856 FZ 29.111
*****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
5 FX -0.052 FY -180.866 FZ -29.171
6 FX 0.052 FY 180.866 FZ -29.109
*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
5 FX 1.515 FY 0 FZ 0
6 FX 1.515 FY 0 FZ 0
*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
5 FX -1.515 FY 0 FZ 0
6 FX -1.515 FY 0 FZ 0
*****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
5 FX -0.007 FY 0.109 FZ -9.437
6 FX 0.007 FY -0.109 FZ 9.417
*****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
5 FX 0.007 FY -0.109 FZ 9.437
6 FX -0.007 FY 0.109 FZ -9.417
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
5 FX 0.021 FY 0.025 FZ 0.004
6 FX 0.021 FY -0.075 FZ 0.006
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
```

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JOINT LOAD
5 FX -0.003 FY 6.428 FZ 0.951
6 FX 0.003 FY -6.428 FZ 0.949
*****************************************
*****Load combination for bearing capacity check
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 6 0.75 2 0.75 3 0.75
LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 7 0.75 2 0.75 3 0.75
LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 0.525 2 0.75 3 0.75
LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 0.525 2 0.75 3 0.75
LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD 312 0.6 DL + 1.0 WL(+Z)
REPEAT LOAD
1 0.6 4 1.0
LOAD 313 0.6 DL + 1.0 WL(-Z)
REPEAT LOAD
1 0.6 5 1.0
LOAD 314 0.6 DL + 1.0 WL(+X)
REPEAT LOAD
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
```

```
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
****************************************
*****Load combination for foundation design
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 5 1.6
LOAD 414 0.9 DL + 1.6 WL(+X)
REPEAT LOAD
1 0.9 6 1.6
LOAD 415 0.9 DL + 1.6 WL(-X)
REPEAT LOAD
```

```
1 0.9 7 1.6
LOAD 416 0.9 DL + 1.0 EL(+X)
REPEAT LOAD
1 0.9 10 1.0
LOAD 417 0.9 DL + 1.0 EL(-X)
REPEAT LOAD
1 0.9 10 -1.0
LOAD 418 0.9 DL + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 11 1.0
LOAD 419 0.9 DL + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 11 -1.0
**************************************
PERFORM ANALYSIS
**************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 5
FYMAIN 420000 MEMB 1 TO 5
TRACK 1 ALL
CLB 0.05 MEMB 1 TO 5
CLT 0.05 MEMB 1 TO 5
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  7, 2020                *
*           Time=    12:30:38                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3110-CNV-002\Staad\3110-CNV-002_TR2.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 06-APR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART TRESTLE TR2 FOUNDATION MKD. F4
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3110-CNV-002
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 -3 0; 2 0 -3 2; 3 0 1.49012E-008 0; 4 0 1.49012E-008 2; 5 0 0.35 0
   18. 6 0 0.35 2; 7 0 -3 -1.7; 8 0 -3 3.7; 9 1.6 -3 -1.7; 10 1.6 -3 3.7
   19. 11 -1.6 -3 -1.7; 12 -1.6 -3 3.7; 13 -0.75 -3 3.7; 14 -0.75 -3 2.75
   20. 15 -1.6 -3 2.75; 16 -0.5 -3 3.7; 17 -0.5 -3 2.75; 18 -0.25 -3 3.7
   21. 19 -0.25 -3 2.75; 20 0 -3 2.75; 21 0.25 -3 3.7; 22 0.25 -3 2.75; 23 0.5 -3 3.7
   22. 24 0.5 -3 2.75; 25 0.75 -3 3.7; 26 0.75 -3 2.75; 27 1.6 -3 2.75
   23. 28 -0.75 -3 2.5; 29 -1.6 -3 2.5; 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5
   24. 33 0.25 -3 2.5; 34 0.5 -3 2.5; 35 0.75 -3 2.5; 36 1.6 -3 2.5; 37 -0.75 -3 2.25
   25. 38 -1.6 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25; 41 0 -3 2.25
   26. 42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 45 1.6 -3 2.25
   27. 46 -0.75 -3 2; 47 -1.6 -3 2; 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2
   28. 51 0.5 -3 2; 52 0.75 -3 2; 53 1.6 -3 2; 54 -0.75 -3 1.75; 55 -1.6 -3 1.75
   29. 56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75; 59 0.25 -3 1.75
   30. 60 0.5 -3 1.75; 61 0.75 -3 1.75; 62 1.6 -3 1.75; 63 -0.75 -3 1.5
   31. 64 -1.6 -3 1.5; 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5
   32. 69 0.5 -3 1.5; 70 0.75 -3 1.5; 71 1.6 -3 1.5; 72 -0.75 -3 1.25
   33. 73 -1.6 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25; 76 0 -3 1.25
   34. 77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 80 1.6 -3 1.25
   35. 81 -0.75 -3 1; 82 -1.6 -3 1; 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1
   36. 86 0.25 -3 1; 87 0.5 -3 1; 88 0.75 -3 1; 89 1.6 -3 1; 90 -0.75 -3 0.75
   37. 91 -1.6 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75; 94 0 -3 0.75
   38. 95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75; 98 1.6 -3 0.75
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39. 99 -0.75 -3 0.5; 100 -1.6 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5
40. 103 0 -3 0.5; 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 107 1.6 -3 0.5
41. 108 -0.75 -3 0.25; 109 -1.6 -3 0.25; 110 -0.5 -3 0.25; 111 -0.25 -3 0.25
42. 112 0 -3 0.25; 113 0.25 -3 0.25; 114 0.5 -3 0.25; 115 0.75 -3 0.25
43. 116 1.6 -3 0.25; 117 -0.75 -3 0; 118 -1.6 -3 0; 119 -0.5 -3 0; 120 -0.25 -3 0
44. 121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 124 1.6 -3 0; 125 -0.75 -3 -0.25
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46. 130 0.25 -3 -0.25; 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 133 1.6 -3 -0.25
47. 134 -0.75 -3 -0.5; 135 -1.6 -3 -0.5; 136 -0.5 -3 -0.5; 137 -0.25 -3 -0.5
48. 138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5; 141 0.75 -3 -0.5
49. 142 1.6 -3 -0.5; 143 -0.75 -3 -0.75; 144 -1.6 -3 -0.75; 145 -0.5 -3 -0.75
50. 146 -0.25 -3 -0.75; 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75
51. 150 0.75 -3 -0.75; 151 1.6 -3 -0.75; 152 -0.75 -3 -1.7; 153 -0.5 -3 -1.7
52. 154 -0.25 -3 -1.7; 155 0.25 -3 -1.7; 156 0.5 -3 -1.7; 157 0.75 -3 -1.7
53. 158 -0.75 -3 -1; 159 -1.6 -3 -1; 160 -0.75 -3 -1.25; 161 -1.6 -3 -1.25
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61. 192 1.6 -3 3.25; 193 1.6 -3 3; 194 -0.75 -3 -1.475; 195 -1.6 -3 -1.475
62. 196 -0.5 -3 -1.475; 197 -0.25 -3 -1.475; 198 0 -3 -1.475; 199 0.25 -3 -1.475
63. 200 0.5 -3 -1.475; 201 0.75 -3 -1.475; 202 1.6 -3 -1.475; 203 -0.75 -3 3.475
64. 204 -1.6 -3 3.475; 205 -0.5 -3 3.475; 206 -0.25 -3 3.475; 207 0 -3 3.475
65. 208 0.25 -3 3.475; 209 0.5 -3 3.475; 210 0.75 -3 3.475; 211 1.6 -3 3.475
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67. 215 1.31667 -3 2.5; 216 1.03333 -3 2.25; 217 1.31667 -3 2.25; 218 1.03333 -3 2
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77. 251 1.31667 -3 -1.475; 252 1.03333 -3 -1.7; 253 1.31667 -3 -1.7
78. 254 1.03333 -3 3.7; 255 1.03333 -3 3.475; 256 1.31667 -3 3.7
79. 257 1.31667 -3 3.475; 258 -1.31667 -3 2.75; 259 -1.31667 -3 2.5
80. 260 -1.03333 -3 2.75; 261 -1.03333 -3 2.5; 262 -1.31667 -3 2.25
81. 263 -1.03333 -3 2.25; 264 -1.31667 -3 2; 265 -1.03333 -3 2
82. 266 -1.31667 -3 1.75; 267 -1.03333 -3 1.75; 268 -1.31667 -3 1.5
83. 269 -1.03333 -3 1.5; 270 -1.31667 -3 1.25; 271 -1.03333 -3 1.25
84. 272 -1.31667 -3 1; 273 -1.03333 -3 1; 274 -1.31667 -3 0.75
85. 275 -1.03333 -3 0.75; 276 -1.31667 -3 0.5; 277 -1.03333 -3 0.5
86. 278 -1.31667 -3 0.25; 279 -1.03333 -3 0.25; 280 -1.31667 -3 0
87. 281 -1.03333 -3 0; 282 -1.31667 -3 -0.25; 283 -1.03333 -3 -0.25
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93. 299 -1.03333 -3 -1.7; 300 -1.31667 -3 3.7; 301 -1.31667 -3 3.475
94. 302 -1.03333 -3 3.7; 303 -1.03333 -3 3.475

95. MEMBER INCIDENCES
96. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
97. ELEMENT INCIDENCES SHELL
98. 15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33
99. 20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41
100. 27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49
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102. 41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63
103. 48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69
104. 55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77
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106. 67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93
107. 74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99
108. 80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104
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110. 90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117
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112. 100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127
113. 105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131
114. 111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137
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133. 218 213 215 217 216; 219 215 36 45 217; 220 44 216 218 52; 221 216 217 219 218
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135. 226 61 220 222 70; 227 220 221 223 222; 228 221 62 71 223; 229 70 222 224 79
136. 230 222 223 225 224; 231 223 71 80 225; 232 79 224 226 88; 233 224 225 227 226
137. 234 225 80 89 227; 235 88 226 228 97; 236 226 227 229 228; 237 227 89 98 229
138. 238 97 228 230 106; 239 228 229 231 230; 240 229 98 107 231
139. 241 106 230 232 115; 242 230 231 233 232; 243 231 107 116 233
140. 244 115 232 234 123; 245 232 233 235 234; 246 233 116 124 235
141. 247 123 234 236 132; 248 234 235 237 236; 249 235 124 133 237
142. 250 132 236 238 141; 251 236 237 239 238; 252 237 133 142 239
143. 253 141 238 240 150; 254 238 239 241 240; 255 239 142 151 241
144. 256 150 240 242 172; 257 240 241 243 242; 258 241 151 174 243
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146. 262 190 246 247 191; 263 246 248 249 247; 264 248 192 193 249
147. 265 191 247 212 26; 266 247 249 214 212; 267 249 193 27 214
148. 268 173 244 250 201; 269 244 245 251 250; 270 245 175 202 251
149. 271 201 250 252 157; 272 250 251 253 252; 273 251 202 9 253
150. 274 25 254 255 210; 275 254 256 257 255; 276 256 10 211 257

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152. 280 15 258 259 29; 281 258 260 261 259; 282 260 14 28 261; 283 29 259 262 38
153. 284 259 261 263 262; 285 261 28 37 263; 286 38 262 264 47; 287 262 263 265 264
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157. 300 271 72 81 273; 301 82 272 274 91; 302 272 273 275 274; 303 273 81 90 275
158. 304 91 274 276 100; 305 274 275 277 276; 306 275 90 99 277
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162. 316 126 282 284 135; 317 282 283 285 284; 318 283 125 134 285
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164. 322 144 286 288 159; 323 286 287 289 288; 324 287 143 158 289
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168. 334 161 290 296 195; 335 290 291 297 296; 336 291 160 194 297
169. 337 195 296 298 11; 338 296 297 299 298; 339 297 194 152 299
170. 340 12 300 301 204; 341 300 302 303 301; 342 302 13 203 303
171. 343 204 301 292 177; 344 301 303 294 292; 345 303 203 176 294
172. START GROUP DEFINITION
173. ELEMENT
174. _MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
175. 63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
176. 108 111 112 115 116 119 120 123 124 137 138 140 141 149 150 152 153 162 163 -
177. 165 166 174 175 177 178 184 TO 187 192 TO 195 200 TO 203 208 TO 211 -
178. 214 TO 345
179. _MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
180. 111 TO 116 119 TO 124 137 138 140 141 143 144 146 147 149 150 152 153 162 -
181. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
182. 214 TO 219 226 TO 243 250 TO 285 292 TO 309 316 TO 345
183. _SHEAR 63 TO 68 71 TO 76 137 138 140 141 143 144 146 147 149 150 152 153 162 -
184. 163 165 166 168 169 171 172 174 175 177 178 184 TO 195 200 TO 211 -
185. 214 TO 345
186. END GROUP DEFINITION
187. ELEMENT PROPERTY
188. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
189. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
190. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
191. 178 184 TO 195 200 TO 211 214 TO 345 THICKNESS 0.55
192. DEFINE MATERIAL START
193. ISOTROPIC CONCRETE
194. E 2.17185E+007
195. POISSON 0.17
196. DENSITY 23.5616
197. ALPHA 1E-005
198. DAMP 0.05
199. TYPE CONCRETE
200. STRENGTH FCU 27579
201. END DEFINE MATERIAL
202. MEMBER PROPERTY AMERICAN
203. 1 TO 4 PRIS YD 0.7 ZD 0.5
204. 5 PRIS YD 0.5 ZD 0.3
205. CONSTANTS
206. MATERIAL CONCRETE ALL
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207. ****************************************
208. SUPPORTS
209. 1 2 7 TO 303 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
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JOINT	INFLUENCE AREA
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9	0.015937
10	0.015937
11	0.015937
12	0.015937
13	0.030000
14	0.066666
15	0.035416
16	0.028125
17	0.062500
18	0.028125
19	0.062500
20	0.062500
21	0.028125
22	0.062500
23	0.028125
24	0.062500
25	0.030000
26	0.066666
27	0.035416
28	0.066666
29	0.035416
30	0.062500
31	0.062500
32	0.062500
33	0.062500
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35	0.066666
36	0.035416
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53	0.035416

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146	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.066666
151	0.035416
152	0.030000
153	0.028125
154	0.028125
155	0.028125
156	0.028125
157	0.030000
158	0.066666
159	0.035416
160	0.063333
161	0.033645
162	0.062500
163	0.059375
164	0.062500
165	0.059375

166	0.062500
167	0.059375
168	0.062500
169	0.059375
170	0.062500
171	0.059375
172	0.066666
173	0.063333
174	0.035416
175	0.033645
176	0.063333
177	0.033645
178	0.066666
179	0.035416
180	0.059375
181	0.062500
182	0.059375
183	0.062500
184	0.059375
185	0.062500
186	0.059375
187	0.062500
188	0.059375
189	0.062500
190	0.063333
191	0.066666
192	0.033645
193	0.035416
194	0.060000
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199	0.056250
200	0.056250
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203	0.060000
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228	0.070834
229	0.070834
230	0.070834
231	0.070834
232	0.070834
233	0.070834
234	0.070834
235	0.070834
236	0.070834
237	0.070834
238	0.070834
239	0.070834
240	0.070834
241	0.070834
242	0.070834
243	0.070834
244	0.067292
245	0.067292
246	0.067292
247	0.070834
248	0.067292
249	0.070834
250	0.063750
251	0.063750
252	0.031875
253	0.031875
254	0.031875
255	0.063750
256	0.031875
257	0.063750
258	0.070834
259	0.070834
260	0.070834
261	0.070834
262	0.070834
263	0.070834
264	0.070834
265	0.070834
266	0.070834
267	0.070834
268	0.070834
269	0.070834
270	0.070834
271	0.070834
272	0.070834
273	0.070834
274	0.070834
275	0.070834
276	0.070834
277	0.070834

278	0.070834
279	0.070834
280	0.070834
281	0.070834
282	0.070834
283	0.070834
284	0.070834
285	0.070834
286	0.070834
287	0.070834
288	0.070834
289	0.070834
290	0.067292
291	0.067292
292	0.067292
293	0.070834
294	0.067292
295	0.070834
296	0.063750
297	0.063750
298	0.031875
299	0.031875
300	0.031875
301	0.063750
302	0.031875
303	0.063750
TOTAL	17.280014

```
210. ****************************************
211. MEMBER OFFSET
212. 5 START 0 -0.25 0
213. 5 END 0 -0.25 0
214. LOAD 1 DEAD LOAD DL
215. SELFWEIGHT Y -1
216. JOINT LOAD
217. 5 FX -0.015 FY -92.861 FZ -2.092
218. 6 FX 0.002 FY -168.361 FZ 10.092
219. ELEMENT LOAD
220. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
221. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 138 140 141 -
222. 143 144 146 147 149 150 152 153 162 163 165 166 168 169 171 172 174 175 177 -
223. 178 184 TO 195 200 TO 211 214 TO 345 PR GY -44.1
224. ****************************************
225. LOAD 2 LIVE LOAD LL
226. JOINT LOAD
227. 5 FX -0.002 FY -48.1 FZ -2.342
228. 6 FX 0.002 FY -47.9 FZ 2.342
229. ****************************************
230. LOAD 3 MATERIAL LOAD LL
231. JOINT LOAD
232. 5 FX -0.005 FY -49.2 FZ -2.456
233. 6 FX 0.005 FY -49.2 FZ 2.456
234. ****************************************
235. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
236. JOINT LOAD
237. 5 FX 0.052 FY 180.856 FZ 29.169
238. 6 FX -0.052 FY -180.856 FZ 29.111
```

```
239. ****************************************
240. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
241. JOINT LOAD
242. 5 FX -0.052 FY -180.866 FZ -29.171
243. 6 FX 0.052 FY 180.866 FZ -29.109
244. ****************************************
245. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
246. JOINT LOAD
247. 5 FX 1.515 FY 0 FZ 0
248. 6 FX 1.515 FY 0 FZ 0
249. ****************************************
250. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
251. JOINT LOAD
252. 5 FX -1.515 FY 0 FZ 0
253. 6 FX -1.515 FY 0 FZ 0
254. ****************************************
255. LOAD 8 TEMP LOAD   TL1
256. JOINT LOAD
257. 5 FX -0.007 FY 0.109 FZ -9.437
258. 6 FX 0.007 FY -0.109 FZ 9.417
259. ****************************************
260. LOAD 9 TEMP LOAD   TL2
261. JOINT LOAD
262. 5 FX 0.007 FY -0.109 FZ 9.437
263. 6 FX -0.007 FY 0.109 FZ -9.417
264. ****************************************
265. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
266. JOINT LOAD
267. 5 FX 0.021 FY 0.025 FZ 0.004
268. 6 FX 0.021 FY -0.075 FZ 0.006
269. ****************************************
270. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
271. JOINT LOAD
272. 5 FX -0.003 FY 6.428 FZ 0.951
273. 6 FX 0.003 FY -6.428 FZ 0.949
274. ****************************************
275. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
276. LOAD 301 1.0 DL
277. REPEAT LOAD
278. 1 1.0
279. LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
280. REPEAT LOAD
281. 1 1.0 8 1.0 2 1.0 3 1.0
282. LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
283. REPEAT LOAD
284. 1 1.0 9 1.0 2 1.0 3 1.0
285. LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
286. REPEAT LOAD
287. 1 1.0 4 0.75 2 0.75 3 0.75
288. LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
289. REPEAT LOAD
290. 1 1.0 5 0.75 2 0.75 3 0.75
291. LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
292. REPEAT LOAD
293. 1 1.0 6 0.75 2 0.75 3 0.75
294. LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
```

```
295. REPEAT LOAD
296. 1 1.0 7 0.75 2 0.75 3 0.75
297. LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
298. REPEAT LOAD
299. 1 1.0 10 0.525 2 0.75 3 0.75
300. LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
301. REPEAT LOAD
302. 1 1.0 10 -0.525 2 0.75 3 0.75
303. LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
304. REPEAT LOAD
305. 1 1.0 11 0.525 2 0.75 3 0.75
306. LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
307. REPEAT LOAD
308. 1 1.0 11 -0.525 2 0.75 3 0.75
309. LOAD 312 0.6 DL + 1.0 WL(+Z)
310. REPEAT LOAD
311. 1 0.6 4 1.0
312. LOAD 313 0.6 DL + 1.0 WL(-Z)
313. REPEAT LOAD
314. 1 0.6 5 1.0
315. LOAD 314 0.6 DL + 1.0 WL(+X)
316. REPEAT LOAD
317. 1 0.6 6 1.0
318. LOAD 315 0.6 DL + 1.0 WL(-X)
319. REPEAT LOAD
320. 1 0.6 7 1.0
321. LOAD 316 0.6 DL + 0.7 EL(+X)
322. REPEAT LOAD
323. 1 0.6 10 0.7
324. LOAD 317 0.6 DL + 0.7 EL(-X)
325. REPEAT LOAD
326. 1 0.6 10 -0.7
327. LOAD 318 0.6 DL + 0.7 EL(+Z)
328. REPEAT LOAD
329. 1 0.6 11 0.7
330. LOAD 319 0.6 DL + 0.7 EL(-Z)
331. REPEAT LOAD
332. 1 0.6 11 -0.7
333. ****************************************
334. *****LOAD COMBINATION FOR FOUNDATION DESIGN
335. LOAD 401 1.4 DL
336. REPEAT LOAD
337. 1 1.4
338. LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
339. REPEAT LOAD
340. 1 1.4 8 1.4 2 1.7 3 1.7
341. LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
342. REPEAT LOAD
343. 1 1.4 9 1.4 2 1.7 3 1.7
344. LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
345. REPEAT LOAD
346. 1 1.2 4 1.6 2 0.5 3 0.5
347. LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
348. REPEAT LOAD
349. 1 1.2 5 1.6 2 0.5 3 0.5
350. LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
```

```
351. REPEAT LOAD
352. 1 1.2 6 1.6 2 0.5 3 0.5
353. LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
354. REPEAT LOAD
355. 1 1.2 7 1.6 2 0.5 3 0.5
356. LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
357. REPEAT LOAD
358. 1 1.2 10 1.0 2 0.5 3 0.5
359. LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
360. REPEAT LOAD
361. 1 1.2 10 -1.0 2 0.5 3 0.5
362. LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
363. REPEAT LOAD
364. 1 1.2 11 1.0 2 0.5 3 0.5
365. LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
366. REPEAT LOAD
367. 1 1.2 11 -1.0 2 0.5 3 0.5
368. LOAD 412 0.9 DL + 1.6 WL(+Z)
369. REPEAT LOAD
370. 1 0.9 4 1.6
371. LOAD 413 0.9 DL + 1.6 WL(-Z)
372. REPEAT LOAD
373. 1 0.9 5 1.6
374. LOAD 414 0.9 DL + 1.6 WL(+X)
375. REPEAT LOAD
376. 1 0.9 6 1.6
377. LOAD 415 0.9 DL + 1.6 WL(-X)
378. REPEAT LOAD
379. 1 0.9 7 1.6
380. LOAD 416 0.9 DL + 1.0 EL(+X)
381. REPEAT LOAD
382. 1 0.9 10 1.0
383. LOAD 417 0.9 DL + 1.0 EL(-X)
384. REPEAT LOAD
385. 1 0.9 10 -1.0
386. LOAD 418 0.9 DL + 1.0 EL(+Z)
387. REPEAT LOAD
388. 1 0.9 11 1.0
389. LOAD 419 0.9 DL + 1.0 EL(-Z)
390. REPEAT LOAD
391. 1 0.9 11 -1.0
392. ****************************************
393. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**303**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**264**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**299**

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 921
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

394. **
395. LOAD LIST 401 TO 419
396. START CONCRETE DESIGN

```
397. CODE ACI 2008
398. FC 35000 MEMB 1 TO 5
399. FYMAIN 420000 MEMB 1 TO 5
400. TRACK 1 ALL
401. CLB 0.05 MEMB 1 TO 5
402. CLT 0.05 MEMB 1 TO 5
403. CLS 0.05 ALL
404. MAXMAIN 25 ALL
405. MINMAIN 16 ALL
406. MINSEC 8 ALL
407. REINF 0 MEMB 1 TO 4
408. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	202.99	0.01	0.00001

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	202.99	9.11	0.01496

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

```
 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    200.01      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    200.01      0.00    0.00000
 ---------------------------------------------------------
```

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 500.0 X 700.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3500.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 20 MM               1.077      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER    8 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    387.30      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    387.30      6.56    0.00564
 ---------------------------------------------------------
```

==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	362.62	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	362.62	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

409. DESIGN BEAM 5

BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

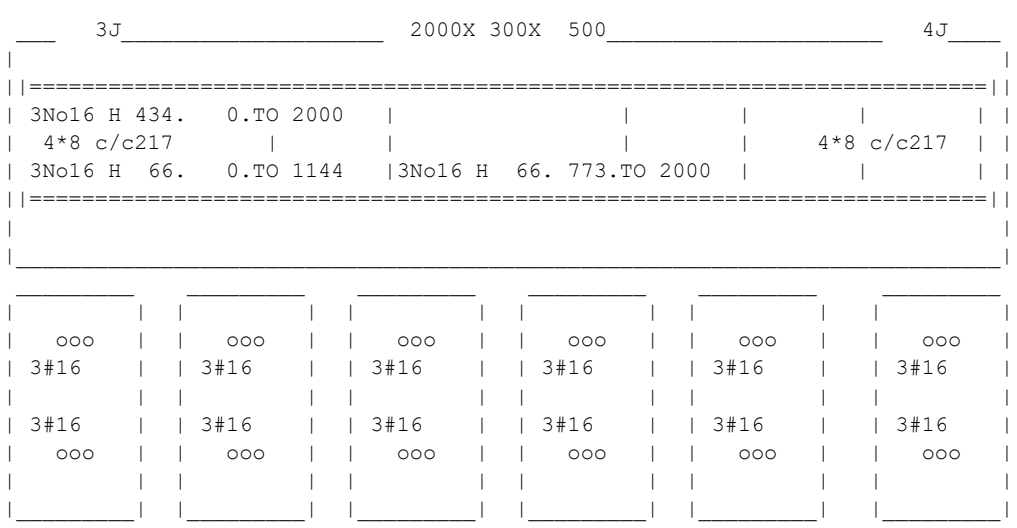
LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	1144.	YES	NO
	CRITICAL POS MOMENT= 64.89 KN-MET AT 0.MM, LOAD 404					
	REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
2	66.	3 - 16MM	773.	2000.	NO	YES
	CRITICAL POS MOMENT= 55.77 KN-MET AT 2000.MM, LOAD 413					
	REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					
3	434.	3 - 16MM	0.	2000.	YES	YES
	CRITICAL NEG MOMENT= 72.64 KN-MET AT 2000.MM, LOAD 404					
	REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 58050.0 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 66.36 KNS Vc= 126.05 KNS Vs= 0.00 KNS
Tu= 0.02 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM

AT END SUPPORT - Vu= 71.16 KNS Vc= 126.05 KNS Vs= 0.00 KNS
Tu= 0.02 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 404
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 8 MM 2-LEGGED STIRRUPS AT 217. MM C/C FOR 566. MM



**********END OF BEAM DESIGN**********

410. END CONCRETE DESIGN
411. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 12:30:45 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	07-04-2020	18-05-2020	12-10-2023				Date	07-04-2020	18-05-2020	12-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 7 **STAAD INPUT & OUTPUT FILES OF FOUNDATION FOR VGTU**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 06-Apr-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART VGTU FDN
JOB REF FOUNDATIONS FOR CONVEYOR 3110-CNV-002
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 5; 3 4 -3 0; 4 4 -3 5; 5 8 -3 0; 6 8 -3 5; 7 0 0.35 0;
8 0 0.35 5; 9 4 0.35 0; 10 4 0.35 5; 11 8 0.35 0; 12 8 0.35 5; 13 0 1 0;
14 0 1 5; 15 4 1 0; 16 4 1 5; 17 8 1 0; 18 8 1 5;
MEMBER INCIDENCES
1 1 7; 2 7 13; 3 2 8; 4 8 14; 5 3 9; 6 9 15; 7 4 10; 8 10 16; 9 5 11; 10 11 17;
11 6 12; 12 12 18; 13 7 8; 14 9 10; 15 11 12; 16 7 9; 17 9 11; 18 8 10;
19 10 12;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 12 PRIS YD 0.6 ZD 0.6
13 TO 15 PRIS YD 0.5 ZD 0.3
MEMBER PROPERTY AMERICAN
16 TO 19 PRIS YD 0.6 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 TO 6 FIXED
*****************************************
MEMBER OFFSET
16 TO 19 START 0 -0.3 0
16 TO 19 END 0 -0.3 0
13 TO 15 START 0 -0.25 0
13 TO 15 END 0 -0.25 0
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
18 FX -25.941 FY -275.436 FZ -12.314
17 FX -26.196 FY -295.582 FZ 16.204
16 FX -23.851 FY -72.76 FZ 1.33
```

15 FX -24.07 FY -105.074 FZ 4.238
14 FX 0 FY -17.052 FZ -0.286
13 FX 0.057 FY -18.904 FZ 0.527
****
LOAD 2 LIVE LOAD LL
JOINT LOAD
18 FX -0.624 FY -162.589 FZ -9.563
17 FX -0.616 FY -157.365 FZ 8.754
16 FX 0.635 FY -216.213 FZ -0.36
15 FX 0.631 FY -220.568 FZ 1.972
14 FX -0.005 FY -33.706 FZ -0.355
13 FX -0.021 FY -30.65 FZ -0.348
****
LOAD 3 MATERIAL LOAD LL
JOINT LOAD
18 FX -0.02 FY -23.389 FZ -1.404
17 FX -0.105 FY -23.236 FZ 1.271
16 FX 0.105 FY -35.761 FZ 0.116
15 FX 0.027 FY -36.613 FZ 0.242
14 FX -0.001 FY -0.217 FZ -0.063
13 FX -0.005 FY 0.217 FZ -0.062
****
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
18 FX 0.324 FY 92.973 FZ 25.208
17 FX -0.327 FY -92.974 FZ 25.433
16 FX -0.117 FY 106.961 FZ 26.177
15 FX 0.112 FY -106.96 FZ 26.168
14 FX 0.001 FY 22.207 FZ 10.472
13 FX 0.008 FY -22.207 FZ 10.468
****
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
18 FX -0.326 FY -92.915 FZ -25.189
17 FX 0.33 FY 92.915 FZ -25.414
16 FX 0.114 FY -106.942 FZ -26.176
15 FX -0.109 FY 106.941 FZ -26.167
14 FX -0.001 FY -22.207 FZ -10.472
13 FX -0.008 FY 22.207 FZ -10.467
****
LOAD 6 WIND LOAD ACROSS THE CONV. WL (+X)
JOINT LOAD
18 FX 51.915 FY 224.702 FZ 8.741
17 FX 51.788 FY 225.901 FZ -9.051
16 FX 51.945 FY -223.957 FZ 0.094
15 FX 53.172 FY -226.642 FZ 0.645
14 FX 1.185 FY -0.745 FZ -0.214
13 FX -0.103 FY 0.74 FZ -0.214
****
LOAD 7 WIND LOAD ACROSS THE CONV. WL (-X)
JOINT LOAD
18 FX -51.915 FY -224.702 FZ -8.741
17 FX -51.788 FY -225.901 FZ 9.051
16 FX -51.945 FY 223.957 FZ -0.094

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15 FX -53.172 FY 226.642 FZ -0.645
14 FX -1.185 FY 0.745 FZ 0.214
13 FX 0.103 FY -0.74 FZ 0.214
*****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
18 FX -1.378 FY -3.013 FZ -38.272
17 FX -1.08 FY 3.013 FZ 37.381
16 FX 1.016 FY 4.084 FZ 0.25
15 FX 0.858 FY -4.084 FZ 1.269
14 FX 0.08 FY -1.094 FZ -0.764
13 FX 0.504 FY 1.094 FZ 0.136
*****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
18 FX 1.378 FY 3.013 FZ 38.272
17 FX 1.08 FY -3.013 FZ -37.381
16 FX -1.016 FY -4.084 FZ -0.25
15 FX -0.858 FY 4.084 FZ -1.269
14 FX -0.08 FY 1.094 FZ 0.764
13 FX -0.504 FY -1.094 FZ -0.136
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
18 FX 1.477 FY 7.247 FZ 0.306
17 FX 1.565 FY 7.624 FZ -0.3
16 FX 1.473 FY -7.382 FZ -0.031
15 FX 1.569 FY -7.558 FZ -0.012
14 FX 0 FY -0.021 FZ -0.006
13 FX -0.006 FY 0.021 FZ -0.006
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
18 FX -0.09 FY 6.376 FZ 1.608
17 FX 0.09 FY -6.376 FZ 1.62
16 FX -0.119 FY 6.204 FZ 1.254
15 FX 0.119 FY -6.204 FZ 1.254
14 FX 0 FY 0.67 FZ 0.201
13 FX 0.001 FY -0.67 FZ 0.201
*****************************************
*****Load combination for bearing capacity check
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 8 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 9 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
```

```
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
1 1.0 10 0.525 2 0.75 3 0.75
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
1 1.0 11 0.525 2 0.75 3 0.75
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
LOAD COMB 314 0.6 DL + 1.0 WL(+X)
1 0.6 6 1.0
LOAD COMB 315 0.6 DL + 1.0 WL(-X)
1 0.6 7 1.0
LOAD COMB 316 0.6 DL + 0.7 EL(+X)
1 0.6 10 0.7
LOAD COMB 317 0.6 DL + 0.7 EL(-X)
1 0.6 10 -0.7
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
1 0.6 11 0.7
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 8 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 9 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
1 1.2 10 1.0 2 0.5 3 0.5
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
1 1.2 11 1.0 2 0.5 3 0.5
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
LOAD COMB 414 0.9 DL + 1.6 WL(+X)
```

```
1 0.9 6 1.6
LOAD COMB 415 0.9 DL + 1.6 WL(-X)
1 0.9 7 1.6
LOAD COMB 416 0.9 DL + 1.0 EL(+X)
1 0.9 10 1.0
LOAD COMB 417 0.9 DL + 1.0 EL(-X)
1 0.9 10 -1.0
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
1 0.9 11 1.0
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
1 0.9 11 -1.0
**************************************
PERFORM ANALYSIS
LOAD LIST 301 TO 319
PRINT SUPPORT REACTION ALL
*******************************************************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 ALL
FYMAIN 420000 ALL
TRACK 1 ALL
CLB 0.05 ALL
CLT 0.05 ALL
CLS 0.05 ALL
MAXMAIN 20 MEMB 1 TO 12
MAXMAIN 20 MEMB 13 TO 19
MINMAIN 16 MEMB 1 TO 12
MINMAIN 16 MEMB 13 TO 19
MINSEC 10 ALL
REINF 0 MEMB 1 TO 12
DESIGN COLUMN 1 TO 12
DESIGN BEAM 13 TO 19
END CONCRETE DESIGN
FINISH
```

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**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  7, 2020                *
*           Time=    12:52:46                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3110-CNV-002\Staad\3110-CNV-002_VGT.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 06-APR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART VGTU FDN
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3110-CNV-002
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 -3 0; 2 0 -3 5; 3 4 -3 0; 4 4 -3 5; 5 8 -3 0; 6 8 -3 5; 7 0 0.35 0
   18. 8 0 0.35 5; 9 4 0.35 0; 10 4 0.35 5; 11 8 0.35 0; 12 8 0.35 5; 13 0 1 0
   19. 14 0 1 5; 15 4 1 0; 16 4 1 5; 17 8 1 0; 18 8 1 5
   20. MEMBER INCIDENCES
   21. 1 1 7; 2 7 13; 3 2 8; 4 8 14; 5 3 9; 6 9 15; 7 4 10; 8 10 16; 9 5 11; 10 11 17
   22. 11 6 12; 12 12 18; 13 7 8; 14 9 10; 15 11 12; 16 7 9; 17 9 11; 18 8 10
   23. 19 10 12
   24. DEFINE MATERIAL START
   25. ISOTROPIC CONCRETE
   26. E 2.17185E+007
   27. POISSON 0.17
   28. DENSITY 23.5616
   29. ALPHA 1E-005
   30. DAMP 0.05
   31. TYPE CONCRETE
   32. STRENGTH FCU 27579
   33. END DEFINE MATERIAL
   34. MEMBER PROPERTY AMERICAN
   35. 1 TO 12 PRIS YD 0.6 ZD 0.6
   36. 13 TO 15 PRIS YD 0.5 ZD 0.3
   37. MEMBER PROPERTY AMERICAN
   38. 16 TO 19 PRIS YD 0.6 ZD 0.3
```

```
39. CONSTANTS
40. MATERIAL CONCRETE ALL
41. SUPPORTS
42. 1 TO 6 FIXED
43. ******************************************
44. MEMBER OFFSET
45. 16 TO 19 START 0 -0.3 0
46. 16 TO 19 END 0 -0.3 0
47. 13 TO 15 START 0 -0.25 0
48. 13 TO 15 END 0 -0.25 0
49. LOAD 1 DEAD LOAD DL
50. SELFWEIGHT Y -1
51. JOINT LOAD
52. 18 FX -25.941 FY -275.436 FZ -12.314
53. 17 FX -26.196 FY -295.582 FZ 16.204
54. 16 FX -23.851 FY -72.76 FZ 1.33
55. 15 FX -24.07 FY -105.074 FZ 4.238
56. 14 FX 0 FY -17.052 FZ -0.286
57. 13 FX 0.057 FY -18.904 FZ 0.527
58. ******************************************
59. LOAD 2 LIVE LOAD LL
60. JOINT LOAD
61. 18 FX -0.624 FY -162.589 FZ -9.563
62. 17 FX -0.616 FY -157.365 FZ 8.754
63. 16 FX 0.635 FY -216.213 FZ -0.36
64. 15 FX 0.631 FY -220.568 FZ 1.972
65. 14 FX -0.005 FY -33.706 FZ -0.355
66. 13 FX -0.021 FY -30.65 FZ -0.348
67. ******************************************
68. LOAD 3 MATERIAL LOAD LL
69. JOINT LOAD
70. 18 FX -0.02 FY -23.389 FZ -1.404
71. 17 FX -0.105 FY -23.236 FZ 1.271
72. 16 FX 0.105 FY -35.761 FZ 0.116
73. 15 FX 0.027 FY -36.613 FZ 0.242
74. 14 FX -0.001 FY -0.217 FZ -0.063
75. 13 FX -0.005 FY 0.217 FZ -0.062
76. ******************************************
77. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
78. JOINT LOAD
79. 18 FX 0.324 FY 92.973 FZ 25.208
80. 17 FX -0.327 FY -92.974 FZ 25.433
81. 16 FX -0.117 FY 106.961 FZ 26.177
82. 15 FX 0.112 FY -106.96 FZ 26.168
83. 14 FX 0.001 FY 22.207 FZ 10.472
84. 13 FX 0.008 FY -22.207 FZ 10.468
85. ******************************************
86. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
87. JOINT LOAD
88. 18 FX -0.326 FY -92.915 FZ -25.189
89. 17 FX 0.33 FY 92.915 FZ -25.414
90. 16 FX 0.114 FY -106.942 FZ -26.176
91. 15 FX -0.109 FY 106.941 FZ -26.167
92. 14 FX -0.001 FY -22.207 FZ -10.472
93. 13 FX -0.008 FY 22.207 FZ -10.467
94. ******************************************
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 95. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
 96. JOINT LOAD
 97. 18 FX 51.915 FY 224.702 FZ 8.741
 98. 17 FX 51.788 FY 225.901 FZ -9.051
 99. 16 FX 51.945 FY -223.957 FZ 0.094
100. 15 FX 53.172 FY -226.642 FZ 0.645
101. 14 FX 1.185 FY -0.745 FZ -0.214
102. 13 FX -0.103 FY 0.74 FZ -0.214
103. ****************************************
104. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
105. JOINT LOAD
106. 18 FX -51.915 FY -224.702 FZ -8.741
107. 17 FX -51.788 FY -225.901 FZ 9.051
108. 16 FX -51.945 FY 223.957 FZ -0.094
109. 15 FX -53.172 FY 226.642 FZ -0.645
110. 14 FX -1.185 FY 0.745 FZ 0.214
111. 13 FX 0.103 FY -0.74 FZ 0.214
112. ****************************************
113. LOAD 8 TEMP LOAD   TL1
114. JOINT LOAD
115. 18 FX -1.378 FY -3.013 FZ -38.272
116. 17 FX -1.08 FY 3.013 FZ 37.381
117. 16 FX 1.016 FY 4.084 FZ 0.25
118. 15 FX 0.858 FY -4.084 FZ 1.269
119. 14 FX 0.08 FY -1.094 FZ -0.764
120. 13 FX 0.504 FY 1.094 FZ 0.136
121. ****************************************
122. LOAD 9 TEMP LOAD   TL2
123. JOINT LOAD
124. 18 FX 1.378 FY 3.013 FZ 38.272
125. 17 FX 1.08 FY -3.013 FZ -37.381
126. 16 FX -1.016 FY -4.084 FZ -0.25
127. 15 FX -0.858 FY 4.084 FZ -1.269
128. 14 FX -0.08 FY 1.094 FZ 0.764
129. 13 FX -0.504 FY -1.094 FZ -0.136
130. ****************************************
131. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
132. JOINT LOAD
133. 18 FX 1.477 FY 7.247 FZ 0.306
134. 17 FX 1.565 FY 7.624 FZ -0.3
135. 16 FX 1.473 FY -7.382 FZ -0.031
136. 15 FX 1.569 FY -7.558 FZ -0.012
137. 14 FX 0 FY -0.021 FZ -0.006
138. 13 FX -0.006 FY 0.021 FZ -0.006
139. ****************************************
140. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
141. JOINT LOAD
142. 18 FX -0.09 FY 6.376 FZ 1.608
143. 17 FX 0.09 FY -6.376 FZ 1.62
144. 16 FX -0.119 FY 6.204 FZ 1.254
145. 15 FX 0.119 FY -6.204 FZ 1.254
146. 14 FX 0 FY 0.67 FZ 0.201
147. 13 FX 0.001 FY -0.67 FZ 0.201
148. ****************************************
149. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
150. LOAD COMB 301 1.0 DL
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151. 1 1.0
152. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
153. 1 1.0 8 1.0 2 1.0 3 1.0
154. LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
155. 1 1.0 9 1.0 2 1.0 3 1.0
156. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
157. 1 1.0 4 0.75 2 0.75 3 0.75
158. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
159. 1 1.0 5 0.75 2 0.75 3 0.75
160. LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
161. 1 1.0 6 0.75 2 0.75 3 0.75
162. LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
163. 1 1.0 7 0.75 2 0.75 3 0.75
164. LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
165. 1 1.0 10 0.525 2 0.75 3 0.75
166. LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
167. 1 1.0 10 -0.525 2 0.75 3 0.75
168. LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
169. 1 1.0 11 0.525 2 0.75 3 0.75
170. LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
171. 1 1.0 11 -0.525 2 0.75 3 0.75
172. LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
173. 1 0.6 4 1.0
174. LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
175. 1 0.6 5 1.0
176. LOAD COMB 314 0.6 DL + 1.0 WL(+X)
177. 1 0.6 6 1.0
178. LOAD COMB 315 0.6 DL + 1.0 WL(-X)
179. 1 0.6 7 1.0
180. LOAD COMB 316 0.6 DL + 0.7 EL(+X)
181. 1 0.6 10 0.7
182. LOAD COMB 317 0.6 DL + 0.7 EL(-X)
183. 1 0.6 10 -0.7
184. LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
185. 1 0.6 11 0.7
186. LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
187. 1 0.6 11 -0.7
188. ****************************************
189. *****LOAD COMBINATION FOR FOUNDATION DESIGN
190. LOAD COMB 401 1.4 DL
191. 1 1.4
192. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
193. 1 1.4 8 1.4 2 1.7 3 1.7
194. LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
195. 1 1.4 9 1.4 2 1.7 3 1.7
196. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
197. 1 1.2 4 1.6 2 0.5 3 0.5
198. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
199. 1 1.2 5 1.6 2 0.5 3 0.5
200. LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
201. 1 1.2 6 1.6 2 0.5 3 0.5
202. LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
203. 1 1.2 7 1.6 2 0.5 3 0.5
204. LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
205. 1 1.2 10 1.0 2 0.5 3 0.5
206. LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
```

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207. 1 1.2 10 -1.0 2 0.5 3 0.5
208. LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
209. 1 1.2 11 1.0 2 0.5 3 0.5
210. LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
211. 1 1.2 11 -1.0 2 0.5 3 0.5
212. LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
213. 1 0.9 4 1.6
214. LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
215. 1 0.9 5 1.6
216. LOAD COMB 414 0.9 DL + 1.6 WL(+X)
217. 1 0.9 6 1.6
218. LOAD COMB 415 0.9 DL + 1.6 WL(-X)
219. 1 0.9 7 1.6
220. LOAD COMB 416 0.9 DL + 1.0 EL(+X)
221. 1 0.9 10 1.0
222. LOAD COMB 417 0.9 DL + 1.0 EL(-X)
223. 1 0.9 10 -1.0
224. LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
225. 1 0.9 11 1.0
226. LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
227. 1 0.9 11 -1.0
228. ****************************************
229. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**18**	**NUMBER OF MEMBERS**	**19**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**6**

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 72
TOTAL LOAD COMBINATION CASES = 38 SO FAR.

```
230. LOAD LIST 301 TO 319
231. PRINT SUPPORT REACTION ALL
```

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	301	19.93	78.29	2.75	1.80	0.33	-39.02
	302	20.70	108.80	3.42	3.26	0.62	-39.22
	303	20.64	111.06	2.66	1.85	0.37	-39.39
	304	20.52	114.39	-6.06	-22.56	1.97	-38.98
	305	20.45	89.66	11.99	27.29	-1.06	-39.48
	306	-7.36	88.51	3.08	2.68	0.51	19.67
	307	48.33	115.53	2.85	2.06	0.41	-98.14
	308	19.91	101.75	2.97	2.38	0.46	-38.01
	309	21.06	102.30	2.96	2.35	0.46	-40.46
	310	20.44	102.28	2.81	1.93	0.53	-39.13
	311	20.53	101.76	3.12	2.81	0.38	-39.34
	312	12.01	63.46	-10.39	-32.16	2.22	-23.08
	313	11.91	30.49	13.69	34.32	-1.82	-23.74
	314	-25.17	28.96	1.80	1.50	0.27	55.13
	315	49.09	64.98	1.50	0.67	0.13	-101.95
	316	11.19	46.61	1.66	1.10	0.20	-21.78
	317	12.73	47.34	1.64	1.07	0.20	-25.04
	318	11.90	47.32	1.44	0.50	0.30	-23.27
	319	12.02	46.63	1.86	1.67	0.09	-23.55
2	301	19.74	76.65	-3.54	-3.97	0.33	-38.80
	302	20.44	112.31	-3.26	-2.84	0.66	-39.11
	303	20.32	110.53	-3.32	-3.68	0.34	-38.74
	304	20.18	90.36	-12.38	-28.36	1.97	-39.15
	305	20.25	115.09	5.67	21.49	-1.06	-38.65
	306	-7.44	90.13	-3.24	-3.12	0.50	19.97
	307	47.88	115.32	-3.47	-3.75	0.42	-97.75
	308	19.67	102.48	-3.35	-3.42	0.46	-37.74
	309	20.76	102.97	-3.36	-3.45	0.46	-40.05
	310	20.26	102.47	-3.51	-3.88	0.54	-39.00
	311	20.17	102.98	-3.20	-3.00	0.38	-38.79
	312	11.80	29.50	-14.16	-35.62	2.22	-23.62
	313	11.89	62.47	9.91	30.86	-1.82	-22.95
	314	-25.04	29.20	-1.97	-1.96	0.26	55.20
	315	48.72	62.78	-2.27	-2.80	0.14	-101.76
	316	11.12	45.66	-2.11	-2.36	0.20	-21.74
	317	12.57	46.31	-2.13	-2.40	0.20	-24.83
	318	11.90	45.64	-2.33	-2.97	0.30	-23.42
	319	11.78	46.33	-1.91	-1.79	0.10	-23.14
3	301	19.38	169.29	1.22	-3.31	0.25	-39.55
	302	19.79	428.27	1.21	-5.59	0.39	-39.48
	303	19.56	420.35	0.92	-3.92	0.16	-39.45
	304	19.51	431.11	-17.22	-54.69	1.54	-39.13
	305	19.69	290.01	19.43	45.90	-1.01	-39.83
	306	-10.08	521.80	0.89	-4.94	0.25	22.59
	307	49.28	199.31	1.31	-3.84	0.29	-101.56
	308	18.98	364.34	1.11	-4.37	0.27	-38.19

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	309	20.22	356.77	1.09	-4.42	0.27	-40.78
	310	19.55	363.47	0.48	-6.09	0.36	-39.37
	311	19.65	357.63	1.72	-2.69	0.18	-39.60
	312	11.50	195.65	-23.70	-69.04	1.85	-23.26
	313	11.75	7.52	25.16	65.07	-1.55	-24.19
	314	-27.95	316.57	0.45	-2.71	0.12	59.03
	315	51.20	-113.42	1.01	-1.25	0.17	-106.49
	316	10.80	106.62	0.74	-1.95	0.15	-22.00
	317	12.45	96.53	0.72	-2.02	0.15	-25.45
	318	11.56	105.47	-0.10	-4.25	0.27	-23.58
	319	11.69	97.68	1.56	0.28	0.03	-23.88
4	301	19.30	139.67	-6.05	-9.91	0.16	-39.42
	302	19.96	386.74	-8.45	-14.25	-0.02	-39.72
	303	19.11	393.81	-6.08	-10.22	0.24	-38.67
	304	19.57	257.07	-25.29	-61.95	1.40	-39.60
	305	19.39	398.17	11.36	38.64	-1.15	-38.90
	306	-10.52	487.20	-7.13	-12.14	0.15	23.19
	307	49.47	168.05	-6.79	-11.16	0.10	-101.69
	308	18.89	331.31	-6.95	-11.63	0.13	-38.02
	309	20.07	323.93	-6.97	-11.68	0.12	-40.48
	310	19.52	324.70	-7.58	-13.35	0.21	-39.36
	311	19.43	330.54	-6.34	-9.95	0.04	-39.14
	312	11.70	-10.28	-28.07	-73.00	1.79	-24.12
	313	11.46	177.86	20.80	61.11	-1.60	-23.19
	314	-28.42	296.57	-3.85	-6.60	0.13	59.60
	315	51.58	-128.97	-3.41	-5.29	0.06	-106.90
	316	10.79	88.72	-3.62	-5.91	0.10	-22.01
	317	12.37	78.88	-3.64	-5.98	0.09	-25.29
	318	11.64	79.91	-4.46	-8.21	0.21	-23.80
	319	11.51	87.69	-2.80	-3.68	-0.02	-23.50
5	301	10.79	331.74	5.66	1.01	-0.04	-32.33
	302	9.44	510.40	22.26	15.66	-0.16	-30.78
	303	10.24	516.11	-3.34	-6.27	-0.22	-31.54
	304	9.72	528.25	-10.58	-48.59	-0.16	-30.89
	305	10.43	407.54	27.59	56.10	-0.14	-32.01
	306	-11.05	320.14	6.39	2.20	-0.23	23.55
	307	31.20	615.61	10.63	5.34	-0.07	-86.45
	308	9.64	464.33	8.46	3.73	-0.15	-30.31
	309	10.52	471.42	8.56	3.82	-0.15	-32.60
	310	10.03	470.84	7.68	1.50	-0.12	-31.34
	311	10.12	464.91	9.34	6.05	-0.18	-31.56
	312	6.01	279.54	-22.05	-69.21	-0.04	-18.65
	313	6.95	118.60	28.83	70.37	-0.01	-20.14
	314	-21.69	2.06	0.57	-1.49	-0.14	53.93
	315	34.64	396.03	6.23	2.70	0.08	-92.73
	316	5.89	194.32	3.33	0.55	-0.02	-17.87

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	317	7.06	203.77	3.47	0.67	-0.03	-20.92
	318	6.42	203.00	2.29	-2.43	0.02	-19.25
	319	6.54	195.09	4.51	3.64	-0.07	-19.54
6	301	10.86	313.61	-9.74	-12.14	-0.16	-32.32
	302	9.67	502.83	-25.07	-23.32	-0.68	-31.05
	303	10.14	497.48	-0.74	-4.85	-0.08	-31.06
	304	10.50	393.15	-31.26	-66.01	-0.33	-31.93
	305	9.79	513.85	7.02	38.77	-0.31	-30.81
	306	-10.97	306.54	-9.84	-11.61	-0.35	23.69
	307	31.26	600.50	-14.39	-15.59	-0.30	-86.43
	308	9.72	450.15	-12.06	-13.56	-0.32	-30.29
	309	10.56	456.89	-12.17	-13.64	-0.33	-32.46
	310	10.19	450.55	-12.95	-15.88	-0.29	-31.48
	311	10.10	456.49	-11.28	-11.32	-0.36	-31.26
	312	6.99	107.67	-31.37	-77.16	-0.11	-20.14
	313	6.05	268.61	19.66	62.54	-0.08	-18.64
	314	-21.63	-7.81	-2.82	-4.64	-0.12	54.03
	315	34.67	384.14	-8.88	-9.93	-0.06	-92.81
	316	5.96	183.67	-5.78	-7.23	-0.09	-17.94
	317	7.08	192.66	-5.92	-7.34	-0.10	-20.84
	318	6.58	184.21	-6.96	-10.32	-0.05	-19.53
	319	6.46	192.12	-4.74	-4.25	-0.14	-19.25

********** END OF LATEST ANALYSIS RESULT **********

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232. ********************************************************************************
233. LOAD LIST 401 TO 419
234. START CONCRETE DESIGN
235. CODE ACI 2008
236. FC 35000 ALL
237. FYMAIN 420000 ALL
238. TRACK 1 ALL
239. CLB 0.05 ALL
240. CLT 0.05 ALL
241. CLS 0.05 ALL
242. MAXMAIN 20 MEMB 1 TO 12
243. MAXMAIN 20 MEMB 13 TO 19
244. MINMAIN 16 MEMB 1 TO 12
245. MINMAIN 16 MEMB 13 TO 19
246. MINSEC 10 ALL
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247. REINF 0 MEMB 1 TO 12
248. DESIGN COLUMN 1 TO 12
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COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	107.42	59.76	0.16607

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	107.42	15.96	0.04435

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COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	40.72	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	40.72	0.00	0.00000

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	103.88	58.84	0.16907

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	103.88	16.96	0.04874

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COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	36.73	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	36.73	0.00	0.00000

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COLUMN NO. 5 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	303.43	54.63	0.05374

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	303.43	15.92	0.01566

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COLUMN NO. 6 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	226.31	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	226.31	0.00	0.00000

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COLUMN NO. 7 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	239.62	54.35	0.06771

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	239.62	22.34	0.02783

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COLUMN NO. 8 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	156.71	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	156.71	0.00	0.00000

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COLUMN NO. 9 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	653.32	8.25	0.00377

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	653.32	38.69	0.01768

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COLUMN NO. 10 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

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 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    636.64      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    636.64      0.00    0.00000
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COLUMN NO. 11 DESIGN PER ACI 318-08 - AXIAL + BENDING

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FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 20 MM               1.047      401      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    614.27      8.78    0.00427
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    614.27     44.16    0.02146
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COLUMN NO. 12 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	593.25	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	593.25	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

249. DESIGN BEAM 13 TO 19

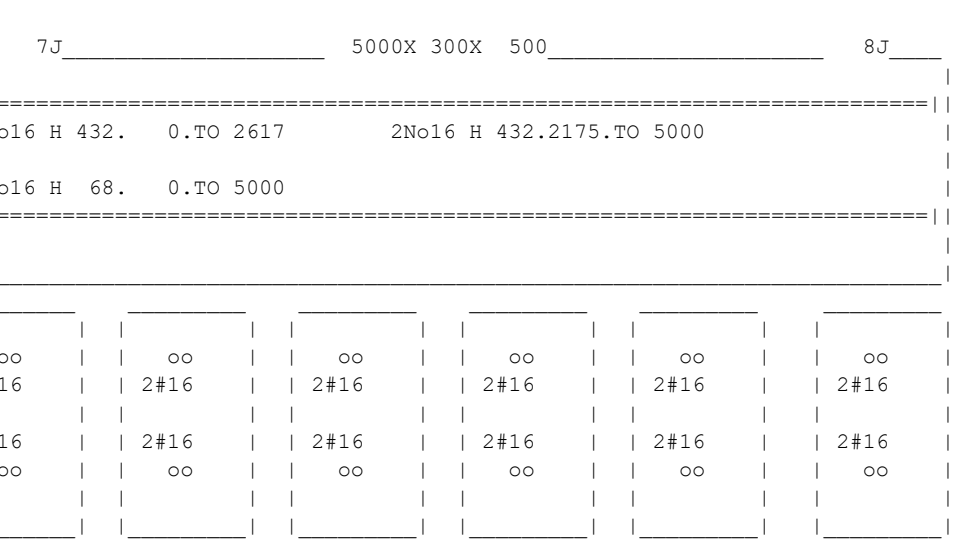
BEAM NO. 13 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	68.	2 - 16MM	0.	5000.	YES	YES
	CRITICAL POS MOMENT= 18.69 KN-MET AT 0.MM, LOAD 412					
	REQD STEEL= 153.MM2, RHO=0.0012, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 164. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 40697.6 cm^4					
2	432.	2 - 16MM	0.	2617.	YES	NO
	CRITICAL NEG MOMENT= 31.35 KN-MET AT 0.MM, LOAD 405					
	REQD STEEL= 258.MM2, RHO=0.0020, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 164. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 40697.6 cm^4					
3	432.	2 - 16MM	2175.	5000.	NO	YES
	CRITICAL NEG MOMENT= 32.18 KN-MET AT 5000.MM, LOAD 404					
	REQD STEEL= 265.MM2, RHO=0.0020, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 164. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 40697.6 cm^4					

B E A M N O. 13 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 9.97 KNS Vc= 125.29 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 10.52 KNS Vc= 124.15 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 14 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	68.	2 - 16MM	0.	5000.	YES	YES

CRITICAL POS MOMENT= 46.50 KN-MET AT 0.MM, LOAD 412
REQD STEEL= 385.MM2, RHO=0.0030, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 164. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 40697.6 cm^4

2	432.	3 - 16MM	0.	2617.	YES	NO

CRITICAL NEG MOMENT= 52.66 KN-MET AT 0.MM, LOAD 405
REQD STEEL= 437.MM2, RHO=0.0034, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 82. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 57472.8 cm^4

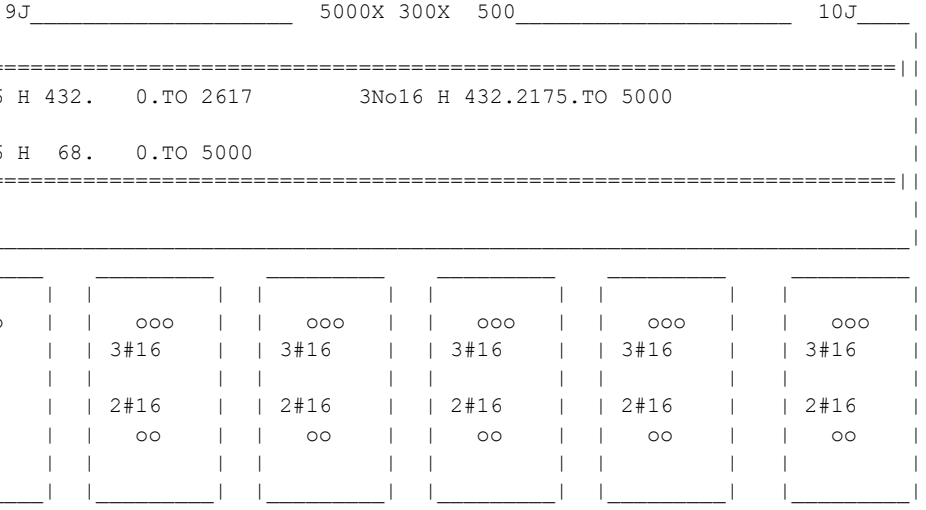
3	432.	3 - 16MM	2175.	5000.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     60.97 KN-MET  AT  5000.MM, LOAD  404|
|   REQD STEEL=   455.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   41./   82. MMS        |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    57472.8 cm^4
```

B E A M N O. 14 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=    8.51 KNS  Vc=  224.37 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   11.98 KNS  Vc=  126.56 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 15 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	68.	3 - 16MM	0.	5000.	YES	YES

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=     48.83 KN-MET  AT     0.MM, LOAD  412|
|   REQD STEEL=   404.MM2, RHO=0.0031, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   41./   82. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    57472.8 cm^4

  2        432.     3 - 16MM         0.         2617.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     53.36 KN-MET  AT     0.MM, LOAD  405|
|   REQD STEEL=   443.MM2, RHO=0.0034, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   41./   82. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                        |
|--------------------------------------------------------------|

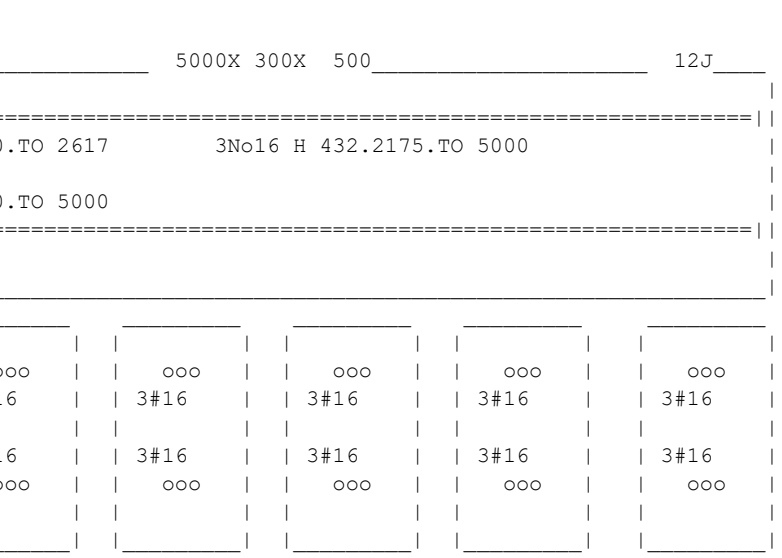
  Cracked Moment of Inertia Iz at above location =    57472.8 cm^4

  3        432.     3 - 16MM      2175.         5000.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     59.07 KN-MET  AT  5000.MM, LOAD  404|
|   REQD STEEL=   455.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   41./   82. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    57472.8 cm^4
```

B E A M N O. 15 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=    8.81 KNS  Vc=  229.35 KNS  Vs=    0.00 KNS
Tu=    0.01 KN-MET  Tc=    5.4 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   11.67 KNS  Vc=  229.35 KNS  Vs=    0.00 KNS
Tu=    0.01 KN-MET  Tc=    5.4 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 16 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

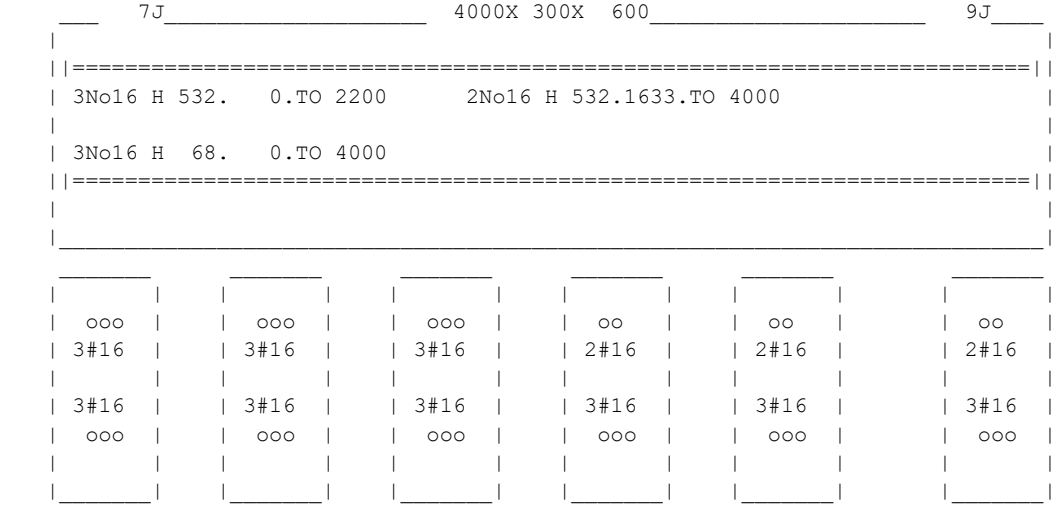
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	68.	3 - 16MM	0.	4000.	YES	YES
	CRITICAL POS MOMENT= 70.70 KN-MET AT 4000.MM, LOAD 407					
	REQD STEEL= 475.MM2, RHO=0.0030, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 82. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90032.8 cm^4					
2	532.	3 - 16MM	0.	2200.	YES	NO
	CRITICAL NEG MOMENT= 82.48 KN-MET AT 0.MM, LOAD 407					
	REQD STEEL= 556.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 82. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90032.8 cm^4					
3	532.	2 - 16MM	1633.	4000.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     45.13 KN-MET  AT  4000.MM, LOAD  414|
|   REQD STEEL=   302.MM2, RHO=0.0019, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   41./  164. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    63391.8 cm^4
```

B E A M N O. 16 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   20.40 KNS  Vc=  156.20 KNS  Vs=    0.00 KNS
 Tu=    0.44 KN-MET  Tc=    6.9 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=    2.94 KNS  Vc=  150.40 KNS  Vs=    0.00 KNS
 Tu=    0.44 KN-MET  Tc=    6.9 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 17 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	68.	4 - 16MM	0.	4000.	YES	YES

```
|-------------------------------------------------------------|
|   CRITICAL POS MOMENT=    128.89 KN-MET  AT  4000.MM, LOAD  407|
|   REQD STEEL=   660.MM2, RHO=0.0041, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   41./   55. MMS         |
|   REQD. DEVELOPMENT LENGTH =   601. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =   114685.2 cm^4

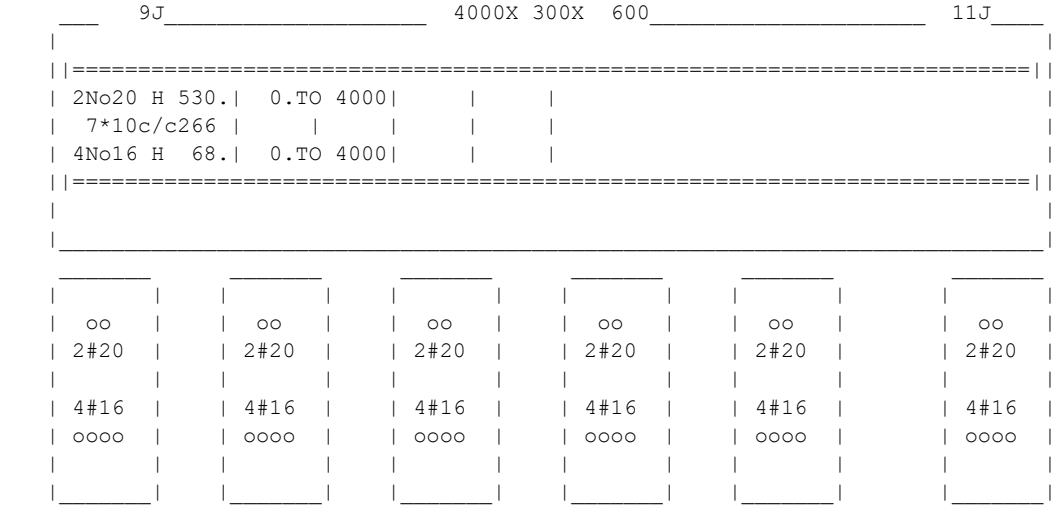
  2        530.     2 - 20MM         0.         4000.        YES  YES
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=    121.88 KN-MET  AT     0.MM, LOAD  407|
|   REQD STEEL=   623.MM2, RHO=0.0039, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   45./  160. MMS         |
|   REQD. DEVELOPMENT LENGTH =   513. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    93186.7 cm^4
```

B E A M N O. 17 D E S I G N R E S U L T S - SHEAR

```
  AT START SUPPORT - Vu=   70.18 KNS  Vc=  154.97 KNS  Vs=    0.00 KNS
  Tu=    1.12 KN-MET  Tc=    7.1 KN-MET  Ts=    0.0 KN-MET  LOAD   407
  NO STIRRUPS ARE REQUIRED FOR TORSION.
  REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
  PROVIDE 10 MM 2-LEGGED STIRRUPS AT 266. MM  C/C FOR 1470. MM

  AT END   SUPPORT - Vu=   10.96 KNS  Vc=  151.56 KNS  Vs=    0.00 KNS
  Tu=    0.54 KN-MET  Tc=    6.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                   STIRRUPS ARE NOT REQUIRED.
```



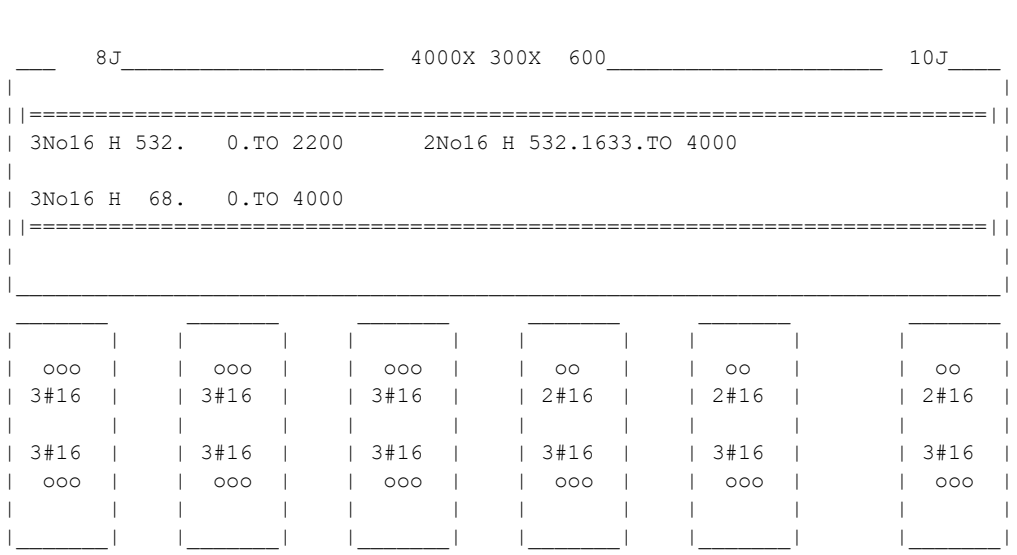
BEAM NO. 18 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	68.	3 - 16MM	0.	4000.	YES	YES
	CRITICAL POS MOMENT= 70.08 KN-MET AT 4000.MM, LOAD 407					
	REQD STEEL= 471.MM2, RHO=0.0030, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 82. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90032.8 cm^4					
2	532.	3 - 16MM	0.	2200.	YES	NO
	CRITICAL NEG MOMENT= 82.76 KN-MET AT 0.MM, LOAD 407					
	REQD STEEL= 558.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 82. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90032.8 cm^4					
3	532.	2 - 16MM	1633.	4000.	NO	YES
	CRITICAL NEG MOMENT= 45.48 KN-MET AT 4000.MM, LOAD 414					
	REQD STEEL= 304.MM2, RHO=0.0019, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 164. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 63391.8 cm^4					

B E A M N O. 18 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 20.14 KNS Vc= 156.15 KNS Vs= 0.00 KNS
Tu= 0.30 KN-MET Tc= 6.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 2.68 KNS Vc= 150.31 KNS Vs= 0.00 KNS
Tu= 0.30 KN-MET Tc= 6.9 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 19 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	68.	4 - 16MM	0.	4000.	YES	YES

CRITICAL POS MOMENT= 128.84 KN-MET AT 4000.MM, LOAD 407
REQD STEEL= 660.MM2, RHO=0.0041, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 55. MMS
REQD. DEVELOPMENT LENGTH = 601. MMS

Cracked Moment of Inertia Iz at above location = 114685.2 cm^4

2	530.	2 - 20MM	0.	4000.	YES	YES

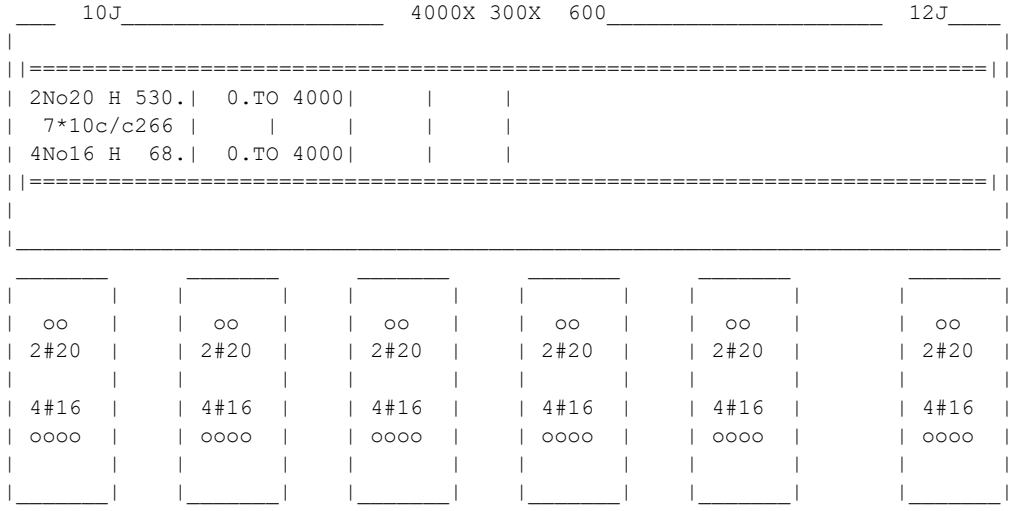
CRITICAL NEG MOMENT= 121.53 KN-MET AT 0.MM, LOAD 407
REQD STEEL= 621.MM2, RHO=0.0039, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 45./ 160. MMS
REQD. DEVELOPMENT LENGTH = 513. MMS

Cracked Moment of Inertia Iz at above location = 93186.7 cm^4

B E A M N O. 19 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 70.07 KNS Vc= 154.98 KNS Vs= 0.00 KNS
Tu= 1.23 KN-MET Tc= 7.1 KN-MET Ts= 0.0 KN-MET LOAD 407
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 266. MM C/C FOR 1470. MM

AT END SUPPORT - Vu= 11.00 KNS Vc= 151.56 KNS Vs= 0.00 KNS
Tu= 0.68 KN-MET Tc= 6.8 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

250. END CONCRETE DESIGN
251. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 12:52:47 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	07-04-2020	18-05-2020	12-10-2023				Date	07-04-2020	18-05-2020	12-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 8 **STAAD INPUT & OUTPUT FILES OF FOUNDATION FOR TR3**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 06-Apr-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART TRESTLE TR3 FOUNDATION MKD. F5
JOB REF FOUNDATIONS FOR CONVEYOR 3110-CNV-002
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 4; 3 0 1.49012e-008 0; 4 0 1.49012e-008 4; 5 0 0.35 0;
6 0 0.35 4;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.7 ZD 0.5
5 PRIS YD 0.6 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 2 FIXED
*****************************************
MEMBER OFFSET
5 START 0 -0.25 0
5 END 0 -0.25 0
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
5 FX -0.01 FY -109.321 FZ -16.683
6 FX 0.002 FY -156.271 FZ 24.683
*****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
5 FX -0.002 FY -48.05 FZ -7.555
6 FX 0.002 FY -47.95 FZ 7.555
*****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
```

```
5 FX -0.003 FY -49.2 FZ -7.773
6 FX 0.003 FY -49.2 FZ 7.773
*****************************************
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
5 FX 0.036 FY 125.096 FZ 31.05
6 FX -0.036 FY -125.096 FZ 31.019
*****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
5 FX -0.036 FY -125.101 FZ -31.051
6 FX 0.036 FY 125.101 FZ -31.018
*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
5 FX 2.225 FY 0 FZ 0
6 FX 2.225 FY 0 FZ 0
*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
5 FX -2.225 FY 0 FZ 0
6 FX -2.225 FY 0 FZ 0
*****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
5 FX -0.005 FY 0.043 FZ -18.937
6 FX 0.005 FY -0.043 FZ 18.917
*****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
5 FX 0.005 FY -0.043 FZ 18.937
6 FX -0.005 FY 0.043 FZ -18.917
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
5 FX 0.018 FY 0.006 FZ 0.001
6 FX 0.018 FY -0.056 FZ 0.009
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
5 FX -0.002 FY 8.744 FZ 2.167
6 FX 0.002 FY -8.744 FZ 2.162
*****************************************
*****Load combination for bearing capacity check
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 8 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 9 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
```

```
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
1 1.0 10 0.525 2 0.75 3 0.75
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
1 1.0 11 0.525 2 0.75 3 0.75
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
LOAD COMB 314 0.6 DL + 1.0 WL(+X)
1 0.6 6 1.0
LOAD COMB 315 0.6 DL + 1.0 WL(-X)
1 0.6 7 1.0
LOAD COMB 316 0.6 DL + 0.7 EL(+X)
1 0.6 10 0.7
LOAD COMB 317 0.6 DL + 0.7 EL(-X)
1 0.6 10 -0.7
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
1 0.6 11 0.7
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
1 0.6 11 -0.7
****************************************
*****Load combination for foundation design
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 8 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 9 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
1 1.2 10 1.0 2 0.5 3 0.5
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
1 1.2 11 1.0 2 0.5 3 0.5
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
```

```
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
LOAD COMB 414 0.9 DL + 1.6 WL(+X)
1 0.9 6 1.6
LOAD COMB 415 0.9 DL + 1.6 WL(-X)
1 0.9 7 1.6
LOAD COMB 416 0.9 DL + 1.0 EL(+X)
1 0.9 10 1.0
LOAD COMB 417 0.9 DL + 1.0 EL(-X)
1 0.9 10 -1.0
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
1 0.9 11 1.0
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
LOAD LIST 301 TO 319
PRINT SUPPORT REACTION ALL
*****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 ALL
FYMAIN 420000 ALL
TRACK 1 ALL
CLB 0.05 ALL
CLT 0.05 ALL
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  7, 2020                *
*           Time=    13: 8:53                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3110-CNV-002\Staad\3110-CNV-002_TR3.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 06-APR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART TRESTLE TR3 FOUNDATION MKD. F5
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3110-CNV-002
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 -3 0; 2 0 -3 4; 3 0 1.49012E-008 0; 4 0 1.49012E-008 4; 5 0 0.35 0
   18. 6 0 0.35 4
   19. MEMBER INCIDENCES
   20. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
   21. DEFINE MATERIAL START
   22. ISOTROPIC CONCRETE
   23. E 2.17185E+007
   24. POISSON 0.17
   25. DENSITY 23.5616
   26. ALPHA 1E-005
   27. DAMP 0.05
   28. TYPE CONCRETE
   29. STRENGTH FCU 27579
   30. END DEFINE MATERIAL
   31. MEMBER PROPERTY AMERICAN
   32. 1 TO 4 PRIS YD 0.7 ZD 0.5
   33. 5 PRIS YD 0.6 ZD 0.3
   34. CONSTANTS
   35. MATERIAL CONCRETE ALL
   36. SUPPORTS
   37. 1 2 FIXED
   38. ****************************************
```

```
39. MEMBER OFFSET
40. 5 START 0 -0.25 0
41. 5 END 0 -0.25 0
42. LOAD 1 DEAD LOAD DL
43. SELFWEIGHT Y -1
44. JOINT LOAD
45. 5 FX -0.01 FY -109.321 FZ -16.683
46. 6 FX 0.002 FY -156.271 FZ 24.683
47. ******************************************
48. LOAD 2 LIVE LOAD LL
49. JOINT LOAD
50. 5 FX -0.002 FY -48.05 FZ -7.555
51. 6 FX 0.002 FY -47.95 FZ 7.555
52. ******************************************
53. LOAD 3 MATERIAL LOAD LL
54. JOINT LOAD
55. 5 FX -0.003 FY -49.2 FZ -7.773
56. 6 FX 0.003 FY -49.2 FZ 7.773
57. ******************************************
58. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
59. JOINT LOAD
60. 5 FX 0.036 FY 125.096 FZ 31.05
61. 6 FX -0.036 FY -125.096 FZ 31.019
62. ******************************************
63. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
64. JOINT LOAD
65. 5 FX -0.036 FY -125.101 FZ -31.051
66. 6 FX 0.036 FY 125.101 FZ -31.018
67. ******************************************
68. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
69. JOINT LOAD
70. 5 FX 2.225 FY 0 FZ 0
71. 6 FX 2.225 FY 0 FZ 0
72. ******************************************
73. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
74. JOINT LOAD
75. 5 FX -2.225 FY 0 FZ 0
76. 6 FX -2.225 FY 0 FZ 0
77. ******************************************
78. LOAD 8 TEMP LOAD   TL1
79. JOINT LOAD
80. 5 FX -0.005 FY 0.043 FZ -18.937
81. 6 FX 0.005 FY -0.043 FZ 18.917
82. ******************************************
83. LOAD 9 TEMP LOAD   TL2
84. JOINT LOAD
85. 5 FX 0.005 FY -0.043 FZ 18.937
86. 6 FX -0.005 FY 0.043 FZ -18.917
87. ******************************************
88. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
89. JOINT LOAD
90. 5 FX 0.018 FY 0.006 FZ 0.001
91. 6 FX 0.018 FY -0.056 FZ 0.009
92. ******************************************
93. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
94. JOINT LOAD
```

```
 95. 5 FX -0.002 FY 8.744 FZ 2.167
 96. 6 FX 0.002 FY -8.744 FZ 2.162
 97. ****************************************
 98. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
 99. LOAD COMB 301 1.0 DL
100. 1 1.0
101. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
102. 1 1.0 8 1.0 2 1.0 3 1.0
103. LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
104. 1 1.0 9 1.0 2 1.0 3 1.0
105. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
106. 1 1.0 4 0.75 2 0.75 3 0.75
107. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
108. 1 1.0 5 0.75 2 0.75 3 0.75
109. LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
110. 1 1.0 6 0.75 2 0.75 3 0.75
111. LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
112. 1 1.0 7 0.75 2 0.75 3 0.75
113. LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
114. 1 1.0 10 0.525 2 0.75 3 0.75
115. LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
116. 1 1.0 10 -0.525 2 0.75 3 0.75
117. LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
118. 1 1.0 11 0.525 2 0.75 3 0.75
119. LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
120. 1 1.0 11 -0.525 2 0.75 3 0.75
121. LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
122. 1 0.6 4 1.0
123. LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
124. 1 0.6 5 1.0
125. LOAD COMB 314 0.6 DL + 1.0 WL(+X)
126. 1 0.6 6 1.0
127. LOAD COMB 315 0.6 DL + 1.0 WL(-X)
128. 1 0.6 7 1.0
129. LOAD COMB 316 0.6 DL + 0.7 EL(+X)
130. 1 0.6 10 0.7
131. LOAD COMB 317 0.6 DL + 0.7 EL(-X)
132. 1 0.6 10 -0.7
133. LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
134. 1 0.6 11 0.7
135. LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
136. 1 0.6 11 -0.7
137. ****************************************
138. *****LOAD COMBINATION FOR FOUNDATION DESIGN
139. LOAD COMB 401 1.4 DL
140. 1 1.4
141. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
142. 1 1.4 8 1.4 2 1.7 3 1.7
143. LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
144. 1 1.4 9 1.4 2 1.7 3 1.7
145. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
146. 1 1.2 4 1.6 2 0.5 3 0.5
147. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
148. 1 1.2 5 1.6 2 0.5 3 0.5
149. LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
150. 1 1.2 6 1.6 2 0.5 3 0.5
```

```
151. LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
152. 1 1.2 7 1.6 2 0.5 3 0.5
153. LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
154. 1 1.2 10 1.0 2 0.5 3 0.5
155. LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
156. 1 1.2 10 -1.0 2 0.5 3 0.5
157. LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
158. 1 1.2 11 1.0 2 0.5 3 0.5
159. LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
160. 1 1.2 11 -1.0 2 0.5 3 0.5
161. LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
162. 1 0.9 4 1.6
163. LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
164. 1 0.9 5 1.6
165. LOAD COMB 414 0.9 DL + 1.6 WL(+X)
166. 1 0.9 6 1.6
167. LOAD COMB 415 0.9 DL + 1.6 WL(-X)
168. 1 0.9 7 1.6
169. LOAD COMB 416 0.9 DL + 1.0 EL(+X)
170. 1 0.9 10 1.0
171. LOAD COMB 417 0.9 DL + 1.0 EL(-X)
172. 1 0.9 10 -1.0
173. LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
174. 1 0.9 11 1.0
175. LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
176. 1 0.9 11 -1.0
177. ****************************************
178. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**6**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**2**

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 38 SO FAR.

```
179. LOAD LIST 301 TO 319
180. PRINT SUPPORT REACTION ALL
```

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	301	0.01	142.73	-6.19	-9.37	-0.00	-0.03
	302	0.02	239.94	-13.32	-14.55	-0.00	-0.06
	303	0.01	240.01	-5.45	-8.84	-0.00	-0.03
	304	-0.01	105.95	-31.86	-57.30	0.00	0.03
	305	0.04	325.38	14.69	35.07	-0.01	-0.11
	306	-1.66	215.66	-8.58	-11.11	-0.00	5.55
	307	1.68	215.66	-8.58	-11.11	-0.00	-5.63
	308	0.00	215.66	-8.59	-11.12	-0.00	-0.01
	309	0.02	215.67	-8.58	-11.11	-0.00	-0.07
	310	0.01	210.30	-9.72	-13.37	-0.00	-0.04
	311	0.01	221.03	-7.45	-8.86	-0.00	-0.04
	312	-0.03	-60.65	-34.74	-67.20	0.01	0.08
	313	0.04	231.93	27.32	55.96	-0.01	-0.12
	314	-2.22	85.64	-3.71	-5.62	-0.00	7.44
	315	2.23	85.64	-3.71	-5.62	-0.00	-7.47
	316	-0.01	85.63	-3.72	-5.63	-0.00	0.02
	317	0.02	85.64	-3.71	-5.61	-0.00	-0.06
	318	0.01	78.48	-5.23	-8.63	-0.00	-0.02
	319	0.00	92.79	-2.20	-2.61	-0.00	-0.01
2	301	-0.00	195.08	-1.81	-6.62	-0.00	0.00
	302	-0.01	292.27	5.34	-1.40	-0.00	0.03
	303	-0.00	292.20	-2.57	-7.19	-0.00	0.00
	304	0.02	377.66	-22.70	-51.06	0.00	-0.06
	305	-0.03	158.22	23.86	41.31	-0.01	0.09
	306	-1.67	267.94	0.58	-4.87	-0.00	5.60
	307	1.66	267.94	0.58	-4.87	-0.00	-5.58
	308	-0.01	267.98	0.58	-4.88	-0.00	0.04
	309	0.01	267.91	0.58	-4.87	-0.00	-0.02
	310	-0.01	273.31	-0.55	-7.13	-0.00	0.02
	311	-0.00	262.58	1.72	-2.62	-0.00	0.01
	312	0.03	263.34	-32.13	-65.56	0.01	-0.10
	313	-0.03	-29.24	29.95	57.61	-0.01	0.10
	314	-2.23	117.05	-1.09	-3.97	-0.00	7.46
	315	2.22	117.05	-1.09	-3.97	-0.00	-7.45
	316	-0.01	117.09	-1.09	-3.98	-0.00	0.04
	317	0.01	117.01	-1.09	-3.97	-0.00	-0.04
	318	-0.00	124.20	-2.60	-6.98	-0.00	0.01
	319	0.00	109.89	0.43	-0.97	-0.00	-0.00

********** END OF LATEST ANALYSIS RESULT **********

```
181. ****************************************
182. LOAD LIST 401 TO 419
183. START CONCRETE DESIGN
184. CODE ACI 2008
185. FC 35000 ALL
186. FYMAIN 420000 ALL
187. TRACK 1 ALL
188. CLB 0.05 ALL
189. CLT 0.05 ALL
190. CLS 0.05 ALL
191. MAXMAIN 25 ALL
192. MINMAIN 16 ALL
193. MINSEC 8 ALL
194. REINF 0 MEMB 1 TO 4
195. DESIGN COLUMN 1 TO 4
```

==========

COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	254.13	0.00	0.00001

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	254.13	19.79	0.02596

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	235.46	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	235.46	0.00	0.00000

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	366.89	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	366.89	2.54	0.00231

==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	336.58	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	336.58	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

196. DESIGN BEAM 5

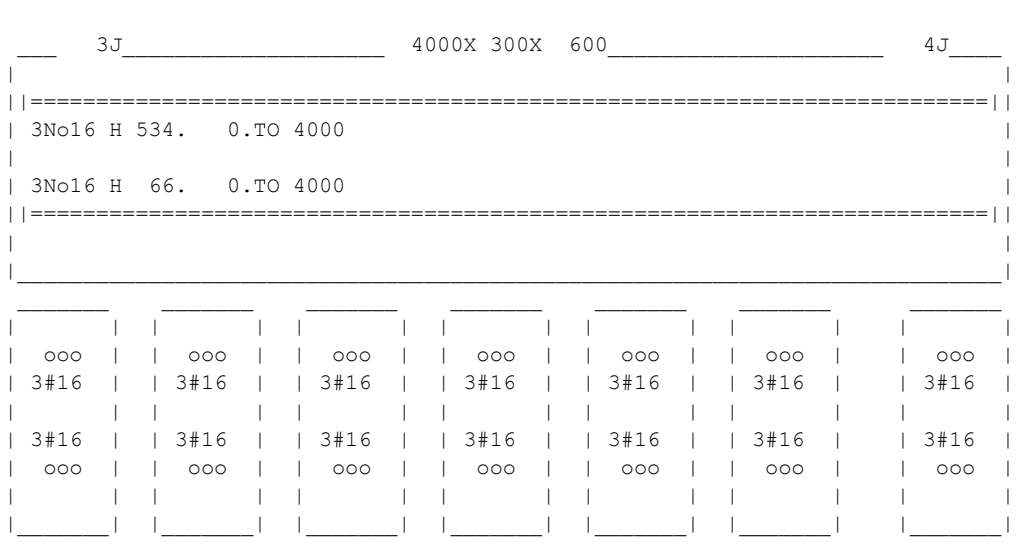
BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	4000.	YES	YES
	CRITICAL POS MOMENT= 65.69 KN-MET AT 0.MM, LOAD 412					
	REQD STEEL= 439.MM2, RHO=0.0027, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					
2	534.	3 - 16MM	0.	4000.	YES	YES
	CRITICAL NEG MOMENT= 84.99 KN-MET AT 4000.MM, LOAD 404					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 90761.5 cm^4					

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 4.92 KNS Vc= 149.27 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.49 KNS Vc= 149.27 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 6.3 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

197. END CONCRETE DESIGN
198. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 13: 8:54 ****

```
************************************************************
*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
*      Copyright (c) 1997-2016 Bentley Systems, Inc.       *
*                http://www.bentley.com                    *
************************************************************
```

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2			
Date	07-04-2020	18-05-2020	12-10-2023				Date	07-04-2020	18-05-2020	12-10-2023			
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT			
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL			
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY			
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI			

ATTACHMENT 9 **STAAD INPUT & OUTPUT FILES OF FOUNDATION FOR TR4**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 06-Apr-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART TRESTLE TR4 FOUNDATION MKD. F5
JOB REF FOUNDATIONS FOR CONVEYOR 3110-CNV-002
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 6; 3 0 1.49012e-008 0; 4 0 1.49012e-008 6; 5 0 0.35 0;
6 0 0.35 6;
MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.7 ZD 0.5
5 PRIS YD 0.6 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
1 2 FIXED
*****************************************
MEMBER OFFSET
5 START 0 -0.25 0
5 END 0 -0.25 0
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
5 FX -0.024 FY -166.03 FZ -31.146
6 FX 0.014 FY -241.697 FZ 42.346
*****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
5 FX -0.008 FY -71.746 FZ -13
6 FX 0.008 FY -72.254 FZ 13.1
*****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
```

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5 FX -0.007 FY -69.513 FZ -12.603
6 FX 0.007 FY -70.088 FZ 12.703
*****************************************
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
5 FX 0.017 FY 184.107 FZ 24.697
6 FX -0.017 FY -184.107 FZ 24.694
*****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
5 FX -0.017 FY -246.635 FZ -49.301
6 FX 0.017 FY 246.635 FZ -49.298
*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
5 FX 4.613 FY 0 FZ 0
6 FX 4.614 FY 0 FZ 0
*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
5 FX -4.613 FY 0 FZ 0
6 FX -4.614 FY 0 FZ 0
*****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
5 FX -0.004 FY -0.013 FZ -58.119
6 FX 0.004 FY 0.013 FZ 58.119
*****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
5 FX 0.004 FY 0.013 FZ 58.119
6 FX -0.004 FY -0.013 FZ -58.119
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
5 FX 0.026 FY -0.02 FZ 0
6 FX 0.03 FY 0.02 FZ 0
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
5 FX -0.013 FY 11.803 FZ 2.19
6 FX 0.013 FY -11.803 FZ 2.19
*****************************************
*****Load combination for bearing capacity check
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 8 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 9 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 4 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 5 0.75 2 0.75 3 0.75
```

```
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
1 1.0 10 0.525 2 0.75 3 0.75
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
1 1.0 11 0.525 2 0.75 3 0.75
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
1 0.6 4 1.0
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
1 0.6 5 1.0
LOAD COMB 314 0.6 DL + 1.0 WL(+X)
1 0.6 6 1.0
LOAD COMB 315 0.6 DL + 1.0 WL(-X)
1 0.6 7 1.0
LOAD COMB 316 0.6 DL + 0.7 EL(+X)
1 0.6 10 0.7
LOAD COMB 317 0.6 DL + 0.7 EL(-X)
1 0.6 10 -0.7
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
1 0.6 11 0.7
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
1 0.6 11 -0.7
****************************************
*****Load combination for foundation design
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 8 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 9 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 4 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 5 1.6 2 0.5 3 0.5
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
1 1.2 10 1.0 2 0.5 3 0.5
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
1 1.2 11 1.0 2 0.5 3 0.5
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
1 0.9 4 1.6
```

```
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
1 0.9 5 1.6
LOAD COMB 414 0.9 DL + 1.6 WL(+X)
1 0.9 6 1.6
LOAD COMB 415 0.9 DL + 1.6 WL(-X)
1 0.9 7 1.6
LOAD COMB 416 0.9 DL + 1.0 EL(+X)
1 0.9 10 1.0
LOAD COMB 417 0.9 DL + 1.0 EL(-X)
1 0.9 10 -1.0
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
1 0.9 11 1.0
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
1 0.9 11 -1.0
*****************************************
PERFORM ANALYSIS
LOAD LIST 301 TO 319
PRINT SUPPORT REACTION ALL
*****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 ALL
FYMAIN 420000 ALL
TRACK 1 ALL
CLB 0.05 ALL
CLT 0.05 ALL
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 16 ALL
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  7, 2020                *
*           Time=    13:11: 6                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3110-CNV-002\Staad\3110-CNV-002_TR4.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 06-APR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART TRESTLE TR4 FOUNDATION MKD. F5
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3110-CNV-002
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 -3 0; 2 0 -3 6; 3 0 1.49012E-008 0; 4 0 1.49012E-008 6; 5 0 0.35 0
   18. 6 0 0.35 6
   19. MEMBER INCIDENCES
   20. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
   21. DEFINE MATERIAL START
   22. ISOTROPIC CONCRETE
   23. E 2.17185E+007
   24. POISSON 0.17
   25. DENSITY 23.5616
   26. ALPHA 1E-005
   27. DAMP 0.05
   28. TYPE CONCRETE
   29. STRENGTH FCU 27579
   30. END DEFINE MATERIAL
   31. MEMBER PROPERTY AMERICAN
   32. 1 TO 4 PRIS YD 0.7 ZD 0.5
   33. 5 PRIS YD 0.6 ZD 0.3
   34. CONSTANTS
   35. MATERIAL CONCRETE ALL
   36. SUPPORTS
   37. 1 2 FIXED
   38. ****************************************
```

```
39. MEMBER OFFSET
40. 5 START 0 -0.25 0
41. 5 END 0 -0.25 0
42. LOAD 1 DEAD LOAD DL
43. SELFWEIGHT Y -1
44. JOINT LOAD
45. 5 FX -0.024 FY -166.03 FZ -31.146
46. 6 FX 0.014 FY -241.697 FZ 42.346
47. ******************************************
48. LOAD 2 LIVE LOAD LL
49. JOINT LOAD
50. 5 FX -0.008 FY -71.746 FZ -13
51. 6 FX 0.008 FY -72.254 FZ 13.1
52. ******************************************
53. LOAD 3 MATERIAL LOAD LL
54. JOINT LOAD
55. 5 FX -0.007 FY -69.513 FZ -12.603
56. 6 FX 0.007 FY -70.088 FZ 12.703
57. ******************************************
58. LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
59. JOINT LOAD
60. 5 FX 0.017 FY 184.107 FZ 24.697
61. 6 FX -0.017 FY -184.107 FZ 24.694
62. ******************************************
63. LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
64. JOINT LOAD
65. 5 FX -0.017 FY -246.635 FZ -49.301
66. 6 FX 0.017 FY 246.635 FZ -49.298
67. ******************************************
68. LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
69. JOINT LOAD
70. 5 FX 4.613 FY 0 FZ 0
71. 6 FX 4.614 FY 0 FZ 0
72. ******************************************
73. LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
74. JOINT LOAD
75. 5 FX -4.613 FY 0 FZ 0
76. 6 FX -4.614 FY 0 FZ 0
77. ******************************************
78. LOAD 8 TEMP LOAD   TL1
79. JOINT LOAD
80. 5 FX -0.004 FY -0.013 FZ -58.119
81. 6 FX 0.004 FY 0.013 FZ 58.119
82. ******************************************
83. LOAD 9 TEMP LOAD   TL2
84. JOINT LOAD
85. 5 FX 0.004 FY 0.013 FZ 58.119
86. 6 FX -0.004 FY -0.013 FZ -58.119
87. ******************************************
88. LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
89. JOINT LOAD
90. 5 FX 0.026 FY -0.02 FZ 0
91. 6 FX 0.03 FY 0.02 FZ 0
92. ******************************************
93. LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
94. JOINT LOAD
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 95. 5 FX -0.013 FY 11.803 FZ 2.19
 96. 6 FX 0.013 FY -11.803 FZ 2.19
 97. ****************************************
 98. *****LOAD COMBINATION FOR BEARING CAPACITY CHECK
 99. LOAD COMB 301 1.0 DL
100. 1 1.0
101. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
102. 1 1.0 8 1.0 2 1.0 3 1.0
103. LOAD COMB 303 1.0 (DL + TL1) + 1.0  (LL+ML)
104. 1 1.0 9 1.0 2 1.0 3 1.0
105. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
106. 1 1.0 4 0.75 2 0.75 3 0.75
107. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
108. 1 1.0 5 0.75 2 0.75 3 0.75
109. LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
110. 1 1.0 6 0.75 2 0.75 3 0.75
111. LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
112. 1 1.0 7 0.75 2 0.75 3 0.75
113. LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
114. 1 1.0 10 0.525 2 0.75 3 0.75
115. LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
116. 1 1.0 10 -0.525 2 0.75 3 0.75
117. LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
118. 1 1.0 11 0.525 2 0.75 3 0.75
119. LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
120. 1 1.0 11 -0.525 2 0.75 3 0.75
121. LOAD COMB 312 0.6 DL + 1.0 WL(+Z)
122. 1 0.6 4 1.0
123. LOAD COMB 313 0.6 DL + 1.0 WL(-Z)
124. 1 0.6 5 1.0
125. LOAD COMB 314 0.6 DL + 1.0 WL(+X)
126. 1 0.6 6 1.0
127. LOAD COMB 315 0.6 DL + 1.0 WL(-X)
128. 1 0.6 7 1.0
129. LOAD COMB 316 0.6 DL + 0.7 EL(+X)
130. 1 0.6 10 0.7
131. LOAD COMB 317 0.6 DL + 0.7 EL(-X)
132. 1 0.6 10 -0.7
133. LOAD COMB 318 0.6 DL + 0.7 EL(+Z)
134. 1 0.6 11 0.7
135. LOAD COMB 319 0.6 DL + 0.7 EL(-Z)
136. 1 0.6 11 -0.7
137. ****************************************
138. *****LOAD COMBINATION FOR FOUNDATION DESIGN
139. LOAD COMB 401 1.4 DL
140. 1 1.4
141. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
142. 1 1.4 8 1.4 2 1.7 3 1.7
143. LOAD COMB 403 1.4 (DL + TL1) + 1.7  (LL+ML)
144. 1 1.4 9 1.4 2 1.7 3 1.7
145. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
146. 1 1.2 4 1.6 2 0.5 3 0.5
147. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
148. 1 1.2 5 1.6 2 0.5 3 0.5
149. LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
150. 1 1.2 6 1.6 2 0.5 3 0.5
```

```
151. LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
152. 1 1.2 7 1.6 2 0.5 3 0.5
153. LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
154. 1 1.2 10 1.0 2 0.5 3 0.5
155. LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
156. 1 1.2 10 -1.0 2 0.5 3 0.5
157. LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
158. 1 1.2 11 1.0 2 0.5 3 0.5
159. LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
160. 1 1.2 11 -1.0 2 0.5 3 0.5
161. LOAD COMB 412 0.9 DL + 1.6 WL(+Z)
162. 1 0.9 4 1.6
163. LOAD COMB 413 0.9 DL + 1.6 WL(-Z)
164. 1 0.9 5 1.6
165. LOAD COMB 414 0.9 DL + 1.6 WL(+X)
166. 1 0.9 6 1.6
167. LOAD COMB 415 0.9 DL + 1.6 WL(-X)
168. 1 0.9 7 1.6
169. LOAD COMB 416 0.9 DL + 1.0 EL(+X)
170. 1 0.9 10 1.0
171. LOAD COMB 417 0.9 DL + 1.0 EL(-X)
172. 1 0.9 10 -1.0
173. LOAD COMB 418 0.9 DL + 1.0 EL(+Z)
174. 1 0.9 11 1.0
175. LOAD COMB 419 0.9 DL + 1.0 EL(-Z)
176. 1 0.9 11 -1.0
177. ****************************************
178. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**6**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**2**

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 11, TOTAL DEGREES OF FREEDOM = 24
TOTAL LOAD COMBINATION CASES = 38 SO FAR.

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179. LOAD LIST 301 TO 319
180. PRINT SUPPORT REACTION ALL
```

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	301	0.02	204.08	-8.58	-13.13	-0.00	-0.07
	302	0.04	345.31	-27.43	-26.03	-0.01	-0.13
	303	0.03	345.29	-1.43	-8.43	-0.01	-0.11
	304	0.02	164.34	-31.49	-55.51	-0.00	-0.07
	305	0.05	510.24	24.00	61.84	-0.01	-0.15
	306	-3.43	309.99	-12.97	-16.20	-0.01	11.48
	307	3.49	309.99	-12.97	-16.20	-0.01	-11.70
	308	0.02	310.01	-12.97	-16.20	-0.01	-0.06
	309	0.05	309.98	-12.97	-16.20	-0.00	-0.15
	310	0.04	303.32	-14.12	-18.63	-0.01	-0.13
	311	0.03	316.67	-11.82	-13.77	-0.00	-0.09
	312	-0.00	-71.76	-29.85	-60.29	0.00	0.01
	313	0.03	389.45	44.15	96.18	-0.00	-0.10
	314	-4.60	122.45	-5.15	-7.88	-0.00	15.41
	315	4.63	122.45	-5.15	-7.88	-0.00	-15.50
	316	-0.00	122.46	-5.15	-7.88	-0.00	0.02
	317	0.03	122.43	-5.15	-7.88	-0.00	-0.11
	318	0.02	113.55	-6.68	-11.12	-0.00	-0.07
	319	0.01	131.34	-3.62	-4.64	-0.00	-0.02
2	301	-0.01	284.34	-2.62	-10.61	-0.00	0.04
	302	-0.03	426.71	16.03	1.87	-0.01	0.10
	303	-0.02	426.74	-9.97	-15.72	-0.01	0.07
	304	-0.01	536.79	-16.90	-47.16	-0.00	0.04
	305	-0.04	190.88	38.59	70.20	-0.01	0.11
	306	-3.48	391.13	1.62	-7.85	-0.01	11.67
	307	3.44	391.13	1.62	-7.85	-0.01	-11.52
	308	-0.04	391.12	1.62	-7.85	-0.01	0.13
	309	-0.01	391.14	1.62	-7.85	-0.00	0.02
	310	-0.03	397.80	0.47	-10.28	-0.01	0.10
	311	-0.02	384.46	2.77	-5.42	-0.00	0.05
	312	0.01	364.82	-26.27	-58.78	0.00	-0.03
	313	-0.02	-96.39	47.73	97.70	-0.00	0.08
	314	-4.62	170.61	-1.57	-6.36	-0.00	15.48
	315	4.61	170.61	-1.57	-6.36	-0.00	-15.43
	316	-0.03	170.59	-1.57	-6.36	-0.00	0.09
	317	0.01	170.62	-1.57	-6.36	-0.00	-0.05
	318	-0.02	179.50	-3.10	-9.60	-0.00	0.05
	319	0.00	161.71	-0.04	-3.13	-0.00	-0.00

********** END OF LATEST ANALYSIS RESULT **********

```
181. ****************************************
182. LOAD LIST 401 TO 419
183. START CONCRETE DESIGN
184. CODE ACI 2008
185. FC 35000 ALL
186. FYMAIN 420000 ALL
187. TRACK 1 ALL
188. CLB 0.05 ALL
189. CLT 0.05 ALL
190. CLS 0.05 ALL
191. MAXMAIN 25 ALL
192. MINMAIN 16 ALL
193. MINSEC 8 ALL
194. REINF 0 MEMB 1 TO 4
195. DESIGN COLUMN 1 TO 4
```

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COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	386.27	0.01	0.00001

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	386.27	27.19	0.02346

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	357.60	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	357.60	0.00	0.00000

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	559.15	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	559.15	5.94	0.00354

==========

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	520.58	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	520.58	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

196. DESIGN BEAM 5

BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 16MM	0.	6000.	YES	YES

CRITICAL POS MOMENT= 77.95 KN-MET AT 6000.MM, LOAD 413
REQD STEEL= 523.MM2, RHO=0.0033, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

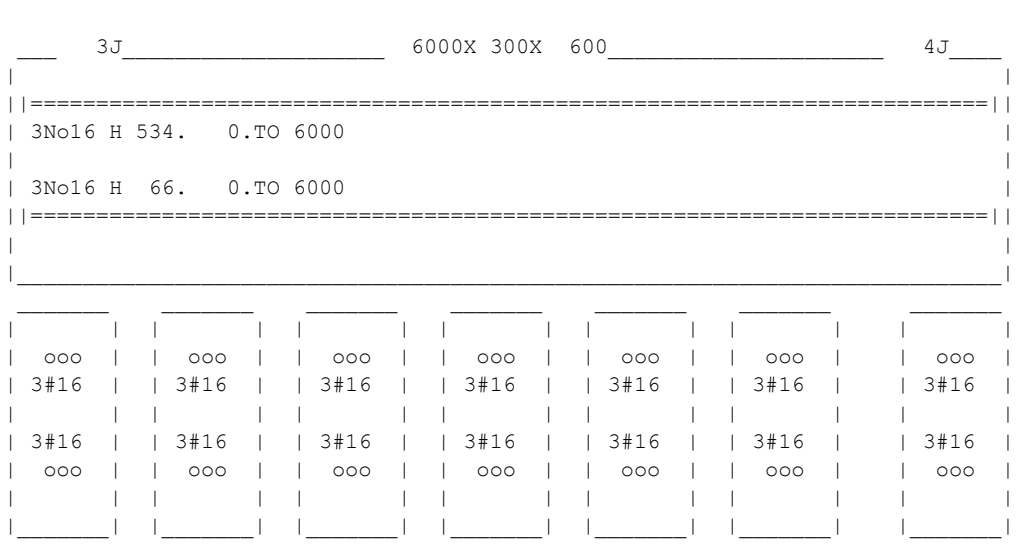
2	534.	3 - 16MM	0.	6000.	YES	YES

CRITICAL NEG MOMENT= 109.29 KN-MET AT 0.MM, LOAD 405
REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 41./ 84. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 90761.5 cm^4

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 11.43 KNS Vc= 143.29 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 6.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 17.86 KNS Vc= 143.29 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 6.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

197. END CONCRETE DESIGN
198. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 7,2020 TIME= 13:11: 6 ****

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*   For technical assistance on STAAD.Pro, please visit     *
*   http://selectservices.bentley.com/en-US/                *
*                                                           *
*   Details about additional assistance from                *
*   Bentley and Partners can be found at program menu       *
*   Help->Technical Support                                 *
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